

Project Data Page:

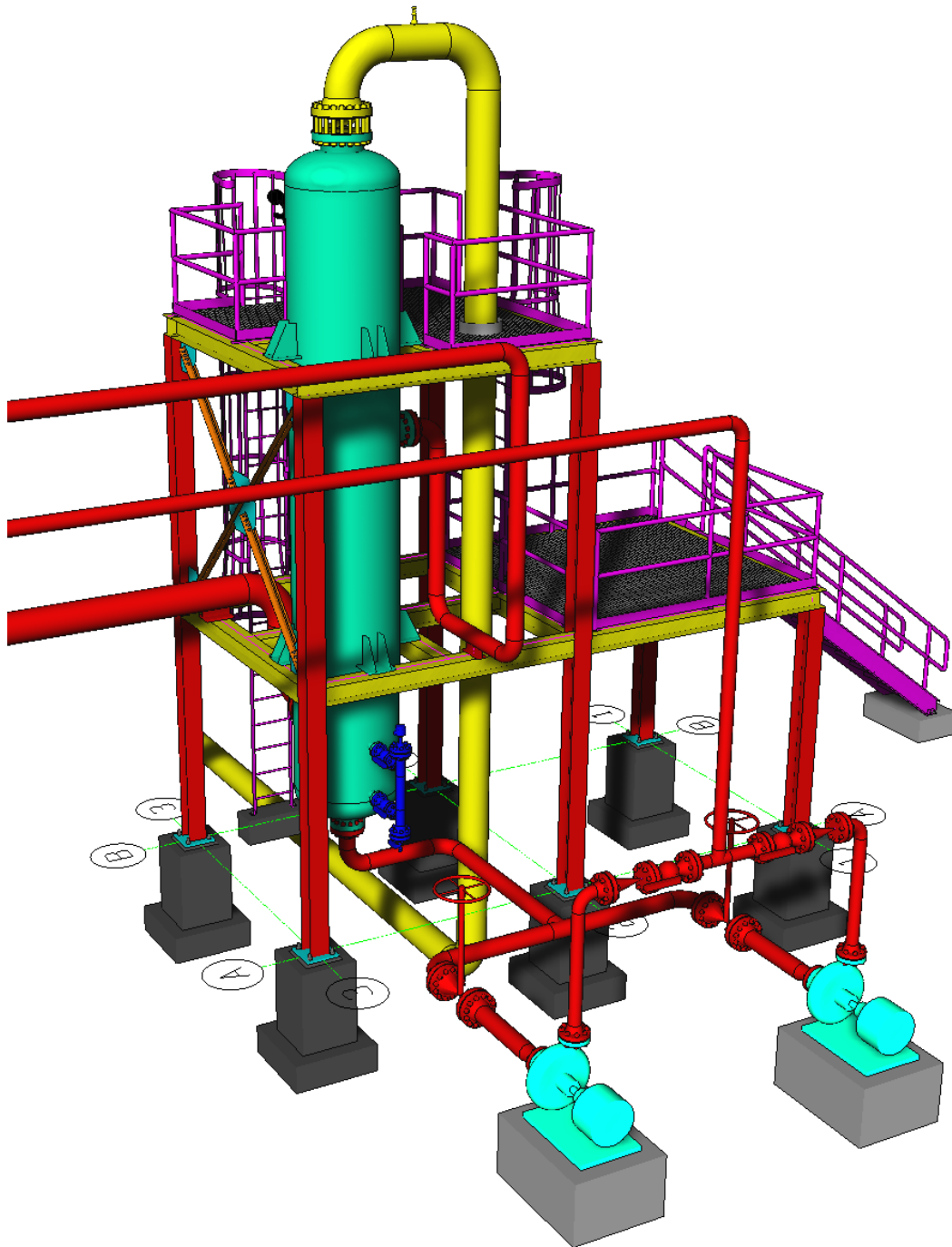




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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 2

ASME Code Version : 2025

Analysis Performed by : SL User (Modify in Config->Save Options)

Job File : ...\\163-D10pe - OverfillRev 01.pvdb

Date of Analysis : Feb 11, 2026 3:20pm

PV Elite 28, January 2026



Vessel Design Summary:

ASME Code, Section VIII Division 2, 2025

Diameter Spec : 1700.000 x 1706.000 mm ID	
Vessel Design Length, Tangent to Tangent	1820.00 cm
Distance of Bottom Tangent above Grade	630.00 cm
Distance of Base above Grade	30.00 cm
Specified Datum Line Distance	600.00 cm
Internal Design Temperature	130 °C
Internal Design Pressure	41.807 kgf/cm ²
External Design Temperature	130 °C
External Design Pressure	1.055 kgf/cm ²
Maximum Allowable Working Pressure	43 kgf/cm ²
External Max. Allowable Working Pressure	6 kgf/cm ²
Shop Test Pressure	54.011 kgf/cm ²
Required Minimum Design Metal Temperature	-12.0 °C
Warmest Computed Minimum Design Metal Temperature	-12.0 °C
Wind Design Code	ASCE/SEI 7-16
Earthquake Design Code	ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	+IT
Head	SA-516 70	...	...	K02700	+IT
Skirt	SA-516 70	...	...	K02700	+IT
Nozzle	SA-105	...	...	K03504	+IT
Nozzle Flg	SA-105	...	...	K03504	+IT
Basering	SA-516 70	...	...	K02700	
Base Bolting	SA-193 B7	...	<= 64	G41400	+IT

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (kgf/cm² & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Skirt-3	...	...	...	1.6000	N/A	No
Skirt 2	...	...	...	1.6000	N/A	No
Skirt 1	...	...	...	1.6000	N/A	No
Bottom head	43.755	1.05	45.000	6.0000	No	No
Shell-1	43.706	1.05	43.000	6.0000	N/A	No
Shell-2	43.400	1.05	44.000	6.0000	N/A	No
Shell-3	43.093	1.05	44.000	6.0000	N/A	No
Shell-4	42.786	1.05	44.000	6.0000	N/A	No
Shell-5	42.397	1.05	43.000	1.0000	N/A	No
Shell-7	42.172	1.05	43.000	1.0000	N/A	No
Top head	41.855	1.05	45.000	1.0000	No	No

Liquid Level: 1905.00 cm Dens.: 1022.530 kg/m³ Sp. Gr.: 1.023

Element Types and Properties:

Element	"To" Elev	Element Length	Nominal Thickness	Finished Thickness	Reqd Thk Internal	Reqd Thk External	Long Eff	Circ Eff
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Type	cm	cm	mm	mm	mm	mm		
Skirt	-400.0	200.0	14.0	14.0	...	...	0.70	0.70
Skirt	-100.0	300.0	14.0	14.0	...	...	0.70	0.70
Skirt	0.0	100.0	14.0	14.0	...	...	0.70	0.50
Ellipse	5.0	5.0	38.0	32.0	30.1	10.1	1.00	1.00
Cylinder	305.0	300.0	34.0	34.0	33.0	20.3	1.00	1.00
Cylinder	605.0	300.0	34.0	34.0	32.8	20.3	1.00	1.00
Cylinder	905.0	300.0	34.0	34.0	32.6	20.3	1.00	1.00
Cylinder	1285.5	380.5	34.0	34.0	32.4	20.3	1.00	1.00
Cylinder	1505.0	219.5	28.0	28.0	27.1	15.2	1.00	1.00
Cylinder	1815.0	310.0	28.0	28.0	27.0	15.2	1.00	1.00
Ellipse	1820.0	5.0	32.0	26.0	24.0	5.1	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
D_150	WRC-107/537	0.956	n/a	Passed
B_150	WRC-297	0.989	Shell	Passed
A_150	WRC-297	0.995	Shell	Passed
E_40	WRC-297	0.993	Shell	Passed
C_100	WRC-107/537	0.951	n/a	Passed

Wind/Earthquake Shear, Bending:

From	To	Distance to Support cm	Cumulative Wind Shear kgf	Earthquake Shear kgf	Wind Bending kgf-m.	Earthquake Bending kgf-m.
10	20	100	5352.41	8327.95	73318.8	147929
20	30	350	5101	8322.94	62865.4	131278
30	40	550	4578.03	8295.24	48346.8	106351
40	50	602.5	4446.59	8282.42	43834.5	98062.1
50	60	755	4182.92	8166.76	43618.8	97650.8
60	70	1055	3768.4	7760.55	31691.8	73759.8
70	80	1355	3107.96	7063.38	21377.2	51523.9
80	90	1695.25	2426.33	6128.03	13075.8	31736.8
90	100	1995.25	1641.33	3968.67	5337.04	12527.8
100	110	2260	1069.7	2636.94	2361.68	5278.11
110	120	2417.5	391.089	661.885	97.458	164.94

Abs Max of the all of the Stress Ratio's : 0.8769

Basing Data : Continuous Top Ring W/Gussets

Thickness of Basing	50.0000	mm
Inside Diameter of Basing	1610.0000	mm
Outside Diameter of Basing	2010.0000	mm
Nominal Diameter of Bolts	36.0000	mm
Diameter of Bolt Circle	1882.0000	mm
Number of Bolts	20	
Thickness of Gusset Plates	16.0000	mm
Average Width of Gusset Plates	118.0000	mm
Height of Gussets	168.0000	mm
Distance between Gussets	76.0000	mm
Thickness of Top Plate or Ring	32.0000	mm
Radial Width of the Top Plate	128.0000	mm

Basing Sketch for Cylindrical Skirts:

Basing Cross Section View

Loads for Foundation/Support Design:



Total Wind Shear on Support	6145. kgf
Total Earthquake Shear on Support	9121. kgf
Wind Moment on Support	100047. kgf-m.
Earthquake Moment on Support	174658. kgf-m.
Maximum Initial Bolt Load	12799. kgf

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	47103.0 kg
Shop Test - Fabricated + water (full)	89676.9 kg
Shipping - Fab. + rem. intls.+ shipping app.	69686.0 kg
Erected - Fab. + rem. intls.+ insul. (etc)	69686.0 kg
Empty - Fab. + intls. + details + wghts.	69686.1 kg
Operating - Empty + operating liquid (no ca)	117393.7 kg
Field Test - Empty weight + test medium (full)	108007.1 kg

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Nozzle Calculation Summary (kgf/cm² & mm):

Description	MAWP	Ext	MAPNC	Tbl 4.5.2	Weld	Pass/Fail
D_150	48	OK	...	OK	OK	Passed
MH4_600	45	OK	...	...	OK	Passed
B_150	45	OK	...	OK	OK	Passed
L1_50	45	OK	...	OK	OK	Passed
L2_50	45	OK	...	OK	OK	Passed
L3_50	45	OK	...	OK	OK	Passed
L4_50	45	OK	...	OK	OK	Passed
L5_50	45	OK	...	OK	OK	Passed
L6_50	45	OK	...	OK	OK	Passed
DP1_50	45	OK	...	OK	OK	Passed
UC2_600	45	OK	...	...	OK	Passed
MH3_600	45	OK	...	...	OK	Passed
DP2_50	45	OK	...	OK	OK	Passed
UC1_600	45	OK	...	...	OK	Passed
A_150	44	OK	...	OK	OK	Passed
MH2_600	44	OK	...	...	OK	Passed
E_40	44	OK	...	OK	OK	Passed
MH1_600	44	OK	...	...	OK	Passed
C_100	45	OK	...	OK	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 44 Nozzle : MH1600
 Minimum MAWP Shells/Flanges : 43 Element : Shell-5
 Minimum MAPnc Shells/Flanges : 45 Element : Shell-7

Computed Vessel M.A.W.P. : 43 kgf/cm²

VIII-2 Nozzle to Gross Structural Discontinuity Check:

No Nozzles were too close to any gross structural discontinuities per 4.5.13.1

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm	Layout Angle deg	Inside Radius mm
40	D_150	0.000	0.000	77.647
50	MH4_600	849.999	270.000	290.000
50	B_150	2849.999	0.000	77.647
50	L1_50	2550.000	180.000	31.400
50	L2_50	249.999	180.000	31.400
50	L3_50	2049.999	180.000	31.400
50	L4_50	1049.999	180.000	31.400
50	L5_50	2049.999	0.000	31.400
50	L6_50	1049.999	0.000	31.400
50	DP1_50	2849.999	180.000	31.400
60	UC2_600	4049.999	0.000	290.000
70	MH3_600	8349.999	180.000	290.000
70	DP2_50	8549.999	0.000	31.400
80	UC1_600	10049.999	0.000	290.000
90	A_150	14974.999	0.000	77.647
90	MH2_600	14749.999	90.000	290.000
100	E_40	15950.000	0.000	26.000
100	MH1_600	16600.002	180.000	290.000
110	C_100	0.000	0.000	49.663

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.



Ic - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia mm	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
E_40	1 in	LWN	WNF	60.0	10.000	...	...	330	900
L1_50	2 in	LWN	WNF	74.8	12.000	...	...	287	300
L2_50	2 in	LWN	WNF	74.8	12.000	...	...	287	300
L3_50	2 in	LWN	WNF	74.8	12.000	...	...	287	300
L4_50	2 in	LWN	WNF	74.8	12.000	...	...	287	300
L5_50	2 in	LWN	WNF	74.8	12.000	...	...	287	300
L6_50	2 in	LWN	WNF	74.8	12.000	...	...	287	300
DP1_50	2 in	LWN	WNF	74.8	12.000	...	...	287	300
DP2_50	50 mm	Actual	WNF	74.8	12.000	...	...	287	300
C_100	100 mm	160	WNF	114.3	13.487	...	...	328	600
D_150	150 mm	120	WNF	168.3	14.275	...	...	336	600
B_150	150 mm	120	WNF	168.3	14.275	...	...	342	600
A_150	150 mm	120	WNF	168.3	14.275	...	...	336	600
MH4_600	600 mm	Actual	WNF	600.0	16.000	...	...	615	300
UC2_600	600 mm	Actual	WNF	600.0	16.000	...	...	615	300
MH3_600	600 mm	Actual	WNF	600.0	16.000	...	...	615	300
UC1_600	600 mm	Actual	WNF	600.0	16.000	...	...	615	300
MH2_600	600 mm	Actual	WNF	600.0	16.000	...	...	609	300
MH1_600	600 mm	Actual	WNF	600.0	16.000	...	...	609	300

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
E_40	SA-105	28.000	10.000	...	...	...
L1_50	SA-105	30.000	10.000	...	...	...
L2_50	SA-105	30.000	10.000	...	...	...
L3_50	SA-105	30.000	10.000	...	...	...
L4_50	SA-105	30.000	10.000	...	...	...
L5_50	SA-105	30.000	10.000	...	...	...
L6_50	SA-105	30.000	10.000	...	...	...
DP1_50	SA-105	30.000	10.000	...	...	...
DP2_50	SA-105	30.000	10.000	...	...	...
C_100	SA-105	26.000	10.000	...	...	...
D_150	SA-105	30.000	10.000	...	...	...
B_150	SA-105	30.000	10.000	...	...	...
A_150	SA-105	28.000	10.000	...	...	...
MH4_600	SA-105	32.000	10.000	...	...	...
UC2_600	SA-105	32.000	10.000	...	...	...
MH3_600	SA-105	32.000	10.000	...	...	...
UC1_600	SA-105	32.000	10.000	...	...	...
MH2_600	SA-105	28.000	10.000	...	...	...
MH1_600	SA-105	28.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum cm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
-------------	-----------------------------	------------------	-----------------	----------------	------------------------



E_40	1595.000	0.0	300.00	0.00	Shell-7
L1_50	255.000	180.0	250.00	0.00	Shell-1
L2_50	25.000	180.0	250.00	0.00	Shell-1
L3_50	205.000	180.0	250.00	0.00	Shell-1
L4_50	105.000	180.0	250.00	0.00	Shell-1
L5_50	205.000	0.0	250.00	0.00	Shell-1
L6_50	105.000	0.0	250.00	0.00	Shell-1
DP1_50	285.000	180.0	250.00	0.00	Shell-1
DP2_50	855.000	0.0	250.00	0.00	Shell-3
C_100	...	0.0	300.00	0.00	Top head
D_150	...	0.0	300.00	0.00	Bottom head
B_150	285.000	0.0	300.00	0.00	Shell-1
A_150	1497.500	0.0	300.00	0.00	Shell-5
MH4_600	85.000	270.0	500.00	0.00	Shell-1
UC2_600	405.000	0.0	500.00	0.00	Shell-2
MH3_600	835.000	180.0	500.00	0.00	Shell-3
UC1_600	1005.000	0.0	500.00	0.00	Shell-4
MH2_600	1475.000	90.0	500.00	0.00	Shell-5
MH1_600	1660.000	180.0	500.00	0.00	Shell-7



Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	ELLIPTICAL HEAD: 2.0 X 1, 38.0mm THK X 1700.0mm ID X 5.0cm	SA-516 70
1	ELLIPTICAL HEAD: 2.0 X 1, 32.0mm THK X 1706.0mm ID X 5.0cm	SA-516 70
7	PLATFORM: 1000mm X 1500mm WIDE	
5	LADDER: 4000.0mm	...
1	INSULATION: 500mm X 80mm THK	
2	LADDER: 3000.0mm	...
2	INSULATION: 475mm X 80mm THK	
3	INSULATION: 3000mm X 80mm THK	
1	PACKING: 3000.000mm	...
1	PACKING: 1275.000mm	...
1	PACKING: 3275.000mm	...
1	INSULATION: 3804mm X 80mm THK	
1	PACKING: 805.000mm	...
1	INSULATION: 2195mm X 80mm THK	
1	CLADDING: 2195.000mm X 3.000mm THK	...
1	INSULATION: 3100mm X 80mm THK	
1	CLADDING: 3100.000mm X 3.000mm THK	...
3	TRAYS # 3 SPACING 12. mm	
1	PLATFORM: 1000mm X 3764mm WIDE	
1	LADDER: 2500.0mm	...
1	CLADDING: 476.500mm X 3.000mm THK	...
6	CLASS 300 GR 1.1, 600.0mm BLIND FLANGE(S)	SA-105
1	BASERING: 2010.0mm OD X 1610.0 ID X 50.0mm	SA-516 70
20	BASERING BOLTS: 36.0mm DIA	SA-193 B7
20	BASERING BOLT NUTS: 36.0mm DIA	...
1	BASERING TOP RING: 2010.0mm OD X 1754.0 ID X 32.0mm	...
40	BASERING GUSSET PLATES: 118.0mm X 118.0mm X 168.0mm X 16.0mm	...
1	CYLINDRICAL SKIRT SEGMENT 14.0mm THK X 3094.6mm X 200.0cm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 14.0mm THK X 2327.7mm X 200.0cm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 14.0mm THK X 3094.6mm X 300.0cm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 14.0mm THK X 2327.7mm X 300.0cm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 14.0mm THK X 3094.6mm X 100.0cm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 14.0mm THK X 2327.7mm X 100.0cm	SA-516 70



2	CYLINDRICAL SEGMENT 34.0mm THK X 2121.4mm X 300.0cm	SA-516 70
1	CYLINDRICAL SEGMENT 34.0mm THK X 1097.8mm X 300.0cm	SA-516 70
2	CYLINDRICAL SEGMENT 34.0mm THK X 3041.2mm X 300.0cm	SA-516 70
2	CYLINDRICAL SEGMENT 34.0mm THK X 2299.4mm X 300.0cm	SA-516 70
1	CYLINDRICAL SEGMENT 34.0mm THK X 3041.2mm X 380.5cm	SA-516 70
1	CYLINDRICAL SEGMENT 34.0mm THK X 2299.4mm X 380.5cm	SA-516 70
1	CYLINDRICAL SEGMENT 28.0mm THK X 2307.5mm X 219.5cm	SA-516 70
1	CYLINDRICAL SEGMENT 28.0mm THK X 3051.9mm X 219.5cm	SA-516 70
1	CYLINDRICAL SEGMENT 28.0mm THK X 2307.5mm X 310.0cm	SA-516 70
1	CYLINDRICAL SEGMENT 28.0mm THK X 3051.9mm X 310.0cm	SA-516 70
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	PWHT	Thickness ratio	Gov Thk mm	E*
Notes							
Bottom head	[10]	!	-12	No			
Bottom head	[7]	!	-12	No			
Shell-1	[8]	!	-12	Yes			
Shell-2	[8]	!	-12	Yes			
Shell-3	[8]	!	-12	Yes			
Shell-4	[8]	!	-12	Yes			
Shell-5	[8]	!	-12	No			
Shell-7	[8]	!	-12	No			
Top head	[10]	!	-12	No			
Top head	[7]	!	-12	No			
D150	[1]	!	-12	No			
Nozzle Flg	[5]	!	-18	-61	No		
MH4600	[1]	!	-12	Yes			
Nozzle Flg	[5]	!	-18	-25	Yes		
B150	[1]	!	-12	Yes			
Nozzle Flg	[5]	!	-18	-61	Yes		
L150	[1]	!	-12	Yes			
Nozzle Flg	[5]	!	-17	-17	Yes		
L250	[1]	!	-12	Yes			
Nozzle Flg	[5]	!	-17	-17	Yes		
L350	[1]	!	-12	Yes			
Nozzle Flg	[5]	!	-17	-17	Yes		
L450	[1]	!	-12	Yes			
Nozzle Flg	[5]	!	-17	-17	Yes		
L550	[1]	!	-12	Yes			
Nozzle Flg	[5]	!	-17	-17	Yes		
L650	[1]	!	-12	Yes			
Nozzle Flg	[5]	!	-17	-17	Yes		
DP150	[1]	!	-12	Yes			
Nozzle Flg	[5]	!	-17	-17	Yes		
UC2600	[1]	!	-12	Yes			
Nozzle Flg	[5]	!	-18	-26	Yes		
MH3600	[1]	!	-12	Yes			
Nozzle Flg	[5]	!	-18	-26	Yes		
DP250	[1]	!	-12	Yes			
Nozzle Flg	[5]	!	-18	-26	Yes		
UC1600	[1]	!	-12	Yes			
Nozzle Flg	[5]	!	-18	-26	Yes		
A150	[1]	!	-12	No			
Nozzle Flg	[5]	!	-18	-64	No		
MH2600	[1]	!	-12	No			
Nozzle Flg	[5]	!	-18	-27	No		
E40	[1]	!	-12	No			
Nozzle Flg	[5]	!	-17	-17	No		
MH1600	[1]	!	-12	No			
Nozzle Flg	[5]	!	-18	-27	No		
C100	[1]	!	-12	No			
Nozzle Flg	[5]	!	-18	-65	No		

Warmest MDMT: -12 -17

Required Minimum Design Metal Temperature -12.0 °C
 Warmest Computed Minimum Design Metal Temperature -12.0 °C

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.1.
- [5] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.3.
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.



- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per 3.11.2.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

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Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.

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Units used in this Analysis (Kgfc/m2):

Name	System Unit	Constant	User Unit
Length	Feet	30.4800	cm
Force	Pounds	0.4536	kgf
Mass	Pounds	0.4536	kg
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	kgf-m.
Stress	lbs./sq.in.	0.0703	kgf/cm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0703	kgf/cm ²
Elast. Modulus	lbs./sq.in.	0.0703	kgf/cm ²
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	16.0183	kg/m ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kgf/cm ²
Inertia	in.4	41.6231	cm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8824	kgf/m ²
Elevation	Feet	0.3048	m
Volume	in.3	16.3871	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	41.807	kgf/cm ²
Design Internal Temperature	130.0	°C
Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Minimum Design Metal Temperature	-12.0	°C
Type of Construction	Welded	
Special Service	None	
Use Higher Longitudinal Stresses (Flag)	N	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	Hydrostatic
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	Not Applied
Pressure Test Position	Vertical
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+1.1WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+1.1WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+1.1WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+1.1WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW



Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	30 cm
Percent Wind for Hydrotect	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 cm
Distance Upwind of Crest Lh	0 cm
Distance from Crest to the Vessel x	0 cm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 cm

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	2.000
Table Value Fa	1.300
Table Value Fv	1.500
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	2.000
Site Class	C
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	2.500
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

M.A.W.P + Static Head to Nozzle	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Using Metric Material Databases, ASME II D	Yes
Using ASME VIII-1 Appendix 46	No
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/11
Element From Node	10
Element To Node	20
Element Type	Skirt Support
Description	Skirt-3
Distance "FROM" to "TO"	200 cm
Skirt Outside Diameter	1754 mm
Diameter of Skirt at Base	1754 mm
Skirt Thickness	14 mm
Internal Corrosion Allowance	0.8 mm



Nominal Thickness	14	mm
External Corrosion Allowance	0.8	mm
Design Temperature Internal Pressure	130	°C
Design Temperature External Pressure	130	°C
Effective Diameter Multiplier	1.3	
Material Name	SA-516 70	+IT
Allowable Stress, Ambient	1407.1	kgf/cm ²
Allowable Stress, Operating	1407.1	kgf/cm ²
Allowable Stress, Hydrotest	2386	kgf/cm ²
Material Density	0.00775	kg/cm ³
P Number Thickness	0	mm
Yield Stress, Operating	2390.1	kgf/cm ²
MDMT Curve	Impact	Tested
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.7	
Weld is pre-Heated	No	
Element From Node	10	
Detail Type	Weight	
Detail ID	Tailing	lug
Dist. from "FROM" Node / Offset dist	0	cm
Miscellaneous Weight	499.99	kgf
Offset from Element Centerline	0	mm

Element#	2/11	
Element From Node	20	
Element To Node	30	
Element Type	Skirt	Support
Description	Skirt	2
Distance "FROM" to "TO"	300	cm
Skirt Outside Diameter	1754	mm
Diameter of Skirt at Base	1754	mm
Skirt Thickness	14	mm
Internal Corrosion Allowance	0.8	mm
Nominal Thickness	14	mm
External Corrosion Allowance	0.8	mm
Design Temperature Internal Pressure	130	°C
Design Temperature External Pressure	130	°C
Effective Diameter Multiplier	1.3	
Material Name	SA-516 70	+IT
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.7	
Weld is pre-Heated	No	
Element From Node	20	
Detail Type	Platform	
Detail ID	PL-8	
Dist. from "FROM" Node / Offset dist	200	cm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	90.0	
Platform Wind Area	15000	cm ²
Platform Weight	842.62	kgf
Platform Railing Weight	0	kgf/cm
Platform Grating Weight	0.017	kgf/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	85	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	cm
Ladder End Elevation	400	cm
Unit Weight of Ladder	0.3773	kgf/cm
Platform Length (top head platform)	0	mm



Element#	3/11
Element From Node	30
Element To Node	40
Element Type	Skirt Support
Description	Skirt 1
Distance "FROM" to "TO"	100 cm
Skirt Outside Diameter	1754 mm
Diameter of Skirt at Base	1754 mm
Skirt Thickness	14 mm
Internal Corrosion Allowance	0.8 mm
Nominal Thickness	14 mm
External Corrosion Allowance	0.8 mm
Design Temperature Internal Pressure	130 °C
Design Temperature External Pressure	130 °C
Effective Diameter Multiplier	1.3
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	0.7
Efficiency, Head-to-Skirt or Circ. Seam	0.5
Weld is pre-Heated	No
Element From Node	30
Detail Type	Insulation
Detail ID	Ins
Dist. from "FROM" Node / Offset dist	50 cm
Height/Length of Insulation	50 cm
Thickness of Insulation	80 mm
Density	0.000352 kg/cm³

Element#	4/11
Element From Node	40
Element To Node	50
Element Type	Elliptical
Description	Bottom head
Distance "FROM" to "TO"	5 cm
Inside Diameter	1700 mm
Element Thickness	32 mm
Internal Corrosion Allowance	6 mm
Nominal Thickness	38 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	41.807 kgf/cm²
Design Temperature Internal Pressure	130 °C
Design External Pressure	1.0546 kgf/cm²
Design Temperature External Pressure	130 °C
Effective Diameter Multiplier	1.3
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No
Element From Node	40
Detail Type	Platform
Detail ID	PL-1
Dist. from "FROM" Node / Offset dist	0 cm
Platform Start Angle (degrees)	0.0
Platform End Angle (degrees)	360.0
Platform Wind Area	30000 cm²
Platform Weight	2913.6 kgf
Platform Railing Weight	0 kgf/cm
Platform Grating Weight	0.017 kgf/cm²
Platform Width	1500 mm
Platform Height	1000 mm
Platform Clearance or End Offset	115 mm
Platform Force Coefficient	1.0
Ladder Layout Angle	0.0
Ladder Start Elevation	0 cm
Ladder End Elevation	300 cm



Unit Weight of Ladder 0.3773 kgf/cm
 Platform Length (top head platform) 0 mm

Element From Node 40
 Detail Type Liquid
 Detail ID Liquid: 40
 Dist. from "FROM" Node / Offset dist -42.5 cm
 Height/Length of Liquid 47.5 cm
 Liquid Density 1022.5 kg/m³

Element From Node 40
 Detail Type Insulation
 Detail ID Ins: 10
 Dist. from "FROM" Node / Offset dist -42.5 cm
 Height/Length of Insulation 47.5 cm
 Thickness of Insulation 80 mm
 Density 0.000352 kg/cm³

Element From Node 40
 Detail Type Nozzle
 Detail ID D150
 Dist. from "FROM" Node / Offset dist 0 mm
 Nozzle Diameter 150 mm
 Nozzle Schedule 120
 Nozzle Class 600
 Layout Angle 0.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 236.34 kgf
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-105 +IT

Element From Node 40
 Detail Type Weight
 Detail ID D FY
 Dist. from "FROM" Node / Offset dist 0 cm
 Miscellaneous Weight 188 kgf
 Offset from Element Centerline 2118 mm

Element From Node 40
 Detail Type Weight
 Detail ID D anti-swirl ba
 Dist. from "FROM" Node / Offset dist -40 cm
 Miscellaneous Weight 9.9998 kgf
 Offset from Element Centerline 0 mm

Element# 5/11
 Element From Node 50
 Element To Node 60
 Element Type Cylinder
 Description Shell-1
 Distance "FROM" to "TO" 300 cm
 Inside Diameter 1700 mm
 Element Thickness 34 mm
 Internal Corrosion Allowance 6 mm
 Nominal Thickness 34 mm
 External Corrosion Allowance 0 mm
 Design Internal Pressure 41.807 kgf/cm²
 Design Temperature Internal Pressure 130 °C
 Design External Pressure 1.0546 kgf/cm²
 Design Temperature External Pressure 130 °C
 Effective Diameter Multiplier 1.3
 Material Name SA-516 70 +IT
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 50
 Detail Type Liquid



Detail ID	Liquid: 50	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	300	cm
Liquid Density	1022.5	kg/m ³
Element From Node	50	
Detail Type	Insulation	
Detail ID	Ins: 20	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	300	cm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm ³
Element From Node	50	
Detail Type	Nozzle	
Detail ID	MH4600	
Dist. from "FROM" Node / Offset dist	80	cm
Nozzle Diameter	600	mm
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	270.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	
Element From Node	50	
Detail Type	Nozzle	
Detail ID	B150	
Dist. from "FROM" Node / Offset dist	280	cm
Nozzle Diameter	150	mm
Nozzle Schedule	120	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	236.34	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	
Element From Node	50	
Detail Type	Nozzle	
Detail ID	L150	
Dist. from "FROM" Node / Offset dist	250	cm
Nozzle Diameter	50.8	mm
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	
Element From Node	50	
Detail Type	Nozzle	
Detail ID	L250	
Dist. from "FROM" Node / Offset dist	20	cm
Nozzle Diameter	50.8	mm
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	
Element From Node	50	
Detail Type	Nozzle	
Detail ID	L350	
Dist. from "FROM" Node / Offset dist	200	cm
Nozzle Diameter	50.8	mm



Nozzle Schedule	None
Nozzle Class	300
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105 +IT

Element From Node	50
Detail Type	Nozzle
Detail ID	L450
Dist. from "FROM" Node / Offset dist	100 cm
Nozzle Diameter	50.8 mm
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105 +IT

Element From Node	50
Detail Type	Nozzle
Detail ID	L550
Dist. from "FROM" Node / Offset dist	200 cm
Nozzle Diameter	50.8 mm
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105 +IT

Element From Node	50
Detail Type	Nozzle
Detail ID	L650
Dist. from "FROM" Node / Offset dist	100 cm
Nozzle Diameter	50.8 mm
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105 +IT

Element From Node	50
Detail Type	Nozzle
Detail ID	DP150
Dist. from "FROM" Node / Offset dist	280 cm
Nozzle Diameter	50.8 mm
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105 +IT

Element From Node	50
Detail Type	Weight
Detail ID	MH4 Davit
Dist. from "FROM" Node / Offset dist	80 cm
Miscellaneous Weight	99.998 kgf
Offset from Element Centerline	0 mm

Element From Node	50
Detail Type	Weight
Detail ID	B FY



Dist. from "FROM" Node / Offset dist	280	cm
Miscellaneous Weight	801.99	kgf
Offset from Element Centerline	2118	mm

Element From Node	50	
Detail Type	Weight	
Detail ID	Bed2 sup	
Dist. from "FROM" Node / Offset dist	295	cm
Miscellaneous Weight	9.9998	kgf
Offset from Element Centerline	0	mm

Element From Node	50	
Detail Type	Weight	
Detail ID	Bed 2 sup grid	
Dist. from "FROM" Node / Offset dist	295	cm
Miscellaneous Weight	180	kgf
Offset from Element Centerline	0	mm

Element#	6/11	
Element From Node	60	
Element To Node	70	
Element Type	Cylinder	
Description	Shell-2	
Distance "FROM" to "TO"	300	cm
Inside Diameter	1700	mm
Element Thickness	34	mm
Internal Corrosion Allowance	6	mm
Nominal Thickness	34	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	41.807	kgf/cm ²
Design Temperature Internal Pressure	130	°C
Design External Pressure	1.0546	kgf/cm ²
Design Temperature External Pressure	130	°C
Effective Diameter Multiplier	1.3	
Material Name	SA-516 70 +IT	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	60	
Detail Type	Platform	
Detail ID	PL-2	
Dist. from "FROM" Node / Offset dist	100	cm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	180.0	
Platform Wind Area	30000	cm ²
Platform Weight	1551.2	kgf
Platform Railing Weight	0	kgf/cm
Platform Grating Weight	0.017	kgf/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	115	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	cm
Ladder End Elevation	400	cm
Unit Weight of Ladder	0.3773	kgf/cm
Platform Length (top head platform)	0	mm

Element From Node	60	
Detail Type	Packing	
Detail ID	Bed 1	
Dist. from "FROM" Node / Offset dist	0.28956E-04	cm
Height of Packed Section	300	cm
Density	0.000224	kg/cm ³
Percent Volume Holdup	0.0	
Specific Gravity of Packing Liquid	1.023	



Element From Node	60	
Detail Type	Liquid	
Detail ID	Liquid: 60	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	300	cm
Liquid Density	1022.5	kg/m ³
Element From Node	60	
Detail Type	Insulation	
Detail ID	Ins: 30	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	300	cm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm ³
Element From Node	60	
Detail Type	Nozzle	
Detail ID	UC2600	
Dist. from "FROM" Node / Offset dist	100	cm
Nozzle Diameter	600	mm
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	

Element#	7/11	
Element From Node	70	
Element To Node	80	
Element Type	Cylinder	
Description	Shell-3	
Distance "FROM" to "TO"	300	cm
Inside Diameter	1700	mm
Element Thickness	34	mm
Internal Corrosion Allowance	6	mm
Nominal Thickness	34	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	41.807	kgf/cm ²
Design Temperature Internal Pressure	130	°C
Design External Pressure	1.0546	kgf/cm ²
Design Temperature External Pressure	130	°C
Effective Diameter Multiplier	1.3	
Material Name	SA-516 70 +IT	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	70	
Detail Type	Platform	
Detail ID	PL-3	
Dist. from "FROM" Node / Offset dist	200	cm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	180.0	
Platform Wind Area	30000	cm ²
Platform Weight	1551.2	kgf
Platform Railing Weight	0	kgf/cm
Platform Grating Weight	0.017	kgf/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	115	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	cm
Ladder End Elevation	400	cm
Unit Weight of Ladder	0.3773	kgf/cm
Platform Length (top head platform)	0	mm



Element From Node	70	
Detail Type	Packing	
Detail ID	Bed 1	
Dist. from "FROM" Node / Offset dist	0	cm
Height of Packed Section	127.5	cm
Density	0.000224	kg/cm ³
Percent Volume Holdup	0.0	
Specific Gravity of Packing Liquid	1.023	

Element From Node	70	
Detail Type	Liquid	
Detail ID	Liquid: 70	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	300	cm
Liquid Density	1022.5	kg/m ³

Element From Node	70	
Detail Type	Insulation	
Detail ID	Ins: 40	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	300	cm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm ³

Element From Node	70	
Detail Type	Nozzle	
Detail ID	MH3600	
Dist. from "FROM" Node / Offset dist	230	cm
Nozzle Diameter	600	mm
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	

Element From Node	70	
Detail Type	Nozzle	
Detail ID	DP250	
Dist. from "FROM" Node / Offset dist	250	cm
Nozzle Diameter	50.8	mm
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	

Element From Node	70	
Detail Type	Weight	
Detail ID	Distributor 2	
Dist. from "FROM" Node / Offset dist	250	cm
Miscellaneous Weight	200	kgf
Offset from Element Centerline	0	mm

Element From Node	70	
Detail Type	Weight	
Detail ID	Dist2 sup	
Dist. from "FROM" Node / Offset dist	245	cm
Miscellaneous Weight	9.9998	kgf
Offset from Element Centerline	0	mm

Element From Node	70	
Detail Type	Weight	
Detail ID	MH3 Davit	
Dist. from "FROM" Node / Offset dist	230	cm
Miscellaneous Weight	99.998	kgf



Offset from Element Centerline 0 mm

Element From Node 70
 Detail Type Weight
 Detail ID Bed2 lim sup
 Dist. from "FROM" Node / Offset dist 130 cm
 Miscellaneous Weight 9.9998 kgf
 Offset from Element Centerline 0 mm

Element From Node 70
 Detail Type Weight
 Detail ID Bed2 limiter
 Dist. from "FROM" Node / Offset dist 0 cm
 Miscellaneous Weight 49.999 kgf
 Offset from Element Centerline 0 mm

Element# 8/11
 Element From Node 80
 Element To Node 90
 Element Type Cylinder
 Description Shell-4
 Distance "FROM" to "TO" 380.5 cm
 Inside Diameter 1700 mm
 Element Thickness 34 mm
 Internal Corrosion Allowance 6 mm
 Nominal Thickness 34 mm
 External Corrosion Allowance 0 mm
 Design Internal Pressure 41.807 kgf/cm²
 Design Temperature Internal Pressure 130 °C
 Design External Pressure 1.0546 kgf/cm²
 Design Temperature External Pressure 130 °C
 Effective Diameter Multiplier 1.3
 Material Name SA-516 70 +IT
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 80
 Detail Type Platform
 Detail ID PL-4
 Dist. from "FROM" Node / Offset dist 200 cm
 Platform Start Angle (degrees) 0.0
 Platform End Angle (degrees) 180.0
 Platform Wind Area 30000 cm²
 Platform Weight 1552.8 kgf
 Platform Railing Weight 0 kgf/cm
 Platform Grating Weight 0.017 kgf/cm²
 Platform Width 1500 mm
 Platform Height 1000 mm
 Platform Clearance or End Offset 115 mm
 Platform Force Coefficient 1.0
 Ladder Layout Angle 0.0
 Ladder Start Elevation 0 cm
 Ladder End Elevation 400 cm
 Unit Weight of Ladder 0.3773 kgf/cm
 Platform Length (top head platform) 0 mm

Element From Node 80
 Detail Type Packing
 Detail ID Bed-1
 Dist. from "FROM" Node / Offset dist 53 cm
 Height of Packed Section 327.5 cm
 Density 0.000224 kg/cm³
 Percent Volume Holdup 0.0
 Specific Gravity of Packing Liquid 1.023

Element From Node 80
 Detail Type Liquid



Detail ID	Liquid: 80	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	380.5	cm
Liquid Density	1022.5	kg/m ³
Element From Node	80	
Detail Type	Insulation	
Detail ID	Ins: 50	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	380.5	cm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm ³
Element From Node	80	
Detail Type	Nozzle	
Detail ID	UC1600	
Dist. from "FROM" Node / Offset dist	100	cm
Nozzle Diameter	600	mm
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	
Element From Node	80	
Detail Type	Weight	
Detail ID	Bed1 sup	
Dist. from "FROM" Node / Offset dist	45	cm
Miscellaneous Weight	9.9998	kgf
Offset from Element Centerline	0	mm
Element From Node	80	
Detail Type	Weight	
Detail ID	Bed 1 sup grid	
Dist. from "FROM" Node / Offset dist	50	cm
Miscellaneous Weight	180	kgf
Offset from Element Centerline	0	mm
Element From Node	80	
Detail Type	Weight	
Detail ID	Clip wt	
Dist. from "FROM" Node / Offset dist	200	cm
Miscellaneous Weight	2800	kgf
Offset from Element Centerline	0	mm
Element From Node	80	
Detail Type	Weight	
Detail ID	Misc wt 10%	
Dist. from "FROM" Node / Offset dist	195	cm
Miscellaneous Weight	4299.9	kgf
Offset from Element Centerline	0	mm

Element#	9/11	
Element From Node	90	
Element To Node	100	
Element Type	Cylinder	
Description	Shell-5	
Distance "FROM" to "TO"	219.5	cm
Inside Diameter	1706	mm
Element Thickness	28	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	28	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	41.807	kgf/cm ²
Design Temperature Internal Pressure	130	°C
Design External Pressure	1.0546	kgf/cm ²



Design Temperature External Pressure	130	°C
Effective Diameter Multiplier	1.3	
Material Name	SA-516 70	+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	90	
Detail Type	Platform	
Detail ID	PL-5	
Dist. from "FROM" Node / Offset dist	200	cm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	180.0	
Platform Wind Area	30000	cm ²
Platform Weight	1550.4	kgf
Platform Railing Weight	0	kgf/cm
Platform Grating Weight	0.017	kgf/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	115	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	cm
Ladder End Elevation	400	cm
Unit Weight of Ladder	0.3773	kgf/cm
Platform Length (top head platform)	0	mm
Element From Node	90	
Detail Type	Packing	
Detail ID	Bed-1	
Dist. from "FROM" Node / Offset dist	0.0001161	cm
Height of Packed Section	80.5	cm
Density	0.000224	kg/cm ³
Percent Volume Holdup	0.0	
Specific Gravity of Packing Liquid	1.023	
Element From Node	90	
Detail Type	Liquid	
Detail ID	Liquid: 90	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	219.5	cm
Liquid Density	1022.5	kg/m ³
Element From Node	90	
Detail Type	Insulation	
Detail ID	Ins: 60	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	219.5	cm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm ³
Element From Node	90	
Detail Type	Lining	
Detail ID	Clad	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Lining	219.5	cm
Thickness of Lining	3	mm
Density	0.008027	kg/cm ³
Element From Node	90	
Detail Type	Nozzle	
Detail ID	A150	
Dist. from "FROM" Node / Offset dist	212	cm
Nozzle Diameter	150	mm
Nozzle Schedule	120	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	kgf
Grade of Attached Flange	GR 1.1	



Nozzle Matl	SA-105 +IT
Element From Node	90
Detail Type	Nozzle
Detail ID	MH2600
Dist. from "FROM" Node / Offset dist	189.5 cm
Nozzle Diameter	600 mm
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	90.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	0 kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105 +IT
Element From Node	90
Detail Type	Weight
Detail ID	Distributor
Dist. from "FROM" Node / Offset dist	200 cm
Miscellaneous Weight	200 kgf
Offset from Element Centerline	0 mm
Element From Node	90
Detail Type	Weight
Detail ID	Distributor sup
Dist. from "FROM" Node / Offset dist	195 cm
Miscellaneous Weight	9.9998 kgf
Offset from Element Centerline	0 mm
Element From Node	90
Detail Type	Weight
Detail ID	Bed limiter sup
Dist. from "FROM" Node / Offset dist	185 cm
Miscellaneous Weight	9.9998 kgf
Offset from Element Centerline	0 mm
Element From Node	90
Detail Type	Weight
Detail ID	Bed limiter
Dist. from "FROM" Node / Offset dist	190 cm
Miscellaneous Weight	49.999 kgf
Offset from Element Centerline	0 mm
Element From Node	90
Detail Type	Weight
Detail ID	A FY
Dist. from "FROM" Node / Offset dist	212 cm
Miscellaneous Weight	1606 kgf
Offset from Element Centerline	2118 mm

Element#	10/11
Element From Node	100
Element To Node	110
Element Type	Cylinder
Description	Shell-7
Distance "FROM" to "TO"	310 cm
Inside Diameter	1706 mm
Element Thickness	28 mm
Internal Corrosion Allowance	1 mm
Nominal Thickness	28 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	41.807 kgf/cm ²
Design Temperature Internal Pressure	130 °C
Design External Pressure	1.0546 kgf/cm ²
Design Temperature External Pressure	130 °C
Effective Diameter Multiplier	1.3
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0



Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	100	
Detail Type	Platform	
Detail ID	PL-6	
Dist. from "FROM" Node / Offset dist	100	cm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	180.0	
Platform Wind Area	30000	cm ²
Platform Weight	1512.6	kgf
Platform Railing Weight	0	kgf/cm
Platform Grating Weight	0.017	kgf/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	115	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	cm
Ladder End Elevation	300	cm
Unit Weight of Ladder	0.3773	kgf/cm
Platform Length (top head platform)	0	mm
Element From Node	100	
Detail Type	Liquid	
Detail ID	Liquid: 100	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	310	cm
Liquid Density	1022.5	kg/m ³
Element From Node	100	
Detail Type	Insulation	
Detail ID	Ins: 70	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	310	cm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm ³
Element From Node	100	
Detail Type	Lining	
Detail ID	Clad	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Lining	310	cm
Thickness of Lining	3	mm
Density	0.008027	kg/cm ³
Element From Node	100	
Detail Type	Nozzle	
Detail ID	E40	
Dist. from "FROM" Node / Offset dist	90	cm
Nozzle Diameter	40	mm
Nozzle Schedule	None	
Nozzle Class	900	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	
Element From Node	100	
Detail Type	Nozzle	
Detail ID	MH1600	
Dist. from "FROM" Node / Offset dist	155	cm
Nozzle Diameter	600	mm
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	kgf
Grade of Attached Flange	GR 1.1	



Nozzle Matl	SA-105 +IT
Element From Node	100
Detail Type	Tray
Detail ID	TRAY 1-3
Dist. from "FROM" Node / Offset dist	24 cm
Tray Spacing	304.8 mm
Number of Trays	3
Tray Weight per Unit Area	0.015 kgf/cm ²
Height of Liquid on Tray	0 mm
Density of Liquid on Tray	0 kg/m ³
Support Ring and Bolting Bar Weight	0 kgf
Element From Node	100
Detail Type	Weight
Detail ID	Demister
Dist. from "FROM" Node / Offset dist	285 cm
Miscellaneous Weight	120 kgf
Offset from Element Centerline	0 mm
Element From Node	100
Detail Type	Weight
Detail ID	Demister sup
Dist. from "FROM" Node / Offset dist	280 cm
Miscellaneous Weight	49.999 kgf
Offset from Element Centerline	0 mm
Element From Node	100
Detail Type	Weight
Detail ID	TSR bolting bar
Dist. from "FROM" Node / Offset dist	20 cm
Miscellaneous Weight	79.999 kgf
Offset from Element Centerline	0 mm
Element From Node	100
Detail Type	Weight
Detail ID	MH1 Davit
Dist. from "FROM" Node / Offset dist	155 cm
Miscellaneous Weight	99.998 kgf
Offset from Element Centerline	0 mm
Element From Node	100
Detail Type	Weight
Detail ID	Trunnion 1
Dist. from "FROM" Node / Offset dist	210 cm
Miscellaneous Weight	499.99 kgf
Offset from Element Centerline	400 mm
Element From Node	100
Detail Type	Weight
Detail ID	Trunnion 2
Dist. from "FROM" Node / Offset dist	210 cm
Miscellaneous Weight	499.99 kgf
Offset from Element Centerline	400 mm

Element#	11/11
Element From Node	110
Element To Node	120
Element Type	Elliptical
Description	Top head
Distance "FROM" to "TO"	5 cm
Inside Diameter	1706 mm
Element Thickness	26 mm
Internal Corrosion Allowance	1 mm
Nominal Thickness	32 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	41.807 kgf/cm ²
Design Temperature Internal Pressure	130 °C



Design External Pressure	1.0546	kgf/cm ²
Design Temperature External Pressure	130	°C
Effective Diameter Multiplier	1.3	
Material Name	SA-516 70	+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	

Element From Node	110	
Detail Type	Platform	
Detail ID	PL-7	
Dist. from "FROM" Node / Offset dist	50	cm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	0.0	
Platform Wind Area	37640	cm ²
Platform Weight	2504	kgf
Platform Railing Weight	0	kgf/cm
Platform Grating Weight	0.017	kgf/cm ²
Platform Width	3764	mm
Platform Height	1000	mm
Platform Clearance or End Offset	1882	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	cm
Ladder End Elevation	250	cm
Unit Weight of Ladder	0.3773	kgf/cm
Platform Length (top head platform)	3764	mm

Element From Node	110	
Detail Type	Liquid	
Detail ID	Liquid: 110	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	47.5	cm
Liquid Density	1022.5	kg/m ³

Element From Node	110	
Detail Type	Insulation	
Detail ID	Ins: 90	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	47.5	cm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm ³

Element From Node	110	
Detail Type	Lining	
Detail ID	Clad	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Lining	47.65	cm
Thickness of Lining	3	mm
Density	0.008027	kg/cm ³

Element From Node	110	
Detail Type	Nozzle	
Detail ID	C100	
Dist. from "FROM" Node / Offset dist	0	mm
Nozzle Diameter	100	mm
Nozzle Schedule	160	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	+IT

Element From Node	110	
Detail Type	Weight	
Detail ID	C FY	
Dist. from "FROM" Node / Offset dist	0	cm
Miscellaneous Weight	564.99	kgf



Offset from Element Centerline	2093	mm
Element From Node	110	
Detail Type	For./Mom.	
Detail ID	C F/M	
Dist. from "FROM" Node / Offset dist	35	cm
Force in X Direction	560.66	kgf
Force in Y Direction	-840.98	kgf
Force in Z Direction	560.66	kgf
Moment about X Axis	156.99	kgf-m.
Moment about Y Axis	0	kgf-m.
Moment about Z Axis	156.99	kgf-m.
Force/Moment Combination Method	SRSS	

XY Coordinate Calculations:

From	To	X (Horiz.) cm	Y (Vert.) cm	DX (Horiz.) cm	DY (Vert.) cm
Skirt-3	...	...	200	...	200
Skirt 2	...	...	500	...	300
Skirt 1	...	...	600	...	100
Bottom head	...	...	605	...	5
Shell-1	...	...	905	...	300
Shell-2	...	...	1205	...	300
Shell-3	...	...	1505	...	300
Shell-4	...	...	1885.5	...	380.5
Shell-5	...	...	2105	...	219.5
Shell-7	...	...	2415	...	310
Top head	...	...	2420	...	5

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd kgf/cm ²	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) kgf/cm ²	Element#
Skirt-3		...	14	1.6	1726	...	1
Skirt 2		...	14	1.6	1726	...	2
Skirt 1		...	14	1.6	1726	...	3
Bottom head		43.755	38	6	1700	1407.1	4
Shell-1		43.706	34	6	1700	1407.1	5
Shell-2		43.4	34	6	1700	1407.1	6
Shell-3		43.093	34	6	1700	1407.1	7
Shell-4		42.786	34	6	1700	1407.1	8
Shell-5		42.397	28	1	1706	1407.1	9
Shell-7		42.172	28	1	1706	1407.1	10
Top head		41.855	32	1	1706	1407.1	11

Element Required Thickness and MAWP :

From	To	Design Pressure kgf/cm ²	M.A.W.P. Corroded kgf/cm ²	M.A.P. New & Cold kgf/cm ²	Minimum Thickness mm	Required Thickness mm	Element#
Skirt-3		...	No Calc	No Calc	14	No Calc	1
Skirt 2		...	No Calc	No Calc	14	No Calc	2
Skirt 1		...	No Calc	No Calc	14	No Calc	3
Bottom head		41.8069	45.2894	58.2511	32	30.0682	4
Shell-1		41.8069	43.3915	55.1885	34	33.0051	5
Shell-2		41.8069	43.6983	55.1885	34	32.8127	6
Shell-3		41.8069	44.005	55.1885	34	32.6202	7
Shell-4		41.8069	44.3118	55.1885	34	32.4279	8
Shell-5		41.8069	43.209	45.4476	28	27.1227	9
Shell-7		41.8069	43.4334	45.4476	28	26.9823	10
Top head		41.8069	45.38	47.2557	26	24.0187	11

Minimum 43 45

MAWP: 43 kgf/cm², limited by: Shell-5.

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025

Internal Pressure Results for: Bottom head

Elliptical Head Design Information:					
Design Pressure	P	43.755 kgf/cm ²	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		130 °C
Length		5.000 cm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	1407.1 kgf/cm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1407.1 kgf/cm ²
Inside Diameter	Di	1700.000 mm	Actual Stress		1303.4 kgf/cm ²
Spec. Min. Thk t _e , or s _a		32.0000 mm	Spec. Nominal Thk.		38.000 mm
Surface Area		36980.5 cm ²	SG of Contents		1.023
Element Volume		756602.1 cm ³	Weight of Contents		773.7 kgf
Empty Weight		1145.1 kgf	Operating Weight		1918.8 kgf
			Comp. All. Ope. Stress		1175.6 kgf/cm ²
			Aspect Ratio		2.00

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 21.250 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 10.625 mm

Material treatment options: +IT

Head Calculation - Section 4.3.6 (and Section 4.3.7 for Ellipsoidal Head)

Computed Minimum Required Thickness [t]:
 tr = 24.0682 mm

$$\begin{aligned}
 t &= tr + ci + co \\
 &= 24.0682 + 6.0 + 0.0 \\
 &= 30.0682 \text{ mm - see below for the derivation}
 \end{aligned}$$

For Elliptical heads: Compute [h, k, r,] per 4.3.7.1

$$\begin{aligned}
 h &= D/(2 \cdot \text{Aspect Ratio}) \\
 &= 1700.0/(2 \cdot 2.0) \\
 &= 425.000 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 k &= D/(2 \cdot h) \\
 &= 1700.0/(2 \cdot 425.0) \\
 &= 2.000
 \end{aligned}$$

$$\begin{aligned}
 r &= D(0.5/k - 0.08) + ca \\
 &= 1700.0(0.5/2.0 - 0.08) + 6.0 \\
 &= 295.000 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 L &= D(0.44 \cdot k + 0.02) + ca \\
 &= 1700.0(0.44 \cdot 2.0 + 0.02) + 6.0 \\
 &= 1536.000 \text{ mm}
 \end{aligned}$$

Step 3: Geometric Constants [βTh, φTh, Rth]:

Equations 4.3.8 to 4.3.11 as appropriate

t (corroded) and P for the calc.: t = 24.0682 mm, P = 43.7549 kgf/cm²

Angle [βTh]:

$$= \arccos[(0.5 \cdot D - r)/(L - r)]$$

$$= \arccos[(0.5 \cdot 1712.0 - 295.0)/(1536.0 - 295.0)]$$

$$= 1.1017 \text{ radians - Equation 4.3.8}$$

$$\phi_{Th} = \sqrt{(L \cdot t) / r}$$

$$= \sqrt{(1536.0 \cdot 24.068) / 295.0}$$

$$= 0.6518 \text{ radians - Equation 4.3.9}$$

$\phi_{Th} < \beta_{Th}$, Thus:

$$R_{th} = (0.5 \cdot D - r) / \cos(\beta_{Th} - \phi_{Th}) + r$$

$$= (0.5 \cdot 1712.0 - 295.0) / \cos(1.102 - 0.652) + 295.0$$

$$= 918.0112 \text{ mm - Equation 4.3.10}$$

Step 4: Coefficients: Equations 4.3.12 to 4.3.15 [C1 and C2]:

$r/D > 0.08$, Thus:

$$C1 = 0.692(r/D) + 0.605$$

$$= 0.692(295.0/1712.0) + 0.605$$

$$= 0.7242 \text{ - Equation 4.3.13}$$

$$C2 = 1.46 - 2.6(r/D) = 1.46 - 2.6(295.0/1712.0)$$

$$= 1.012, \text{ Equation 4.3.15}$$

Step 5: Pressure to produce knuckle elastic buckling [Peth]:

$$= C1 \cdot Et \cdot t^2 / (C2 \cdot R_{th}(0.5 \cdot R_{th} - r))$$

$$= 0.724 \cdot 2001648 \cdot 24.068^2 / (1.012 \cdot 918.011(0.5 \cdot 918.011 - 295.0))$$

$$= 5511.57 \text{ kgf/cm}^2, \text{ Equation 4.3.16}$$

Step 6: Internal Pressure for max stress in knuckle equals yield [Py]:

$C3 = S_y = 2390.102 \text{ kgf/cm}^2$, Not in the creep range - 4.3.6.1(f)

$$P_y = C3 \cdot t / (C2 \cdot R_{th}(0.5 \cdot R_{th} / r - 1))$$

$$= 2390.102 \cdot 24.068 / (1.012 \cdot 918.011(0.5 \cdot 918.011/295.0 - 1))$$

$$= 111.378 \text{ kgf/cm}^2, \text{ Equation 4.3.17}$$

Step 7: Value of Internal Press. resulting in buckling of knuckle [G and Pck]:

$$G = P_{eth}/P_y = 5511.569/111.378 = 49.485$$

$G > 1$, Thus:

Buckling Pressure [Pck]:

$$= ((0.77508 \cdot G - 0.20354 \cdot G^2 + 0.019274 \cdot G^3) /$$

$$(1 + 0.19014 \cdot G - 0.089534 \cdot G^2 + 0.0093965 \cdot G^3)) P_y$$

$$= ((0.77508 \cdot 49.485 - 0.20354 \cdot 49.485^2 + 0.019274 \cdot 49.485^3) /$$

$$(1 + 0.19014 \cdot 49.485 - 0.089534 \cdot 49.485^2 + 0.0093965 \cdot 49.485^3)) \cdot 111.378$$

$$= 224.66 \text{ kgf/cm}^2, \text{ Equation 4.3.19}$$

Step 8: Compute Allowable Pressure base on buckling of knuckle [Pak]:

$$= P_{ck}/1.5 = 224.66/1.5 = 149.77 \text{ kgf/cm}^2 \text{ - Equation 4.3.21}$$

Step 9: Compute allowable pressure based on rupture of crown [Pac]:

$$= 2 \cdot S \cdot E / (L/t + 0.5)$$

$$= 2 \cdot 1407.129 \cdot 1.0 / (1536.0/24.068 + 0.5)$$

$$= 43.75 \text{ kgf/cm}^2, \text{ Equation 4.3.22}$$

Step 10: Compute final pressure [Pa]:

$$= \min(P_{ak}, P_{ac}) = \min(149.774, 43.755)$$

$$= 43.75 \text{ kgf/cm}^2, \text{ Equation 4.3.23: (same as original design pressure)}$$

Allowable pressure in Crown at given thickness [Pac]:

$$= 2 \cdot S \cdot E / (L/t + 0.5)$$

$$= 2 \cdot 1407.129 \cdot 1.0 / (1536.0/26.0 + 0.5)$$

$$= 47.237 \text{ kgf/cm}^2$$

Computed Maximum Allowable Pressure [MAWP]:

$$= \text{MaxPressure} - P_{liq} = 47.2374 - 1.948 = 45.2894 \text{ kgf/cm}^2$$

Computed Maximum Allowable Pressure New and Cold [MAPnc]:

$$= 58.2511 \text{ kgf/cm}^2$$

Computed Membrane Stress Intensity at Design Pressure [Sm]:

= 1303.4 kgf/cm²

Required Thickness of Straight Flange = 33.036 mm

Percent Fiber Elongation per Table 6.1

For Heads formed from Segments $(75 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 9.253 %
 For one Piece doubly Curved Heads $100 \cdot \ln(D_b/(D_f - 2t))$ 24.406 %

Where D_b is the blank diameter for one piece doubly curved surfaces [D_b]:

= square root($4 \cdot \text{Head Surface Area} / 3.14159$)
 = square root($4 \cdot 36980.5 / 3.14159$)
 = 2169.910 mm

MDMT Calculations in the Knuckle Portion:

Note: This Element/Detail was specified as being Impact Tested.

MDMT Calculations in the Head Straight Flange:

Note: This Element/Detail was specified as being Impact Tested.

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 21.250 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 10.625 mm

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: Bottom head

Elliptical Head From 40 to 50 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Elastic Hoop Compressive Membrane Failure Stress [F_{he}]:

= $0.075 \cdot E_y(t / R_o)$
 = $0.075 \cdot 2001648(26.0/1533.685)$
 = 2544.990 kgf/cm²

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [F_r]:

= F_{he} / S_y
 = $2545.0 / 2390.1$
 = 1.065

Engineering Yield to Tensile Stress ratio [R]:

= $\sigma_{ys} / \sigma_{uts}$
 = $2390.1/4921.5$
 = 0.486

Ratio [A_E]:

= F_{he} / E
 = $2545.0/2001648$
 = 0.001271

Factor [m_1]:

= $(\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$
 = $(\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$
 = 0.157

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A_1]:

= $\sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m_1}$
 = $2390.1(1 + 0.002) / \ln(1 + 0.002)^{0.157304}$
 = 6366.646 kgf/cm²

Curve Fitting Exponent for, Ferritic Steel [m_2]:

= $m_2 f(m_2 f_2 - R)$
 = $0.6(1.0 - 0.486)$
 = 0.309

Curve Fitting Constant [A2]:
 $= (\sigma_{uts} \cdot \exp(m_2)) / (m_2 m_2)$
 $= (4921.5 \cdot \exp(0.309)) / (0.309 \cdot 0.309)$
 $= 9631.605 \text{ kgf/cm}^2$

Material Parameter for Stress-Strain Curve Model [K]:
 $= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$
 $= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$
 $= 0.346$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	2460	-1.839	0.307E-05	0.265E-05	0.791E-05	-0.620E-05	0.126E+06	0.182E-01
2	1230	-4.651	0.748E-07	0.748E-07	0.167E-05	-0.167E-05	0.154E+07	0.472E-03
3	1845	-3.245	0.657E-06	0.652E-06	0.415E-05	-0.411E-05	0.539E+06	0.215E-02
4	1537	-3.948	0.247E-06	0.247E-06	0.276E-05	-0.275E-05	0.100E+07	0.267E-03
5	1384	-4.299	0.141E-06	0.141E-06	0.218E-05	-0.218E-05	0.128E+07	0.188E-03
6	1461	-4.124	0.188E-06	0.188E-06	0.246E-05	-0.246E-05	0.114E+07	0.129E-04
7	1422	-4.211	0.163E-06	0.163E-06	0.232E-05	-0.231E-05	0.121E+07	0.932E-04
8	1441	-4.167	0.175E-06	0.175E-06	0.239E-05	-0.238E-05	0.117E+07	0.417E-04
9	1451	-4.146	0.181E-06	0.181E-06	0.242E-05	-0.242E-05	0.116E+07	0.148E-04
10	1456	-4.135	0.185E-06	0.184E-06	0.244E-05	-0.244E-05	0.115E+07	0.106E-05
11	1458	-4.129	0.186E-06	0.186E-06	0.245E-05	-0.245E-05	0.114E+07	0.589E-05
12	1457	-4.132	0.185E-06	0.185E-06	0.245E-05	-0.244E-05	0.114E+07	0.241E-05
13	1456	-4.133	0.185E-06	0.185E-06	0.244E-05	-0.244E-05	0.115E+07	0.671E-06
14	1456	-4.134	0.185E-06	0.185E-06	0.244E-05	-0.244E-05	0.115E+07	0.197E-06
15	1456	-4.133	0.185E-06	0.185E-06	0.244E-05	-0.244E-05	0.115E+07	0.237E-06
16	1456	-4.134	0.185E-06	0.185E-06	0.244E-05	-0.244E-05	0.115E+07	0.201E-07
17	1456	-4.134	0.185E-06	0.185E-06	0.244E-05	-0.244E-05	0.115E+07	0.883E-07

Factor of Safety [FS]:
 Since $Fic > 0.55 F_y$ and $Fic < F_y$
 $= 2.407 - 0.741 \cdot Fic / F_y$
 $= 2.407 - 0.741 \cdot 1456.6/2390.1$
 $= 1.955$

Allowable Circumferential Compressive Stress [Fha]:
 $= Fic/FS$
 $= 1456.599/1.955$
 $= 744.906 \text{ kgf/cm}^2$

Allowable External Pressure at the given Thickness [Pa]:
 $= 2 \cdot Fha[t/R_o]$
 $= 2 \cdot 744.906[26.0/1533.685]$
 $= 25.256 \text{ kgf/cm}^2$

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025

Internal Pressure Results for: Shell-1

Cylindrical Shell Design Information:					
Design Pressure	P	43.706 kgf/cm²	Int. Design Temperature	130 °C	
Material		SA-516 70	Ext. Design Temperature	130 °C	
Length		300.000 cm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	1407.1 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1407.1 kgf/cm²
Inside Diameter	Di	1700.000 mm	Actual Stress	1336.2 kgf/cm²	
Spec. Min. Thk t,e, or sa		34.0000 mm	Spec. Nominal Thk.	34.000 mm	
Surface Area		166630.1 cm²	SG of Contents	1.023	
Element Volume		6809418.5 cm³	Weight of Contents	6962.9 kgf	
Empty Weight		4306.5 kgf	Operating Weight	11269.4 kgf	
			Comp. All. Ope. Stress	1186.5 kgf/cm²	

Radiography Information:				
...		...		
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v	100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 1712.0 \cdot (\exp(43.706/(1407.129 \cdot 1.0)) - 1) + 6.0 + 0.0$
 $= 27.0051 + 6.0 + 0.0 = 33.0051 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1407.129 \cdot 1.0 \cdot \ln((2 \cdot 28.0 + 1712.0) / 1712.0) - 1.899$
 $= 45.291 - 1.899 = 43.3915 \text{ kgf/cm}^2$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 1407.129 \cdot 1.0 \cdot \ln((2 \cdot 34.0 + 1700.0) / 1700.0)$
 $= 55.1885 \text{ kgf/cm}^2$

Computed Stress Intensity at Design Pressure [Si]:
 $= P \cdot D / (E(D_o - D))$
 $= 43.706 \cdot 1712.0 / (1.0(1768.0 - 1712.0))$
 $= 1336.1667 \text{ kgf/cm}^2$

Computed Membrane Stress at Design Pressure [Sm]:
 $= P / (E \cdot \ln((2 \cdot t + D) / D))$
 $= 43.706 / (1.0 \cdot \ln((2 \cdot 28.0 + 1712.0) / 1712.0))$
 $= 1357.8982 \text{ kgf/cm}^2$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.961 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was

specified as being heat treated.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: Shell-1

Cylindrical Shell From 50 to 60 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(Ro \cdot t)}$$

$$= 18483.834 / \sqrt{(884.0 \cdot 28.0)}$$

$$= 117.486$$

Compute the Factor [Ch]:
 Since $Mx > 2 \cdot (Do/t)0.94$, $Ch = 0.55 / (Do/t)$

$$= 0.55 \cdot (1768.0/28.0)$$

$$= 0.0087$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot Ey(t / Do)$$

$$= 1.6 \cdot 0.009 \cdot 2001648(28.0/1768.0)$$

$$= 441.796 \text{ kgf/cm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / Fy$$

$$= 441.8/2390.1$$

$$= 0.185$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 2390.1/4921.5$$

$$= 0.486$$

Ratio [AE]:

$$= Fhe / E$$

$$= 441.8/2001648$$

$$= 0.000221$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 2390.1(1 + 0.002) / \ln(1 + 0.002) 0.157304$$

$$= 6366.646 \text{ kgf/cm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (4921.5 \cdot \exp(0.309)) / (0.309^{0.309})$$

$$= 9631.605 \text{ kgf/cm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3



	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	2460	-1.839	0.307E-05	0.265E-05	0.791E-05	-0.620E-05	0.126E+06	0.193E-01
2	1230	-4.651	0.748E-07	0.748E-07	0.167E-05	-0.167E-05	0.154E+07	0.579E-03
3	615	-6.057	0.183E-08	0.183E-08	0.354E-06	-0.354E-06	0.199E+07	0.889E-04
4	307	-6.760	0.445E-10	0.445E-10	0.750E-07	-0.750E-07	0.200E+07	0.670E-04
5	461	-6.409	0.391E-09	0.391E-09	0.186E-06	-0.186E-06	0.200E+07	0.102E-04
6	384	-6.584	0.147E-09	0.147E-09	0.124E-06	-0.124E-06	0.200E+07	0.285E-04
7	422	-6.496	0.245E-09	0.245E-09	0.153E-06	-0.153E-06	0.200E+07	0.921E-05
8	442	-6.452	0.311E-09	0.311E-09	0.169E-06	-0.169E-06	0.200E+07	0.460E-06
9	432	-6.474	0.277E-09	0.277E-09	0.161E-06	-0.161E-06	0.200E+07	0.438E-05
10	437	-6.463	0.293E-09	0.293E-09	0.165E-06	-0.165E-06	0.200E+07	0.196E-05
11	439	-6.458	0.302E-09	0.302E-09	0.167E-06	-0.167E-06	0.200E+07	0.750E-06
12	440	-6.455	0.307E-09	0.307E-09	0.168E-06	-0.168E-06	0.200E+07	0.145E-06
13	441	-6.454	0.309E-09	0.309E-09	0.169E-06	-0.169E-06	0.200E+07	0.158E-06
14	441	-6.455	0.308E-09	0.308E-09	0.168E-06	-0.168E-06	0.200E+07	0.637E-08

Factor of Safety [FS]:

Since Fic [441.26] < 0.55 * 2390.1, FS = 2

Allowable Circumferential Compressive Stress [Fha]:

= Fic/FS

= 441.264/2.0

= 220.632 kgf/cm²

Allowable External Pressure at the given Thickness [Pa]:

= 2 • Fha(t / Do)

= 2 • 220.632(28.0/1768.0)

= 6.988 kgf/cm²

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025

Internal Pressure Results for: Shell-2

Cylindrical Shell Design Information:					
Design Pressure	P	43.400 kgf/cm²	Int. Design Temperature	130 °C	
Material		SA-516 70	Ext. Design Temperature	130 °C	
Length		300.000 cm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	1407.1 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1407.1 kgf/cm²
Inside Diameter	Di	1700.000 mm	Actual Stress	1326.8 kgf/cm²	
Spec. Min. Thk t,e, or sa		34.0000 mm	Spec. Nominal Thk.	34.000 mm	
Surface Area		166630.1 cm²	SG of Contents	1.023	
Element Volume		6809418.5 cm³	Weight of Contents	6962.9 kgf	
Empty Weight		4306.5 kgf	Operating Weight	11269.4 kgf	
			Comp. All. Ope. Stress	1186.5 kgf/cm²	

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 1712.0 \cdot (\exp(43.4/(1407.129 \cdot 1.0)) - 1) + 6.0 + 0.0$
 $= 26.8127 + 6.0 + 0.0 = 32.8127 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1407.129 \cdot 1.0 \cdot \ln((2 \cdot 28.0 + 1712.0) / 1712.0) - 1.593$
 $= 45.291 - 1.593 = 43.6983 \text{ kgf/cm}^2$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 1407.129 \cdot 1.0 \cdot \ln((2 \cdot 34.0 + 1700.0) / 1700.0)$
 $= 55.1885 \text{ kgf/cm}^2$

Computed Stress Intensity at Design Pressure [Si]:
 $= P \cdot D / (E \cdot (D_o - D))$
 $= 43.4 \cdot 1712.0 / (1.0 \cdot (1768.0 - 1712.0))$
 $= 1326.7885 \text{ kgf/cm}^2$

Computed Membrane Stress at Design Pressure [Sm]:
 $= P / (E \cdot \ln((2 \cdot t + D) / D))$
 $= 43.4 / (1.0 \cdot \ln((2 \cdot 28.0 + 1712.0) / 1712.0))$
 $= 1348.3674 \text{ kgf/cm}^2$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.961 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was

specified as being heat treated.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: Shell-2

Cylindrical Shell From 60 to 70 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$\begin{aligned} &= L / \sqrt{(Ro \cdot t)} \\ &= 18483.834 / \sqrt{(884.0 \cdot 28.0)} \\ &= 117.486 \end{aligned}$$

Compute the Factor [Ch]:

$$\begin{aligned} &\text{Since } Mx > 2 \cdot (Do/t)0.94, Ch = 0.55 / (Do/t) \\ &= 0.55 \cdot (1768.0/28.0) \\ &= 0.0087 \end{aligned}$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$\begin{aligned} &= 1.6 \cdot Ch \cdot Ey(t / Do) \\ &= 1.6 \cdot 0.009 \cdot 2001648(28.0/1768.0) \\ &= 441.796 \text{ kgf/cm}^2 \end{aligned}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$\begin{aligned} &= Fhe / Fy \\ &= 441.8/2390.1 \\ &= 0.185 \end{aligned}$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$\begin{aligned} &= \sigma_{ys} / \sigma_{uts} \\ &= 2390.1/4921.5 \\ &= 0.486 \end{aligned}$$

Ratio [AE]:

$$\begin{aligned} &= Fhe / E \\ &= 441.8/2001648 \\ &= 0.000221 \end{aligned}$$

Factor [m1]:

$$\begin{aligned} &= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys})) \\ &= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002)) \\ &= 0.157 \end{aligned}$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$\begin{aligned} &= \sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1} \\ &= 2390.1(1 + 0.002) / \ln(1 + 0.002) 0.157304 \\ &= 6366.646 \text{ kgf/cm}^2 \end{aligned}$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$\begin{aligned} &= m2f(m2f2 - R) \\ &= 0.6(1.0 - 0.486) \\ &= 0.309 \end{aligned}$$

Curve Fitting Constant [A2]:

$$\begin{aligned} &= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2}) \\ &= (4921.5 \cdot \exp(0.309)) / (0.309^{0.309}) \\ &= 9631.605 \text{ kgf/cm}^2 \end{aligned}$$

Material Parameter for Stress-Strain Curve Model [K]:

$$\begin{aligned} &= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5 \\ &= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5 \\ &= 0.346 \end{aligned}$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	2460	-1.839	0.307E-05	0.265E-05	0.791E-05	-0.620E-05	0.126E+06	0.193E-01
2	1230	-4.651	0.748E-07	0.748E-07	0.167E-05	-0.167E-05	0.154E+07	0.579E-03
3	615	-6.057	0.183E-08	0.183E-08	0.354E-06	-0.354E-06	0.199E+07	0.889E-04
4	307	-6.760	0.445E-10	0.445E-10	0.750E-07	-0.750E-07	0.200E+07	0.670E-04
5	461	-6.409	0.391E-09	0.391E-09	0.186E-06	-0.186E-06	0.200E+07	0.102E-04
6	384	-6.584	0.147E-09	0.147E-09	0.124E-06	-0.124E-06	0.200E+07	0.285E-04
7	422	-6.496	0.245E-09	0.245E-09	0.153E-06	-0.153E-06	0.200E+07	0.921E-05
8	442	-6.452	0.311E-09	0.311E-09	0.169E-06	-0.169E-06	0.200E+07	0.460E-06
9	432	-6.474	0.277E-09	0.277E-09	0.161E-06	-0.161E-06	0.200E+07	0.438E-05
10	437	-6.463	0.293E-09	0.293E-09	0.165E-06	-0.165E-06	0.200E+07	0.196E-05
11	439	-6.458	0.302E-09	0.302E-09	0.167E-06	-0.167E-06	0.200E+07	0.750E-06
12	440	-6.455	0.307E-09	0.307E-09	0.168E-06	-0.168E-06	0.200E+07	0.145E-06
13	441	-6.454	0.309E-09	0.309E-09	0.169E-06	-0.169E-06	0.200E+07	0.158E-06
14	441	-6.455	0.308E-09	0.308E-09	0.168E-06	-0.168E-06	0.200E+07	0.637E-08

Factor of Safety [FS]:

Since Fic [441.26] < 0.55 * 2390.1, FS = 2

Allowable Circumferential Compressive Stress [Fha]:

= Fic/FS

= 441.264/2.0

= 220.632 kgf/cm²

Allowable External Pressure at the given Thickness [Pa]:

= 2 • Fha(t / Do)

= 2 • 220.632(28.0/1768.0)

= 6.988 kgf/cm²

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025

Internal Pressure Results for: Shell-3

Cylindrical Shell Design Information:					
Design Pressure	P	43.093 kgf/cm²	Int. Design Temperature	130 °C	
Material		SA-516 70	Ext. Design Temperature	130 °C	
Length		300.000 cm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	1407.1 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1407.1 kgf/cm²
Inside Diameter	Di	1700.000 mm	Actual Stress	1317.4 kgf/cm²	
Spec. Min. Thk t,e, or sa		34.0000 mm	Spec. Nominal Thk.	34.000 mm	
Surface Area		166630.1 cm²	SG of Contents	1.023	
Element Volume		6809418.5 cm³	Weight of Contents	6962.9 kgf	
Empty Weight		4306.5 kgf	Operating Weight	11269.4 kgf	
			Comp. All. Ope. Stress	1186.5 kgf/cm²	

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 1712.0 \cdot (\exp(43.093/(1407.129 \cdot 1.0)) - 1) + 6.0 + 0.0$
 $= 26.6202 + 6.0 + 0.0 = 32.6202 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1407.129 \cdot 1.0 \cdot \ln((2 \cdot 28.0 + 1712.0) / 1712.0) - 1.286$
 $= 45.291 - 1.286 = 44.005 \text{ kgf/cm}^2$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 1407.129 \cdot 1.0 \cdot \ln((2 \cdot 34.0 + 1700.0) / 1700.0)$
 $= 55.1885 \text{ kgf/cm}^2$

Computed Stress Intensity at Design Pressure [Si]:
 $= P \cdot D / (E(D_o - D))$
 $= 43.093 \cdot 1712.0 / (1.0(1768.0 - 1712.0))$
 $= 1317.4104 \text{ kgf/cm}^2$

Computed Membrane Stress at Design Pressure [Sm]:
 $= P / (E \cdot \ln((2 \cdot t + D) / D))$
 $= 43.093 / (1.0 \cdot \ln((2 \cdot 28.0 + 1712.0) / 1712.0))$
 $= 1338.8367 \text{ kgf/cm}^2$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.961 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was

specified as being heat treated.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: Shell-3

Cylindrical Shell From 70 to 80 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 18483.834 / \sqrt{(884.0 \cdot 28.0)}$$

$$= 117.486$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 2 \cdot (D_o/t)0.94, Ch = 0.55 / (D_o/t)$$

$$= 0.55 \cdot (1768.0/28.0)$$

$$= 0.0087$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y(t / D_o)$$

$$= 1.6 \cdot 0.009 \cdot 2001648(28.0/1768.0)$$

$$= 441.796 \text{ kgf/cm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / F_y$$

$$= 441.8/2390.1$$

$$= 0.185$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 2390.1/4921.5$$

$$= 0.486$$

Ratio [AE]:

$$= Fhe / E$$

$$= 441.8/2001648$$

$$= 0.000221$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 2390.1(1 + 0.002) / \ln(1 + 0.002) 0.157304$$

$$= 6366.646 \text{ kgf/cm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (4921.5 \cdot \exp(0.309)) / (0.309^{0.309})$$

$$= 9631.605 \text{ kgf/cm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3



	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	2460	-1.839	0.307E-05	0.265E-05	0.791E-05	-0.620E-05	0.126E+06	0.193E-01
2	1230	-4.651	0.748E-07	0.748E-07	0.167E-05	-0.167E-05	0.154E+07	0.579E-03
3	615	-6.057	0.183E-08	0.183E-08	0.354E-06	-0.354E-06	0.199E+07	0.889E-04
4	307	-6.760	0.445E-10	0.445E-10	0.750E-07	-0.750E-07	0.200E+07	0.670E-04
5	461	-6.409	0.391E-09	0.391E-09	0.186E-06	-0.186E-06	0.200E+07	0.102E-04
6	384	-6.584	0.147E-09	0.147E-09	0.124E-06	-0.124E-06	0.200E+07	0.285E-04
7	422	-6.496	0.245E-09	0.245E-09	0.153E-06	-0.153E-06	0.200E+07	0.921E-05
8	442	-6.452	0.311E-09	0.311E-09	0.169E-06	-0.169E-06	0.200E+07	0.460E-06
9	432	-6.474	0.277E-09	0.277E-09	0.161E-06	-0.161E-06	0.200E+07	0.438E-05
10	437	-6.463	0.293E-09	0.293E-09	0.165E-06	-0.165E-06	0.200E+07	0.196E-05
11	439	-6.458	0.302E-09	0.302E-09	0.167E-06	-0.167E-06	0.200E+07	0.750E-06
12	440	-6.455	0.307E-09	0.307E-09	0.168E-06	-0.168E-06	0.200E+07	0.145E-06
13	441	-6.454	0.309E-09	0.309E-09	0.169E-06	-0.169E-06	0.200E+07	0.158E-06
14	441	-6.455	0.308E-09	0.308E-09	0.168E-06	-0.168E-06	0.200E+07	0.637E-08

Factor of Safety [FS]:

Since Fic [441.26] < 0.55 * 2390.1, FS = 2

Allowable Circumferential Compressive Stress [Fha]:

= Fic/FS

= 441.264/2.0

= 220.632 kgf/cm²

Allowable External Pressure at the given Thickness [Pa]:

= 2 • Fha(t / Do)

= 2 • 220.632(28.0/1768.0)

= 6.988 kgf/cm²

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025

Internal Pressure Results for: Shell-4

Cylindrical Shell Design Information:					
Design Pressure	P	42.786 kgf/cm²	Int. Design Temperature	130 °C	
Material		SA-516 70	Ext. Design Temperature	130 °C	
Length		380.500 cm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	1407.1 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1407.1 kgf/cm²
Inside Diameter	Di	1700.000 mm	Actual Stress	1308.0 kgf/cm²	
Spec. Min. Thk t,e, or sa		34.0000 mm	Spec. Nominal Thk.	34.000 mm	
Surface Area		211342.5 cm²	SG of Contents	1.023	
Element Volume		8636612.0 cm³	Weight of Contents	8831.2 kgf	
Empty Weight		5462.0 kgf	Operating Weight	14293.3 kgf	
			Comp. All. Ope. Stress	1186.5 kgf/cm²	

Radiography Information:				
...		...		
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v	100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 1712.0 \cdot (\exp(42.786/(1407.129 \cdot 1.0)) - 1) + 6.0 + 0.0$
 $= 26.4279 + 6.0 + 0.0 = 32.4279 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1407.129 \cdot 1.0 \cdot \ln((2 \cdot 28.0 + 1712.0) / 1712.0) - 0.979$
 $= 45.291 - 0.979 = 44.3118 \text{ kgf/cm}^2$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 1407.129 \cdot 1.0 \cdot \ln((2 \cdot 34.0 + 1700.0) / 1700.0)$
 $= 55.1885 \text{ kgf/cm}^2$

Computed Stress Intensity at Design Pressure [Si]:
 $= P \cdot D / (E (D_o - D))$
 $= 42.786 \cdot 1712.0 / (1.0 (1768.0 - 1712.0))$
 $= 1308.0322 \text{ kgf/cm}^2$

Computed Membrane Stress at Design Pressure [Sm]:
 $= P / (E \cdot \ln((2 \cdot t + D) / D))$
 $= 42.786 / (1.0 \cdot \ln((2 \cdot 28.0 + 1712.0) / 1712.0))$
 $= 1329.3060 \text{ kgf/cm}^2$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.961 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was

specified as being heat treated.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: Shell-4

Cylindrical Shell From 80 to 90 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 18483.834 / \sqrt{(884.0 \cdot 28.0)}$$

$$= 117.486$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 2 \cdot (D_o/t)0.94, Ch = 0.55 / (D_o/t)$$

$$= 0.55 \cdot (1768.0/28.0)$$

$$= 0.0087$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y(t / D_o)$$

$$= 1.6 \cdot 0.009 \cdot 2001648(28.0/1768.0)$$

$$= 441.796 \text{ kgf/cm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / F_y$$

$$= 441.8/2390.1$$

$$= 0.185$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 2390.1/4921.5$$

$$= 0.486$$

Ratio [AE]:

$$= Fhe / E$$

$$= 441.8/2001648$$

$$= 0.000221$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 2390.1(1 + 0.002) / \ln(1 + 0.002) 0.157304$$

$$= 6366.646 \text{ kgf/cm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (4921.5 \cdot \exp(0.309)) / (0.309^{0.309})$$

$$= 9631.605 \text{ kgf/cm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3



	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	2460	-1.839	0.307E-05	0.265E-05	0.791E-05	-0.620E-05	0.126E+06	0.193E-01
2	1230	-4.651	0.748E-07	0.748E-07	0.167E-05	-0.167E-05	0.154E+07	0.579E-03
3	615	-6.057	0.183E-08	0.183E-08	0.354E-06	-0.354E-06	0.199E+07	0.889E-04
4	307	-6.760	0.445E-10	0.445E-10	0.750E-07	-0.750E-07	0.200E+07	0.670E-04
5	461	-6.409	0.391E-09	0.391E-09	0.186E-06	-0.186E-06	0.200E+07	0.102E-04
6	384	-6.584	0.147E-09	0.147E-09	0.124E-06	-0.124E-06	0.200E+07	0.285E-04
7	422	-6.496	0.245E-09	0.245E-09	0.153E-06	-0.153E-06	0.200E+07	0.921E-05
8	442	-6.452	0.311E-09	0.311E-09	0.169E-06	-0.169E-06	0.200E+07	0.460E-06
9	432	-6.474	0.277E-09	0.277E-09	0.161E-06	-0.161E-06	0.200E+07	0.438E-05
10	437	-6.463	0.293E-09	0.293E-09	0.165E-06	-0.165E-06	0.200E+07	0.196E-05
11	439	-6.458	0.302E-09	0.302E-09	0.167E-06	-0.167E-06	0.200E+07	0.750E-06
12	440	-6.455	0.307E-09	0.307E-09	0.168E-06	-0.168E-06	0.200E+07	0.145E-06
13	441	-6.454	0.309E-09	0.309E-09	0.169E-06	-0.169E-06	0.200E+07	0.158E-06
14	441	-6.455	0.308E-09	0.308E-09	0.168E-06	-0.168E-06	0.200E+07	0.637E-08

Factor of Safety [FS]:

Since Fic [441.26] < 0.55 * 2390.1, FS = 2

Allowable Circumferential Compressive Stress [Fha]:

= Fic/FS

= 441.264/2.0

= 220.632 kgf/cm²

Allowable External Pressure at the given Thickness [Pa]:

= 2 • Fha(t / Do)

= 2 • 220.632(28.0/1768.0)

= 6.988 kgf/cm²

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025

Internal Pressure Results for: Shell-5

Cylindrical Shell Design Information:				
Design Pressure	P	42.397 kgf/cm²	Int. Design Temperature	130 °C
Material		SA-516 70	Ext. Design Temperature	130 °C
Length		219.500 cm	External Pressure Chart	CS-2
			UNS Number	K02700
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope) S	1407.1 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb) Sa	1407.1 kgf/cm²
Inside Diameter	Di	1706.000 mm	Actual Stress	1341.0 kgf/cm²
Spec. Min. Thk t,e, or sa		28.0000 mm	Spec. Nominal Thk.	28.000 mm
Surface Area		121503.9 cm²	SG of Contents	1.023
Element Volume		5017454.5 cm³	Weight of Contents	5094.5 kgf
Empty Weight		2594.9 kgf	Operating Weight	7689.4 kgf
			Comp. All. Ope. Stress	1181.6 kgf/cm²

Radiography Information:			
...			...
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 1708.0 \cdot (\exp(42.397/(1407.129 \cdot 1.0)) - 1) + 1.0 + 0.0$
 $= 26.1227 + 1.0 + 0.0 = 27.1227 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1407.129 \cdot 1.0 \cdot \ln((2 \cdot 27.0 + 1708.0) / 1708.0) - 0.59$
 $= 43.799 - 0.59 = 43.209 \text{ kgf/cm}^2$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 1407.129 \cdot 1.0 \cdot \ln((2 \cdot 28.0 + 1706.0) / 1706.0)$
 $= 45.4476 \text{ kgf/cm}^2$

Computed Stress Intensity at Design Pressure [Si]:
 $= P \cdot D / (E(D_o - D))$
 $= 42.397 \cdot 1708.0 / (1.0(1762.0 - 1708.0))$
 $= 1340.9990 \text{ kgf/cm}^2$

Computed Membrane Stress at Design Pressure [Sm]:
 $= P / (E \cdot \ln((2 \cdot t + D) / D))$
 $= 42.397 / (1.0 \cdot \ln((2 \cdot 27.0 + 1708.0) / 1708.0))$
 $= 1362.0854 \text{ kgf/cm}^2$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.615 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: Shell-5

Cylindrical Shell From 90 to 100 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 18483.836 / \sqrt{(881.0 \cdot 27.0)}$$

$$= 119.846$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 2 \cdot (D_o/t)^{0.94}, Ch = 0.55 / (D_o/t)$$

$$= 0.55 / (1762.0/27.0)$$

$$= 0.0084$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y(t / D_o)$$

$$= 1.6 \cdot 0.008 \cdot 2001648(27.0/1762.0)$$

$$= 413.605 \text{ kgf/cm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / F_y$$

$$= 413.6/2390.1$$

$$= 0.173$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_y / \sigma_{ts}$$

$$= 2390.1/4921.5$$

$$= 0.486$$

Ratio [AE]:

$$= Fhe / E$$

$$= 413.6/2001648$$

$$= 0.000207$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{ps} - \epsilon_{sys})) / (\ln(1 + \epsilon_{ps}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / (\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_y(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 2390.1(1 + 0.002) / \ln(1 + 0.002)^{0.157304}$$

$$= 6366.646 \text{ kgf/cm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{ts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (4921.5 \cdot \exp(0.309)) / (0.309^{0.309})$$

$$= 9631.605 \text{ kgf/cm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	2460	-1.839	0.307E-05	0.265E-05	0.791E-05	-0.620E-05	0.126E+06	0.193E-01



2	1230	-4.651	0.748E-07	0.748E-07	0.167E-05	-0.167E-05	0.154E+07	0.593E-03
3	615	-6.057	0.183E-08	0.183E-08	0.354E-06	-0.354E-06	0.199E+07	0.103E-03
4	307	-6.760	0.445E-10	0.445E-10	0.750E-07	-0.750E-07	0.200E+07	0.529E-04
5	461	-6.409	0.391E-09	0.391E-09	0.186E-06	-0.186E-06	0.200E+07	0.242E-04
6	384	-6.584	0.147E-09	0.147E-09	0.124E-06	-0.124E-06	0.200E+07	0.144E-04
7	422	-6.496	0.245E-09	0.245E-09	0.153E-06	-0.153E-06	0.200E+07	0.487E-05
8	403	-6.540	0.191E-09	0.191E-09	0.138E-06	-0.138E-06	0.200E+07	0.479E-05
9	413	-6.518	0.217E-09	0.217E-09	0.145E-06	-0.145E-06	0.200E+07	0.412E-07
10	408	-6.529	0.204E-09	0.204E-09	0.142E-06	-0.142E-06	0.200E+07	0.237E-05
11	410	-6.524	0.210E-09	0.210E-09	0.143E-06	-0.143E-06	0.200E+07	0.117E-05
12	412	-6.521	0.213E-09	0.213E-09	0.144E-06	-0.144E-06	0.200E+07	0.562E-06
13	412	-6.520	0.215E-09	0.215E-09	0.145E-06	-0.145E-06	0.200E+07	0.261E-06
14	413	-6.519	0.216E-09	0.216E-09	0.145E-06	-0.145E-06	0.200E+07	0.110E-06
15	413	-6.519	0.216E-09	0.216E-09	0.145E-06	-0.145E-06	0.200E+07	0.342E-07
16	413	-6.519	0.217E-09	0.217E-09	0.145E-06	-0.145E-06	0.200E+07	0.351E-08

Factor of Safety [FS]:

Since $Fic [413.25] < 0.55 * 2390.1$, $FS = 2$

Allowable Circumferential Compressive Stress [Fha]:

= Fic/FS

= $413.253/2.0$

= 206.627 kgf/cm^2

Allowable External Pressure at the given Thickness [Pa]:

= $2 * Fha (t / Do)$

= $2 * 206.627 (27.0/1762.0)$

= 6.332 kgf/cm^2

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025

Internal Pressure Results for: Shell-7

Cylindrical Shell Design Information:					
Design Pressure	P	42.172 kgf/cm²	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		130 °C
Length		310.000 cm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	1407.1 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1407.1 kgf/cm²
Inside Diameter	Di	1706.000 mm	Actual Stress		1333.9 kgf/cm²
Spec. Min. Thk t,e, or sa		28.0000 mm	Spec. Nominal Thk.		28.000 mm
Surface Area		171600.1 cm²	SG of Contents		1.023
Element Volume		7086154.0 cm³	Weight of Contents		7195.0 kgf
Empty Weight		3664.7 kgf	Operating Weight		10859.7 kgf
			Comp. All. Ope. Stress		1181.6 kgf/cm²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 1708.0 \cdot (\exp(42.172/(1407.129 \cdot 1.0)) - 1) + 1.0 + 0.0$
 $= 25.9823 + 1.0 + 0.0 = 26.9823 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1407.129 \cdot 1.0 \cdot \ln((2 \cdot 27.0 + 1708.0) / 1708.0) - 0.366$
 $= 43.799 - 0.366 = 43.4334 \text{ kgf/cm}^2$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 1407.129 \cdot 1.0 \cdot \ln((2 \cdot 28.0 + 1706.0) / 1706.0)$
 $= 45.4476 \text{ kgf/cm}^2$

Computed Stress Intensity at Design Pressure [Si]:
 $= P \cdot D / (E(D_o - D))$
 $= 42.172 \cdot 1708.0 / (1.0(1762.0 - 1708.0))$
 $= 1333.8999 \text{ kgf/cm}^2$

Computed Membrane Stress at Design Pressure [Sm]:
 $= P / (E \cdot \ln((2 \cdot t + D) / D))$
 $= 42.172 / (1.0 \cdot \ln((2 \cdot 27.0 + 1708.0) / 1708.0))$
 $= 1354.8746 \text{ kgf/cm}^2$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.615 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: Shell-7

Cylindrical Shell From 100 to 110 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 18483.836 / \sqrt{(881.0 \cdot 27.0)}$$

$$= 119.846$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 2 \cdot (D_o/t)^{0.94}, Ch = 0.55 / (D_o/t)$$

$$= 0.55 / (1762.0/27.0)$$

$$= 0.0084$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y(t / D_o)$$

$$= 1.6 \cdot 0.008 \cdot 2001648(27.0/1762.0)$$

$$= 413.605 \text{ kgf/cm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / F_y$$

$$= 413.6/2390.1$$

$$= 0.173$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_y / \sigma_{ts}$$

$$= 2390.1/4921.5$$

$$= 0.486$$

Ratio [AE]:

$$= Fhe / E$$

$$= 413.6/2001648$$

$$= 0.000207$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{ps} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{ps}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.157$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_y(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 2390.1(1 + 0.002) / \ln(1 + 0.002)^{0.157304}$$

$$= 6366.646 \text{ kgf/cm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.486)$$

$$= 0.309$$

Curve Fitting Constant [A2]:

$$= (\sigma_{ts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (4921.5 \cdot \exp(0.309)) / (0.309^{0.309})$$

$$= 9631.605 \text{ kgf/cm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$$

$$= 0.346$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	2460	-1.839	0.307E-05	0.265E-05	0.791E-05	-0.620E-05	0.126E+06	0.193E-01



2	1230	-4.651	0.748E-07	0.748E-07	0.167E-05	-0.167E-05	0.154E+07	0.593E-03
3	615	-6.057	0.183E-08	0.183E-08	0.354E-06	-0.354E-06	0.199E+07	0.103E-03
4	307	-6.760	0.445E-10	0.445E-10	0.750E-07	-0.750E-07	0.200E+07	0.529E-04
5	461	-6.409	0.391E-09	0.391E-09	0.186E-06	-0.186E-06	0.200E+07	0.242E-04
6	384	-6.584	0.147E-09	0.147E-09	0.124E-06	-0.124E-06	0.200E+07	0.144E-04
7	422	-6.496	0.245E-09	0.245E-09	0.153E-06	-0.153E-06	0.200E+07	0.487E-05
8	403	-6.540	0.191E-09	0.191E-09	0.138E-06	-0.138E-06	0.200E+07	0.479E-05
9	413	-6.518	0.217E-09	0.217E-09	0.145E-06	-0.145E-06	0.200E+07	0.412E-07
10	408	-6.529	0.204E-09	0.204E-09	0.142E-06	-0.142E-06	0.200E+07	0.237E-05
11	410	-6.524	0.210E-09	0.210E-09	0.143E-06	-0.143E-06	0.200E+07	0.117E-05
12	412	-6.521	0.213E-09	0.213E-09	0.144E-06	-0.144E-06	0.200E+07	0.562E-06
13	412	-6.520	0.215E-09	0.215E-09	0.145E-06	-0.145E-06	0.200E+07	0.261E-06
14	413	-6.519	0.216E-09	0.216E-09	0.145E-06	-0.145E-06	0.200E+07	0.110E-06
15	413	-6.519	0.216E-09	0.216E-09	0.145E-06	-0.145E-06	0.200E+07	0.342E-07
16	413	-6.519	0.217E-09	0.217E-09	0.145E-06	-0.145E-06	0.200E+07	0.351E-08

Factor of Safety [FS]:

Since $Fic [413.25] < 0.55 * 2390.1$, $FS = 2$

Allowable Circumferential Compressive Stress [Fha]:

= Fic/FS

= $413.253/2.0$

= 206.627 kgf/cm^2

Allowable External Pressure at the given Thickness [Pa]:

= $2 * Fha(t / Do)$

= $2 * 206.627(27.0/1762.0)$

= 6.332 kgf/cm^2

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025

Internal Pressure Results for: Top head

Elliptical Head Design Information:					
Design Pressure	P	41.850 kgf/cm²	Int. Design Temperature	130 °C	
Material		SA-516 70	Ext. Design Temperature	130 °C	
Length		5.000 cm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	1407.1 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1407.1 kgf/cm²
Inside Diameter	Di	1706.000 mm	Actual Stress	1296.4 kgf/cm²	
Spec. Min. Thk t _e , or s _a		26.0000 mm	Spec. Nominal Thk.	32.000 mm	
Surface Area		36740.5 cm²	SG of Contents	1.023	
Element Volume		764237.8 cm³	Weight of Contents	773.7 kgf	
Empty Weight		964.3 kgf	Operating Weight	1738.0 kgf	
			Comp. All. Ope. Stress	1170.0 kgf/cm²	
			Aspect Ratio	2.00	

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 21.325 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 10.662 mm

Material treatment options: +IT

Head Calculation - Section 4.3.6 (and Section 4.3.7 for Ellipsoidal Head)

Computed Minimum Required Thickness [t]:
 tr = 23.0187 mm

$$\begin{aligned}
 t &= t_r + c_i + c_o \\
 &= 23.0187 + 1.0 + 0.0 \\
 &= 24.0187 \text{ mm - see below for the derivation}
 \end{aligned}$$

For Elliptical heads: Compute [h, k, r,] per 4.3.7.1

$$\begin{aligned}
 h &= D/(2 \cdot \text{Aspect Ratio}) \\
 &= 1706.0/(2 \cdot 2.0) \\
 &= 426.500 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 k &= D/(2 \cdot h) \\
 &= 1706.0/(2 \cdot 426.5) \\
 &= 2.000
 \end{aligned}$$

$$\begin{aligned}
 r &= D(0.5/k - 0.08) + c_a \\
 &= 1706.0(0.5/2.0 - 0.08) + 1.0 \\
 &= 291.020 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 L &= D(0.44 \cdot k + 0.02) + c_a \\
 &= 1706.0(0.44 \cdot 2.0 + 0.02) + 1.0 \\
 &= 1536.400 \text{ mm}
 \end{aligned}$$

Step 3: Geometric Constants [β_{Th} , ϕ_{Th} , R_{th}]:

Equations 4.3.8 to 4.3.11 as appropriate

t (corroded) and P for the calc.: t = 23.0187 mm, P = 41.8504 kgf/cm²

Angle [β_{Th}]:

$$= \arccos[(0.5 \cdot D - r)/(L - r)]$$

$$= \arccos[(0.5 \cdot 1708.0 - 291.02)/(1536.4 - 291.02)]$$

$$= 1.1017 \text{ radians} - \text{Equation 4.3.8}$$

$$\phi_{Th} = \sqrt{(L \cdot t) / r}$$

$$= \sqrt{(1536.4 \cdot 23.019) / 291.02}$$

$$= 0.6462 \text{ radians} - \text{Equation 4.3.9}$$

$\phi_{Th} < \beta_{Th}$, Thus:

$$R_{th} = (0.5 \cdot D - r) / \cos(\beta_{Th} - \phi_{Th}) + r$$

$$= (0.5 \cdot 1708.0 - 291.02) / \cos(1.102 - 0.646) + 291.02$$

$$= 917.9252 \text{ mm} - \text{Equation 4.3.10}$$

Step 4: Coefficients: Equations 4.3.12 to 4.3.15 [C1 and C2]:

$r/D > 0.08$, Thus:

$$C1 = 0.692(r/D) + 0.605$$

$$= 0.692(291.02/1708.0) + 0.605$$

$$= 0.7229 - \text{Equation 4.3.13}$$

$$C2 = 1.46 - 2.6(r/D) = 1.46 - 2.6(291.02/1708.0)$$

$$= 1.017, \text{Equation 4.3.15}$$

Step 5: Pressure to produce knuckle elastic buckling [Peth]:

$$= C1 \cdot Et \cdot t^2 / (C2 \cdot R_{th}(0.5 \cdot R_{th} - r))$$

$$= 0.723 \cdot 2001648 \cdot 23.019^2 / (1.017 \cdot 917.925(0.5 \cdot 917.925 - 291.02))$$

$$= 4890.41 \text{ kgf/cm}^2, \text{Equation 4.3.16}$$

Step 6: Internal Pressure for max stress in knuckle equals yield [Py]:

$C3 = S_y = 2390.102 \text{ kgf/cm}^2$, Not in the creep range - 4.3.6.1(f)

$$P_y = C3 \cdot t / (C2 \cdot R_{th}(0.5 \cdot R_{th} / r - 1))$$

$$= 2390.102 \cdot 23.019 / (1.017 \cdot 917.925(0.5 \cdot 917.925 / 291.02 - 1))$$

$$= 102.125 \text{ kgf/cm}^2, \text{Equation 4.3.17}$$

Step 7: Value of Internal Press. resulting in buckling of knuckle [G and Pck]:

$$G = P_{eth}/P_y = 4890.407/102.125 = 47.886$$

$G > 1$, Thus:

Buckling Pressure [Pck]:

$$= ((0.77508 \cdot G - 0.20354 \cdot G^2 + 0.019274 \cdot G^3) /$$

$$(1 + 0.19014 \cdot G - 0.089534 \cdot G^2 + 0.0093965 \cdot G^3)) P_y$$

$$= ((0.77508 \cdot 47.886 - 0.20354 \cdot 47.886^2 + 0.019274 \cdot 47.886^3) /$$

$$(1 + 0.19014 \cdot 47.886 - 0.089534 \cdot 47.886^2 + 0.0093965 \cdot 47.886^3)) \cdot 102.125$$

$$= 205.91 \text{ kgf/cm}^2, \text{Equation 4.3.19}$$

Step 8: Compute Allowable Pressure base on buckling of knuckle [Pak]:

$$= P_{ck}/1.5 = 205.91/1.5 = 137.27 \text{ kgf/cm}^2 - \text{Equation 4.3.21}$$

Step 9: Compute allowable pressure based on rupture of crown [Pac]:

$$= 2 \cdot S \cdot E / (L/t + 0.5)$$

$$= 2 \cdot 1407.129 \cdot 1.0 / (1536.4/23.019 + 0.5)$$

$$= 41.85 \text{ kgf/cm}^2, \text{Equation 4.3.22}$$

Step 10: Compute final pressure [Pa]:

$$= \min(P_{ak}, P_{ac}) = \min(137.274, 41.85)$$

$$= 41.85 \text{ kgf/cm}^2, \text{Equation 4.3.23: (same as original design pressure)}$$

Allowable pressure in Crown at given thickness [Pac]:

$$= 2 \cdot S \cdot E / (L/t + 0.5)$$

$$= 2 \cdot 1407.129 \cdot 1.0 / (1536.4/25.0 + 0.5)$$

$$= 45.423 \text{ kgf/cm}^2$$

Computed Maximum Allowable Pressure [MAWP]:

$$= \text{MaxPressure} - P_{liq} = 45.4235 - 0.043 = 45.38 \text{ kgf/cm}^2$$

Computed Maximum Allowable Pressure New and Cold [MAPnc]:

$$= 47.2557 \text{ kgf/cm}^2$$

Computed Membrane Stress Intensity at Design Pressure [Sm]:

= 1296.4 kgf/cm²

Required Thickness of Straight Flange = 26.781 mm

Percent Fiber Elongation per Table 6.1

For Heads formed from Segments $(75 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 7.843 %
 For one Piece doubly Curved Heads $100 \cdot \ln(D_b/(D_f - 2t))$ 23.728 %

Where D_b is the blank diameter for one piece doubly curved surfaces [D_b]:

= square root($4 \cdot \text{Head Surface Area} / 3.14159$)
 = square root($4 \cdot 36740.5 / 3.14159$)
 = 2162.855 mm

MDMT Calculations in the Knuckle Portion:

Note: This Element/Detail was specified as being Impact Tested.

MDMT Calculations in the Head Straight Flange:

Note: This Element/Detail was specified as being Impact Tested.

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 21.325 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 10.662 mm

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: Top head

Elliptical Head From 110 to 120 Ext. Chart: CS-2 at 130 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Elastic Hoop Compressive Membrane Failure Stress [F_{he}]:

= $0.075 \cdot E_y(t / R_o)$
 = $0.075 \cdot 2001648(25.0/1538.227)$
 = 2439.881 kgf/cm²

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [F_r]:

= F_{he} / S_y
 = $2439.9 / 2390.1$
 = 1.021

Engineering Yield to Tensile Stress ratio [R]:

= $\sigma_{ys} / \sigma_{uts}$
 = $2390.1/4921.5$
 = 0.486

Ratio [AE]:

= F_{he} / E
 = $2439.9/2001648$
 = 0.001219

Factor [m_1]:

= $(\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$
 = $(\ln(0.486) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$
 = 0.157

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A_1]:

= $\sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m_1}$
 = $2390.1(1 + 0.002) / \ln(1 + 0.002)^{0.157304}$
 = 6366.646 kgf/cm²

Curve Fitting Exponent for, Ferritic Steel [m_2]:

= $m_2 f(m_2 f_2 - R)$
 = $0.6(1.0 - 0.486)$
 = 0.309



Curve Fitting Constant [A2]:
 $= (\sigma_{uts} \cdot \exp(m_2)) / (m_2 m_2)$
 $= (4921.5 \cdot \exp(0.309)) / (0.309 \cdot 0.309)$
 $= 9631.605 \text{ kgf/cm}^2$

Material Parameter for Stress-Strain Curve Model [K]:
 $= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$
 $= 1.5 \cdot 0.4861.5 - 0.5 \cdot 0.4862.5 - 0.4863.5$
 $= 0.346$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	2460	-1.839	0.307E-05	0.265E-05	0.791E-05	-0.620E-05	0.126E+06	0.183E-01
2	1230	-4.651	0.748E-07	0.748E-07	0.167E-05	-0.167E-05	0.154E+07	0.419E-03
3	1845	-3.245	0.657E-06	0.652E-06	0.415E-05	-0.411E-05	0.539E+06	0.220E-02
4	1537	-3.948	0.247E-06	0.247E-06	0.276E-05	-0.275E-05	0.100E+07	0.319E-03
5	1384	-4.299	0.141E-06	0.141E-06	0.218E-05	-0.218E-05	0.128E+07	0.135E-03
6	1461	-4.124	0.188E-06	0.188E-06	0.246E-05	-0.246E-05	0.114E+07	0.654E-04
7	1422	-4.211	0.163E-06	0.163E-06	0.232E-05	-0.231E-05	0.121E+07	0.407E-04
8	1441	-4.167	0.175E-06	0.175E-06	0.239E-05	-0.238E-05	0.117E+07	0.108E-04
9	1432	-4.189	0.169E-06	0.169E-06	0.235E-05	-0.235E-05	0.119E+07	0.154E-04
10	1437	-4.178	0.172E-06	0.172E-06	0.237E-05	-0.237E-05	0.118E+07	0.239E-05
11	1439	-4.173	0.173E-06	0.173E-06	0.238E-05	-0.238E-05	0.118E+07	0.417E-05
12	1438	-4.176	0.173E-06	0.172E-06	0.237E-05	-0.237E-05	0.118E+07	0.886E-06
13	1437	-4.177	0.172E-06	0.172E-06	0.237E-05	-0.237E-05	0.118E+07	0.751E-06
14	1437	-4.176	0.172E-06	0.172E-06	0.237E-05	-0.237E-05	0.118E+07	0.670E-07
15	1437	-4.177	0.172E-06	0.172E-06	0.237E-05	-0.237E-05	0.118E+07	0.342E-06
16	1437	-4.177	0.172E-06	0.172E-06	0.237E-05	-0.237E-05	0.118E+07	0.138E-06
17	1437	-4.176	0.172E-06	0.172E-06	0.237E-05	-0.237E-05	0.118E+07	0.353E-07

Factor of Safety [FS]:
 Since $Fic > 0.55 F_y$ and $Fic < F_y$
 $= 2.407 - 0.741 \cdot Fic / F_y$
 $= 2.407 - 0.741 \cdot 1437.9/2390.1$
 $= 1.961$

Allowable Circumferential Compressive Stress [Fha]:
 $= Fic/FS$
 $= 1437.9/1.961$
 $= 733.170 \text{ kgf/cm}^2$

Allowable External Pressure at the given Thickness [Pa]:
 $= 2 \cdot Fha[t/R_o]$
 $= 2 \cdot 733.17[25.0/1538.227]$
 $= 23.832 \text{ kgf/cm}^2$

Hydrostatic Test Pressure Results:

Test Pressure at high point in vessel per Equation 8.1
 $= \gamma_{St/s} \cdot MAWP(St/S)$
 $= 1.25 \cdot 43.209(1.0)$
 $= 54.011 \text{ kgf/cm}^2$

Horizontal Test performed per: $\gamma_{St/s} \cdot MAWP(St/S)$

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kgf/cm²):

From To	Stress	Allowable	Ratio	Pressure
Bottom head	1308.8	2386.0	0.549	54.18
Shell-1	1381.4	2386.0	0.579	54.18
Shell-2	1381.4	2386.0	0.579	54.18
Shell-3	1381.4	2386.0	0.579	54.18
Shell-4	1381.4	2386.0	0.579	54.18
Shell-5	1677.6	2386.0	0.703	54.18
Shell-7	1677.6	2386.0	0.703	54.18
Top head	1613.4	2386.0	0.676	54.18

Stress ratios for Nozzle and Pad Materials (kgf/cm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
D_150	Nozzle	1682.00	1505.00	1.118
MH4_600	Nozzle	1682.00	1505.00	1.118
B_150	Nozzle	1682.00	1505.00	1.118
L1_50	Nozzle	1682.00	1505.00	1.118
L2_50	Nozzle	1682.00	1505.00	1.118
L3_50	Nozzle	1682.00	1505.00	1.118
L4_50	Nozzle	1682.00	1505.00	1.118
L5_50	Nozzle	1682.00	1505.00	1.118
L6_50	Nozzle	1682.00	1505.00	1.118
DP1_50	Nozzle	1682.00	1505.00	1.118
UC2_600	Nozzle	1682.00	1505.00	1.118
MH3_600	Nozzle	1682.00	1505.00	1.118
DP2_50	Nozzle	1682.00	1505.00	1.118
UC1_600	Nozzle	1682.00	1505.00	1.118
A_150	Nozzle	1682.00	1505.00	1.118
MH2_600	Nozzle	1682.00	1505.00	1.118
E_40	Nozzle	1682.00	1505.00	1.118
MH1_600	Nozzle	1682.00	1505.00	1.118
C_100	Nozzle	1682.00	1505.00	1.118

Minimum 1.118

Stress ratios for Pressurized Vessel Elements (kgf/cm²):

Description	Ambient	Operating	Ratio
Bottom head	1407.13	1407.13	1.000
Shell-1	1407.13	1407.13	1.000
Shell-2	1407.13	1407.13	1.000
Shell-3	1407.13	1407.13	1.000
Shell-4	1407.13	1407.13	1.000
Shell-5	1407.13	1407.13	1.000
Shell-7	1407.13	1407.13	1.000
Top head	1407.13	1407.13	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (kgf/cm²):



Description	Ambient	Operating	Ratio
D_150	337.16	2422.00	0.139
MH4_600	988.56	2422.00	0.408
B_150	337.16	2422.00	0.139
L1_50	140.03	2422.00	0.058
L2_50	140.03	2422.00	0.058
L3_50	140.03	2422.00	0.058
L4_50	140.03	2422.00	0.058
L5_50	140.03	2422.00	0.058
L6_50	140.03	2422.00	0.058
DP1_50	140.03	2422.00	0.058
UC2_600	988.56	2422.00	0.408
MH3_600	988.56	2422.00	0.408
DP2_50	140.03	2422.00	0.058
UC1_600	988.56	2422.00	0.408
A_150	337.16	2422.00	0.139
MH2_600	988.57	2422.00	0.408
E_40	133.63	2422.00	0.055
MH1_600	988.57	2422.00	0.408
C_100	201.28	2422.00	0.083

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 2, 2025

External Pressure Calculations:

From	To	Section Length cm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B kgf/cm ²
10	20	600	...	...	No Calc	No Calc
20	30	600	...	...	No Calc	No Calc
30	40	600	...	...	No Calc	No Calc
40	50	No Calc	1764	26	No Calc	No Calc
50	60	1848.38	1768	28	No Calc	No Calc
60	70	1848.38	1768	28	No Calc	No Calc
70	80	1848.38	1768	28	No Calc	No Calc
80	90	1848.38	1768	28	No Calc	No Calc
90	100	1848.38	1762	27	No Calc	No Calc
100	110	1848.38	1762	27	No Calc	No Calc
110	120	No Calc	1758	25	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure kgf/cm ²	External M.A.W.P. kgf/cm ²
10	20	...	No Calc	...	No Calc
20	30	...	No Calc	...	No Calc
30	40	...	No Calc	...	No Calc
40	50	32	10.0665	1.05461	25.2562
50	60	34	20.2612	1.05461	6.98834
60	70	34	20.2612	1.05461	6.98834
70	80	34	20.2612	1.05461	6.98834
80	90	34	20.2612	1.05461	6.98834
90	100	28	15.2334	1.05461	6.33248
100	110	28	15.2334	1.05461	6.33248
110	120	26	5.07849	1.05461	23.8317

Minimum

6

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners cm	Allowable Length Bet. Stiffeners cm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	600	No Calc	No Calc	No Calc
20	30	600	No Calc	No Calc	No Calc
30	40	600	No Calc	No Calc	No Calc
40	50	No Calc	No Calc	No Calc	No Calc
50	60	1848.38	2539997	No Calc	No Calc
60	70	1848.38	2539997	No Calc	No Calc
70	80	1848.38	2539997	No Calc	No Calc
80	90	1848.38	2539997	No Calc	No Calc
90	100	1848.38	2539997	No Calc	No Calc
100	110	1848.38	2539997	No Calc	No Calc
110	120	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg	Element ID Volume cm ³	Corroded Metal Wgt. kg	Corroded ID Volume cm ³	Extra due Misc % kg
10	20	1912.99	...	1777.41	...	...
20	30	1779.39	...	1576.02	...	...
30	40	593.129	...	525.339	...	...
40	50	1145.15	756602	964.337	771925	...
50	60	4306.47	6809418	3558.77	6905891	...
60	70	4306.47	6809418	3558.77	6905891	...
70	80	4306.47	6809418	3558.77	6905891	...
80	90	5462.03	8636612	4513.71	8758972	...
90	100	2594.86	5017454	2503.63	5029225	...
100	110	3664.73	7086154	3535.88	7102778	...
110	120	964.337	764238	934.201	766795	...
Total		31036	42689312.00	27006	43147372.00	0

Weight of Details:

From	Type	Weight of Detail kg	X Offset, Dtl. Cent. cm	Y Offset, Dtl. Cent. cm	Z Offset Dtl. Cent. cm	Description
10	Wght	499.991	...	...	...	Tailing lug
20	Plat	842.623	115.128	200	-115.128	PL-8
30	Insl	81.218	...	75	...	Ins
40	Plat	2913.65	0.16154E-05	...	...	PL-1
40	Liqd	773.654	...	-21.25	...	Liquid: 40
40	Insl	138.591	...	-18.75	...	Ins: 10
40	Nozl	236.34	...	-60.9	...	D_150
40	Wght	187.997	211.8	...	...	D FY
40	Wght	9.99982	...	-40	...	D anti-swirl ba
50	Liqd	6962.89	...	150	...	Liquid: 50
50	Insl	491.028	...	150	...	Ins: 20
50	Nozl	899.132	...	80	110	MH4_600
50	Nozl	236.34	100	280	...	B_150
50	Nozl	24.8189	-97.5006	250	...	L1_50
50	Nozl	24.8189	-97.5006	20	...	L2_50
50	Nozl	24.8189	-97.5006	200	...	L3_50
50	Nozl	24.8189	-97.5006	100	...	L4_50
50	Nozl	24.8189	97.5006	200	...	L5_50
50	Nozl	24.8189	97.5006	100	...	L6_50
50	Nozl	24.8189	-97.5006	280	...	DP1_50
50	Wght	99.9982	...	80	...	MH4 Davit
50	Wght	801.986	211.8	280	...	B FY
50	Wght	9.99982	...	295	...	Bed2 sup
50	Wght	179.997	...	295	...	Bed 2 sup grid
60	Plat	1551.16	...	100	-116.14	PL-2
60	Pack	1527.07	...	150	...	Bed 1
60	Liqd	6962.89	...	150	...	Liquid: 60
60	Insl	491.028	...	150	...	Ins: 30
60	Nozl	899.132	110	100	...	UC2_600
70	Plat	1551.16	...	200	-116.14	PL-3
70	Pack	649.003	...	63.75	...	Bed 1
70	Liqd	6962.89	...	150	...	Liquid: 70
70	Insl	491.028	...	150	...	Ins: 40
70	Nozl	899.132	-110	230	...	MH3_600
70	Nozl	24.8189	97.5006	250	...	DP2_50
70	Wght	199.996	...	250	...	Distributor 2
70	Wght	9.99982	...	245	...	Dist2 sup
70	Wght	99.9982	...	230	...	MH3 Davit
70	Wght	9.99982	...	130	...	Bed2 lim sup
70	Wght	49.9991	...	...	...	Bed2 limiter
80	Plat	1552.76	...	200	-116.14	PL-4
80	Pack	1667.05	...	216.75	...	Bed-1
80	Liqd	8831.24	...	190.25	...	Liquid: 80



80	Insl	622.787	...	190.25	...	Ins: 50
80	Nozl	899.132	110	100	...	UC1_600
80	Wght	9.99982	...	45	...	Bed1 sup
80	Wght	179.997	...	50	...	Bed 1 sup grid
80	Wght	2799.95	...	200	...	Clip wt
80	Wght	4299.92	...	195	...	Misc wt 10%
90	Plat	1550.35	...	200	-116.319	PL-5
90	Pack	409.762	...	40.2501	...	Bed-1
90	Liqd	5094.51	...	109.75	...	Liquid: 90
90	Insl	358.102	...	109.75	...	Ins: 60
90	Lini	283.124	...	109.75	...	Clad
90	Nozl	80.0362	100.3	212	...	A_150
90	Nozl	891.559	...	189.5	-110.3	MH2_600
90	Wght	199.996	...	200	...	Distributor
90	Wght	9.99982	...	195	...	Distributor sup
90	Wght	9.99982	...	185	...	Bed limiter sup
90	Wght	49.9991	...	190	...	Bed limiter
90	Wght	1605.97	211.8	212	...	A FY
100	Plat	1512.62	...	100	-116.319	PL-6
100	Liqd	7194.98	...	155	...	Liquid: 100
100	Insl	505.748	...	155	...	Ins: 70
100	Lini	399.856	...	155	...	Clad
100	Nozl	24.701	100.3	90	...	E_40
100	Nozl	891.559	-110.3	155	...	MH1_600
100	Tray	1029.14	...	54.48	...	TRAY 1-3
100	Wght	119.998	...	285	...	Demister
100	Wght	49.9991	...	280	...	Demister sup
100	Wght	79.9986	...	20	...	TSR bolting bar
100	Wght	99.9982	...	155	...	MH1 Davit
100	Wght	499.991	40	210	...	Trunnion 1
100	Wght	499.991	40	210	...	Trunnion 2
110	Plat	2504.03	...	50	...	PL-7
110	Liqd	773.653	...	26.25	...	Liquid: 110
110	Insl	143.174	...	23.75	...	Ins: 90
110	Lini	90.1338	...	23.825	...	Clad
110	Nozl	48.3922	...	61.35	...	C_100
110	Wght	564.99	209.3	...	...	C FY
110	Forc	...	...	35	...	C F/M

Note: The individual tray liquid weights are listed below, but these weights are included in the tray weights above.

From	Tliq	Weight	X Offset	Y Offset
100	Tliq	0.00	0.000	54.480

Total Weight of Each Detail Type:

Platforms	13978.3
Packing	4252.9
Liquid	43556.7
Insulation	3322.7
Cladding & Overlay	773.1
Nozzles	6204.0
Trays	1029.1
Weights	13240.8

Sum of the Detail Weights 86357.6 kg

Weight Summation Results: (kg)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	31036.0	31036.0	31036.0	31036.0	31036.0	31036.0
Cladding	773.1	773.1	773.1	773.1	773.1	773.1
Nozzles	6204.0	6204.0	6204.0	6204.0	6204.0	6204.0
Wld Weights	9089.8	9089.8	9089.8	9089.8	9089.8	9089.8
Platforms	...	...	13978.3	13978.3	13978.3	13978.3
Packing	...	...	4252.9	4252.9	4252.9	4252.9



Insulation	...	...	3322.7	3322.7	3322.7	3322.7
Trays	...	...	1029.1	1029.1	1029.1	1029.1
Ope Weights	...	...	...	...	...	4150.9
Ope. Liquid	...	...	...	...	...	43556.7
Test Liquid	...	42573.9	...	...	...	...
Totals	47103.0	89676.9	69686.0	69686.0	69686.1	117393.7

Field Installation Options:

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	47103.0 kg
Shop Test Wt.	- Fabricated weight + water (full)	89676.9 kg
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	69686.0 kg
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	69686.0 kg
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	69686.1 kg
Operating Wt.	- Empty weight + operating liq. uncorroded	117393.7 kg
Field Test Wt.	- Empty weight + test medium (full)	108007.1 kg

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	54676.6 kg
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	48796.4 kg
Mass of the Upper 1/3 of the Vertical Vessel (empty)	33579.5 kg

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	110207	108448
20	30	165311	162672
30	40	55103.5	54223.9
40	50	36980.5	33997.5
50	60	166630	160221
60	70	166630	160221
70	80	166630	160221
80	90	211342	203214
90	100	121504	117642
100	110	171600	166146
110	120	36740.5	34228.5

Total	1408679.0	1361235.6 cm ²
Total	1516.3	1465.2 ft ²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kg	Total. Ele. Oper. Wgt. kg	Total. Ele. Hydro. Wgt. kg	Total Dtl. Offset Mom. kgf-m.	Oper. Wgt. No Liquid kg
10	20	2412.98	2412.98	2412.98	...	2412.98
20	30	2622.01	2622.01	2622.01	970.095	2622.01
30	40	674.347	674.347	674.347	...	674.347
40	50	4443.73	5405.38	5199.99	...	4631.72
50	60	6216.7	14161.6	13023	989.046	7198.68
60	70	8774.85	15737.7	14054.1	1801.52	8774.85
70	80	8041.6	15254.5	14198.9	1801.52	8291.6
80	90	17313.6	26324.9	24279.3	1803.38	17493.6
90	100	6177.8	13138.3	10783.2	2786.75	8043.77
100	110	9258.33	16573.3	16341.3	1759.47	9378.33
110	120	3750.07	5088.71	4513.96	...	4315.06

Empty Support Force + the Sum of the Y forces	70527.03	kgf
Operating Support Force + the Sum of the Y forces	118234.65	kgf
Hydro Support Force + the Sum of the Y forces	108944.21	kgf



Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kg	Cumulative Oper. Wgt. kg	Cumulative Hydro. Wgt. kg
10	20	73837	117394	108103
20	30	71424	114981	105690
30	40	68802	112359	103068
40	50	68127.6	111684	102394
50	60	63495.9	106279	97193.9
60	70	56297.2	92117.4	84170.9
70	80	47522.4	76379.6	70116.7
80	90	39230.8	61125.2	55917.8
90	100	21737.2	34800.3	31638.5
100	110	13693.4	21662	20855.3
110	120	4315.06	5088.71	4513.96

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. kgf-m.	Cumulative Oper. Mom. kgf-m.	Cumulative Hydro. Mom. kgf-m.
10	20	11911.8	11911.8	11911.8
20	30	11911.8	11911.8	11911.8
30	40	10941.7	10941.7	10941.7
40	50	10941.7	10941.7	10941.7
50	60	10941.7	10941.7	10941.7
60	70	9952.64	9952.64	9952.64
70	80	8151.12	8151.12	8151.12
80	90	6349.6	6349.6	6349.6
90	100	4546.22	4546.22	4546.22
100	110	1759.47	1759.47	1759.47
110	120	...	...	...

Nozzle Flange MAWP Results: (kgf/cm² & °C)

Nozzle Description	Flange Ope.	Rating Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
D_150	93.23	104.01	130	600	GR 1.1	36.554	93.225	93.225	93.225
MH4_600	46.64	52.01	130	300	GR 1.1	...	...	...	...
B_150	93.23	104.01	130	600	GR 1.1	36.554	93.225	93.225	93.225
L1_50	46.64	52.01	130	300	GR 1.1	...	...	...	...
L2_50	46.64	52.01	130	300	GR 1.1	...	...	...	...
L3_50	46.64	52.01	130	300	GR 1.1	...	...	...	...
L4_50	46.64	52.01	130	300	GR 1.1	...	...	...	...
L5_50	46.64	52.01	130	300	GR 1.1	...	...	...	...
L6_50	46.64	52.01	130	300	GR 1.1	...	...	...	...
DP1_50	46.64	52.01	130	300	GR 1.1	...	...	...	...
UC2_600	46.64	52.01	130	300	GR 1.1	...	...	...	...
MH3_600	46.64	52.01	130	300	GR 1.1	...	...	...	...
DP2_50	46.64	52.01	130	300	GR 1.1	...	...	...	...
UC1_600	46.64	52.01	130	300	GR 1.1	...	...	...	...
A_150	93.23	104.01	130	600	GR 1.1	36.554	93.225	93.225	93.225
MH2_600	46.64	52.01	130	300	GR 1.1	...	...	...	...
E_40	139.87	156.12	130	900	GR 1.1	140.477	69.326	39.869	69.326
MH1_600	46.64	52.01	130	300	GR 1.1	...	...	...	...
C_100	93.23	104.01	130	600	GR 1.1	47.119	92.719	93.225	92.719
Min Rating	46.643	52.007, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME 4.16.12

The 4.16.12 Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the 4.16.12 method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME kgf-m.	Axial Force FE kgf	MAWP PD kgf/cm ²	Maximum Pressure Pmax kgf/cm ²	Pass Fail
D_150	192.151	0.50	454	1147	43.209	93.225	Pass
B_150	192.151	0.50	454	1147	43.209	93.225	Pass
A_150	192.151	0.50	454	1147	43.209	93.225	Pass
E_40	58.801	0.50	50	382	43.209	69.326	Pass
C_100	135.001	0.50	202	765	43.209	92.719	Pass

Evaluating Flange on Nozzle [D150]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(454) + 4 \cdot \text{abs}(1147) \cdot 192.151 \leq \pi \cdot 192.151^3 \cdot [(93.225 - 43.209) + .500 \cdot 93.225]$$

$$= 0.81474E+04 \leq 0.21537E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle D150 [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(93.225, 0.500 \cdot 93.225 + 93.225 - 8147.4 / (\pi \cdot 192.151^3))$$

$$= \min(93.225, 103.284)$$

$$= 93.225 \text{ kgf/cm}^2$$

Evaluating Flange on Nozzle [B150]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(454) + 4 \cdot \text{abs}(1147) \cdot 192.151 \leq \pi \cdot 192.151^3 \cdot [(93.225 - 43.209) + .500 \cdot 93.225]$$

$$= 0.81474E+04 \leq 0.21537E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle B150 [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$



$$= \min(93.225, 0.500 \cdot 93.225 + 93.225 - 8147.4 / (\pi \cdot 192.151^3))$$

$$= \min(93.225, 103.284)$$

$$= 93.225 \text{ kgf/cm}^2$$

Evaluating Flange on Nozzle [A150]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$$

$$= 16(454) + 4 \cdot \text{abs}(1147) 192.151 \leq \pi \cdot 192.151^3 [(93.225 - 43.209) + .500 \cdot 93.225]$$

$$= 0.81474E+04 \leq 0.21537E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle A150 [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(93.225, 0.500 \cdot 93.225 + 93.225 - 8147.4 / (\pi \cdot 192.151^3))$$

$$= \min(93.225, 103.284)$$

$$= 93.225 \text{ kgf/cm}^2$$

Evaluating Flange on Nozzle [E40]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$$

$$= 16(50) + 4 \cdot \text{abs}(382) 58.801 \leq \pi \cdot 58.801^3 [(139.869 - 43.209) + .500 \cdot 139.869]$$

$$= 0.89724E+03 \leq 0.10641E+04 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle E40 [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(139.869, 0.500 \cdot 139.869 + 139.869 - 897.2 / (\pi \cdot 58.801^3))$$

$$= \min(139.869, 69.326)$$

$$= 69.326 \text{ kgf/cm}^2$$

Evaluating Flange on Nozzle [C100]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$$

$$= 16(202) + 4 \cdot \text{abs}(765) 135.001 \leq \pi \cdot 135.001^3 [(93.225 - 43.209) + .500 \cdot 93.225]$$

$$= 0.36422E+04 \leq 0.74691E+04 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle C100 [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(93.225, 0.500 \cdot 93.225 + 93.225 - 3642.2 / (\pi \cdot 135.001^3))$$

$$= \min(93.225, 92.719)$$

$$= 92.719 \text{ kgf/cm}^2$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 1.14763 Hz.

The Natural Frequency for the Vessel (Ope...) is 0.91078 Hz.

The Natural Frequency for the Vessel (Filled) is 0.96284 Hz.

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Forces/Moments Applied to Vessel (Combined w/Wind Loads)

From	To	X and Z Dir Force Res. kgf	X,Z Moment and For Res kgf-m.
10	20	...	...
20	30	...	...
30	40	...	...
40	50	...	...
50	60	...	...
60	70	...	...
70	80	...	...
80	90	...	...
90	100	...	...
100	110	...	...
110	120	792.888	499.527

Forces/Moments Applied to Vessel (Combined w/Seismic Loads)

From	To	X and Z Dir Force Res. kgf	X,Z Moment and For Res kgf-m.
10	20	...	...
20	30	...	...
30	40	...	...
40	50	...	...
50	60	...	...
60	70	...	...
70	80	...	...
80	90	...	...
90	100	...	...
100	110	...	...
110	120	792.888	499.527

User Input Forces and Moments:

From Node	Distance From	- - - - Fx	Forces - - - - Fy	- - - - Fz	- - - - Mx	Moments - - - - My	- - - - Mz
110	35.00	561.	-841.	561.	157.		157.

Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	30 cm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 cm
Distance Upwind of Crest Lh	0 cm
Distance from Crest to the Vessel x	0 cm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 cm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Dynamic Gust-Effect Factor, Operating Case [Gf]:

$$= 0.925[(1+1.7 \cdot I_z \bar{v} \sqrt{(gQ^2 + gR^2 \cdot R^2)}) / (1+1.7 \cdot gV \cdot I_z \bar{v})]$$

$$= 0.925[(1+1.7 \cdot 0.188 \sqrt{(3.4^2 + 0.914^2 + 4.167^2 \cdot 0.608^2)}) / (1+1.7 \cdot 3.4 \cdot 0.188)]$$

$$= 1.011$$

Natural Frequency of Vessel (Operating)	0.911 Hz
Natural Frequency of Vessel (Empty)	1.148 Hz
Natural Frequency of Vessel (Test)	0.963 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.634
Structure Height to Diameter ratio	13.192
Height to top of Structure	2473.250 cm

This is classified as a flexible structure. Dynamic analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:
 Exposure Category: C from Table 26.11-1
 $\alpha = 9.5$: $Zg = 27432.0$ cm

Effective Height $[z]$:
 $=$ Centroid Height + Vessel Base Elevation
 $= 100.0 + 30.0 = 130.0$ cm
 $= 4.265$ ft. Imperial Units

Velocity Pressure coefficient evaluated at height z $[Kz]$:
 Because z (4.265 ft.) $<$ 15 ft.
 $= 2.01(15 / Zg)^{2/\alpha}$
 $= 2.01(15/900.0)^{2/9.5}$
 $= 0.849$

Type of Hill: No Hill

Wind Directionality Factor $[Kd]$:
 $= 1$, per Table 26.6-1

As there is No Hill Present: $[Kzt]$:

K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$$

$$= 1.0$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 0.849 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 60.399^2)$$

$$= 16.0 \text{ psf } [78.119 \text{ kgf/m}^2]$$

Force on the first element [F]:

$$= qz \cdot Gf \cdot Cf \cdot \text{WindArea}$$

$$= 16.0 \cdot 1.011 \cdot 0.634 \cdot 49.088$$

$$= 503.9 \text{ lbs. } [228.6 \text{ kgf}]$$

Element	Hgt (z) cm	K1	K2	K3	Kz	Kzt	qz kgf/m ²
Skirt-3	130.0	0.000	0.000	0.000	0.849	1.000	78.119
Skirt 2	380.0	0.000	0.000	0.000	0.849	1.000	78.119
Skirt 1	580.0	0.000	0.000	0.000	0.892	1.000	78.119
Bottom head	632.5	0.000	0.000	0.000	0.909	1.000	78.119
Shell-1	785.0	0.000	0.000	0.000	0.951	1.000	78.119
Shell-2	1085.0	0.000	0.000	0.000	1.018	1.000	78.119
Shell-3	1385.0	0.000	0.000	0.000	1.072	1.000	78.119
Shell-4	1725.2	0.000	0.000	0.000	1.123	1.000	78.119
Shell-5	2025.3	0.000	0.000	0.000	1.161	1.000	78.119
Shell-7	2290.0	0.000	0.000	0.000	1.192	1.000	78.119
Top head	2469.9	0.000	0.000	0.000	1.211	1.000	78.119

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

Cf - Correction factor for natural frequency
 D - Average outer insulated diameter of vessel, cm
 Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
 Dr - Average outer insulated of top half of vessel, cm
 f - Natural frequency of vibration (Hertz)
 f1 - Natural frequency of bare vessel based on a unit value of $(D/L^2)(10^4)$
 L - Total height of structure cm
 Lc - Total length of conical section(s) of vessel, cm
 tb - Uncorroded plate thickness at bottom of vessel, mm
 V30 - Design Wind Speed provided by user, km/hr
 Vc - Critical wind velocity, km/hr
 Vw - Maximum wind speed at top of structure, km/hr
 W - Total corroded weight of structure, kgf
 Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. kgf
 Z - Maximum amplitude of vibration at top of vessel, mm
 D1 - Logarithmic decrement (taken as 0.03 for Welded Structures)
 Vp - Vib. Chance, <= 0.00032 (High); 0.00032 < 0.0004 (Probable)
 P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5
 :: 0.0/2420.0 = 0.0

:: $[D / L^2] \cdot 10^4 < 8$ (English Units, ft.)
 :: $[6.19/79.4^2] \cdot 10^4 = 9.82$ [Geometry Violation]

Compute the vibration possibility. If Vp > 0.0004 no chance. [Vp]:

$$= W / (L \cdot Dr^2)$$

$$= 113364 / (2420.0 \cdot 187.2^2)$$

$$= 0.13367\text{E-}02$$



Since Vp is > 0.0004 no further vibration analysis is required !

Platform Load Calculations

ID	Wind Area cm ²	Elevation cm	Pressure kgf/m ²	Force kgf	Cf
PL-8	15000.00	430.00	78.12	117.18	1.00
PL-1	30000.00	630.00	78.12	234.36	1.00
PL-2	30000.00	1035.00	78.12	234.36	1.00
PL-3	30000.00	1435.00	78.12	234.36	1.00
PL-4	30000.00	1735.00	78.12	234.36	1.00
PL-5	30000.00	2115.50	78.12	234.36	1.00
PL-6	30000.00	2235.00	78.12	234.36	1.00
PL-7	37640.00	2495.00	78.12	294.04	1.00

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation cm	Pressure kgf/m ²	Force kgf
Tailing lug	0.00	30.00	78.12	0.00
D FY	0.00	630.00	78.12	0.00
D anti-swirl	0.00	590.00	78.12	0.00
MH4 Davit	0.00	715.00	78.12	0.00
B FY	0.00	915.00	78.12	0.00
Bed2 sup	0.00	930.00	78.12	0.00
Bed 2 sup gri	0.00	930.00	78.12	0.00
Distributor 2	0.00	1485.00	78.12	0.00
Dist2 sup	0.00	1480.00	78.12	0.00
MH3 Davit	0.00	1465.00	78.12	0.00
Bed2 lim sup	0.00	1365.00	78.12	0.00
Bed2 limiter	0.00	1235.00	78.12	0.00
Bed1 sup	0.00	1580.00	78.12	0.00
Bed 1 sup gri	0.00	1585.00	78.12	0.00
Clip wt	0.00	1735.00	78.12	0.00
Misc wt 10%	0.00	1730.00	78.12	0.00
Distributor	0.00	2115.50	78.12	0.00
Distributor s	0.00	2110.50	78.12	0.00
Bed limiter s	0.00	2100.50	78.12	0.00
Bed limiter	0.00	2105.50	78.12	0.00
A FY	0.00	2127.50	78.12	0.00
Demister	0.00	2420.00	78.12	0.00
Demister sup	0.00	2415.00	78.12	0.00
TSR bolting b	0.00	2155.00	78.12	0.00
MH1 Davit	0.00	2290.00	78.12	0.00
Trunnion 1	0.00	2345.00	78.12	0.00
Trunnion 2	0.00	2345.00	78.12	0.00
C FY	0.00	2445.00	78.12	0.00

The Natural Frequency for the Vessel (Ope...) is 0.91078 Hz.

Wind Load Calculation:

From	To	Wind Height cm	Wind Diameter cm	Wind Area cm ²	Wind Pressure kgf/m ²	Element Wind Load kgf
10	20	130	228.02	45604	78.1189	251.41
20	30	380	228.02	68406	78.1189	522.971
30	40	580	238.42	23842	78.1189	131.438
40	50	632.5	250.12	1250.6	78.1189	263.667
50	60	785	250.64	75192	78.1189	414.525
60	70	1085	250.64	75192	78.1189	660.438
70	80	1385	250.64	75192	78.1189	681.624
80	90	1725.25	250.64	95368.5	78.1189	785.005
90	100	2025.25	249.86	54844.3	78.1189	571.63
100	110	2290	249.86	77456.6	78.1189	678.61



110	120	2469.92	249.34	11674.7	78.1189	391.089
-----	-----	---------	--------	---------	---------	---------

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	2.000
Table Value Fa	1.300
Table Value Fv	1.500
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	2.000
Site Class	C
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	2.500
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned} SMS &= Fa * Ss = 1.3 * 0.147 = 0.191 \\ SM1 &= Fv * S1 = 1.5 * 0.065 = 0.097 \\ SDS &= 2/3 * Sms = 2/3 * 0.191 = 0.127 \\ SD1 &= 2/3 * Sm1 = 2/3 * 0.097 = 0.065 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 $= Ct \cdot h_{nx}$ where $Ct = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 $= 0.020(80.79560.75)$
 $= 0.539$ seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 $= (1/\text{Natural Frequency}) = (1/0.911)$
 $= 1.098$

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 $= \text{Minimum Value of } (1.7 \cdot 0.539, 1.098)$ per 12.8.2
 $= 0.916$

Seismic Response Coefficient per equation 12.8-2 [Cs]:
 $= Sds / (R / I)$
 $= 0.127 / (2.0/2.0)$
 $= 0.127$

Maximum value of C_s per equation 12.8-3 [Cs]:
 $= Sd1 / (T(R / I_e))$
 $= 0.065 / (0.916(2.0/2.0))$
 $= 0.071$

Minimum value of C_s per equation 15.4-1 [Cs]:
 $= \max(0.044 \cdot SDS \cdot I_e, 0.030)$
 $= \max(0.044 \cdot 0.1274 \cdot 2.0, 0.030)$
 $= 0.030$

Total Base Shear [V]:
 $= C_s \cdot W$ (Equation 12.8-1):
 $= 0.0709 \cdot 117393.7$
 $= 8327.953$ kgf

Vertical load per 12.4-4, [Ev]:
 $= 0.2 \cdot Sds \cdot W$
 $= 0.2 \cdot 0.127 \cdot 117394$



= 2991.2 kgf

Distribute the Base shear force to each element according to the equations

$F_x = C_{vx} * V$ (eqn. 12.8-11) and the vertical distribution factor

$C_{vx} = W_x * h_x^k / (\text{Sum of } W_i * h_i^k)$ and k is an exponent which is related to the period of Vibration.

In this case, the value of k was 1.299

The Natural Frequency for the Vessel (Ope...) is 0.91078 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height cm	Earthquake Weight kgf	Element Ope Load kgf	Element Emp Load kgf	Element Vertical Load kgf
10	20	100	2412.98	5.00914	7.40738	2150.21
20	30	350	2622.01	27.7062	35.5497	2088.73
30	40	550	674.347	12.8177	15.6253	2021.92
40	50	602.5	5405.38	115.66	114.72	2004.74
50	60	755	14161.6	406.215	209.717	1867.01
60	70	1055	15737.7	697.165	440.15	1506.17
70	80	1355	15254.5	935.348	542.714	1105.17
80	90	1695.25	26324.9	2159.36	1523.97	716.487
90	100	1995.25	13138.3	1331.74	659.668	45.7297
100	110	2260	16573.3	1975.05	1146	-289.033
110	120	2417.5	5088.71	661.885	502.782	-711.321



Bending Moments due to user defined forces and moments.

User Force/Moment Shear and Bending

From	To	Distance to Support cm	Cumulative Shr Wind Cas kgf	Cumulative Shr Eqk Cas kgf	Wind Bending kgf-m.	Earthquake Bending kgf-m.
10	20	100	792.888	792.888	19647.8	19647.8
20	30	350	792.888	792.888	18062	18062
30	40	550	792.888	792.888	15683.3	15683.3
40	50	602.5	792.888	792.888	14890.5	14890.5
50	60	755	792.888	792.888	14850.8	14850.8
60	70	1055	792.888	792.888	12472.1	12472.1
70	80	1355	792.888	792.888	10093.5	10093.5
80	90	1695.25	792.888	792.888	7714.81	7714.81
90	100	1995.25	792.888	792.888	4697.87	4697.87
100	110	2260	792.888	792.888	2957.48	2957.48
110	120	2417.5	792.888	792.888	499.527	499.527

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

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The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support cm	Cumulative Wind Shear kgf	Earthquake Shear kgf	Wind Bending kgf-m.	Earthquake Bending kgf-m.
10	20	100	5352.41	8327.95	73318.8	147929
20	30	350	5101	8322.94	62865.4	131278
30	40	550	4578.03	8295.24	48346.8	106351
40	50	602.5	4446.59	8282.42	43834.5	98062.1
50	60	755	4182.92	8166.76	43618.8	97650.8
60	70	1055	3768.4	7760.55	31691.8	73759.8
70	80	1355	3107.96	7063.38	21377.2	51523.9
80	90	1695.25	2426.33	6128.03	13075.8	31736.8
90	100	1995.25	1641.33	3968.67	5337.04	12527.8
100	110	2260	1069.7	2636.94	2361.68	5278.11
110	120	2417.5	391.089	661.885	97.458	164.94

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Applied Forces Case.

Wind Load Deflection:

From	To	Cumulative Shear kgf	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	792.888	0.01725	0.068061	0.67120E-04
20	30	792.888	0.20412	0.40774	0.00015722
30	40	792.888	0.48981	0.57868	0.00018443
40	50	792.888	0.58329	0.58792	0.0001851
50	60	792.888	0.87921	1.19629	0.00021948
60	70	792.888	1.53691	1.89883	0.00024788
70	80	792.888	2.2798	2.67758	0.00027029
80	90	792.888	3.20275	3.74677	0.0002901
90	100	792.888	4.06749	4.39225	0.00029747
100	110	792.888	4.85602	5.32343	0.00030218
110	120	792.888	5.33099	5.33854	0.00030218

Allowable deflection at the Tower Top (For)(6.000"/100ft. Criteria)
 Allowable deflection : 121.000 Actual deflection : 5.339 mm

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear kgf	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	5352.41	0.065442	0.25536	0.00024898
20	30	5101	0.75303	1.47891	0.00055344
30	40	4578.03	1.76651	2.07519	0.00063774
40	50	4446.59	2.09114	2.10713	0.00063978
50	60	4182.92	3.1071	4.17987	0.00073691
60	70	3768.4	5.3142	6.4996	0.00080512
70	80	3107.96	7.72655	8.98652	0.00084924
80	90	2426.33	10.6203	12.2819	0.00087863
90	100	1641.33	13.2487	14.2197	0.00088605
100	110	1069.7	15.5951	16.9726	0.00088899
110	120	391.089	16.9948	17.0171	0.00088899

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed km/hr	2nd Crit. Wind Speed km/hr
10	20	37.281	233.006
20	30	37.281	233.006
30	40	38.9813	243.633
40	50	40.8943	255.589
50	60	40.9793	256.121
60	70	40.9793	256.121
70	80	40.9793	256.121
80	90	40.9793	256.121
90	100	40.8518	255.324
100	110	40.8518	255.324
110	120	40.7667	254.792

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
 Allowable deflection : 121.000 Actual deflection : 17.017 mm

PV Elite 28 Licensee: SL User (Modify in Config->Save Options)
FileName : 163-D1_Ope - Overfill_Rev 01 Page 80 of 245
Wind/Earthquake Deflection: Step: 22 3:21pm Feb 11,2026



HEXAGON

Total Deflection in the Operating Condition + Applied Forces:
Allowable deflection : 121.000 Actual deflection : 22.356 mm

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Longitudinal Stress Constants:

From	To	Metal Area New cm ²	Metal Area Corroded cm ²	Section Modulus New mm ³	Section Modulus Corroded mm ³
10	20	765.292	677.825	33026588	29278256
20	30	765.292	677.825	33026588	29278256
30	40	765.292	677.825	33026588	29278256
40	50	1741.19	1419.62	74051240	60786932
50	60	1852.15	1530.58	78777088	65542800
60	70	1852.15	1530.58	78777088	65542800
70	80	1852.15	1530.58	78777088	65542800
80	90	1852.15	1530.58	78777088	65542800
90	100	1525.31	1471.68	65088336	62871224
100	110	1525.31	1471.68	65088336	62871224
110	120	1414.72	1361.09	60364948	58142800



Longitudinal Allowable Stresses:

From	To	Tensile kgf/cm ²	Hydrotest Tensile kgf/cm ²	Compressive kgf/cm ²	Hydrotest Compressive kgf/cm ²
10	20	984.99	2386.01	-1028.9	-1028.9
20	30	984.99	2386.01	-1028.9	-1028.9
30	40	703.565	2386.01	-1028.9	-1028.9
40	50	1407.13	2386.01	-1175.61	-1175.61
50	60	1407.13	2386.01	-1186.47	-1186.47
60	70	1407.13	2386.01	-1186.47	-1186.47
70	80	1407.13	2386.01	-1186.47	-1186.47
80	90	1407.13	2386.01	-1186.47	-1186.47
90	100	1407.13	2386.01	-1181.56	-1181.56
100	110	1407.13	2386.01	-1181.56	-1181.56
110	120	1407.13	2386.01	-1169.96	-1169.96



Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure kgf/cm ²	Longitudinal Stress External Pressure kgf/cm ²	Longitudinal Stress Hydrotest Pressure kgf/cm ²
10	20	...	...	...
20	30	...	...	...
30	40	...	...	...
40	50	677.912	-18.1554	875.809
50	60	628.766	-16.9156	812.316
60	70	628.766	-16.9156	812.316
70	80	628.766	-16.9156	812.316
80	90	628.766	-16.9156	812.316
90	100	650.879	-17.4734	840.883
100	110	650.879	-17.4734	840.883
110	120	703.763	-18.8075	909.206

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty kgf/cm ²	Wght. Str. Operating kgf/cm ²	Wght. Str. Hydrotest kgf/cm ²	Wght. Str. Emp. Mom. kgf/cm ²	Wght. Str. Opr. Mom. kgf/cm ²
10	20	-108.932	-173.192	-159.486	40.6848	40.6848
20	30	-105.372	-169.632	-155.926	40.6848	40.6848
30	40	-101.504	-165.764	-152.057	37.3714	37.3714
40	50	-47.9901	-47.9901	-47.9901	18	18
50	60	-41.4849	-41.4849	-41.4849	16.694	16.694
60	70	-36.7817	-36.7817	-36.7817	15.185	15.185
70	80	-31.0486	-31.0486	-31.0486	12.4363	12.4363
80	90	-25.6313	-25.6313	-25.6313	9.68772	9.68772
90	100	-14.7703	-14.7703	-14.7703	7.231	7.231
100	110	-9.30461	-9.30461	-9.30461	2.79853	2.79853
110	120	-3.1703	-3.1703	-3.1703	...	...

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. kgf/cm ²	Bend. Str. Oper. Wind kgf/cm ²	Bend. Str. Oper. Equ. kgf/cm ²	Bend. Str. Hyd. Wind kgf/cm ²	Bend. Str. Hyd. Equ. kgf/cm ²
10	20	40.6848	250.421	505.253	125.21	...
20	30	40.6848	214.717	448.381	107.358	...
30	40	37.3714	165.129	363.242	82.5644	...
40	50	18	72.1117	161.321	36.0559	...
50	60	16.694	66.5501	148.988	33.275	...
60	70	15.185	48.3528	112.537	24.1764	...
70	80	12.4363	32.6157	78.6111	16.3078	...
80	90	9.68772	19.95	48.4215	9.97499	...
90	100	7.231	8.48884	19.9261	4.24442	...
100	110	2.79853	3.75638	8.39512	1.87819	...
110	120	...	0.16762	0.28368	0.083809	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case	Vortex Shedding Empty Case	Vortex Shedding Test Case	Earthquake Empty Case
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		kgf/cm ²	kgf/cm ²	kgf/cm ²	kgf/cm ²
10	20	...	...	...	313.044
20	30	...	...	...	277.56
30	40	...	...	...	224.553
40	50	...	...	...	99.6889
50	60	...	...	...	92.0676
60	70	...	...	...	69.5473
70	80	...	...	...	48.5143
80	90	...	...	...	29.7306
90	100	...	...	...	12.4115
100	110	...	...	...	5.50365
110	120	...	...	...	0.21549

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind kgf/cm ²	Longitudinal Stress Y Forces Seismic kgf/cm ²
10	20	-1.24071	3.17222
20	30	-1.24071	3.08152
30	40	-1.24071	2.98295
40	50	-0.5924	1.41217
50	60	-0.54945	1.2198
60	70	-0.54945	0.98405
70	80	-0.54945	0.72206
80	90	-0.54945	0.46812
90	100	-0.57144	0.031073
100	110	-0.57144	-0.1964
110	120	-0.61787	-0.52261

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded kgf/cm ²	Earthquake For/Mom Corroded kgf/cm ²	Wind For/Mom No Corrosion kgf/cm ²	Earthquake For/Mom No Corrosion kgf/cm ²
10	20	67.1071	67.1071	59.4908	59.4908
20	30	61.6909	61.6909	54.6893	54.6893
30	40	53.5665	53.5665	47.487	47.487
40	50	24.4962	24.4962	20.1083	20.1083
50	60	22.6582	22.6582	18.8517	18.8517
60	70	19.029	19.029	15.8322	15.8322
70	80	15.3998	15.3998	12.8127	12.8127
80	90	11.7707	11.7707	9.79322	9.79322
90	100	7.47221	7.47221	7.21769	7.21769
100	110	4.70403	4.70403	4.5438	4.5438
110	120	0.85914	0.85914	0.82751	0.82751

Bending Stresses due to Blast Load:

From	To	Bending Stress kgf/cm ²
10	20	...
20	30	...
30	40	...
40	50	...
50	60	...
60	70	...
70	80	...
80	90	...
90	100	...
100	110	...
110	120	...



Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

For Division 2, the overall stress intensity is computed in accordance with the Section VIII Division II Example problem manual PTB-3, example 4.3.6, stress due to combined loads.
 The stresses are combined per equation 4.3.44 to determine the overall stress intensity.
 The unity check is from Section 4.4.12 equations 4.4.85 and 4.4.90.

An asterisk (*) in the final column denotes overstress.

Note: Performing Combined Stress Analysis per VIII-2 paragraph 4.3.10.2.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	468.39	984.99	-468.39	1130.12	0.4755	0.4145	0.3056
20	423.71	984.99	-423.71	1130.12	0.4302	0.3749	0.2705
30	358.81	703.56	-358.81	1130.12	0.5100	0.3175	0.2184
40	163.19	1407.13	-163.19	1413.95	0.1160	0.1154	0.2401
50	147.94	1407.13	-147.94	1683.17	0.1051	0.0879	0.0629
60	119.90	1407.13	-119.90	1683.17	0.0852	0.0712	0.0491
70	92.05	1407.13	-92.05	1683.17	0.0654	0.0547	0.0359
80	67.59	1407.13	-67.59	1683.17	0.0480	0.0402	0.0246
90	38.53	1407.13	-38.53	1669.20	0.0274	0.0231	0.0139
100	21.13	1407.13	-21.13	1669.20	0.0150	0.0127	0.0096
110	4.81	1407.13	-4.81	733.17	0.0034	0.0066	...

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	526.60	984.99	-526.60	1130.12	0.5346	0.4660	0.3590



20	482.23	984.99	-482.23	1130.12	0.4896	0.4267	0.3241
30	414.01	703.56	-414.01	1130.12	0.5885	0.3663	0.2691
40	188.76	1407.13	-188.76	1415.20	0.1341	0.1334	0.2789
50	171.68	1407.13	-171.68	1683.17	0.1220	0.1020	0.0781
60	139.56	1407.13	-139.56	1683.17	0.0992	0.0829	0.0616
70	106.68	1407.13	-106.68	1683.17	0.0758	0.0634	0.0454
80	76.35	1407.13	-76.35	1683.17	0.0543	0.0454	0.0304
90	41.85	1407.13	-41.85	1669.20	0.0297	0.0251	0.0162
100	22.51	1407.13	-22.51	1669.20	0.0160	0.0135	0.0140
110	4.77	1407.13	-4.77	1397.03	0.0034	0.0034	0.0141

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	532.65	984.99	-532.65	1130.12	0.5408	0.4713	0.3056
20	487.97	984.99	-487.97	1130.12	0.4954	0.4318	0.2705
30	423.07	703.56	-423.07	1130.12	0.6013	0.3744	0.2184
40	163.19	1407.13	-163.19	1412.65	0.1160	0.1155	0.2858
50	147.94	1407.13	-147.94	1683.17	0.1051	0.0879	0.0629
60	119.90	1407.13	-119.90	1683.17	0.0852	0.0712	0.0491
70	92.05	1407.13	-92.05	1683.17	0.0654	0.0547	0.0359
80	67.59	1407.13	-67.59	1683.17	0.0480	0.0402	0.0246
90	38.53	1407.13	-38.53	1669.20	0.0274	0.0231	0.0139
100	21.13	1407.13	-21.13	1669.20	0.0150	0.0127	0.0096
110	4.81	1407.13	-4.81	733.17	0.0034	0.0066	...

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	783.06	984.99	-783.06	1130.12	0.7950	0.6929	0.5229
20	717.31	984.99	-717.31	1130.12	0.7282	0.6347	0.4698
30	616.96	703.56	-616.96	1130.12	0.8769	0.5459	0.3874
40	250.40	1407.13	-250.40	1416.64	0.1779	0.1768	0.4218
50	228.61	1407.13	-228.61	1683.17	0.1625	0.1358	0.1119
60	182.55	1407.13	-182.55	1683.17	0.1297	0.1085	0.0872
70	136.77	1407.13	-136.77	1683.17	0.0972	0.0813	0.0632
80	95.04	1407.13	-95.04	1683.17	0.0675	0.0565	0.0415
90	49.37	1407.13	-49.37	1669.20	0.0351	0.0296	0.0207
100	25.40	1407.13	-25.40	1669.20	0.0181	0.0152	0.0140
110	4.84	1407.13	-4.84	1397.10	0.0034	0.0035	0.0162

Analysis of Load Case 5 : NP+HW+HI

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	325.38	2386.01	-325.38	1028.90	0.1364	0.3162	0.1733
20	303.97	2386.01	-303.97	1028.90	0.1274	0.2954	0.1695
30	271.99	2386.01	-271.99	1028.90	0.1140	0.2644	0.1653
40	102.05	2386.01	-102.05	1175.61	0.0428	0.0868	...
50	91.45	2386.01	-91.45	1186.47	0.0383	0.0771	0.0326
60	76.14	2386.01	-76.14	1186.47	0.0319	0.0642	0.0289
70	59.79	2386.01	-59.79	1186.47	0.0251	0.0504	0.0244
80	45.29	2386.01	-45.29	1186.47	0.0190	0.0382	0.0201
90	26.25	2386.01	-26.25	1181.56	0.0110	0.0222	0.0117
100	13.98	2386.01	-13.98	1181.56	0.0059	0.0118	0.0095
110	3.25	2386.01	-3.25	1169.96	0.0014	0.0028	0.0100

Analysis of Load Case 6 : NP+HW+HE

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	200.17	2386.01	-200.17	1028.90	0.0839	0.1945	0.1733
20	196.61	2386.01	-196.61	1028.90	0.0824	0.1911	0.1695
30	189.43	2386.01	-189.43	1028.90	0.0794	0.1841	0.1653
40	65.99	2386.01	-65.99	1175.61	0.0277	0.0561	0.1863
50	58.18	2386.01	-58.18	1186.47	0.0244	0.0490	0.0326
60	51.97	2386.01	-51.97	1186.47	0.0218	0.0438	0.0289
70	43.48	2386.01	-43.48	1186.47	0.0182	0.0367	0.0244



80	35.32	2386.01	-35.32	1186.47	0.0148	0.0298	0.0201
90	22.00	2386.01	-22.00	1181.56	0.0092	0.0186	0.0117
100	12.10	2386.01	-12.10	1181.56	0.0051	0.0102	0.0074
110	3.17	2386.01	-3.17	1169.96	0.0013	0.0027	...

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	532.65	984.99	-532.65	1130.12	0.5408	0.4713	0.3056
20	487.97	984.99	-487.97	1130.12	0.4954	0.4318	0.2705
30	423.07	703.56	-423.07	1130.12	0.6013	0.3744	0.2184
40	1137.05	1407.13		1412.65	0.8081		0.2858
50	1172.21	1407.13		1683.17	0.8330		0.0629
60	1159.38	1407.13		1683.17	0.8239		0.0491
70	1147.41	1407.13		1683.17	0.8154		0.0359
80	1136.59	1407.13		1683.17	0.8077		0.0246
90	1162.79	1407.13		1669.20	0.8264		0.0139
100	1155.79	1407.13		1669.20	0.8214		0.0096
110	1123.99	1407.13		1169.96	0.7988		...

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	783.06	984.99	-783.06	1130.12	0.7950	0.6929	0.5229
20	717.31	984.99	-717.31	1130.12	0.7282	0.6347	0.4698
30	616.96	703.56	-616.96	1130.12	0.8769	0.5459	0.3874
40	1150.81	1407.13		1416.64	0.8178		0.4218
50	1187.77	1407.13		1683.17	0.8441		0.1119
60	1169.38	1407.13		1683.17	0.8310		0.0872
70	1153.03	1407.13		1683.17	0.8194		0.0632
80	1139.16	1407.13		1683.17	0.8096		0.0415
90	1163.39	1407.13		1669.20	0.8268		0.0207
100	1155.94	1407.13		1669.20	0.8215		0.0140
110	1124.00	1407.13		1169.96	0.7988		...

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	532.65	984.99	-532.65	1130.12	0.5408	0.4713	0.3056
20	487.97	984.99	-487.97	1130.12	0.4954	0.4318	0.2705
30	423.07	703.56	-423.07	1130.12	0.6013	0.3744	0.2184
40	181.35	1407.13	-181.35	1412.65	0.1289	0.1284	0.2858
50	164.85	1407.13	-164.85	1683.17	0.1172	0.0979	0.2362
60	136.81	1407.13	-136.81	1683.17	0.0972	0.0813	0.2304
70	108.97	1407.13	-108.97	1683.17	0.0774	0.0647	0.2241
80	84.50	1407.13	-84.50	1683.17	0.0601	0.0502	0.2178
90	56.01	1407.13	-56.01	1669.20	0.0398	0.0336	0.2226
100	38.61	1407.13	-38.61	1669.20	0.0274	0.0231	0.2096
110	23.62	1407.13	-23.62	1397.38	0.0168	0.0169	...

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	783.06	984.99	-783.06	1130.12	0.7950	0.6929	0.5229
20	717.31	984.99	-717.31	1130.12	0.7282	0.6347	0.4698
30	616.96	703.56	-616.96	1130.12	0.8769	0.5459	0.3874
40	268.55	1407.13	-268.55	1416.64	0.1908	0.1896	0.4218
50	245.52	1407.13	-245.52	1683.17	0.1745	0.1459	0.2519
60	199.46	1407.13	-199.46	1683.17	0.1418	0.1185	0.2389
70	153.69	1407.13	-153.69	1683.17	0.1092	0.0913	0.2281
80	111.96	1407.13	-111.96	1683.17	0.0796	0.0665	0.2191
90	66.84	1407.13	-66.84	1669.20	0.0475	0.0400	0.2221
100	42.87	1407.13	-42.87	1669.20	0.0305	0.0257	0.2088
110	23.64	1407.13	-23.64	1397.49	0.0168	0.0169	0.0037

Analysis of Load Case 11 : HP+HW+HI



From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	325.38	2386.01	-325.38	1028.90	0.1364	0.3162	0.1733
20	303.97	2386.01	-303.97	1028.90	0.1274	0.2954	0.1695
30	271.99	2386.01	-271.99	1028.90	0.1140	0.2644	0.1653
40	1156.06	2386.01		1175.61	0.4845		...
50	1203.42	2386.01		1186.47	0.5044		0.0326
60	1202.79	2386.01		1186.47	0.5041		0.0289
70	1202.30	2386.01		1186.47	0.5039		0.0244
80	1201.94	2386.01		1186.47	0.5037		0.0201
90	1453.00	2386.01		1181.56	0.6090		0.0117
100	1452.85	2386.01		1181.56	0.6089		0.0095
110	1400.75	2386.01		1169.96	0.5871		...

Analysis of Load Case 12 : HP+HW+HE

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	200.17	2386.01	-200.17	1028.90	0.0839	0.1945	0.1733
20	196.61	2386.01	-196.61	1028.90	0.0824	0.1911	0.1695
30	189.43	2386.01	-189.43	1028.90	0.0794	0.1841	0.1653
40	1149.51	2386.01		1175.61	0.4818		0.1863
50	1200.27	2386.01		1186.47	0.5030		0.0326
60	1200.54	2386.01		1186.47	0.5032		0.0289
70	1200.79	2386.01		1186.47	0.5033		0.0244
80	1201.02	2386.01		1186.47	0.5034		0.0201
90	1452.94	2386.01		1181.56	0.6089		0.0117
100	1452.84	2386.01		1181.56	0.6089		0.0074
110	1400.74	2386.01		1169.96	0.5871		...

Analysis of Load Case 13 : IP+WE+EW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	149.62	984.99	-149.62	1130.12	0.1519	0.1324	0.1191
20	146.06	984.99	-146.06	1130.12	0.1483	0.1292	0.1152
30	138.88	703.56	-138.88	1130.12	0.1974	0.1229	0.1110
40	1129.47	1407.13		744.91	0.8027		0.1143
50	1161.25	1407.13		1683.17	0.8253		0.0329
60	1152.29	1407.13		1683.17	0.8189		0.0292
70	1143.27	1407.13		1683.17	0.8125		0.0247
80	1134.41	1407.13		1683.17	0.8062		0.0204
90	1162.08	1407.13		1669.20	0.8259		0.0118
100	1155.53	1407.13		1669.20	0.8212		0.0074
110	1123.97	1407.13		1169.96	0.7988		...

Analysis of Load Case 14 : IP+WF+CW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	153.40	984.99	-153.40	1130.12	0.1557	0.1357	0.1677
20	150.24	984.99	-150.24	1130.12	0.1525	0.1329	0.1643
30	146.82	703.56	-146.82	1130.12	0.2087	0.1299	0.1606
40	1128.84	1407.13		744.91	0.8022		...
50	1159.49	1407.13		1683.17	0.8240		0.0272
60	1150.87	1407.13		1683.17	0.8179		0.0241
70	1142.26	1407.13		1683.17	0.8118		0.0204
80	1133.74	1407.13		1683.17	0.8057		0.0168
90	1161.83	1407.13		1669.20	0.8257		0.0114
100	1155.46	1407.13		1669.20	0.8211		0.0072
110	1123.98	1407.13		1169.96	0.7988		...

Analysis of Load Case 15 : IP+VO+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	213.88	984.99	-213.88	1130.12	0.2171	0.1893	0.1894
20	210.32	984.99	-210.32	1130.12	0.2135	0.1861	0.1855
30	203.14	703.56	-203.14	1130.12	0.2887	0.1797	0.1813



40	1129.47	1407.13	744.91	0.8027	0.2066
50	1161.25	1407.13	1683.17	0.8253	0.0329
60	1152.29	1407.13	1683.17	0.8189	0.0292
70	1143.27	1407.13	1683.17	0.8125	0.0247
80	1134.41	1407.13	1683.17	0.8062	0.0204
90	1162.08	1407.13	1669.20	0.8259	0.0118
100	1155.53	1407.13	1669.20	0.8212	0.0074
110	1123.97	1407.13	1169.96	0.7988	...

Analysis of Load Case 16 : IP+VE+EW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	149.62	984.99	-149.62	1130.12	0.1519	0.1324	0.1191
20	146.06	984.99	-146.06	1130.12	0.1483	0.1292	0.1152
30	138.88	703.56	-138.88	1130.12	0.1974	0.1229	0.1110
40	1129.47	1407.13		744.91	0.8027		0.1143
50	1161.25	1407.13		1683.17	0.8253		0.0329
60	1152.29	1407.13		1683.17	0.8189		0.0292
70	1143.27	1407.13		1683.17	0.8125		0.0247
80	1134.41	1407.13		1683.17	0.8062		0.0204
90	1162.08	1407.13		1669.20	0.8259		0.0118
100	1155.53	1407.13		1669.20	0.8212		0.0074
110	1123.97	1407.13		1169.96	0.7988		...

Analysis of Load Case 17 : NP+VO+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	213.88	984.99	-213.88	1130.12	0.2171	0.1893	0.1894
20	210.32	984.99	-210.32	1130.12	0.2135	0.1861	0.1855
30	203.14	703.56	-203.14	1130.12	0.2887	0.1797	0.1813
40	65.99	1407.13	-65.99	744.91	0.0469	0.0886	0.2066
50	58.18	1407.13	-58.18	1683.17	0.0413	0.0346	0.0329
60	51.97	1407.13	-51.97	1683.17	0.0369	0.0309	0.0292
70	43.48	1407.13	-43.48	1683.17	0.0309	0.0258	0.0247
80	35.32	1407.13	-35.32	1683.17	0.0251	0.0210	0.0204
90	22.00	1407.13	-22.00	1669.20	0.0156	0.0132	0.0118
100	12.10	1407.13	-12.10	1669.20	0.0086	0.0073	0.0074
110	3.17	1407.13	-3.17	733.17	0.0023	0.0043	...

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	277.81	984.99	-277.81	1130.12	0.2820	0.2458	0.1859
20	268.93	984.99	-268.93	1130.12	0.2730	0.2380	0.1821
30	253.72	703.56	-253.72	1130.12	0.3606	0.2245	0.1780
40	1130.52	1407.13		744.91	0.8034		...
50	1163.25	1407.13		1683.17	0.8267		0.0320
60	1153.79	1407.13		1683.17	0.8200		0.0284
70	1144.31	1407.13		1683.17	0.8132		0.0241
80	1135.07	1407.13		1683.17	0.8067		0.0200
90	1162.38	1407.13		1669.20	0.8261		0.0118
100	1155.66	1407.13		1669.20	0.8213		0.0076
110	1123.99	1407.13		1169.96	0.7988		...

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	277.81	984.99	-277.81	1130.12	0.2820	0.2458	0.1859
20	268.93	984.99	-268.93	1130.12	0.2730	0.2380	0.1821
30	253.72	703.56	-253.72	1130.12	0.3606	0.2245	0.1780
40	107.23	1407.13	-107.23	744.91	0.0762	0.1440	...
50	96.53	1407.13	-96.53	1683.17	0.0686	0.0574	0.2273
60	86.93	1407.13	-86.93	1683.17	0.0618	0.0516	0.2242
70	75.08	1407.13	-75.08	1683.17	0.0534	0.0446	0.2199
80	63.54	1407.13	-63.54	1683.17	0.0452	0.0377	0.2149
90	46.92	1407.13	-46.92	1669.20	0.0333	0.0281	0.2206



100	34.48	1407.13	-34.48	1669.20	0.0245	0.0207	0.2083
110	23.36	1407.13	-23.36	1399.29	0.0166	0.0167	0.0232

Analysis of Load Case 20 : BL+IP+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	213.88	984.99	-213.88	1130.12	0.2171	0.1893	0.1894
20	210.32	984.99	-210.32	1130.12	0.2135	0.1861	0.1855
30	203.14	703.56	-203.14	1130.12	0.2887	0.1797	0.1813
40	1129.47	1407.13		744.91	0.8027		0.2066
50	1161.25	1407.13		1683.17	0.8253		0.0329
60	1152.29	1407.13		1683.17	0.8189		0.0292
70	1143.27	1407.13		1683.17	0.8125		0.0247
80	1134.41	1407.13		1683.17	0.8062		0.0204
90	1162.08	1407.13		1669.20	0.8259		0.0118
100	1155.53	1407.13		1669.20	0.8212		0.0074
110	1123.97	1407.13		1169.96	0.7988		...

Absolute Maximum of the all of the Stress Ratio's 0.8769

Governing Element: Skirt 1

Governing Load Case 4 : NP+OW+EQ+FS+BS

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
Skirt-3	...	-170.0	40.7	572.4	443.0	+ Tens
Skirt-3	...	-170.0	-40.7	-572.4	-783.1	- Comp
Skirt 2	...	-166.6	40.7	510.1	384.2	+ Tens
Skirt 2	...	-166.6	-40.7	-510.1	-717.3	- Comp
Skirt 1	...	-162.8	37.4	416.8	291.4	+ Tens
Skirt 1	...	-162.8	-37.4	-416.8	-617.0	- Comp
Bottom head	...	-46.6	18.0	185.8	157.2	+ Tens
Bottom head	...	-46.6	-18.0	-185.8	-250.4	- Comp
Shell-1	...	-40.3	16.7	171.6	148.1	+ Tens
Shell-1	...	-40.3	-16.7	-171.6	-228.6	- Comp
Shell-2	...	-35.8	15.2	131.6	111.0	+ Tens
Shell-2	...	-35.8	-15.2	-131.6	-182.5	- Comp
Shell-3	...	-30.3	12.4	94.0	76.1	+ Tens
Shell-3	...	-30.3	-12.4	-94.0	-136.8	- Comp
Shell-4	...	-25.2	9.7	60.2	44.7	+ Tens
Shell-4	...	-25.2	-9.7	-60.2	-95.0	- Comp
Shell-5	...	-14.7	7.2	27.4	19.9	+ Tens
Shell-5	...	-14.7	-7.2	-27.4	-49.4	- Comp
Shell-7	...	-9.5	2.8	13.1	6.4	+ Tens
Shell-7	...	-9.5	-2.8	-13.1	-25.4	- Comp
Top head	...	-3.7	...	1.1	...	+ Tens
Top head	...	-3.7	...	-1.1	-4.8	- Comp

Additional Results per Section 4.4, see 4.4.15

Nomenclature :

- L - Section Length cm
- Fxa - Allowable comp mem stress due to compression with lamc <= 0.15, kgf/cm²
- Fba - Allowable all comp mem stress due to bending moment, kgf/cm²
- Fva - Allowable shear stress when only shear is present, kgf/cm²
- Fha - Allowable hoop comp memb stress under ext pressure alone, kgf/cm²



Fxha - Allowable axial comp w/hoop compression and lamc <= 0.15, kgf/cm²
 Fhxa - Allowable hoop comp w/axial compression and lamc <= 0.15, kgf/cm²
 Faha - Allowable axial comp w/hoop compression and lamc > 0.15, kgf/cm²
 Fbha - Allowable axial comp from bending w/hoop compression, kgf/cm²
 Fhba - Allowable hoop comp w/long compression due to moment, kgf/cm²

Allowable Stress Results for Case 1 : NP+EW+1.1WI+FW+BW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.834	6000	1130	1172	707		70307	1130	70307	70307	70307
20	0.834	6000	1130	1172	707		70307	1130	70307	70307	70307
30	0.834	6000	1130	1172	707		70307	1130	70307	70307	70307
40						1413	70307		70307	70307	70307
50	0.987	18483	1683	1683	830	220	70307	1683	70307	70307	70307
60	0.987	18483	1683	1683	830	220	70307	1683	70307	70307	70307
70	0.987	18483	1683	1683	830	220	70307	1683	70307	70307	70307
80	0.987	18483	1683	1683	830	220	70307	1683	70307	70307	70307
90	0.987	18483	1669	1669	818	206	70307	1669	70307	70307	70307
100	0.987	18483	1669	1669	818	206	70307	1669	70307	70307	70307
110						733	70307		70307	70307	70307

Allowable Stress Results for Case 2 : NP+EW+EE+FS+BS

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.834	6000	1130	1172	707		70307	1130	70307	70307	70307
20	0.834	6000	1130	1172	707		70307	1130	70307	70307	70307
30	0.834	6000	1130	1172	707		70307	1130	70307	70307	70307
40						1415	70307		70307	70307	70307
50	0.987	18483	1683	1683	830	220	70307	1683	70307	70307	70307
60	0.987	18483	1683	1683	830	220	70307	1683	70307	70307	70307
70	0.987	18483	1683	1683	830	220	70307	1683	70307	70307	70307
80	0.987	18483	1683	1683	830	220	70307	1683	70307	70307	70307
90	0.987	18483	1669	1669	818	206	70307	1669	70307	70307	70307
100	0.987	18483	1669	1669	818	206	70307	1669	70307	70307	70307
110						1397	70307		70307	70307	70307

Allowable Stress Results for Case 3 : NP+OW+1.1WI+FW+BW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.834	6000	1130	1172	707		70307	1130	70307	70307	70307
20	0.834	6000	1130	1172	707		70307	1130	70307	70307	70307
30	0.834	6000	1130	1172	707		70307	1130	70307	70307	70307
40						1412	70307		70307	70307	70307
50	0.987	18483	1683	1683	830	220	70307	1683	70307	70307	70307
60	0.987	18483	1683	1683	830	220	70307	1683	70307	70307	70307
70	0.987	18483	1683	1683	830	220	70307	1683	70307	70307	70307
80	0.987	18483	1683	1683	830	220	70307	1683	70307	70307	70307
90	0.987	18483	1669	1669	818	206	70307	1669	70307	70307	70307
100	0.987	18483	1669	1669	818	206	70307	1669	70307	70307	70307
110						733	70307		70307	70307	70307

Allowable Stress Results for Case 4 : NP+OW+EQ+FS+BS

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.834	6000	1130	1172	707		70307	1130	70307	70307	70307
20	0.834	6000	1130	1172	707		70307	1130	70307	70307	70307
30	0.834	6000	1130	1172	707		70307	1130	70307	70307	70307
40						1416	70307		70307	70307	70307
50	0.987	18483	1683	1683	830	220	70307	1683	70307	70307	70307
60	0.987	18483	1683	1683	830	220	70307	1683	70307	70307	70307
70	0.987	18483	1683	1683	830	220	70307	1683	70307	70307	70307
80	0.987	18483	1683	1683	830	220	70307	1683	70307	70307	70307
90	0.987	18483	1669	1669	818	206	70307	1669	70307	70307	70307
100	0.987	18483	1669	1669	818	206	70307	1669	70307	70307	70307
110						1397	70307		70307	70307	70307

Allowable Stress Results for Case 5 : NP+HW+HI

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.823	6000	1131	1173	712		70307	1131	70307	70307	70307
20	0.823	6000	1131	1173	712		70307	1131	70307	70307	70307
30	0.823	6000	1131	1173	712		70307	1131	70307	70307	70307
40						749	70307		70307	70307	70307
50	0.976	18483	1691	1691	833	226	70307	1691	70307	70307	70307
60	0.976	18483	1691	1691	833	226	70307	1691	70307	70307	70307



70	0.976	18483	1691	1691	833	226	70307	1691	70307	70307	70307
80	0.976	18483	1691	1691	833	226	70307	1691	70307	70307	70307
90	0.976	18483	1678	1678	822	212	70307	1678	70307	70307	70307
100	0.976	18483	1678	1678	822	212	70307	1678	70307	70307	70307
110						737	70307		70307	70307	70307

Allowable Stress Results for Case 6 : NP+HW+HE

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.823	6000	1131	1173	712		70307	1131	70307	70307	70307
20	0.823	6000	1131	1173	712		70307	1131	70307	70307	70307
30	0.823	6000	1131	1173	712		70307	1131	70307	70307	70307
40						749	70307		70307	70307	70307
50	0.976	18483	1691	1691	833	226	70307	1691	70307	70307	70307
60	0.976	18483	1691	1691	833	226	70307	1691	70307	70307	70307
70	0.976	18483	1691	1691	833	226	70307	1691	70307	70307	70307
80	0.976	18483	1691	1691	833	226	70307	1691	70307	70307	70307
90	0.976	18483	1678	1678	822	212	70307	1678	70307	70307	70307
100	0.976	18483	1678	1678	822	212	70307	1678	70307	70307	70307
110						737	70307		70307	70307	70307

Allowable Stress Results for Case 7 : IP+OW+1.1WI+FW+BW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.834	6000	1130	1172	707		70307	1130	70307	70307	70307
20	0.834	6000	1130	1172	707		70307	1130	70307	70307	70307
30	0.834	6000	1130	1172	707		70307	1130	70307	70307	70307
40						1412	70307		70307	70307	70307
50	0.987	18483	1683	1683	830	220	70307	1683	70307	70307	70307
60	0.987	18483	1683	1683	830	220	70307	1683	70307	70307	70307
70	0.987	18483	1683	1683	830	220	70307	1683	70307	70307	70307
80	0.987	18483	1683	1683	830	220	70307	1683	70307	70307	70307
90	0.987	18483	1669	1669	818	206	70307	1669	70307	70307	70307
100	0.987	18483	1669	1669	818	206	70307	1669	70307	70307	70307
110							70307		70307	70307	70307

Allowable Stress Results for Case 8 : IP+OW+EQ+FS+BS

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.834	6000	1130	1172	707		70307	1130	70307	70307	70307
20	0.834	6000	1130	1172	707		70307	1130	70307	70307	70307
30	0.834	6000	1130	1172	707		70307	1130	70307	70307	70307
40						1416	70307		70307	70307	70307
50	0.987	18483	1683	1683	830	220	70307	1683	70307	70307	70307
60	0.987	18483	1683	1683	830	220	70307	1683	70307	70307	70307
70	0.987	18483	1683	1683	830	220	70307	1683	70307	70307	70307
80	0.987	18483	1683	1683	830	220	70307	1683	70307	70307	70307
90	0.987	18483	1669	1669	818	206	70307	1669	70307	70307	70307
100	0.987	18483	1669	1669	818	206	70307	1669	70307	70307	70307
110							70307		70307	70307	70307

Allowable Stress Results for Case 9 : EP+OW+1.1WI+FW+BW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.834	6000	1130	1172	707		70307	1130	70307	70307	70307
20	0.834	6000	1130	1172	707		70307	1130	70307	70307	70307
30	0.834	6000	1130	1172	707		70307	1130	70307	70307	70307
40						1412	70307		70307	70307	70307
50	0.987	18483	1683	1683	830	220	389	223	289	663	208
60	0.987	18483	1683	1683	830	220	358	224	266	526	212
70	0.987	18483	1683	1683	830	220	319	224	237	390	215
80	0.987	18483	1683	1683	830	220	283	224	210	270	217
90	0.987	18483	1669	1669	818	206	193	209	143	138	205
100	0.987	18483	1669	1669	818	206	159	209	118	67	206
110						1397	70307		70307	70307	70307

Allowable Stress Results for Case 10 : EP+OW+EQ+FS+BS

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.834	6000	1130	1172	707		70307	1130	70307	70307	70307
20	0.834	6000	1130	1172	707		70307	1130	70307	70307	70307
30	0.834	6000	1130	1172	707		70307	1130	70307	70307	70307
40						1416	70307		70307	70307	70307
50	0.987	18483	1683	1683	830	220	377	224	280	1069	189
60	0.987	18483	1683	1683	830	220	347	224	258	882	200



70	0.987	18483	1683	1683	830	220	311	224	231	666	208
80	0.987	18483	1683	1683	830	220	276	224	205	449	214
90	0.987	18483	1669	1669	818	206	189	209	140	205	204
100	0.987	18483	1669	1669	818	206	157	209	117	95	205
110						1397	70307		70307	70307	70307

Allowable Stress Results for Case 11 : HP+HW+HI

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.823	6000	1131	1173	712		70307	1131	70307	70307	70307
20	0.823	6000	1131	1173	712		70307	1131	70307	70307	70307
30	0.823	6000	1131	1173	712		70307	1131	70307	70307	70307
40						749	70307		70307	70307	70307
50	0.976	18483	1691	1691	833	226	70307	1691	70307	70307	70307
60	0.976	18483	1691	1691	833	226	70307	1691	70307	70307	70307
70	0.976	18483	1691	1691	833	226	70307	1691	70307	70307	70307
80	0.976	18483	1691	1691	833	226	70307	1691	70307	70307	70307
90	0.976	18483	1678	1678	822	212	70307	1678	70307	70307	70307
100	0.976	18483	1678	1678	822	212	70307	1678	70307	70307	70307
110							70307		70307	70307	70307

Allowable Stress Results for Case 12 : HP+HW+HE

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.823	6000	1131	1173	712		70307	1131	70307	70307	70307
20	0.823	6000	1131	1173	712		70307	1131	70307	70307	70307
30	0.823	6000	1131	1173	712		70307	1131	70307	70307	70307
40						749	70307		70307	70307	70307
50	0.976	18483	1691	1691	833	226	70307	1691	70307	70307	70307
60	0.976	18483	1691	1691	833	226	70307	1691	70307	70307	70307
70	0.976	18483	1691	1691	833	226	70307	1691	70307	70307	70307
80	0.976	18483	1691	1691	833	226	70307	1691	70307	70307	70307
90	0.976	18483	1678	1678	822	212	70307	1678	70307	70307	70307
100	0.976	18483	1678	1678	822	212	70307	1678	70307	70307	70307
110							70307		70307	70307	70307

Allowable Stress Results for Case 13 : IP+WE+EW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.834	6000	1130	1172	707		70307	1130	70307	70307	70307
20	0.834	6000	1130	1172	707		70307	1130	70307	70307	70307
30	0.834	6000	1130	1172	707		70307	1130	70307	70307	70307
40						744	70307		70307	70307	70307
50	0.987	18483	1683	1683	830	220	70307	1683	70307	70307	70307
60	0.987	18483	1683	1683	830	220	70307	1683	70307	70307	70307
70	0.987	18483	1683	1683	830	220	70307	1683	70307	70307	70307
80	0.987	18483	1683	1683	830	220	70307	1683	70307	70307	70307
90	0.987	18483	1669	1669	818	206	70307	1669	70307	70307	70307
100	0.987	18483	1669	1669	818	206	70307	1669	70307	70307	70307
110							70307		70307	70307	70307

Allowable Stress Results for Case 14 : IP+WF+CW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.834	6000	1130	1172	707		70307	1130	70307	70307	70307
20	0.834	6000	1130	1172	707		70307	1130	70307	70307	70307
30	0.834	6000	1130	1172	707		70307	1130	70307	70307	70307
40						744	70307		70307	70307	70307
50	0.987	18483	1683	1683	830	220	70307	1683	70307	70307	70307
60	0.987	18483	1683	1683	830	220	70307	1683	70307	70307	70307
70	0.987	18483	1683	1683	830	220	70307	1683	70307	70307	70307
80	0.987	18483	1683	1683	830	220	70307	1683	70307	70307	70307
90	0.987	18483	1669	1669	818	206	70307	1669	70307	70307	70307
100	0.987	18483	1669	1669	818	206	70307	1669	70307	70307	70307
110							70307		70307	70307	70307

Allowable Stress Results for Case 15 : IP+VO+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.834	6000	1130	1172	707		70307	1130	70307	70307	70307
20	0.834	6000	1130	1172	707		70307	1130	70307	70307	70307
30	0.834	6000	1130	1172	707		70307	1130	70307	70307	70307
40						744	70307		70307	70307	70307
50	0.987	18483	1683	1683	830	220	70307	1683	70307	70307	70307
60	0.987	18483	1683	1683	830	220	70307	1683	70307	70307	70307



70	0.987	18483	1683	1683	830	220	70307	1683	70307	70307	70307
80	0.987	18483	1683	1683	830	220	70307	1683	70307	70307	70307
90	0.987	18483	1669	1669	818	206	70307	1669	70307	70307	70307
100	0.987	18483	1669	1669	818	206	70307	1669	70307	70307	70307
110							70307		70307	70307	70307

Allowable Stress Results for Case 16 : IP+VE+EW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.834	6000	1130	1172	707		70307	1130	70307	70307	70307
20	0.834	6000	1130	1172	707		70307	1130	70307	70307	70307
30	0.834	6000	1130	1172	707		70307	1130	70307	70307	70307
40						744	70307		70307	70307	70307
50	0.987	18483	1683	1683	830	220	70307	1683	70307	70307	70307
60	0.987	18483	1683	1683	830	220	70307	1683	70307	70307	70307
70	0.987	18483	1683	1683	830	220	70307	1683	70307	70307	70307
80	0.987	18483	1683	1683	830	220	70307	1683	70307	70307	70307
90	0.987	18483	1669	1669	818	206	70307	1669	70307	70307	70307
100	0.987	18483	1669	1669	818	206	70307	1669	70307	70307	70307
110							70307		70307	70307	70307

Allowable Stress Results for Case 17 : NP+VO+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.834	6000	1130	1172	707		70307	1130	70307	70307	70307
20	0.834	6000	1130	1172	707		70307	1130	70307	70307	70307
30	0.834	6000	1130	1172	707		70307	1130	70307	70307	70307
40						744	70307		70307	70307	70307
50	0.987	18483	1683	1683	830	220	70307	1683	70307	70307	70307
60	0.987	18483	1683	1683	830	220	70307	1683	70307	70307	70307
70	0.987	18483	1683	1683	830	220	70307	1683	70307	70307	70307
80	0.987	18483	1683	1683	830	220	70307	1683	70307	70307	70307
90	0.987	18483	1669	1669	818	206	70307	1669	70307	70307	70307
100	0.987	18483	1669	1669	818	206	70307	1669	70307	70307	70307
110						733	70307		70307	70307	70307

Allowable Stress Results for Case 18 : FS+BS+IP+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.834	6000	1130	1172	707		70307	1130	70307	70307	70307
20	0.834	6000	1130	1172	707		70307	1130	70307	70307	70307
30	0.834	6000	1130	1172	707		70307	1130	70307	70307	70307
40						744	70307		70307	70307	70307
50	0.987	18483	1683	1683	830	220	70307	1683	70307	70307	70307
60	0.987	18483	1683	1683	830	220	70307	1683	70307	70307	70307
70	0.987	18483	1683	1683	830	220	70307	1683	70307	70307	70307
80	0.987	18483	1683	1683	830	220	70307	1683	70307	70307	70307
90	0.987	18483	1669	1669	818	206	70307	1669	70307	70307	70307
100	0.987	18483	1669	1669	818	206	70307	1669	70307	70307	70307
110							70307		70307	70307	70307

Allowable Stress Results for Case 19 : FS+BS+EP+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.834	6000	1130	1172	707		70307	1130	70307	70307	70307
20	0.834	6000	1130	1172	707		70307	1130	70307	70307	70307
30	0.834	6000	1130	1172	707		70307	1130	70307	70307	70307
40						744	70307		70307	70307	70307
50	0.987	18483	1683	1683	830	220	377	224	280	149	219
60	0.987	18483	1683	1683	830	220	347	224	258	125	219
70	0.987	18483	1683	1683	830	220	311	224	231	101	219
80	0.987	18483	1683	1683	830	220	276	224	205	77	219
90	0.987	18483	1669	1669	818	206	189	209	140	44	206
100	0.987	18483	1669	1669	818	206	157	209	117	28	206
110						1399	70307		70307	70307	70307

Allowable Stress Results for Case 20 : BL+IP+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.834	6000	1130	1172	707		70307	1130	70307	70307	70307
20	0.834	6000	1130	1172	707		70307	1130	70307	70307	70307
30	0.834	6000	1130	1172	707		70307	1130	70307	70307	70307
40						744	70307		70307	70307	70307
50	0.987	18483	1683	1683	830	220	70307	1683	70307	70307	70307
60	0.987	18483	1683	1683	830	220	70307	1683	70307	70307	70307



70	0.987	18483	1683	1683	830	220	70307	1683	70307	70307	70307
80	0.987	18483	1683	1683	830	220	70307	1683	70307	70307	70307
90	0.987	18483	1669	1669	818	206	70307	1669	70307	70307	70307
100	0.987	18483	1669	1669	818	206	70307	1669	70307	70307	70307
110							70307		70307	70307	70307

Note:

Load cases involving Wind and Seismic are assumed to cause a net section bending moment. These load case components are: EQ, EE, WI, WE, VO, VE, VF, HE, HI, WF. Static bending moments due to the weights of nozzles etc. are considered to be local in nature and will not cause a net section bending moment. This is an important consideration when computing the unity check per section 4.4.12 equations 4.4.86, 4.4.87, 4.4.90, 4.4.91 and 4.4.92.

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Skirt Support Analysis Per EN 13445-3 Section 16.12:

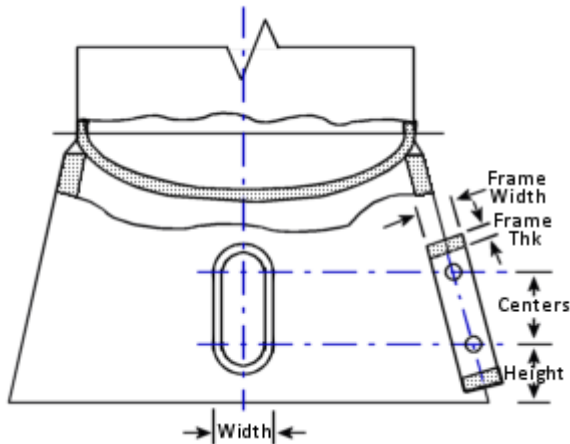
The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

The access opening(s) is/are in section 4-4 only. The other calculations are for areas where no access openings exist. Please be aware of this when you review the results.

The skirt comprises more than one piece. If this is the case, the Openings will only be considered in this Section of the Skirt! Otherwise, make the whole skirt all one piece.

Item: Skirt 2

Total vessel (empty) weight	Fg	69551.477 kgf
Weight of vessel contents (liquid)	Ff	43556.695 kgf
Vessel Gross Weight	Fg + Ff	113108.172 kgf
Skirt Mean Diameter	Dz	1740.000 mm
Skirt Analysis Thickness (Corroded)	ez	12.400 mm
Skirt Height	H	3000.000 mm
Additional Axial Force at section 1-1	F1	-840.982 kgf
Moment at section 1-1	M1	122034.242 kgf-m.
Applied Moment - Bottom of Skirt	Mb	149340.203 kgf-m.
Weight at Bottom of Skirt	W	114980.680 kgf



Skirt Access Opening Input Data:

Access #	Diameter mm	Layout Angle deg	Cntr. Dist. From Bottom mm	Obround Cntr-Cntr mm	Frame Width mm	Frame Thickness mm
1	300.0000	0.0	2000.00	0.00	80.000	16.000

Section properties at least metal area [A]:

Number of Openings At Critical Height = 1

Access No	Layout Angle 1 Deg.	Layout Angle 2 Deg.
1	-9.9282	9.9282

The points for integrating Ixx, Iyy are as follows:

FROM THETA1 Deg	TO THETA2 Deg
-350.072	-9.928

Height of Weakest Section	H	2000.009	mm
Rotation of Principle Axes	a	-0.000	deg.
Centroid Position in X direction	g	-15.653	mm
Centroid Position in Y direction	h	0.000	mm
Principle Moment of Inertia	Ixx	2567597.750	cm ⁴
Principle Moment of Inertia	Iyy	2471591.500	cm ⁴
Distance to Extreme Fiber	ybar	876.200	mm
Distance to Extreme Fiber	xbar	891.853	mm
Minimum Section Modulus	W4	27713024.000	mm ³
Metal Area at the Cutting Plane	A4	666.043	cm ²

Half Opening Width Subtended Angle at Weakest Section [δ]:

$$= \sin^{-1}(\text{Largest Opening Width} / \text{Dz})$$

$$= \sin^{-1}(300.0/1740.0)$$

$$= 0.1733 \text{ rad (9.9282 deg)}$$

Skirt Mean Radius at Section 4-4 [r]:

$$= \text{Dz} / 2$$

$$= 1740.0/2$$

$$= 870.0 \text{ mm}$$

This calculation of ratio below is per an earlier version of EN 13445.

Opening ratio at weakest section: EN 13445 equation 16.12-76 [ratio]:

$$= \delta \cdot \sqrt{(r / e_z)}$$

$$= 0.173 \cdot \sqrt{(870.0/12.4)}$$

$$= 1.451$$

Ok as ratio is < 2 - See EN 13445 16.12.76(a).

Analysis of the Section at Weakest Section Modulus [Section 4-4]:

Skirt Corroded Thickness [ez]:

$$= \text{ex}(\text{new}) - \text{ci} - \text{cext}$$

$$= 14.0 - 0.8 - 0.8$$

$$= 12.4 \text{ mm}$$

Find the moment acting at the Critical Height [M4]:

$$= (\text{Mbottom} - \text{Mtop}) \cdot (\text{H} - \text{CutHeight}) / \text{H} + \text{Mtop}$$

$$= (149340.188 - 122034.242) \cdot (3000.0 - 2000.009) / 3000.0 + 122034.242$$

$$= 131136.141 \text{ kgf-m.}$$

Compute the section modulus [Zxx and Zyy]:

$$\text{Zxx} = \text{Ixx}/\text{ybar} = 2567598/876.2 = 29303812 \text{ mm}^3$$

$$\text{Zyy} = \text{Iyy}/\text{xbar} = 2471592/891.853 = 27713024 \text{ mm}^3$$

$$\text{Z} = \min(\text{Zxx}, \text{Zyy}) = \min(29303812, 27713024) = 27713024 \text{ mm}^3$$

The balance of the calculations are per EN 13445-3, and they may be treated as academic.

Radius of Gyration of Skirt at 4-4 [rg]:

$$= \sqrt{(\min(\text{Ixx}, \text{Iyy}) / \text{Area}(\text{A4}))}$$

$$= \sqrt{(\min(2567598, 2471592) / 666.043)}$$

$$= 609.168 \text{ mm}$$

Note: rg is not included in the analysis.

Skirt Slenderness Ratio [rg/L]:

$$= \text{rg} / \text{L}$$

$$= 609.168/3000.0$$

$$= 0.203$$

Location of the Skirt Centroid at critical height [e]:

$$= \sqrt{(\text{ex}^2 + \text{ey}^2)}$$

$$= \sqrt{(-15.653^2 + 0.^2)}$$

$$= 0.000 \text{ mm}$$

Where $\text{ex} = -15.653$: $\text{ey} = 0$.

Moment Increment at Centroid [ΔM4]:

$$\begin{aligned}
 &= \text{abs}(F1 - Ff - Fg) \cdot e = F4 \cdot e \\
 &= \text{abs}(-840.98 - 69551.48 - 43556.7) \cdot 15.653 \\
 &= 1783.633 \text{ kgf-m.}
 \end{aligned}$$

Check of the Skirt in Regions with Openings, per 16.12.4.3:

Protrusion, Figure 16.12-5 [ht]:

$$\begin{aligned}
 &= \text{abs}(lt - ea3) / 2 \\
 &= \text{abs}(80.0 - 12.4) / 2 \\
 &= 33.800 \text{ mm}
 \end{aligned}$$

Mean Diameter of Opening, Figure 16.12-5 [d]:

$$\begin{aligned}
 &= \text{opening width} - et \\
 &= 300.0 - 16.0 \\
 &= 284.000 \text{ mm}
 \end{aligned}$$

Angle, 16.12-72 [δ]:

$$\begin{aligned}
 &= \sin^{-1}(d / D3) \\
 &= \sin^{-1}(284.0 / 1740.0) \\
 &= 0.164 \text{ radians}
 \end{aligned}$$

Area in the Frame [At]:

$$\begin{aligned}
 &= 2 \cdot lt \cdot et \\
 &= 2 \cdot 80.0 \cdot 16.0 \\
 &= 25.600 \text{ cm}^2
 \end{aligned}$$

Cross Sectional Shell Area at Opening, 16.12-73(b) [AS]:

$$\begin{aligned}
 &= (\pi - \delta) \cdot D3 \cdot ea3 \\
 &= (\pi - 0.164) \cdot 1740.0 \cdot 12.4 \\
 &= 642.456 \text{ cm}^2
 \end{aligned}$$

Combined Area at Opening, 16.12-73 [A4]:

$$\begin{aligned}
 &= AS + AT \\
 &= 642.456 + 25.6 \\
 &= 668.056 \text{ cm}^2
 \end{aligned}$$

Computed Offset, 16.12-74(b) [yt]:

$$\begin{aligned}
 &= 0.5 \cdot D3 \cdot \cos(\delta) + ht - 0.5 \cdot lt \\
 &= 0.5 \cdot 1740.0 \cdot \cos(0.164) + 33.8 - 0.5 \cdot 80.0 \\
 &= 852.133 \text{ mm}
 \end{aligned}$$

Distance to Neutral Axis, 16.12-74 [yG]:

$$\begin{aligned}
 &= (0.5 \cdot D3 \cdot ea3 \cdot d - 2 \cdot lt \cdot eat \cdot yt) / A4 \\
 &= (0.5 \cdot 1740.0 \cdot 12.4 \cdot 284.0 - 2 \cdot 80.0 \cdot 16.0 \cdot 852.133) / 668.056 \\
 &= 13.207 \text{ mm}
 \end{aligned}$$

Computed Offset [ymax]:

$$\begin{aligned}
 &= \max[\text{abs}(0.5 \cdot d3 \cdot \cos(\delta) + ht + yG); \text{abs}(0.5 \cdot D3 - yG)] \\
 &= \max[\text{abs}(0.5 \cdot 1740.0 \cdot \cos(0.164) + 33.8 + 13.207); \text{abs}(0.5 \cdot 1740.0 - 13.207)] \\
 &= 905.341 \text{ mm}
 \end{aligned}$$

Inertia Value, 16.12-75(b) [IS]:

$$\begin{aligned}
 &= (\pi - \delta - \sin(\delta) \cdot \cos(\delta)) \cdot ea3 \cdot (D3/2)^3 \\
 &= (\pi - 0.164 - \sin(0.164) \cdot \cos(0.164)) \cdot 12.4 \cdot (1740.0/2)^3 \\
 &= 2299884.2 \text{ cm}^4
 \end{aligned}$$

Section Modulus [W4]:

$$\begin{aligned}
 &= (Is + At(yt^2 + lt^2/12) - A4 \cdot yG^2) / ymax \\
 &= (2299884 + 25.6(852.133^2 + 80.0^2/12) - 668.056 \cdot 13.207^2) / 905.341 \\
 &= 27445438.000 \text{ mm}^3
 \end{aligned}$$

Weakening Factor, 16.12-70(b) [psi1]:

$$\begin{aligned}
 &= \min(1; A4 / (\pi \cdot D3 \cdot ea3)) \\
 &= \min(1; 668.056 / (\pi \cdot 1740.0 \cdot 12.4)) \\
 &= 0.986
 \end{aligned}$$

Weakening Factor, 16.12-70(c) [psi2]:

$$\begin{aligned}
 &= \min(1; 4 \cdot W4 / (\pi \cdot D^2 / 3 / (\pi \cdot D3 \cdot ea3))) \\
 &= \min(1; 4.0 \cdot 27445438 / (\pi \cdot 1740.0^2 \cdot 12.4)) \\
 &= 0.931
 \end{aligned}$$



Maximum Bending Moment [Mmax]:

$$\begin{aligned} &= \pi/4 \cdot d^3 \cdot e a^3 \cdot \sigma_{call} \\ &= \pi/4 \cdot 1740.0^2 \cdot 12.4 \cdot 1028.9 \\ &= 303377.8 \text{ kgf-m.} \end{aligned}$$

Maximum Compressive Force [Fc,max]:

$$\begin{aligned} &= \pi \cdot d^3 \cdot e a^3 \cdot \sigma_{call} \\ &= \pi \cdot 1740.0 \cdot 12.4 \cdot 1028.9 \\ &= 697419.4 \text{ kgf} \end{aligned}$$

Unity Check of Weakened Section, 16.12-70 [UC1]:

$$\begin{aligned} &= F4 / (\psi_{i1} \cdot F_{c,max}) + (\text{abs}(M4) + F4 \cdot \text{abs}(y_q)) / (\psi_{i2} \cdot M_{max}) \\ &= 113949.1 / (0.986 \cdot 697419) + (\text{abs}(131136.1) + 113949.1 \cdot \text{abs}(13.207)) / \\ &\quad (0.931 \cdot 303377.8) \\ &= 0.635 \text{ must be } \leq \text{ to } 1, \text{ Passed} \end{aligned}$$

The stresses are satisfactory in the region with the opening(s).

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Skirt Support Analysis Per EN 13445-3 Section 16.12:

The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

The access opening(s) is/are in section 4-4 only. The other calculations are for areas where no access openings exist. Please be aware of this when you review the results.

Item: Skirt 1

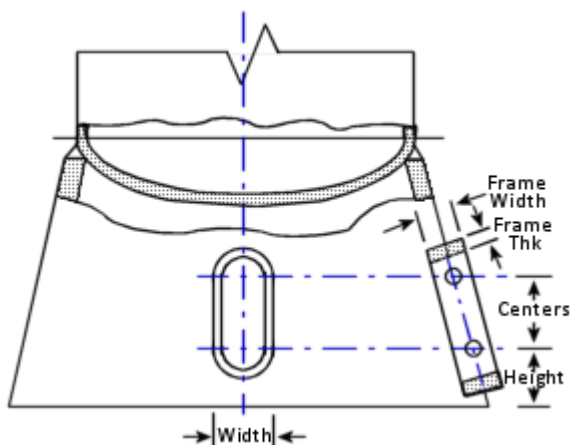
Total vessel (empty) weight	Fg	69551.477 kgf
Head Weight below section 2-2	DeltaFg	4631.723 kgf
Weight of vessel contents (liquid)	Ff	43556.695 kgf
Vessel Gross Weight	Fg + Ff	113108.172 kgf
Skirt Mean Diameter	Dz	1740.000 mm
Skirt Analysis Thickness (Corroded)	ez	12.400 mm
Skirt Height	H	1000.000 mm
Additional Axial Force at section 1-1	F1	-840.982 kgf
Moment at section 1-1	M1	112952.500 kgf-m.
Applied Moment - Bottom of Skirt	Mb	122034.242 kgf-m.
Weight at Bottom of Skirt	W	112358.672 kgf

Structural Type B Skirt per EN 13445 Figure 16.12-2:

Type of Head:	Elliptical
Head Mean Diameter at Skirt Top	Db 1738.000 mm
Head Corroded Thickness	eb 26.000 mm

Per EN 13445 Section 7.5.4 [R, r]:

$$\begin{aligned}
 h_i &= 431.0 \text{ mm} \\
 K &= D_i / (2 * h_i) = 1712.0 / (2 * 431.0) = 1.9861 \\
 R &= D_i (0.44 * K + 0.02) = 1712.0 * (0.44 * 1.986 + 0.02) = 1530.3136 \text{ mm} \\
 r &= D_i ((0.5 / K) - 0.08) = 1712.0 * ((0.5 / 1.986) - 0.08) = 294.04 \text{ mm}
 \end{aligned}$$



Skirt Access Opening Input Data:

Access #	Diameter mm	Layout Angle deg	Cntr. Dist. From Bottom mm	Obround Cntr-Cntr mm	Frame Width mm	Frame Thickness mm
1	50.0000	0.0	800.00	0.00	80.000	16.000
2	50.0000	90.0	800.00	0.00	80.000	16.000
3	50.0000	180.0	800.00	0.00	80.000	16.000
4	50.0000	270.0	800.00	0.00	80.000	16.000

Interference check:



Access	Access	Pitch mm	Ligament mm
1	2	1366.5699	1316.5699
2	3	1366.5699	1316.5699
3	4	1366.5699	1316.5699
1	4	1366.5699	1316.5699

Interference Check: No Overlap Detected between Openings

Minimum Pitch between openings is: 1366.5699 mm
 Minimum Ligament between Openings is: 1316.5699 mm

Recommended minimum Ligament [lMin]:
 $= 2 \cdot \sqrt{(D \cdot t)}$
 $= 2 \cdot \sqrt{(1740.0 \cdot 12.4)} \text{ mm}$
 $= 293.775 \text{ mm}$

Carefully evaluate this acceptance criteria.

Section properties at least metal area [A]:

Number of Openings At Critical Height = 4

Access No	Layout Angle 1 Deg.	Layout Angle 2 Deg.
1	-1.6467	1.6467
2	88.3533	91.6467
3	178.3533	181.6467
4	268.3533	271.6467

The points for integrating lxx, lyy are as follows:

FROM THETA1 Deg	TO THETA2 Deg
1.647	88.353
91.647	178.353
181.647	268.353
-88.353	-1.647

Height of Weakest Section	H	799.970	mm
Rotation of Principle Axes	a	-1.077	deg.
Centroid Position in X direction	g	-0.000	mm
Centroid Position in Y direction	h	0.000	mm
Principle Moment of Inertia	Ixx	2859021.250	cm ⁴
Principle Moment of Inertia	Iyy	2859021.750	cm ⁴
Distance to Extreme Fiber	ybar	876.200	mm
Distance to Extreme Fiber	xbar	876.200	mm
Minimum Section Modulus	W4	32629808.000	mm ³
Metal Area at the Cutting Plane	A4	755.427	cm ²

Half Opening Width Subtended Angle at Weakest Section [δ]:
 $= \sin^{-1}(\text{Largest Opening Width} / D_z)$
 $= \sin^{-1}(50.0/1740.0)$
 $= 0.0287 \text{ rad (1.6467 deg)}$

Skirt Mean Radius at Section 4-4 [r]:
 $= D_z / 2$
 $= 1740.0/2$
 $= 870.0 \text{ mm}$

Because the Code uses the words 'may be' in the evaluation below, the results should be considered optional.

Check areas for openings per EN 13445 Equation 16.12-76 [A,check]:

At,i	Adelta,i	Status	lt,i <= 8*eat,i
------	----------	--------	-----------------

25.6	4.2	...	...
25.6	4.2	...	...
25.6	4.2	...	...
25.6	4.2	...	...

This calculation of ratio below is per an earlier version of EN 13445.

Opening ratio at weakest section: EN 13445 equation 16.12-76 [ratio]:
 $= \delta \cdot \sqrt{(r / e_z)}$
 $= 0.029 \cdot \sqrt{(870.0/12.4)}$
 $= 0.241$

Ok as ratio is < 2 - See EN 13445 16.12.76(a).

Analysis of the Section at Weakest Section Modulus [Section 4-4]:

Skirt Corroded Thickness [ez]:
 $= ex(new) - ci - cext$
 $= 14.0 - 0.8 - 0.8$
 $= 12.4 \text{ mm}$

Find the moment acting at the Critical Height [M4]:
 $= (M_{bottom} - M_{top}) \cdot (H - CutHeight) / H + M_{top}$
 $= (122034.242 - 112952.5) \cdot (1000.0 - 799.97) / 1000.0 + 112952.5$
 $= 114769.125 \text{ kgf-m.}$

Compute the section modulus [Zxx and Zyy]:
 $Z_{xx} = I_{xx} / y_{bar} = 2859021 / 876.2 = 32629808 \text{ mm}^3$
 $Z_{yy} = I_{yy} / x_{bar} = 2859022 / 876.2 = 32629816 \text{ mm}^3$
 $Z = \min(Z_{xx}, Z_{yy}) = \min(32629808, 32629816) = 32629808 \text{ mm}^3$

The balance of the calculations are per EN 13445-3, and they may be treated as academic.

Radius of Gyration of Skirt at 4-4 [rg]:
 $= \sqrt{(\min(I_{xx}, I_{yy}) / Area(A4))}$
 $= \sqrt{(\min(2859021, 2859022) / 755.427)}$
 $= 615.195 \text{ mm}$

Note: rg is not included in the analysis.

Skirt Slenderness Ratio [rg/L]:
 $= rg / L$
 $= 615.195 / 1000.0$
 $= 0.615$

Location of the Skirt Centroid at critical height [e]:
 $= \sqrt{ex^2 + ey^2}$
 $= \sqrt{(-0.0^2 + 0.0^2)}$
 $= 15.653 \text{ mm}$

Where $ex = -0.0$: $ey = 0.0$

Moment Increment at Centroid [$\Delta M4$]:
 $= \text{abs}(F1 - Ff - Fg) \cdot e = F4 \cdot e$
 $= \text{abs}(-840.98 - 69551.48 - 43556.7) \cdot 0.0$
 $= 0.000 \text{ kgf-m.}$

16.12.3.4 Checking at connection areas:

Force at the joint, 16.12-1 [Fzp]:
 $= F1 - Fg - Ff + 4 \cdot M1 / Dz$
 $= -840.982 - 69551.477 - 43556.695 + 4 \cdot 112952.5 / 1740.0$
 $= 145711.5 \text{ kgf}$

Force at the joint, 16.12-2 [Fzq]:
 $= F1 - Fg - Ff - 4 \cdot M1 / Dz$
 $= -840.982 - 69551.477 - 43556.695 - 4 \cdot 112952.5 / 1740.0$
 $= -373609.8 \text{ kgf}$

16.12.3.4.1 Membrane stresses:

Membrane stress at 1-1, 16.12-3 & 4:

$$\begin{aligned} \text{Stress } [\sigma_{1pm}]: \\ &= (F_{zp} + \Delta F_g) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b) \\ &= (145711.5 + 4631.723 + 43556.695) / (\pi \cdot 1738.0 \cdot 26.0) + 41.807 \cdot 1738.0 / (4 \cdot 26.0) \\ &= 835.24 \text{ kgf/cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{1qm}]: \\ &= (F_{zq} + \Delta F_q + F_f) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b) \\ &= (-0.4E+06 + 4631.723 + 43556.695) / (\pi \cdot 1738.0 \cdot 26.0) + 41.807 \cdot 1738.0 / (4 \cdot 26.0) \\ &= 469.43 \text{ kgf/cm}^2 \end{aligned}$$

Membrane stress at 2-2, 16.12-9:

$$\begin{aligned} \text{Stress } [\sigma_{2pm} \text{ \& } \sigma_{2qm}]: \\ &= (F_f + \Delta F_g) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b) \\ &= (43556.695 + 4631.723) / (\pi \cdot 1738.0 \cdot 26.0) + 41.807 \cdot 1738.0 / (4 \cdot 26.0) \\ &= 732.60 \text{ kgf/cm}^2 \end{aligned}$$

Membrane stress at 3-3, 16.12 12 & 13:

$$\begin{aligned} \text{Stress } [\sigma_{3pm}]: \\ &= F_{zp} / (\pi \cdot D_z \cdot e_z) \\ &= 145711.5 / (\pi \cdot 1740.0 \cdot 12.4) \\ &= 214.97 \text{ kgf/cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{3qm}]: \\ &= F_{zq} / (\pi \cdot D_z \cdot e_z) \\ &= -373609.781 / (\pi \cdot 1740.0 \cdot 12.4) \\ &= -551.19 \text{ kgf/cm}^2 \end{aligned}$$

16.12.3.4.2 Bending stresses:

$$\begin{aligned} \text{Cosine of angle, 16.12-25 } [\cos(\gamma)]: \\ &= 1 - 0.5(D_b + e_b - D_z + e_z) / (r + e_b) \\ &= 1 - 0.5(1738.0 + 26.0 - 1740.0 + 12.4) / (294.04 + 26.0) \\ &= 0.9431 \end{aligned}$$

$$\gamma = \arccos(\cos(\gamma)) = \arccos(0.943) = 19.416 \text{ deg}$$

$$\begin{aligned} \text{Dimension } [a]: \\ &= 0.5 \cdot \sqrt{e_b^2 + e_z^2 + 2 \cdot e_b \cdot e_z \cdot \cos(\gamma)} \\ &= 0.5 \cdot \sqrt{26.0^2 + 12.4^2 + 2 \cdot 26.0 \cdot 12.4 \cdot 0.943} \\ &= 18.96 \text{ mm} \end{aligned}$$

Bending moment at verification points p and q [Mp & Mq]:

$$\begin{aligned} M_p &= a \cdot F_{zp} = 18.96 \cdot 145711.5 = 2762.66 \text{ kgf-m.} \\ M_q &= a \cdot F_{zq} = 18.96 \cdot -373609.8 = -7083.56 \text{ kgf-m.} \end{aligned}$$

$$\begin{aligned} \text{Correction factor } [C]: \\ &= 0.63 - 0.057(e_b/e_z)^2 \\ &= 0.63 - 0.057(26.0/12.4)^2 \\ &= 0.379 \end{aligned}$$

Bending stresses in sections 1-1 to 3-3 at the outer surface (o), 16.12-26 to 29:

$$\begin{aligned} \text{Stress } [\sigma_{1pb(o)} \text{ \& } \sigma_{2pb(o)}]: \\ &= C \cdot 6 \cdot M_p / (\pi \cdot D_b \cdot e_b^2) \\ &= 0.379 \cdot 6 \cdot 2762.66 / (\pi \cdot 1738.0 \cdot 26.0^2) \\ &= 170.39 \text{ kgf/cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{1qb(o)} \text{ \& } \sigma_{2qb(o)}]: \\ &= C \cdot 6 \cdot M_q / (\pi \cdot D_b \cdot e_b^2) \\ &= 0.379 \cdot 6 \cdot -7083.564 / (\pi \cdot 1738.0 \cdot 26.0^2) \\ &= -436.87 \text{ kgf/cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{3pb(o)}]: \\ &= C \cdot 6 \cdot M_p / (\pi \cdot D_z \cdot e_z^2) \\ &= 0.379 \cdot 6 \cdot 2762.66 / (\pi \cdot 1740.0 \cdot 12.4^2) \end{aligned}$$



$$= 748.23 \text{ kgf/cm}^2$$

Stress [$\sigma_{3qb(o)}$]:

$$\begin{aligned} &= C \cdot 6 \cdot M_q / (\pi \cdot D_z \cdot e_z^2) \\ &= 0.379 \cdot 6 \cdot -7083.564 / (\pi \cdot 1740.0 \cdot 12.4^2) \\ &= -1918.49 \text{ kgf/cm}^2 \end{aligned}$$

Geometry variables, 16.12-32 to 36:

$$\begin{aligned} eb/Db &= 1.024/68.425 = 0.015 \\ y &= 125 \cdot eb/Db = 125 \cdot 26.0/1738.0 = 1.87 \end{aligned}$$

As this an elliptical head, it will be treated a a Korbogen type head

Geometric value [α]:

$$= 4.2 - 0.2 \cdot y = 4.2 - 0.2 \cdot 1.87 = 3.826$$

Bending stress in sections 1-1 and 2-2 Equation 16.12-31 [$\sigma_{1b(p)}$ & $\sigma_{2b(p)}$]:

$$\begin{aligned} &= (P + Ph) \cdot Db ((y / y_A) \cdot \alpha - 1) / (4 \cdot eb) \\ &= (41.807 + 0.0) \cdot 1738.0 ((19.42/40.0) \cdot 3.826 - 1) / (4 \cdot 26.0) \\ &= 598.83 \text{ kgf/cm}^2 \end{aligned}$$

16.12.3.4.3 Total stresses and strength conditions:

Structural shape B and C:

Stress [σ_{1pitot}]:

$$\begin{aligned} &= \sigma_{1pm} - \sigma_{1pb(o)} + \sigma_{1b(p)} \\ &= 835.244 - 170.385 + 598.826 \\ &= 1263.68 \text{ kgf/cm}^2 \end{aligned}$$

Stress [σ_{1potot}]:

$$\begin{aligned} &= \sigma_{1pm} + \sigma_{1pb(o)} - \sigma_{1b(p)} \\ &= 835.244 + 170.385 - 598.826 \\ &= 406.80 \text{ kgf/cm}^2 \end{aligned}$$

Stress [σ_{1qitot}]:

$$\begin{aligned} &= \sigma_{1qm} - \sigma_{1qb(o)} + \sigma_{1b(p)} \\ &= 469.427 - -436.874 + 598.826 \\ &= 1505.13 \text{ kgf/cm}^2 \end{aligned}$$

Stress [σ_{1qotot}]:

$$\begin{aligned} &= \sigma_{1qm} + \sigma_{1qb(o)} - \sigma_{1b(p)} \\ &= 469.427 + -436.874 - 598.826 \\ &= -566.27 \text{ kgf/cm}^2 \end{aligned}$$

Stress [σ_{2pitot}]:

$$\begin{aligned} &= \sigma_{2pm} + \sigma_{2pb(o)} + \sigma_{2b(p)} \\ &= 732.603 + 170.385 + 598.826 \\ &= 1501.81 \text{ kgf/cm}^2 \end{aligned}$$

Stress [σ_{2potot}]:

$$\begin{aligned} &= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)} \\ &= 732.603 - 170.385 - 598.826 \\ &= -36.61 \text{ kgf/cm}^2 \end{aligned}$$

Stress [σ_{2qitot}]:

$$\begin{aligned} &= \sigma_{2qm} + \sigma_{2qb(o)} + \sigma_{2b(p)} \\ &= 732.603 + -436.874 + 598.826 \\ &= 894.55 \text{ kgf/cm}^2 \end{aligned}$$

Stress [σ_{2qotot}]:

$$\begin{aligned} &= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)} \\ &= 732.603 - -436.874 - 598.826 \\ &= 570.65 \text{ kgf/cm}^2 \end{aligned}$$

Stress [σ_{3pitot}]:

$$\begin{aligned} &= \sigma_{3pm} - \sigma_{3pb} \\ &= 214.968 - 748.231 \\ &= -533.26 \text{ kgf/cm}^2 \end{aligned}$$



Stress [σ_{3potot}]:
 $= \sigma_{3pm} + \sigma_{3pb}$
 $= 214.968 + 748.231$
 $= 963.20 \text{ kgf/cm}^2$

Stress [σ_{3qitot}]:
 $= \sigma_{3qm} - \sigma_{3qb}$
 $= -551.186 - -1918.492$
 $= 1367.31 \text{ kgf/cm}^2$

Stress [σ_{3qotot}]:
 $= \sigma_{3qm} + \sigma_{3qb}$
 $= -551.186 + -1918.492$
 $= -2469.68 \text{ kgf/cm}^2$

Allowable stresses:

For σ_{1pitot} and σ_{1potot} , the stress must not exceed:
 $= fs(3 - (1 / 1.5)) \cdot (\sigma_{1pm} / f)^2$
 $= 1407.129(3 - (1/1.5)) \cdot (835.244/1407.129)^2$
 $= 3890.86 \text{ kgf/cm}^2$

For σ_{1qitot} and σ_{1qotot} , the stress must not exceed:
 $= fs(3 - (1 / 1.5)) \cdot (\sigma_{1qm} / f)^2$
 $= 1407.129(3 - (1/1.5)) \cdot (469.427/1407.129)^2$
 $= 4116.98 \text{ kgf/cm}^2$

For σ_{2pitot} and σ_{2potot} , the stress must not exceed:
 $= fs(3 - (1 / 1.5)) \cdot (\sigma_{2pm} / f)^2$
 $= 1407.129(3 - (1/1.5)) \cdot (732.603/1407.129)^2$
 $= 3967.11 \text{ kgf/cm}^2$

For σ_{2qitot} and σ_{2qotot} , the stress must not exceed:
 $= fs(3 - (1 / 1.5)) \cdot (\sigma_{2qm} / f)^2$
 $= 1407.129(3 - (1/1.5)) \cdot (732.603/1407.129)^2$
 $= 3967.11 \text{ kgf/cm}^2$

For σ_{3pitot} & σ_{3potot} , the stress must not exceed:
 $= fs(3 - (1 / 1.5)) \cdot (\sigma_{3pm} / f)^2$
 $= 1407.129(3 - (1/1.5)) \cdot (214.968/1407.129)^2$
 $= 4199.49 \text{ kgf/cm}^2$

For σ_{3qitot} & σ_{3qotot} , the stress must not exceed:
 $= fs(3 - (1 / 1.5)) \cdot (\sigma_{3qm} / f)^2$
 $= 1407.129(3 - (1/1.5)) \cdot (-551.186/1407.129)^2$
 $= 4077.45 \text{ kgf/cm}^2$

Check of the Skirt in Regions with Openings, per 16.12.4.3:

Weakening Factor, 16.12-70(b) [psi1]:
 $= \min(1; A4 / (\pi \cdot D3 \cdot ea3))$
 $= \min(1; 755.427 / (\pi \cdot 1740.0 \cdot 12.4))$
 $= 1.000$

Weakening Factor, 16.12-70(c) [psi2]:
 $= \min(1; 4 \cdot W4 / (\pi \cdot D^2 / 3 / (\pi \cdot D3 \cdot ea3)))$
 $= \min(1; 4.0 \cdot 32629816 / (\pi \cdot 1740.0^2 \cdot 12.4))$
 $= 1.000$

Maximum Bending Moment [Mmax]:
 $= \pi / 4 \cdot d3^2 \cdot ea3 \cdot \sigma_{call}$
 $= \pi / 4 \cdot 1740.0^2 \cdot 12.4 \cdot 1028.9$
 $= 303377.8 \text{ kgf-m}$

Maximum Compressive Force [Fc,max]:
 $= \pi \cdot d3 \cdot ea3 \cdot \sigma_{call}$
 $= \pi \cdot 1740.0 \cdot 12.4 \cdot 1028.9$
 $= 697419.4 \text{ kgf}$

Unity Check of Weakened Section, 16.12-70 [UC1]:



$$= F4 / (\psi_1 \cdot F_{c,max}) + (\text{abs}(M_4) + F_4 \cdot \text{abs}(y_q)) / (\psi_2 \cdot M_{max})$$

$$= 113949.1 / (1.0 \cdot 697419) + (\text{abs}(114769.1) + 113949.1 \cdot \text{abs}(0.0)) /$$

$$(1.0 \cdot 303377.8)$$

$$= 0.542 \text{ must be } \leq \text{ to } 1, \text{ Passed}$$

Please note: None of the following stresses in the table below are in the region where there is an access opening.

Summary of Stresses: (kgf/cm²)

Stress	Actual	Allowed	Result	Equation
Stot1pi	1263.68	3890.86	Pass	46
Stot1po	406.80	3890.86	Pass	47
Stot1qi	1505.13	4116.98	Pass	48
Stot1qo	-566.27	4116.98	Pass	49
Stot2pi	1501.81	3967.11	Pass	50
Stot2po	-36.61	3967.11	Pass	51
Stot2qi	894.55	3967.11	Pass	52
Stot2qo	570.65	3967.11	Pass	53
Stot3pi	-533.26	4199.49	Pass	54
Stot3po	963.20	4199.49	Pass	55
Stot3qi	1367.31	4077.45	Pass	56
Stot3qo	-2469.68	4077.45	Pass	57

The stresses are satisfactory in the region without openings.
 The stresses are satisfactory in the region with the opening(s).

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Shop/Field Installation Options :

Platform(s) installed in the Shop.
Packing is installed in the Shop.
Trays are installed in the Shop.
Insulation is installed in the Shop.
Lining is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Platforms	1517.490 cm
Center of Gravity of the Packing	1432.868 cm
Center of Gravity of the Liquid	1510.000 cm
Center of Gravity of the Insulation	1487.884 cm
Center of Gravity of the Lining	2183.894 cm
Center of Gravity of the Nozzles	1445.206 cm
Center of Gravity of the Trays	2159.480 cm
Center of Gravity of the Added Weights (Operating)	1690.368 cm
Center of Gravity of the Added Weights (Empty)	1672.707 cm
Center of Gravity of Bare Shell New and Cold	1305.161 cm
Center of Gravity of Bare Shell Corroded	1324.078 cm
Vessel CG in the Operating Condition	1491.102 cm
Vessel CG in the Fabricated (Shop/Empty) Condition	1447.035 cm
Vessel CG in the Test Condition	1471.497 cm

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Skirt Data :

Skirt Outside Diameter at Base	SOD, D6	1754.0001	mm
Skirt Thickness	STHK	14.0000	mm
Skirt Internal Corrosion Allowance	SCA	0.8000	mm
Skirt External Corrosion Allowance		0.8000	mm
Skirt Material		SA-516 70	[Impact Tested]

Basing Input: Type of Geometry: Continuous Top Ring W/Gussets

Thickness of Basing	TBA, ea4	50.0000	mm
Design Temperature of the Basing		38.00	°C
Basing Material		SA-516 70	
Basing Ope. All. Stress	BASOPE, f4	1784.41	kgf/cm²
Basing Yield Stress		2671.52	kgf/cm²
Inside Diameter of Basing	DI, D4	1610.0000	mm
Outside Diameter of Basing	DOU, D5	2010.0000	mm
Nominal Diameter of Bolts	BND, d(b0)	36.0000	mm
Bolt Corrosion Allowance	BCA	1.5000	mm
Bolt Material		SA-193 B7	
Bolt Material Class/Thickness Range		<= 64	
Bolt Operating Allowable Stress	SA, fB	1407.16	kgf/cm²
Number of Bolts	RN, nb	20	
Diameter of Bolt Circle	DC, DBC	1882.0000	mm
Bolt Allowable Shear Stress		844.296	kgf/cm²
Ultimate Comp. Strength of Concrete	FPC	418.1	kgf/cm²
Allowable Comp. Strength of Concrete	FC	226.4	kgf/cm²
Modular ratio Steel/Concrete		9.833	
Thickness of Gusset Plates	TGA, e7	16.0000	mm
Width of Gussets at Top Plate	TWDT	118.0000	mm
Width of Gussets at Base Plate	BWDT	118.0000	mm
Gusset Plate Elastic Modulus	E	2057170.1	kgf/cm²
Gusset Plate Yield Stress	SY	2671.5	kgf/cm²
Height of Gussets	HG, h1	168.0000	mm
Distance between Gussets	RG, b6	76.0000	mm
Dist. from Bolt Center to Gusset (Rg/2)	CG	38.0000	mm
Number of Gussets per bolt	NG	2	
Thickness of Top Plate or Ring	TTA, e5	32.0000	mm
Radial Width of the Top Plate	TOPWTH, b5	128.0000	mm
Anchor Bolt Hole Dia. in Top Plate	dia,H	50.0000	mm
External Corrosion Allowance	CA	6.0000	mm
Dead Weight of Vessel	DW	70527.0	kgf
Operating Weight of Vessel	ROW	118234.6	kgf
Earthquake Moment on Basing	EQMOM	155009.9	kgf-m.
Wind Moment on Basing	WIMOM	80399.6	kgf-m.
User Moment at the Base [Earthquake Case]		19647.8	kgf-m. (UEQMOM)
User Moment at the Base [WindLoad Case]		19647.8	kgf-m. (UWIMOM)
Percent Bolt Preload	ppl	100.0	
Use AISC A5.2 Increase in Fc and Bolt Stress		No	
Use Allowable Weld Stress per AISC J2.5		No	
Factor for Increase of Allowables	Fact	1.0000	

Results for Brownell and Young Basing Analysis : Analyze Option

Note: This analysis is based on Neutral Axis shift method for Steel on Concrete (or a material with significantly different Young's modulus).

Calculation of Bolt Loads for Multiple Load Conditions:

Bolt Load, Earthquake Operating Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 174658/1882.0) - 115243)/20$$

$$= 12798.7 \text{ kgf}$$

Bolt Load, Windload Empty Case:

$$= ((4 \cdot M/DC) - W) / nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 100047/1882.0) - 70527)/20$$

$$= 7105.7 \text{ kgf}$$

Bolt Load, Windload Vortex Shedding Case:

$$= ((4 \cdot M/DC) - W) / nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0/1882.0) - 118235)/20$$

$$= -5911.7 \text{ kgf}$$

Governing Bolt Load Condition, Earthquake + Operating Condition:

Area Available in 1 Bolt (corroded) Abss : 6.0063 cm²
 Area Available in all the Bolts Abss * RN : 120.1260 cm²

Trial#	k	knew	Cc	Ct	z	j	Ft	Fc	Fs
2	0.299	0.448	1.506	2.445	0.438	0.781	54200.0	169443.4	7481.0
4	0.373	0.336	1.699	2.282	0.422	0.784	56428.1	171671.5	8343.9
6	0.317	0.308	1.556	2.404	0.434	0.782	54744.1	169987.6	7683.7
8	0.313	0.310	1.543	2.414	0.435	0.782	54607.3	169850.8	7632.3
10	0.309	0.310	1.534	2.422	0.436	0.782	54505.0	169748.5	7594.1
12	0.310	0.310	1.536	2.420	0.435	0.782	54530.5	169774.0	7603.6

The Actual Stress in a Single Bolt [Sbolt]:

$$= 2 \cdot Ft / (T1 \cdot Dc \cdot Ct)$$

$$= 2 \cdot 54530.5 / (2.032 \cdot 1882.0 \cdot 2.42)$$

$$= 1178.561 \text{ kgf/cm}^2, \text{ Should be } \leq 1407.2 \text{ kgf/cm}^2$$

Thickness of the Band of Bolting Steel [T1]

$$= RN \cdot \text{Bolt Area} / (\pi \cdot Dc)$$

$$= 20 \cdot 6.006 / (\pi \cdot 1882.0)$$

$$= 2.032 \text{ mm}$$

Check the Bearing Stress in the Concrete [fc(max)]

$$= fc[(2kd + t3) / (2kd)]$$

$$= 53.879[(2 \cdot 0.3101 \cdot 1882.0 + 200.0) / (2 \cdot 0.3101 \cdot 1882.0)]$$

$$= 63.112 \text{ kgf/cm}^2, \text{ Should be } \leq 226.4 \text{ kgf/cm}^2$$

Values for table 10.3, l = 128.0, b = 147.812, l/b = 0.865965

Maximum Moment per unit width [Mmax]:

$$= \max(Mx, My) = \max(1110.734, 1679.52) = 1679.52 \text{ kgf}$$

Reqd Thickness of Basing, Brownell & Young Method [T]:

$$= \sqrt{(6 \cdot Mmax / fallow)} + Ca$$

$$= \sqrt{(6 \cdot 1679.52/1763.2)} + 6.0$$

$$= 29.907 \text{ mm}$$

Nomenclature:

a = (Dc-Ds)/2	Skirt Distance to Bolt Circle
P = Sa * Abss	Maximum Load on one Bolt
l	Radial Top Plate Width
lavg	Average Gusset Plate Width
g1 = Gamma 1	Constant Term f(b/l)
g2 = Gamma 2	Constant Term f(b/l)
g = Flat distance / 2	Nut 1/2 Dimension (from Tema)
Fb	Allowable Bending Stress

Values for table 10.6, l = 128.0, b = 76.0, b/l = 0.59375
 As b/l (0.594) is less than 1, inverting b/l = 1.684.

Moment Term, based on geometry [Mo]:

$$= P/(4 \cdot \pi) [1.3(\ln((2l \sin(\pi \cdot a/l))/(\pi \cdot g))) + 1] - [(0.7-g2)P/(4\pi)]$$

$$= 8451.81/(4 \cdot \pi) [1.3(\ln((2 \cdot 128.0 \cdot \sin(\pi \cdot 64.0/128.0))/(\pi \cdot 30.0))) + 1] - [(0.7 - 0.051) \cdot 8451.81/(4 \cdot \pi)]$$

$$= 1109.7666 \text{ kgf}$$

Required Thickness of Continuous Top Ring [Tc]:

$$= \sqrt{(6 \cdot \text{Abs}(\text{Mo}) / \text{Fb}) + \text{Ca}}$$

$$= \sqrt{(6 \cdot \text{Abs}(1109.77) / 1781.01) + 6.0}$$

$$= 25.3357 \text{ mm}$$

Required Gusset Plate Thickness [tg]:

$$= P / (\text{Stress Term} \cdot \text{lavg}) + \text{Ca}$$

$$= 8451.81 / (1265.526 \cdot 118.0) + 6.0$$

$$= 15.525 \text{ (not less than } 9.525 + 6.0 \text{) mm}$$

Bolt spacing [m]:

$$= \pi \cdot \text{Bolt Circle Diameter} / \text{number of Bolts}$$

$$= \pi \cdot 1882.0 / 20 = 295.624 \text{ mm}$$

Req. Skirt Thk. to withstand Local Bending, (Brownell and Young) [t]:

$$= 1.76 (P \cdot a / (m \cdot (h + tba) \cdot 1.5 \cdot \text{Sktope}))^{2/3} \cdot r^{1/3} + \text{Ca}$$

$$= 1.76 (8452 \cdot 64.0 / (295.62 \cdot 218.0 \cdot 2111))^{2/3} \cdot 877.01^{1/3} + \text{Ca}$$

$$= 9.11 + 1.6 = 10.71 \text{ mm}$$

Shear Stress in a Single Bolt [tb]:

$$= \text{Shear Force} / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 9120.8 / (6.006 \cdot 20)$$

$$= 75.9 \text{ kgf/cm}^2. \text{ Must be less than } 844.3 \text{ kgf/cm}^2.$$

Summary of Basing Thickness Calculations

Required Basing Thickness (tension)	29.9066	mm
Actual Basing Thickness as entered by user	50.0000	mm
Required Thickness of Chair Cap	25.3357	mm
Actual Top Ring Thickness as entered by user	32.0000	mm
Required Gusset thickness, + CA	15.5250	mm
Actual Gusset Thickness as entered by user	16.0000	mm
Required Thickness of Skirt for Local Stress	10.7100	mm
Given Thickness of Skirt	14.0000	mm
Required Gusset Height to meet local stress	214.9106	mm

Weld Size Calculations per Steel Plate Engineering Data - Vol. 2

Compute the Weld load at the Skirt/Base Junction [W]:

$$= \text{SkirtStress} \cdot (\text{SkirtThickness} - \text{CA})$$

$$= 783.064 \cdot (14.0 - 1.6)$$

$$= 97.1 \text{ kgf/mm}$$

Results for Computed Minimum Basing Weld Size [BWeld]:

$$= W / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 97 / [(0.4 \cdot 2390) \cdot 2 \cdot 0.707]$$

$$= 7.183 \text{ mm}$$

Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size

Vertical Plate Load [Wv]:

$$= \text{Bolt Load} / (\text{Cmwth} + 2 \cdot (\text{Hg} + \text{Tta}))$$

$$= 8451.8 / (160.0 + 2 \cdot (168.0 + 32.0))$$

$$= 15.093 \text{ kgf/mm}$$

Horizontal Plate Load [Wh]:

$$= \text{Bolt Load} \cdot e / (\text{Cmwth} \cdot (\text{Hg} + \text{Tta}) + 0.6667 \cdot (\text{Hg} + \text{Tta})^2)$$

$$= 8451.8 \cdot 64.0 / (160.0 \cdot (200.0) + 0.6667 \cdot (200.0)^2)$$

$$= 9.22 \text{ kgf/mm}$$

Resultant Weld Load [Wr]:

$$= \sqrt{Wv^2 + Wh^2}$$

$$= \sqrt{15.09^2 + 9.22^2}$$

$$= 17.686 \text{ kgf/mm}$$

Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:

$$= Wr / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 17.69 / [(0.4 \cdot 2390) \cdot 2 \cdot 0.707]$$
$$= 1.308 \text{ mm}$$

Results for Computed Minimum Gusset to Top Plate Weld Size

Weld Load [Wv]:

$$= \text{Bolt Load} / (2 \cdot \text{TopWth})$$
$$= 8451.8 / (2 \cdot 118.0)$$
$$= 35.813 \text{ kgf/mm}$$

Weld Load [Wh]:

$$= \text{Bolt Load} \cdot e / (2 \cdot \text{Hgt} \cdot \text{TopWth})$$
$$= 8451.8 \cdot 64.0 / (2 \cdot 200.0 \cdot 118.0)$$
$$= 11.46 \text{ kgf/mm}$$

Resultant Weld Load [Wr]:

$$= \sqrt{Wv^2 + Wh^2}$$
$$= \sqrt{35.81^2 + 11.46^2}$$
$$= 37.602 \text{ kgf/mm}$$

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:

$$= Wr / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$
$$= 37.6 / [(0.4 \cdot 2390) \cdot 2 \cdot 0.707]$$
$$= 2.782 \text{ mm}$$

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

Summary of Required Weld Sizes:

Required Basing to Skirt Double Fillet Weld Size	7.1828	mm
Required Gusset to Skirt Double Fillet Weld Size	6.3500	mm
Required Top Plate to Skirt Weld Size	6.3500	mm
Required Gusset to Top Plate Double Fillet Weld Size	2.7815	mm

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: D150

From: 40

Pressure for Reinforcement Calculations	P	45.157	kgf/cm ²
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²

Inside Diameter of Elliptical Head	D	1700.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	32.0000	mm
Head Internal Corrosion Allowance	c	6.0000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	0.0000	mm
-------------------------------	----	--------	----

User Entered Minimum Design Metal Temperature		-12.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1505.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1682.00	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		150.0000	mm

Size and Thickness Basis		Minimum	
Nominal Thickness		120	

Flange Material		SA-105	+IT
-----------------	--	--------	-----

Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	38.5105	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm

Corrosion Allowance	can	6.0000	mm
---------------------	-----	--------	----

Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

Flange Class		600	
Flange Grade		GR 1.1	

Piping Exit Angle [North Clockwise reference]		90.0	deg
Bend Radius Multiplier		1.50	
Horizontal Run Length		1500.0	mm
Centerline Distance to Tangent		2000.0	mm

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: D150

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		143.294	mm.
Actual Thickness Used in Calculation		12.490	mm

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 155.294 \cdot (\exp(45.157/(1505.0 \cdot 1.0)) - 1) + 6.0 + 0.0$
 $= 2.3651 + 6.0 + 0.0 = 8.3651 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1505.0 \cdot 1.0 \cdot \ln((2 \cdot 6.49 + 155.294) / 155.294) - 0.0$
 $= 120.82 - 0.0 = 120.8195 \text{ kgf/cm}^2$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	12.220 mm
Min. required shell thickness, internal p + c :	36.019 mm
Min. required shell thickness, external p + c :	16.066 mm

Nozzle neck minimum required thickness	: 12.220 mm
--	-------------

Minimum nozzle neck thickness 12.497 mm , is greater than required thickness 12.220 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 578.0,	Allowable	: 1505.0 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 3405.8 kgf/cm ²	Passed
Occasional	: 259.3,	Allowable	: 2001.7 kgf/cm ²	Passed
Shear	: 254.9,	Allowable	: 1053.5 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[1505.0/1407.1, 1]$
 $= 1.000$

Thickness of Nozzle at Shell [tn]:
 $= \text{hub thickness} - \text{corrosion allowance}$
 $= 50.0 - 6.0$
 $= 44.000 \text{ mm}$

Thickness of Nozzle at Top [tn2]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 12.49 - 6.0$
 $= 6.490 \text{ mm}$

Effective Pressure Radius [Reff]:
 $= 0.9/6 \cdot D_i [2 + (D_i/(2h))^2]$
 $= 0.9/6 \cdot 1712.0 [2 + (1712.0/(2 \cdot 431.0))^2]$
 $= 1526.550 \text{ mm}$

Effective Length of Vessel Wall [LR]:
 $= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$
 $= \min(\sqrt{1526.55 \cdot 26.0}, 2 \cdot 77.647)$
 $= 155.294 \text{ mm}$

Intermediate value [Cn]:
 $= \min(((t + t_e)/t_n)^{0.35}, 1)$
 $= \min(((26.0 + 0.0)/44.0)^{0.35}, 1)$
 $= \min(0.832, 1)$
 $= 0.832$

Intermediate value [Cp]:
 $= \exp((0.35 \cdot d_i - X_o)/(16t))$



$$= \exp((0.35 \cdot 1712.0 - 121.647) / (16 \cdot 26.0))$$

$$= 3.152$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min(t + t_e + F_p \cdot \sqrt{R_n \cdot t_n}, L_{pr1} + t)$$

$$= \min(26.0+0.0+0.832 \cdot \sqrt{77.647 \cdot 44.0}, 300.0+26.0)$$

$$= 74.621 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A_5 \cdot f_{rp}) / LR)$$

$$= 26.0 + ((0.0 \cdot 1.0) / 155.294)$$

$$= 26.000 \text{ mm}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR$$

$$= 26.0 \cdot 155.294$$

$$= 40.376 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$

$$= 44.0 \cdot 74.621$$

$$= 32.833 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}_1^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A_1 + f_{rn}(A_2 + A_3) + A_{41} + A_{42} + A_{43}$$

$$= 40.376 + 1.0(32.833 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 73.710 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n / R_n]$$

$$= 44.0 / \ln[1 + 44.0 / 77.647]$$

$$= 98.006 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff} / R_{eff}]$$

$$= 26.0 / \ln[1 + 26.0 / 1526.55]$$

$$= 1539.510 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 1407.129 \cdot 1.0$$

$$= 2110.7 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$

$$= 45.157 \cdot 98.006 \cdot 74.621$$

$$= 3302.5 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} \cdot (LR + t_n) / 2$$

$$= 45.157 \cdot 1539.51 \cdot (155.294 + 44.0) / 2$$

$$= 69274.0 \text{ kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= (P \cdot R_{xs} \cdot R_{nc}) / 2$$

$$= (45.157 \cdot 1539.51 \cdot 77.647) / 2$$

$$= 26989.9 \text{ kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (69274.0 + 26989.9 + 3302.5) / 45.157$$

$$= 2204.9 \text{ cm}^2$$



Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= \text{Sallow} / (2 \cdot \text{Ap}/\text{AT} - \text{Rxs}/(2 \cdot \text{teff}))$$

$$= 2110.694 / (2 \cdot 2204.901/73.71 - 1539.51/(2 \cdot 26.0))$$

$$= 69.8 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot \text{S}/[\text{t}/\text{Rxs}]$$

$$= 2 \cdot 1407.129[26.0/1539.51]$$

$$= 47.5 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(\text{Pmax1}, \text{Pmax2})$$

$$= \min(69.842, 47.529)$$

$$= 47.529 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (\text{fN} + \text{fS} + \text{fY}) / \text{AT}$$

$$= (3302.458 + 69274.031 + 26989.883) / 73.71$$

$$= 1350.795 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= \text{P} \cdot \text{Rxs} / (2 \cdot \text{teff})$$

$$= 45.157 \cdot 1539.51 / (2 \cdot 26.0)$$

$$= 1336.9 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}})$$

$$= \max(2 \cdot 1350.795 - 1336.913, 1336.913)$$

$$= 1364.7 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2110.69	kgf/cm ²
Local Primary Membrane Stress	PL	1364.68	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	47.53	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (\text{Rnc} + \text{tn}) / \text{Rnc}$$

$$= (77.647 + 44.0) / 77.647$$

$$= 1.567 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [LT]:

$$= \pi/2 \cdot (\text{Rn} + \text{tn})$$

$$= \pi/2 \cdot (77.647 + 44.0)$$

$$= 191.083 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$\text{tc}, \text{L41T} = 7.071, \text{ must be } \geq \min(0.7 \cdot 44.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(\text{fy} \cdot \text{ky}, 1.5 \cdot \text{Sn}(\text{A2} + \text{A3}), \pi/4 \cdot \text{P} \cdot \text{Rn}^2 \cdot \text{ky}^2)$$

$$= \min(26990 \cdot 1.57, 1.5 \cdot 1505.0(37.158 + 0.0), \pi/4 \cdot 45.2 \cdot 77.65^2 \cdot 1.57^2)$$

$$= 5248.275 \text{ kgf}$$

Weld Stress Value [τ]:

$$= \text{fwelds} / (\text{LT}(0.49 \cdot \text{L41T} + 0.6 \cdot \text{tw1} + 0.49 \cdot \text{L43T}))$$

$$= 5248.275 / (191.083(0.49 \cdot 7.071 + 0.6 \cdot 26.0 + 0.49 \cdot 0.0))$$

$$= 144.067 < \text{or} = \text{to } 1407.129 \text{ Weld Stress Passed}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

$$= \text{tn} / \ln[1 + \text{tn}/\text{Rn}]$$

$$= 44.0 / \ln[1 + 44.0/77.647]$$

$$= 98.006 \text{ mm}$$



Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 26.0 / \ln[1 + 26.0/1526.55]$$

$$= 1539.510 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S$$

$$= 1.5 \cdot 1407.129$$

$$= 2110.7 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 1.055 \cdot 98.006 \cdot 74.621$$

$$= 77.1 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} \cdot (L_R + t_n)/2$$

$$= 1.055 \cdot 1539.51 \cdot (155.294 + 44.0)/2$$

$$= 1617.8 \text{ kgf}$$

Discontinuity Force from External Pressure [fY]:

$$= (P \cdot R_{xs} \cdot R_{nc})/2$$

$$= (1.055 \cdot 1539.51 \cdot 77.647)/2$$

$$= 630.3 \text{ kgf}$$

Area Resisting External Pressure [Ap]:

$$= (f_S + f_Y + f_N)/P$$

$$= (1617.8 + 630.3 + 77.1)/1.055$$

$$= 2204.9 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / (2 \cdot t_{eff}))$$

$$= 2110.694 / (2 \cdot 2204.901 / 73.71 - 1539.51 / (2 \cdot 26.0))$$

$$= 69.8 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot S / [t / R_{xs}]$$

$$= 2 \cdot 1407.129 [26.0 / 1539.51]$$

$$= 47.5 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(69.842, 47.529)$$

$$= 47.529 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σAvg]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (77.126 + 1617.842 + 630.328) / 73.71$$

$$= 31.547 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot R_{xs} / (2 \cdot t_{eff})$$

$$= 1.055 \cdot 1539.51 / (2 \cdot 26.0)$$

$$= 31.2 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 31.547 - 31.223, 31.223)$$

$$= 31.9 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2110.69	kgf/cm ²
Local Primary Membrane Stress	PL	31.87	kgf/cm ²
Maximum Allowable External Pressure	Pmax	47.53	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.



Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Flange MDMT including temperature reduction per 3.11:
 Flange MDMT with Temp reduction -61 °C

The Drop for this Nozzle is : 4.7637 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 336.7636 mm

Input Echo, WRC107/537 Item 1, Description: D150 :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	6.0000	mm
Vessel Diameter	Dv	3060.000	mm
Vessel Thickness	Tv	32.000	mm
Design Temperature	T1	130.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	1407.13	kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1407.13	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	6.0000	mm
Nozzle Diameter	Dn	143.294	mm
Nozzle Thickness	Tn	50.000	mm
Nozzle Material		SA-105	
Nozzle UNS Number		K03504	
Nozzle Cold S.I. Allowable	SNmc	1682.00	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1505.00	kgf/cm ²
Design Internal Pressure	Dp	45.157	kgf/cm ²
Include Pressure Thrust		No	

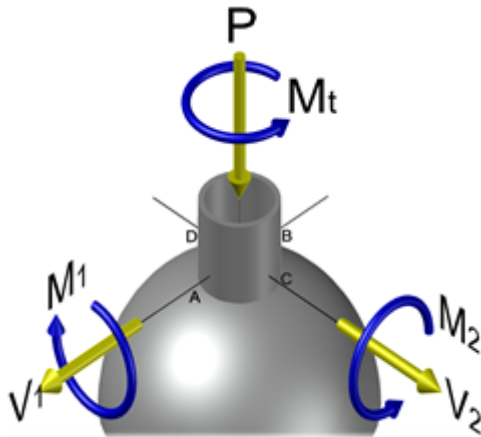
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-1146.8	kgf
Longitudinal Shear (SUS)	(V1) V1	764.5	kgf
Circumferential Shear (SUS)	(Vc) V2	764.5	kgf
Circumferential Moment (SUS)	(Mc) M1	321.1	kgf-m.
Longitudinal Moment (SUS)	(Ml) M2	321.1	kgf-m.
Torsional Moment (SUS)	Mt	458.7	kgf-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 243.294 + 2 \cdot 1.65 \cdot \sqrt{(1549.0 (32.0 - 6.0))}$$

$$= 905.551 \text{ mm}$$

WRC 107 Stress Calculation for SUSTained loads:

Radial Load	P	-1146.8	kgf
Circumferential Shear	(VC) V2	764.5	kgf
Longitudinal Shear	(VL) V1	764.5	kgf
Circumferential Moment	(MC) M1	321.1	kgf-m.
Longitudinal Moment	(ML) M2	321.1	kgf-m.
Torsional Moment	MT	458.7	kgf-m.

Dimensionless Param: U = 0.61 TAU = 5.00 (2.26) RHO = 0.59

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 1	0.06388	(A,B,C,D)
$M(x) / P$	SP 1	0.07523	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.08984	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 1	0.17566	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.08984	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 1	0.17566	(A,B,C,D)
$N(y) * T / P$	SP 1	0.10810	(A,B,C,D)
$M(y) / P$	SP 1	0.02389	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.09572	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 1	0.06521	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.09572	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 1	0.06521	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (kgf/cm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb.	P	10.8	10.8	10.8	10.8	10.8	10.8	10.8	10.8
Rad. Bend.	P	76.6	-76.6	76.6	-76.6	76.6	-76.6	76.6	-76.6
Rad. Memb.	MC	0.0	0.0	0.0	0.0	-21.3	-21.3	21.3	21.3
Rad. Bend.	MC	0.0	0.0	0.0	0.0	-249.5	249.5	249.5	-249.5
Rad. Memb.	ML	-21.3	-21.3	21.3	21.3	0.0	0.0	0.0	0.0
Rad. Bend.	ML	-249.5	249.5	249.5	-249.5	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-183.3	162.5	358.1	-293.9	-183.3	162.5	358.1	-293.9



Tang. Memb. P	18.3	18.3	18.3	18.3	18.3	18.3	18.3	18.3
Tang. Bend. P	24.3	-24.3	24.3	-24.3	24.3	-24.3	24.3	-24.3
Tang. Memb. MC	0.0	0.0	0.0	0.0	-22.7	-22.7	22.7	22.7
Tang. Bend. MC	0.0	0.0	0.0	0.0	-92.6	92.6	92.6	-92.6
Tang. Memb. ML	-22.7	-22.7	22.7	22.7	0.0	0.0	0.0	0.0
Tang. Bend. ML	-92.6	92.6	92.6	-92.6	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-72.6	64.0	157.9	-75.9	-72.6	64.0	157.9	-75.9
Shear VC	7.7	7.7	-7.7	-7.7	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-7.7	-7.7	7.7	7.7
Shear MT	19.0	19.0	19.0	19.0	19.0	19.0	19.0	19.0
Tot. Shear	26.7	26.7	11.3	11.3	11.3	11.3	26.7	26.7
Str. Int.	189.4	169.2	358.8	294.5	184.5	163.8	361.6	297.2

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (kgf/cm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		1345.1	1345.1	1345.1	1345.1	1345.1	1345.1	1345.1	1345.1
Rad. Pl (SUS)		-10.4	-10.4	32.1	32.1	-10.4	-10.4	32.1	32.1
Rad. Q (SUS)		-172.9	172.9	326.0	-326.0	-172.9	172.9	326.0	-326.0
Long. Pm (SUS)		1345.1	1345.1	1345.1	1345.1	1345.1	1345.1	1345.1	1345.1
Long. Pl (SUS)		-4.3	-4.3	41.0	41.0	-4.3	-4.3	41.0	41.0
Long. Q (SUS)		-68.3	68.3	116.9	-116.9	-68.3	68.3	116.9	-116.9
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		7.7	7.7	-7.7	-7.7	-7.7	-7.7	7.7	7.7
Shear Q (SUS)		19.0	19.0	19.0	19.0	19.0	19.0	19.0	19.0
Pm (SUS)		1345.1	1345.1	1345.1	1345.1	1345.1	1345.1	1345.1	1345.1
Pm+Pl (SUS)		1346.0	1346.0	1390.6	1390.6	1346.0	1346.0	1390.6	1390.6
Pm+Pl+Q (Total)		1278.6	1514.4	1703.9	1269.8	1273.7	1508.9	1706.8	1272.4

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1345.13	1407.13	Passed
Pm+Pl (SUS)	1390.57	2110.69	Passed
Pm+Pl+Q (TOTAL)	1706.77	4780.43	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

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Input, Nozzle Desc: MH4600

From: 50

Pressure for Reinforcement Calculations	P	45.027	kgf/cm ²
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	1700.00	mm
Design Length of Section	L	18483.8340	mm
Shell Finished (Minimum) Thickness	t	34.0000	mm
Shell Internal Corrosion Allowance	c	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		85.00	cm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1505.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1682.00	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		568.0000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm

Flange Material		SA-105	+IT
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Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	64.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		80.0000	mm

Corrosion Allowance	can	6.0000	mm
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Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	32.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

This is a Manway, Coupling or Access Opening.

Flange Class		300
Flange Grade		GR 1.1

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: MH4600

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	568.000	mm.
Actual Thickness Used in Calculation	16.000	mm

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight



of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 580.0 \cdot (\exp(45.027/(1505.0 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 8.8073 + 6.0 + 0.0 = 14.8073 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1505.0 \cdot 1.0 \cdot \ln((2 \cdot 10.0 + 580.0) / 580.0) - 0.0$$

$$= 51.022 - 0.0 = 51.0218 \text{ kgf/cm}^2$$

This nozzle or its diameter limit intersects weld seam # [3] on this element. The distance between the nozzle and seam is, 239.04 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 518.82 mm

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1505.0/1407.1, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 80.0 - 6.0$$

$$= 74.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 16.0 - 6.0$$

$$= 10.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 1712.0/28.0$$

$$= 61.143 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 1700.0/2 + 6.0$$

$$= 856.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{856.0 \cdot 28.0}, 2 \cdot 290.0)$$

$$= 154.816 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 28.0, 0.0] + \sqrt{290.0 \cdot 74.0}$$

$$= 146.492 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 500.0$$

$$= 500.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(28.0 + 0.0)$$

$$= 224.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:
 $= \min[LH1, LH2, LH3] + t$ for set-in nozzles
 $= \min[146.492, 500.0, 224.0] + 28.0$
 $= 174.492 \text{ mm}$

Effective Vessel Thickness [teff]:
 $= t + ((A5 \cdot frp) / LR)$
 $= 28.0 + ((0.0 \cdot 1.0) / 154.816)$
 $= 28.000 \text{ mm}$

Parameter [λ]:
 $= \min(12, (2 \cdot Rn + tn) / (\sqrt{(Di + teff) \cdot teff}))$
 $= \min(12, (2 \cdot 290.0 + 74.0) / (\sqrt{(1712.0 + 28.0) \cdot 28.0}))$
 $= 2.963$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:
 $= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$
 $= 28.0 \cdot 154.816 \cdot \max((2.963/5)^{0.85}, 1)$
 $= 43.348 \text{ cm}^2$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:
 $= A2A + A2B$
 $= 94.72 + 23.597$
 $= 118.317 \text{ cm}^2$

Area Contributed by the Outside Fillet Weld [A41]:
 $= 0.5 \cdot Leg41^2$
 $= 0.5 \cdot 10.0^2$
 $= 0.500 \text{ cm}^2$

The total area contributed by A1 through A43 [AT]:
 $= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$
 $= 43.348 + 1.0(118.317 + 0.0) + 0.5 + 0.0 + 0.0$
 $= 162.165 \text{ cm}^2$

Nozzle Radius for Force Calculation [Rxn]:
 $= tn / \ln[1 + tn/Rn]$
 $= 74.0 / \ln[1 + 74.0/290.0]$
 $= 325.600 \text{ mm}$

Shell Radius for Force Calculation [Rxs]:
 $= teff / \ln[1 + teff/Reff]$
 $= 28.0 / \ln[1 + 28.0/856.0]$
 $= 869.924 \text{ mm}$

Allowable Local Primary Membrane Stress [Sallow]:
 $= 1.5 \cdot S \cdot E$
 $= 1.5 \cdot 1407.129 \cdot 1.0$
 $= 2110.7 \text{ kgf/cm}^2$

Force acting on the Nozzle [fN]:
 $= P \cdot Rxn \cdot LH$
 $= 45.027 \cdot 325.6 \cdot 174.492$
 $= 25581.6 \text{ kgf}$

Force acting on the Shell [fS]:
 $= P \cdot Rxs(LR + tn)$
 $= 45.027 \cdot 869.924(154.816 + 74.0)$
 $= 89626.3 \text{ kgf}$

Discontinuity Force from Internal Pressure [fY]:
 $= P \cdot Rxs \cdot Rnc$
 $= 45.027 \cdot 869.924 \cdot 290.0$
 $= 113591.9 \text{ kgf}$

Area Resisting Internal Pressure [Ap]:
 $= (fS + fY + fN) / P$



$$= (89626.3 + 113591.9 + 25581.6) / 45.027$$

$$= 5081.5 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= \text{Sallow} / (2 \cdot \text{Ap}/\text{AT} - \text{Rxs}/\text{teff})$$

$$= 2110.694 / (2 \cdot 5081.452 / 162.165 - 869.924 / 28.0)$$

$$= 66.8 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= \text{S}[t/\text{Rxs}]$$

$$= 1407.129 [28.0 / 869.924]$$

$$= 45.3 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(\text{Pmax1}, \text{Pmax2})$$

$$= \min(66.791, 45.291)$$

$$= 45.291 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / \text{AT}$$

$$= (25581.635 + 89626.336 + 113591.859) / 162.165$$

$$= 1410.909 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot \text{Rxs} / \text{teff}$$

$$= 45.027 \cdot 869.924 / 28.0$$

$$= 1398.9 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}})$$

$$= \max(2 \cdot 1410.909 - 1398.916, 1398.916)$$

$$= 1422.9 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2110.69	kgf/cm ²
Local Primary Membrane Stress	PL	1422.90	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	45.29	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (\text{Rnc} + \text{tn}) / \text{Rnc}$$

$$= (290.0 + 74.0) / 290.0$$

$$= 1.255 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi / 2 \cdot (\text{Rn} + \text{tn})$$

$$= \pi / 2 \cdot (290.0 + 74.0)$$

$$= 571.770 \text{ mm}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$\text{tc}, \text{L41T} = 7.071, \text{ must be } \geq \min(0.7 \cdot 74.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot \text{Sn}(\text{A2} + \text{A3}), \pi / 4 \cdot P \cdot \text{Rn}^2 \cdot k_y^2)$$

$$= \min(113592 \cdot 1.26, 1.5 \cdot 1505.0 (118.317 + 0.0), \pi / 4 \cdot 45.0 \cdot 290.0^2 \cdot 1.26^2)$$

$$= 46855.508 \text{ kgf}$$

Weld Stress Value [τ]:

$$= \text{fwelds} / (\text{Lt} (0.49 \cdot \text{L41T} + 0.6 \cdot \text{tw1} + 0.49 \cdot \text{L43T}))$$

$$= 46855.508 / (571.77 (0.49 \cdot 7.071 + 0.6 \cdot 28.0 + 0.49 \cdot 0.0))$$

$$= 404.388 < \text{ or } = \text{ to } 1407.129 \text{ Weld Stress Passed}$$

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: External Pressure Case:



Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 74.0 / \ln[1 + 74.0/290.0]$$

$$= 325.600 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 28.0 / \ln[1 + 28.0/856.0]$$

$$= 869.924 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S$$

$$= 1.5 \cdot 1407.129$$

$$= 2110.7 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 1.055 \cdot 325.6 \cdot 174.492$$

$$= 599.2 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 1.055 \cdot 869.924 (154.816 + 74.0)$$

$$= 2099.2 \text{ kgf}$$

Discontinuity Force from External Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 1.055 \cdot 869.924 \cdot 290.0$$

$$= 2660.5 \text{ kgf}$$

Area Resisting External Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (2099.2 + 2660.5 + 599.2) / 1.055$$

$$= 5081.5 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 2110.694 / (2 \cdot 5081.452 / 162.165 - 869.924 / 28.0)$$

$$= 66.8 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / (t / R_{xs})$$

$$= 1407.129 [28.0 / 869.924]$$

$$= 45.3 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(66.791, 45.291)$$

$$= 45.291 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σAvg]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (599.169 + 2099.215 + 2660.532) / 162.165$$

$$= 33.046 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 1.055 \cdot 869.924 / 28.0$$

$$= 32.8 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 33.046 - 32.765, 32.765)$$

$$= 33.3 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2110.69	kgf/cm ²
Local Primary Membrane Stress	PL	33.33	kgf/cm ²
Maximum Allowable External Pressure	Pmax	45.29	kgf/cm ²



Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -25 °C

The Drop for this Nozzle is : 81.8828 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 615.8828 mm

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Input, Nozzle Desc: B150

From: 50

Pressure for Reinforcement Calculations	P	44.822	kgf/cm ²
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	1700.00	mm
Design Length of Section	L	18483.8340	mm
Shell Finished (Minimum) Thickness	t	34.0000	mm
Shell Internal Corrosion Allowance	c	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		285.00	cm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1505.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1682.00	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		150.0000	mm

Size and Thickness Basis		Minimum	
Nominal Thickness		120	

Flange Material		SA-105	+IT
-----------------	--	--------	-----

Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	38.5105	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm

Corrosion Allowance	can	6.0000	mm
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Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: B150

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		143.294	mm.
Actual Thickness Used in Calculation		12.490	mm

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 155.294 \cdot (\exp(44.822/(1505.0 \cdot 1.0)) - 1) + 6.0 + 0.0$
 $= 2.3473 + 6.0 + 0.0 = 8.3473 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1505.0 \cdot 1.0 \cdot \ln((2 \cdot 6.49 + 155.294) / 155.294) - 0.0$
 $= 120.82 - 0.0 = 120.8195 \text{ kgf/cm}^2$

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 276.46 mm

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 12.220 mm
 Min. required shell thickness, internal p + c : 33.708 mm
 Min. required shell thickness, external p + c : 26.261 mm

Nozzle neck minimum required thickness : 12.220 mm

Minimum nozzle neck thickness 12.497 mm , is greater than required thickness 12.220 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	576.0,	Allowable	:	1505.0 kgf/cm ²	Passed
Expansion	:	0.0,	Allowable	:	3407.7 kgf/cm ²	Passed
Occasional	:	257.4,	Allowable	:	2001.7 kgf/cm ²	Passed
Shear	:	254.9,	Allowable	:	1053.5 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[1505.0/1407.1, 1]$
 $= 1.000$

Thickness of Nozzle at Shell [tn]:
 $= \text{hub thickness} - \text{corrosion allowance}$
 $= 50.0 - 6.0$
 $= 44.000 \text{ mm}$

Thickness of Nozzle at Top [tn2]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 12.49 - 6.0$
 $= 6.490 \text{ mm}$

Shell Diameter to Thickness ratio [D/t]:
 $= D_i/t$
 $= 1712.0/28.0$
 $= 61.143 \text{ must be less than } 400.$

Effective Pressure Radius [Reff]:
 $= D_i/2 + \text{corrosion allowance}$
 $= 1700.0/2 + 6.0$
 $= 856.000 \text{ mm}$

Effective Length of Vessel Wall [LR]:
 $= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$
 $= \min(\sqrt{856.0 \cdot 28.0}, 2 \cdot 77.647)$
 $= 154.816 \text{ mm}$

Thickness Limit Candidate [LH1]:
 $= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$
 $= \min[1.5 \cdot 28.0, 0.0] + \sqrt{77.647 \cdot 44.0}$
 $= 58.451 \text{ mm}$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 300.0$$

$$= 300.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(28.0 + 0.0)$$

$$= 224.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[58.451, 300.0, 224.0] + 28.0$$

$$= 86.451 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 28.0 + ((0.0 \cdot 1.0) / 154.816)$$

$$= 28.000 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}}))$$

$$= \min(12, (2 \cdot 77.65 + 44.0) / (\sqrt{(1712.0 + 28.0) \cdot 28.0}))$$

$$= 0.903$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 28.0 \cdot 154.816 \cdot \max((0.903/5)^{0.85}, 1)$$

$$= 43.348 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$

$$= 44.0 \cdot 86.451$$

$$= 38.038 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + frn(A2 + A3) + A41 + A42 + A43$$

$$= 43.348 + 1.0(38.038 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 81.887 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 44.0 / \ln[1 + 44.0/77.647]$$

$$= 98.006 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 28.0 / \ln[1 + 28.0/856.0]$$

$$= 869.924 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 1407.129 \cdot 1.0$$

$$= 2110.7 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$

$$= 44.822 \cdot 98.006 \cdot 86.451$$

$$= 3797.6 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs}(LR + t_n)$$



$$= 44.822 \cdot 869.924 (154.816 + 44.0)$$

$$= 77521.8 \text{ kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot Rxs \cdot Rnc$$

$$= 44.822 \cdot 869.924 \cdot 77.647$$

$$= 30275.9 \text{ kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P$$

$$= (77521.8 + 30275.9 + 3797.6) / 44.822$$

$$= 2489.7 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap / AT - Rxs / teff)$$

$$= 2110.694 / (2 \cdot 2489.746 / 81.887 - 869.924 / 28.0)$$

$$= 71.0 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/Rxs]$$

$$= 1407.129 [28.0 / 869.924]$$

$$= 45.3 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(70.97, 45.291)$$

$$= 45.291 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σAvg]:

$$= (fN + fS + fY) / AT$$

$$= (3797.635 + 77521.75 + 30275.906) / 81.887$$

$$= 1362.802 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot Rxs / teff$$

$$= 44.822 \cdot 869.924 / 28.0$$

$$= 1392.6 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ)$$

$$= \max(2 \cdot 1362.802 - 1392.562, 1392.562)$$

$$= 1392.6 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2110.69	kgf/cm ²
Local Primary Membrane Stress	PL	1392.56	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	45.29	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$

$$= (77.647 + 44.0) / 77.647$$

$$= 1.567 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi / 2 \cdot (Rn + tn)$$

$$= \pi / 2 \cdot (77.647 + 44.0)$$

$$= 191.083 \text{ mm}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 44.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi / 4 \cdot P \cdot Rn^2 \cdot ky^2)$$

$$= \min(30276 \cdot 1.57, 1.5 \cdot 1505.0(38.038 + 0.0), \pi / 4 \cdot 44.8 \cdot 77.65^2 \cdot 1.57^2)$$

$$= 5209.354 \text{ kgf}$$

Weld Stress Value [τ]:
= $f_{welds} / (L_t(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$
= $5209.354 / (191.083(0.49 \cdot 7.071 + 0.6 \cdot 28.0 + 0.49 \cdot 0.0))$
= $134.531 < \text{or} = \text{to } 1407.129$ Weld Stress Passed

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:
= $t_n / \ln[1 + t_n/R_n]$
= $44.0 / \ln[1 + 44.0/77.647]$
= 98.006 mm

Shell Radius for Force Calculation [R_{xs}]:
= $t_{eff} / \ln[1 + t_{eff}/R_{eff}]$
= $28.0 / \ln[1 + 28.0/856.0]$
= 869.924 mm

Allowable Local Primary Membrane Stress [S_{allow}]:
= $1.5 \cdot S$
= $1.5 \cdot 1407.129$
= 2110.7 kgf/cm^2

Force acting on the Nozzle [f_N]:
= $P \cdot R_{xn} \cdot L_H$
= $1.055 \cdot 98.006 \cdot 86.451$
= 89.4 kgf

Force acting on the Shell [f_S]:
= $P \cdot R_{xs} (L_R + t_n)$
= $1.055 \cdot 869.924 (154.816 + 44.0)$
= 1824.0 kgf

Discontinuity Force from External Pressure [f_Y]:
= $P \cdot R_{xs} \cdot R_{nc}$
= $1.055 \cdot 869.924 \cdot 77.647$
= 712.4 kgf

Area Resisting External Pressure [A_p]:
= $(f_S + f_Y + f_N) / P$
= $(1824.0 + 712.4 + 89.4) / 1.055$
= 2489.7 cm^2

Maximum Allowable Working Pressure Candidate [P_{max1}]:
= $S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$
= $2110.694 / (2 \cdot 2489.746 / 81.887 - 869.924 / 28.0)$
= 71.0 kgf/cm^2

Maximum Allowable Working Pressure Candidate [P_{max2}]:
= $S / [t / R_{xs}]$
= $1407.129 [28.0 / 869.924]$
= 45.3 kgf/cm^2

Maximum Allowable Working Pressure [P_{max}]:
= $\min(P_{max1}, P_{max2})$
= $\min(70.97, 45.291)$
= 45.291 kgf/cm^2

Average Primary Membrane Stress [σ_{Avg}]:
= $(f_N + f_S + f_Y) / A_T$
= $(89.354 + 1823.988 + 712.353) / 81.887$
= 32.065 kgf/cm^2

General Primary Membrane Stress [σ_{Circ}]:
= $P \cdot R_{xs} / t_{eff}$
= $1.055 \cdot 869.924 / 28.0$
= 32.8 kgf/cm^2

Maximum Local Primary Membrane Stress [PL]:
= $\max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$



$$= \max(2 \cdot 32.065 - 32.765, 32.765)$$

$$= 32.8 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2110.69	kgf/cm ²
Local Primary Membrane Stress	PL	32.77	kgf/cm ²
Maximum Allowable External Pressure	Pmax	45.29	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -61 °C

The Drop for this Nozzle is : 8.7498 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 342.7497 mm

Input Echo, WRC297 Item 1, Description: B150 :

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	6.0000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	1700.000	mm	
Shell Thickness	34.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-516 70		
Vessel Cold S.I. Allowable	Smc	1407.13	kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1407.13	kgf/cm ²
Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance	6.0000	mm	
Nozzle Diameter	143.294	mm	
Nozzle Thickness	50.0000	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-105		
Nozzle Cold S.I. Allowable	SNmc	1682.00	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1505.00	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	44.82	kgf/cm ²
Radial Load	P	-1146.80	kgf
Circumferential Shear	Vc	764.53	kgf
Longitudinal Shear	Vl	764.53	kgf
Circumferential Moment	Mc	321.11	kgf-m.
Longitudinal Moment	Ml	321.11	kgf-m.
Torsional Moment	Mt	458.73	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (4.866) is < 10
 check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:



Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$$

$$= 243.294 + 2 \cdot 1.65 \cdot \sqrt{(870.0(34.0 - 6.0))}$$

$$= 758.348 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1740.000 mm
Nozzle Outside Diameter	(d) =	243.294 mm
Vessel Thickness used	(T) =	28.000 mm
Nozzle Thickness used	(t) =	44.000 mm
T / t	=	0.636
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.102$$

Nr/P	=	0.105
Mr/P	=	0.201
M0/P	=	0.061
N0/P	=	0.125
MrD/Mc	=	0.298
NrDL/Mc	=	0.107
M0d/Mc	=	0.098
N0DL/Mc	=	0.042
MrD/Ml	=	0.184
NrDL/Ml	=	0.091
M0D/Ml	=	0.062
N0DL/Ml	=	0.124

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	18	18	18	18
Outplane Bending (P)	53	-53	53	-53
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-20	-20	20	20
Outplane Bending (ML)	-62	62	62	-62
Normal Pressure Stress	1348	1392	1348	1392
Outplane Stress Summary	1336	1399	1503	1315

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	15	15	15	15
Inplane Bending (P)	176	-176	176	-176
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-15	-15	15	15
Inplane Bending (ML)	-186	186	186	-186
Inplane Pressure Stress	674	674	674	674
Inplane Stress Summary	664	683	1067	342

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	7	7	-7	-7
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	17	17	17	17
Shear Stress Summary	24	24	10	10



Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	1337	1400	1504	1315

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	18	18	18	18
Outplane Bending (P)	53	-53	53	-53
Outplane Membrane (Mc)	-7	-7	7	7
Outplane Bending (Mc)	-99	99	99	-99
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	674	674	674	674
Outplane Stress Summary	640	730	852	546

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	15	15	15	15
Inplane Bending (P)	176	-176	176	-176
Inplane Membrane (Mc)	-17	-17	17	17
Inplane Bending (Mc)	-300	300	300	-300
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1348	1392	1348	1392
Inplane Stress Summary	1221	1514	1858	949

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-7	-7	7	7
Torsional Shear (Mt)	17	17	17	17
Shear Stress Summary	10	10	24	24

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	1221	1514	1859	950

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	18	18	18	18
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-20	-20	20	20
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	73	127	73	127
Hoop Stress Summary	71	125	113	166



Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	4	4	4	4
Axial Bending (P)	52	-52	52	-52
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-27	-27	27	27
Axial Bending (ML)	-56	56	56	-56
Axial Pressure Stress	36	36	36	36
Axial Stress Summary	10	17	177	-41

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	4	4	-4	-4
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	11	11	11	11
Shear Stress Summary	15	15	6	6

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	75	127	178	208

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	18	18	18	18
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-7	-7	7	7
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	73	127	73	127
Hoop Stress Summary	85	138	99	153

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	4	4	4	4
Axial Bending (P)	52	-52	52	-52
Axial Membrane (Mc)	-27	-27	27	27
Axial Bending (Mc)	-99	99	99	-99
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	36	36	36	36
Axial Stress Summary	-33	60	220	-84

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-4	-4	4	4
Shear due to Torsion	11	11	11	11



Shear Stress Summary 6 6 15 15

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	118	139	222	239

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1348	1392	1348	1392	1348	1392	1348	1392
Circ. Pl (SUS)	-2	-2	38	38	-2	-2	32	32
Circ. Q (SUS)	-9	9	115	-115	-124	124	476	-476
Long. Pm (SUS)	674	674	674	674	674	674	674	674
Long. Pl (SUS)	0	0	30	30	11	11	25	25
Long. Q (SUS)	-10	10	362	-362	-46	46	152	-152
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	7	7	-7	-7	-7	-7	7	7
Shear Q (SUS)	17	17	17	17	17	17	17	17
Pm (SUS)	1348	1392	1348	1392	1348	1392	1348	1392
Pm+Pl (SUS)	1346	1390	1386	1430	1346	1390	1380	1424
Pm+Pl+Q (Total)	1337	1399	1501	1315	1222	1514	1856	949

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1392.00	1407.13	Passed
Pm+Pl (SUS)	1430.07	2110.69	Passed
Pm+Pl+Q (TOTAL)	1856.57	4221.39	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	73	127	73	127	73	127	73	127
Circ. Pl (SUS)	-2	-2	38	38	11	11	25	25
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	36	36	36	36	36	36	36	36
Long. Pl (SUS)	-23	-23	31	31	-23	-23	31	31
Long. Q (SUS)	-4	4	108	-108	-47	47	151	-151
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	4	4	-4	-4	-4	-4	4	4
Shear Q (SUS)	11	11	11	11	11	11	11	11
Pm (SUS)	73.0	127.0	73.0	127.0	73.0	127.0	73.0	127.0
Pm+Pl (SUS)	71.3	125.1	111.4	165.2	84.2	138.1	98.5	152.2
Pm+Pl+Q (Total)	74.4	127.0	175.8	206.5	118.8	138.6	219.8	237.9



Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	127.00	1505.00	Passed
Pm+Pl (SUS)	165.16	2257.50	Passed
Pm+Pl+Q (TOTAL)	237.90	4515.00	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: L150

From: 50

Pressure for Reinforcement Calculations	P	44.853	kgf/cm ²
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	1700.00	mm
Design Length of Section	L	18483.8340	mm
Shell Finished (Minimum) Thickness	t	34.0000	mm
Shell Internal Corrosion Allowance	c	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		255.00	cm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1505.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1682.00	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		50.8000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm

Flange Material		SA-105	+IT
Flange Type		N/A	

Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	33.2360	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Style of FVC Nozzle		Long Weld Neck	

Corrosion Allowance	can	6.0000	mm
---------------------	-----	--------	----

Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

Flange Class		300
Flange Grade		GR 1.1

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: L150

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		50.800	mm.
Actual Thickness Used in Calculation		12.000	mm

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 62.8 \cdot (\exp(44.853/(1505.0 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 0.9499 + 6.0 + 0.0 = 6.9499 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1505.0 \cdot 1.0 \cdot \ln((2 \cdot 6.0 + 62.8) / 62.8) - 0.0$$

$$= 263.169 - 0.0 = 263.1686 \text{ kgf/cm}^2$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	10.800 mm
Min. required shell thickness, internal p + c	: 33.727 mm
Min. required shell thickness, external p + c	: 26.261 mm

Nozzle neck minimum required thickness	: 10.800 mm
--	-------------

Minimum nozzle neck thickness 11.989 mm , is greater than required thickness 10.800 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1505.0/1407.1, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 50.0 - 6.0$$

$$= 44.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 12.0 - 6.0$$

$$= 6.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D/t$$

$$= 1712.0/28.0$$

$$= 61.143 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 1700.0/2 + 6.0$$

$$= 856.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{856.0 \cdot 28.0}, 2 \cdot 31.4)$$

$$= 62.800 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 28.0, 0.0] + \sqrt{31.4 \cdot 44.0}$$

$$= 37.170 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 250.0$$

$$= 250.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(28.0 + 0.0)$$

$$= 224.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[37.17, 250.0, 224.0] + 28.0$$

$$= 65.170 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 28.0 + ((0.0 \cdot 1.0) / 62.8)$$

$$= 28.000 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot Rn + tn) / (\sqrt{(Di + teff) \cdot teff}))$$

$$= \min(12, (2 \cdot 31.4 + 44.0) / (\sqrt{(1712.0 + 28.0) \cdot 28.0}))$$

$$= 0.484$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 28.0 \cdot 62.8 \cdot \max((0.484/5)^{0.85}, 1)$$

$$= 17.584 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= tn \cdot LH$$

$$= 44.0 \cdot 65.17$$

$$= 28.675 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot Leg^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$

$$= 17.584 + 1.0(28.675 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 46.759 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= tn / \ln[1 + tn/Rn]$$

$$= 44.0 / \ln[1 + 44.0/31.4]$$

$$= 50.228 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= teff / \ln[1 + teff/Reff]$$

$$= 28.0 / \ln[1 + 28.0/856.0]$$

$$= 869.924 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 1407.129 \cdot 1.0$$

$$= 2110.7 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot Rxn \cdot LH$$

$$= 44.853 \cdot 50.228 \cdot 65.17$$

$$= 1468.2 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot Rxs \cdot (LR + tn)$$

$$= 44.853 \cdot 869.924 \cdot (62.8 + 44.0)$$

$$= 41671.6 \text{ kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot Rxs \cdot Rnc$$

$$= 44.853 \cdot 869.924 \cdot 31.4$$

$$= 12251.8 \text{ kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P$$

$$= (41671.6 + 12251.8 + 1468.2) / 44.853$$

$$= 1235.0 \text{ cm}^2$$



Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= \text{Sallow} / (2 \cdot \text{Ap} / \text{AT} - \text{Rxs} / \text{teff})$$

$$= 2110.694 / (2 \cdot 1234.969 / 46.759 - 869.924 / 28.0)$$

$$= 97.0 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= \text{S} / \text{Rxs}$$

$$= 1407.129 / [28.0 / 869.924]$$

$$= 45.3 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(\text{Pmax1}, \text{Pmax2})$$

$$= \min(97.024, 45.291)$$

$$= 45.291 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (fN + fS + fY) / \text{AT}$$

$$= (1468.195 + 41671.637 + 12251.773) / 46.759$$

$$= 1184.627 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot \text{Rxs} / \text{teff}$$

$$= 44.853 \cdot 869.924 / 28.0$$

$$= 1393.5 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}})$$

$$= \max(2 \cdot 1184.627 - 1393.516, 1393.516)$$

$$= 1393.5 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2110.69	kgf/cm ²
Local Primary Membrane Stress	PL	1393.52	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	45.29	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (31.4 + 44.0) / 31.4$$

$$= 2.401 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [LT]:

$$= \pi / 2 \cdot (R_n + t_n)$$

$$= \pi / 2 \cdot (31.4 + 44.0)$$

$$= 118.438 \text{ mm}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 44.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi / 4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(12252 \cdot 2.4, 1.5 \cdot 1505.0(28.675 + 0.0), \pi / 4 \cdot 44.9 \cdot 31.4^2 \cdot 2.4^2)$$

$$= 2002.722 \text{ kgf}$$

Weld Stress Value [τ]:

$$= \text{fwelds} / (L_T(0.49 \cdot L41T + 0.6 \cdot t_{w1} + 0.49 \cdot L43T))$$

$$= 2002.722 / (118.438(0.49 \cdot 7.071 + 0.6 \cdot 28.0 + 0.49 \cdot 0.0))$$

$$= 83.443 < \text{or} = \text{to } 1407.129 \text{ Weld Stress Passed}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n / R_n]$$

$$= 44.0 / \ln[1 + 44.0 / 31.4]$$

$$= 50.228 \text{ mm}$$



Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 28.0 / \ln[1 + 28.0/856.0]$$

$$= 869.924 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S$$

$$= 1.5 \cdot 1407.129$$

$$= 2110.7 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 1.055 \cdot 50.228 \cdot 65.17$$

$$= 34.5 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 1.055 \cdot 869.924 (62.8 + 44.0)$$

$$= 979.8 \text{ kgf}$$

Discontinuity Force from External Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 1.055 \cdot 869.924 \cdot 31.4$$

$$= 288.1 \text{ kgf}$$

Area Resisting External Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (979.8 + 288.1 + 34.5) / 1.055$$

$$= 1235.0 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 2110.694 / (2 \cdot 1234.969 / 46.759 - 869.924 / 28.0)$$

$$= 97.0 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / [t / R_{xs}]$$

$$= 1407.129 [28.0 / 869.924]$$

$$= 45.3 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(97.024, 45.291)$$

$$= 45.291 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σAvg]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (34.521 + 979.81 + 288.071) / 46.759$$

$$= 27.854 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 1.055 \cdot 869.924 / 28.0$$

$$= 32.8 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 27.854 - 32.765, 32.765)$$

$$= 32.8 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2110.69	kgf/cm ²
Local Primary Membrane Stress	PL	32.77	kgf/cm ²
Maximum Allowable External Pressure	Pmax	45.29	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.



Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

The Drop for this Nozzle is : 3.3508 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 287.3507 mm

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Input, Nozzle Desc: L250

From: 50

Pressure for Reinforcement Calculations	P	45.088	kgf/cm ²
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	1700.00	mm
Design Length of Section	L	18483.8340	mm
Shell Finished (Minimum) Thickness	t	34.0000	mm
Shell Internal Corrosion Allowance	c	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		25.00	cm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1505.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1682.00	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		50.8000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm

Flange Material		SA-105	+IT
Flange Type		N/A	

Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	33.2360	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Style of FVC Nozzle		Long Weld Neck	

Corrosion Allowance	can	6.0000	mm
---------------------	-----	--------	----

Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

Flange Class		300
Flange Grade		GR 1.1

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: L250

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	50.800	mm.
Actual Thickness Used in Calculation	12.000	mm

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 62.8 \cdot (\exp(45.088/(1505.0 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 0.9549 + 6.0 + 0.0 = 6.9549 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1505.0 \cdot 1.0 \cdot \ln((2 \cdot 6.0 + 62.8) / 62.8) - 0.0$$

$$= 263.169 - 0.0 = 263.1686 \text{ kgf/cm}^2$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	10.800 mm
Min. required shell thickness, internal p + c :	33.875 mm
Min. required shell thickness, external p + c :	26.261 mm

Nozzle neck minimum required thickness	: 10.800 mm
--	-------------

Minimum nozzle neck thickness 11.989 mm , is greater than required thickness 10.800 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1505.0/1407.1, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 50.0 - 6.0$$

$$= 44.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 12.0 - 6.0$$

$$= 6.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D/t$$

$$= 1712.0/28.0$$

$$= 61.143 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 1700.0/2 + 6.0$$

$$= 856.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{856.0 \cdot 28.0}, 2 \cdot 31.4)$$

$$= 62.800 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 28.0, 0.0] + \sqrt{31.4 \cdot 44.0}$$

$$= 37.170 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 250.0$$

$$= 250.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(28.0 + 0.0)$$

$$= 224.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$



$$= \min[37.17, 250.0, 224.0] + 28.0$$

$$= 65.170 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 28.0 + ((0.0 \cdot 1.0) / 62.8)$$

$$= 28.000 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot Rn + tn) / (\sqrt{((Di + teff) \cdot teff)}))$$

$$= \min(12, (2 \cdot 31.4 + 44.0) / (\sqrt{((1712.0 + 28.0) \cdot 28.0)}))$$

$$= 0.484$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 28.0 \cdot 62.8 \cdot \max((0.484/5)^{0.85}, 1)$$

$$= 17.584 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= tn \cdot LH$$

$$= 44.0 \cdot 65.17$$

$$= 28.675 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot Leg^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + frn(A2 + A3) + A41 + A42 + A43$$

$$= 17.584 + 1.0(28.675 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 46.759 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= tn / \ln[1 + tn/Rn]$$

$$= 44.0 / \ln[1 + 44.0/31.4]$$

$$= 50.228 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= teff / \ln[1 + teff/Reff]$$

$$= 28.0 / \ln[1 + 28.0/856.0]$$

$$= 869.924 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 1407.129 \cdot 1.0$$

$$= 2110.7 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot Rxn \cdot LH$$

$$= 45.088 \cdot 50.228 \cdot 65.17$$

$$= 1475.9 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot Rxs(LR + tn)$$

$$= 45.088 \cdot 869.924(62.8 + 44.0)$$

$$= 41890.1 \text{ kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot Rxs \cdot Rnc$$

$$= 45.088 \cdot 869.924 \cdot 31.4$$

$$= 12316.0 \text{ kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P$$

$$= (41890.1 + 12316.0 + 1475.9) / 45.088$$

$$= 1235.0 \text{ cm}^2$$



Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= \text{Sallow} / (2 \cdot \text{Ap} / \text{AT} - \text{Rxs} / \text{teff})$$

$$= 2110.694 / (2 \cdot 1234.969 / 46.759 - 869.924 / 28.0)$$

$$= 97.0 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= \text{S} / \text{Rxs}$$

$$= 1407.129 / [28.0 / 869.924]$$

$$= 45.3 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(\text{Pmax1}, \text{Pmax2})$$

$$= \min(97.024, 45.291)$$

$$= 45.291 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (fN + fS + fY) / \text{AT}$$

$$= (1475.894 + 41890.141 + 12316.015) / 46.759$$

$$= 1190.839 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot \text{Rxs} / \text{teff}$$

$$= 45.088 \cdot 869.924 / 28.0$$

$$= 1400.8 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}})$$

$$= \max(2 \cdot 1190.839 - 1400.823, 1400.823)$$

$$= 1400.8 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2110.69	kgf/cm ²
Local Primary Membrane Stress	PL	1400.82	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	45.29	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (31.4 + 44.0) / 31.4$$

$$= 2.401 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [LT]:

$$= \pi / 2 \cdot (R_n + t_n)$$

$$= \pi / 2 \cdot (31.4 + 44.0)$$

$$= 118.438 \text{ mm}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 44.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi / 4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(12316 \cdot 2.4, 1.5 \cdot 1505.0(28.675 + 0.0), \pi / 4 \cdot 45.1 \cdot 31.4^2 \cdot 2.4^2)$$

$$= 2013.223 \text{ kgf}$$

Weld Stress Value [τ]:

$$= \text{fwelds} / (L_T(0.49 \cdot L41T + 0.6 \cdot t_{w1} + 0.49 \cdot L43T))$$

$$= 2013.223 / (118.438(0.49 \cdot 7.071 + 0.6 \cdot 28.0 + 0.49 \cdot 0.0))$$

$$= 83.880 < \text{or } = \text{to } 1407.129 \text{ Weld Stress Passed}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n / R_n]$$

$$= 44.0 / \ln[1 + 44.0 / 31.4]$$

$$= 50.228 \text{ mm}$$



Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 28.0 / \ln[1 + 28.0/856.0]$$

$$= 869.924 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S$$

$$= 1.5 \cdot 1407.129$$

$$= 2110.7 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 1.055 \cdot 50.228 \cdot 65.17$$

$$= 34.5 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 1.055 \cdot 869.924 (62.8 + 44.0)$$

$$= 979.8 \text{ kgf}$$

Discontinuity Force from External Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 1.055 \cdot 869.924 \cdot 31.4$$

$$= 288.1 \text{ kgf}$$

Area Resisting External Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (979.8 + 288.1 + 34.5) / 1.055$$

$$= 1235.0 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 2110.694 / (2 \cdot 1234.969 / 46.759 - 869.924 / 28.0)$$

$$= 97.0 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / [t / R_{xs}]$$

$$= 1407.129 [28.0 / 869.924]$$

$$= 45.3 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(97.024, 45.291)$$

$$= 45.291 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σAvg]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (34.521 + 979.81 + 288.071) / 46.759$$

$$= 27.854 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 1.055 \cdot 869.924 / 28.0$$

$$= 32.8 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 27.854 - 32.765, 32.765)$$

$$= 32.8 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2110.69	kgf/cm ²
Local Primary Membrane Stress	PL	32.77	kgf/cm ²
Maximum Allowable External Pressure	Pmax	45.29	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.



Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

The Drop for this Nozzle is : 3.3508 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 287.3507 mm

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Input, Nozzle Desc: L350

From: 50

Pressure for Reinforcement Calculations	P	44.904	kgf/cm ²
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	1700.00	mm
Design Length of Section	L	18483.8340	mm
Shell Finished (Minimum) Thickness	t	34.0000	mm
Shell Internal Corrosion Allowance	c	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		205.00	cm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1505.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1682.00	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		50.8000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm

Flange Material		SA-105	+IT
Flange Type		N/A	

Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	33.2360	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Style of FVC Nozzle		Long Weld Neck	

Corrosion Allowance	can	6.0000	mm
---------------------	-----	--------	----

Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

Flange Class		300
Flange Grade		GR 1.1

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: L350

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		50.800	mm.
Actual Thickness Used in Calculation		12.000	mm

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 62.8 \cdot (\exp(44.904/(1505.0 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 0.951 + 6.0 + 0.0 = 6.951 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1505.0 \cdot 1.0 \cdot \ln((2 \cdot 6.0 + 62.8) / 62.8) - 0.0$$

$$= 263.169 - 0.0 = 263.1686 \text{ kgf/cm}^2$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	10.800 mm
Min. required shell thickness, internal p + c	: 33.759 mm
Min. required shell thickness, external p + c	: 26.261 mm

Nozzle neck minimum required thickness	: 10.800 mm
--	-------------

Minimum nozzle neck thickness 11.989 mm , is greater than required thickness 10.800 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1505.0/1407.1, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 50.0 - 6.0$$

$$= 44.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 12.0 - 6.0$$

$$= 6.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D/t$$

$$= 1712.0/28.0$$

$$= 61.143 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 1700.0/2 + 6.0$$

$$= 856.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{856.0 \cdot 28.0}, 2 \cdot 31.4)$$

$$= 62.800 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 28.0, 0.0] + \sqrt{31.4 \cdot 44.0}$$

$$= 37.170 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 250.0$$

$$= 250.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(28.0 + 0.0)$$

$$= 224.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[37.17, 250.0, 224.0] + 28.0$$

$$= 65.170 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 28.0 + ((0.0 \cdot 1.0) / 62.8)$$

$$= 28.000 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot Rn + tn) / (\sqrt{(Di + teff) \cdot teff}))$$

$$= \min(12, (2 \cdot 31.4 + 44.0) / (\sqrt{(1712.0 + 28.0) \cdot 28.0}))$$

$$= 0.484$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 28.0 \cdot 62.8 \cdot \max((0.484/5)^{0.85}, 1)$$

$$= 17.584 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= tn \cdot LH$$

$$= 44.0 \cdot 65.17$$

$$= 28.675 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot Leg^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + frn(A2 + A3) + A41 + A42 + A43$$

$$= 17.584 + 1.0(28.675 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 46.759 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= tn / \ln[1 + tn/Rn]$$

$$= 44.0 / \ln[1 + 44.0/31.4]$$

$$= 50.228 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= teff / \ln[1 + teff/Reff]$$

$$= 28.0 / \ln[1 + 28.0/856.0]$$

$$= 869.924 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 1407.129 \cdot 1.0$$

$$= 2110.7 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot Rxn \cdot LH$$

$$= 44.904 \cdot 50.228 \cdot 65.17$$

$$= 1469.9 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot Rxs \cdot (LR + tn)$$

$$= 44.904 \cdot 869.924 \cdot (62.8 + 44.0)$$

$$= 41719.1 \text{ kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot Rxs \cdot Rnc$$

$$= 44.904 \cdot 869.924 \cdot 31.4$$

$$= 12265.7 \text{ kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P$$

$$= (41719.1 + 12265.7 + 1469.9) / 44.904$$

$$= 1235.0 \text{ cm}^2$$



Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= \text{Sallow} / (2 \cdot \text{Ap} / \text{AT} - \text{Rxs} / \text{teff})$$

$$= 2110.694 / (2 \cdot 1234.969 / 46.759 - 869.924 / 28.0)$$

$$= 97.0 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= \text{S} / \text{Rxs}$$

$$= 1407.129 [28.0 / 869.924]$$

$$= 45.3 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(\text{Pmax1}, \text{Pmax2})$$

$$= \min(97.024, 45.291)$$

$$= 45.291 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (fN + fS + fY) / \text{AT}$$

$$= (1469.869 + 41719.137 + 12265.739) / 46.759$$

$$= 1185.977 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot \text{Rxs} / \text{teff}$$

$$= 44.904 \cdot 869.924 / 28.0$$

$$= 1395.1 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}})$$

$$= \max(2 \cdot 1185.977 - 1395.104, 1395.104)$$

$$= 1395.1 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2110.69	kgf/cm ²
Local Primary Membrane Stress	PL	1395.10	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	45.29	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (31.4 + 44.0) / 31.4$$

$$= 2.401 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [LT]:

$$= \pi / 2 \cdot (R_n + t_n)$$

$$= \pi / 2 \cdot (31.4 + 44.0)$$

$$= 118.438 \text{ mm}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 44.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi / 4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(12266 \cdot 2.4, 1.5 \cdot 1505.0(28.675 + 0.0), \pi / 4 \cdot 44.9 \cdot 31.4^2 \cdot 2.4^2)$$

$$= 2005.004 \text{ kgf}$$

Weld Stress Value [τ]:

$$= \text{fwelds} / (L_T(0.49 \cdot L41T + 0.6 \cdot t_{w1} + 0.49 \cdot L43T))$$

$$= 2005.004 / (118.438(0.49 \cdot 7.071 + 0.6 \cdot 28.0 + 0.49 \cdot 0.0))$$

$$= 83.538 < \text{or } = \text{to } 1407.129 \text{ Weld Stress Passed}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n / R_n]$$

$$= 44.0 / \ln[1 + 44.0 / 31.4]$$

$$= 50.228 \text{ mm}$$



Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 28.0 / \ln[1 + 28.0/856.0]$$

$$= 869.924 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S$$

$$= 1.5 \cdot 1407.129$$

$$= 2110.7 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 1.055 \cdot 50.228 \cdot 65.17$$

$$= 34.5 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 1.055 \cdot 869.924 (62.8 + 44.0)$$

$$= 979.8 \text{ kgf}$$

Discontinuity Force from External Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 1.055 \cdot 869.924 \cdot 31.4$$

$$= 288.1 \text{ kgf}$$

Area Resisting External Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (979.8 + 288.1 + 34.5) / 1.055$$

$$= 1235.0 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 2110.694 / (2 \cdot 1234.969 / 46.759 - 869.924 / 28.0)$$

$$= 97.0 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / [t / R_{xs}]$$

$$= 1407.129 [28.0 / 869.924]$$

$$= 45.3 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(97.024, 45.291)$$

$$= 45.291 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σAvg]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (34.521 + 979.81 + 288.071) / 46.759$$

$$= 27.854 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 1.055 \cdot 869.924 / 28.0$$

$$= 32.8 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 27.854 - 32.765, 32.765)$$

$$= 32.8 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2110.69	kgf/cm ²
Local Primary Membrane Stress	PL	32.77	kgf/cm ²
Maximum Allowable External Pressure	Pmax	45.29	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.



Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

The Drop for this Nozzle is : 3.3508 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 287.3507 mm

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Input, Nozzle Desc: L450

From: 50

Pressure for Reinforcement Calculations	P	45.006	kgf/cm ²
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	1700.00	mm
Design Length of Section	L	18483.8340	mm
Shell Finished (Minimum) Thickness	t	34.0000	mm
Shell Internal Corrosion Allowance	c	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		105.00	cm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1505.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1682.00	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		50.8000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm

Flange Material		SA-105	+IT
Flange Type		N/A	

Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	33.2360	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Style of FVC Nozzle		Long Weld Neck	

Corrosion Allowance	can	6.0000	mm
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Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

Flange Class		300
Flange Grade		GR 1.1

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: L450

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	50.800	mm.
Actual Thickness Used in Calculation	12.000	mm

Nozzle input data check completed without errors.



Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 62.8 \cdot (\exp(45.006/(1505.0 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 0.9532 + 6.0 + 0.0 = 6.9532 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1505.0 \cdot 1.0 \cdot \ln((2 \cdot 6.0 + 62.8) / 62.8) - 0.0$$

$$= 263.169 - 0.0 = 263.1686 \text{ kgf/cm}^2$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	10.800 mm
Min. required shell thickness, internal p + c	: 33.824 mm
Min. required shell thickness, external p + c	: 26.261 mm

Nozzle neck minimum required thickness	: 10.800 mm
--	-------------

Minimum nozzle neck thickness 11.989 mm , is greater than required thickness 10.800 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1505.0/1407.1, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 50.0 - 6.0$$

$$= 44.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 12.0 - 6.0$$

$$= 6.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D/t$$

$$= 1712.0/28.0$$

$$= 61.143 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 1700.0/2 + 6.0$$

$$= 856.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{856.0 \cdot 28.0}, 2 \cdot 31.4)$$

$$= 62.800 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 28.0, 0.0] + \sqrt{31.4 \cdot 44.0}$$

$$= 37.170 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 250.0$$

$$= 250.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(28.0 + 0.0)$$

$$= 224.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[37.17, 250.0, 224.0] + 28.0$$

$$= 65.170 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 28.0 + ((0.0 \cdot 1.0) / 62.8)$$

$$= 28.000 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot Rn + tn) / (\sqrt{(Di + teff) \cdot teff}))$$

$$= \min(12, (2 \cdot 31.4 + 44.0) / (\sqrt{(1712.0 + 28.0) \cdot 28.0}))$$

$$= 0.484$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 28.0 \cdot 62.8 \cdot \max((0.484/5)^{0.85}, 1)$$

$$= 17.584 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= tn \cdot LH$$

$$= 44.0 \cdot 65.17$$

$$= 28.675 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot Leg^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + frn(A2 + A3) + A41 + A42 + A43$$

$$= 17.584 + 1.0(28.675 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 46.759 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= tn / \ln[1 + tn/Rn]$$

$$= 44.0 / \ln[1 + 44.0/31.4]$$

$$= 50.228 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= teff / \ln[1 + teff/Reff]$$

$$= 28.0 / \ln[1 + 28.0/856.0]$$

$$= 869.924 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 1407.129 \cdot 1.0$$

$$= 2110.7 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot Rxn \cdot LH$$

$$= 45.006 \cdot 50.228 \cdot 65.17$$

$$= 1473.2 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot Rxs \cdot (LR + tn)$$

$$= 45.006 \cdot 869.924 \cdot (62.8 + 44.0)$$

$$= 41814.1 \text{ kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot Rxs \cdot Rnc$$

$$= 45.006 \cdot 869.924 \cdot 31.4$$

$$= 12293.7 \text{ kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P$$

$$= (41814.1 + 12293.7 + 1473.2) / 45.006$$

$$= 1235.0 \text{ cm}^2$$



Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= \text{Sallow} / (2 \cdot \text{Ap}/\text{AT} - \text{Rxs}/\text{teff})$$

$$= 2110.694 / (2 \cdot 1234.969/46.759 - 869.924/28.0)$$

$$= 97.0 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= \text{S}/[\text{t}/\text{Rxs}]$$

$$= 1407.129[28.0/869.924]$$

$$= 45.3 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(\text{Pmax1}, \text{Pmax2})$$

$$= \min(97.024, 45.291)$$

$$= 45.291 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (fN + fS + fY) / \text{AT}$$

$$= (1473.216 + 41814.137 + 12293.67) / 46.759$$

$$= 1188.678 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot \text{Rxs} / \text{teff}$$

$$= 45.006 \cdot 869.924 / 28.0$$

$$= 1398.3 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}})$$

$$= \max(2 \cdot 1188.678 - 1398.281, 1398.281)$$

$$= 1398.3 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2110.69	kgf/cm ²
Local Primary Membrane Stress	PL	1398.28	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	45.29	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (31.4 + 44.0) / 31.4$$

$$= 2.401 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [LT]:

$$= \pi/2 \cdot (R_n + t_n)$$

$$= \pi/2 \cdot (31.4 + 44.0)$$

$$= 118.438 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 44.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(12294 \cdot 2.4, 1.5 \cdot 1505.0(28.675 + 0.0), \pi/4 \cdot 45.0 \cdot 31.4^2 \cdot 2.4^2)$$

$$= 2009.570 \text{ kgf}$$

Weld Stress Value [τ]:

$$= \text{fwelds} / (L_T(0.49 \cdot L41T + 0.6 \cdot t_{w1} + 0.49 \cdot L43T))$$

$$= 2009.57 / (118.438(0.49 \cdot 7.071 + 0.6 \cdot 28.0 + 0.49 \cdot 0.0))$$

$$= 83.728 < \text{or} = \text{to } 1407.129 \text{ Weld Stress Passed}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 44.0 / \ln[1 + 44.0/31.4]$$

$$= 50.228 \text{ mm}$$



Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 28.0 / \ln[1 + 28.0/856.0]$$

$$= 869.924 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S$$

$$= 1.5 \cdot 1407.129$$

$$= 2110.7 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 1.055 \cdot 50.228 \cdot 65.17$$

$$= 34.5 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 1.055 \cdot 869.924 (62.8 + 44.0)$$

$$= 979.8 \text{ kgf}$$

Discontinuity Force from External Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 1.055 \cdot 869.924 \cdot 31.4$$

$$= 288.1 \text{ kgf}$$

Area Resisting External Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (979.8 + 288.1 + 34.5) / 1.055$$

$$= 1235.0 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 2110.694 / (2 \cdot 1234.969 / 46.759 - 869.924 / 28.0)$$

$$= 97.0 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / [t / R_{xs}]$$

$$= 1407.129 [28.0 / 869.924]$$

$$= 45.3 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(97.024, 45.291)$$

$$= 45.291 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (34.521 + 979.81 + 288.071) / 46.759$$

$$= 27.854 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 1.055 \cdot 869.924 / 28.0$$

$$= 32.8 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 27.854 - 32.765, 32.765)$$

$$= 32.8 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2110.69	kgf/cm ²
Local Primary Membrane Stress	PL	32.77	kgf/cm ²
Maximum Allowable External Pressure	Pmax	45.29	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.



Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

The Drop for this Nozzle is : 3.3508 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 287.3507 mm

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Input, Nozzle Desc: L550

From: 50

Pressure for Reinforcement Calculations	P	44.904	kgf/cm ²
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	1700.00	mm
Design Length of Section	L	18483.8340	mm
Shell Finished (Minimum) Thickness	t	34.0000	mm
Shell Internal Corrosion Allowance	c	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		205.00	cm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1505.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1682.00	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		50.8000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm

Flange Material		SA-105	+IT
Flange Type		N/A	

Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	33.2360	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Style of FVC Nozzle		Long Weld Neck	

Corrosion Allowance	can	6.0000	mm
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Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

Flange Class		300
Flange Grade		GR 1.1

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: L550

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	50.800	mm.
Actual Thickness Used in Calculation	12.000	mm

Nozzle input data check completed without errors.



Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 62.8 \cdot (\exp(44.904/(1505.0 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 0.951 + 6.0 + 0.0 = 6.951 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1505.0 \cdot 1.0 \cdot \ln((2 \cdot 6.0 + 62.8) / 62.8) - 0.0$$

$$= 263.169 - 0.0 = 263.1686 \text{ kgf/cm}^2$$

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 138.20 mm

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	10.800 mm
Min. required shell thickness, internal p + c :	33.759 mm
Min. required shell thickness, external p + c :	26.261 mm
Nozzle neck minimum required thickness :	10.800 mm

Minimum nozzle neck thickness 11.989 mm , is greater than required thickness 10.800 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1505.0/1407.1, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 50.0 - 6.0$$

$$= 44.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 12.0 - 6.0$$

$$= 6.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 1712.0/28.0$$

$$= 61.143 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 1700.0/2 + 6.0$$

$$= 856.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{856.0 \cdot 28.0}, 2 \cdot 31.4)$$

$$= 62.800 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 28.0, 0.0] + \sqrt{31.4 \cdot 44.0}$$

$$= 37.170 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 250.0$$

$$= 250.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(28.0 + 0.0)$$

$$= 224.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles}$$

$$= \min[37.17, 250.0, 224.0] + 28.0$$

$$= 65.170 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 28.0 + ((0.0 \cdot 1.0) / 62.8)$$

$$= 28.000 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot Rn + tn) / (\sqrt{(Di + teff) \cdot teff}))$$

$$= \min(12, (2 \cdot 31.4 + 44.0) / (\sqrt{(1712.0 + 28.0) \cdot 28.0}))$$

$$= 0.484$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 28.0 \cdot 62.8 \cdot \max((0.484/5)^{0.85}, 1)$$

$$= 17.584 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= tn \cdot LH$$

$$= 44.0 \cdot 65.17$$

$$= 28.675 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot Leg^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + frn(A2 + A3) + A41 + A42 + A43$$

$$= 17.584 + 1.0(28.675 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 46.759 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= tn / \ln[1 + tn/Rn]$$

$$= 44.0 / \ln[1 + 44.0/31.4]$$

$$= 50.228 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= teff / \ln[1 + teff/Reff]$$

$$= 28.0 / \ln[1 + 28.0/856.0]$$

$$= 869.924 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 1407.129 \cdot 1.0$$

$$= 2110.7 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot Rxn \cdot LH$$

$$= 44.904 \cdot 50.228 \cdot 65.17$$

$$= 1469.9 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot Rxs(LR + tn)$$

$$= 44.904 \cdot 869.924(62.8 + 44.0)$$

$$= 41719.1 \text{ kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot Rxs \cdot Rnc$$

$$= 44.904 \cdot 869.924 \cdot 31.4$$

$$= 12265.7 \text{ kgf}$$

Area Resisting Internal Pressure [Ap]:



$$= (fS + fY + fN) / P$$

$$= (41719.1 + 12265.7 + 1469.9) / 44.904$$

$$= 1235.0 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap / AT - Rxs / teff)$$

$$= 2110.694 / (2 \cdot 1234.969 / 46.759 - 869.924 / 28.0)$$

$$= 97.0 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t / Rxs]$$

$$= 1407.129 [28.0 / 869.924]$$

$$= 45.3 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(97.024, 45.291)$$

$$= 45.291 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (fN + fS + fY) / AT$$

$$= (1469.869 + 41719.137 + 12265.739) / 46.759$$

$$= 1185.977 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot Rxs / teff$$

$$= 44.904 \cdot 869.924 / 28.0$$

$$= 1395.1 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 1185.977 - 1395.104, 1395.104)$$

$$= 1395.1 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2110.69	kgf/cm ²
Local Primary Membrane Stress	PL	1395.10	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	45.29	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$

$$= (31.4 + 44.0) / 31.4$$

$$= 2.401 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi / 2 \cdot (Rn + tn)$$

$$= \pi / 2 \cdot (31.4 + 44.0)$$

$$= 118.438 \text{ mm}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 44.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi / 4 \cdot P \cdot Rn^2 \cdot ky^2)$$

$$= \min(12266 \cdot 2.4, 1.5 \cdot 1505.0(28.675 + 0.0), \pi / 4 \cdot 44.9 \cdot 31.4^2 \cdot 2.4^2)$$

$$= 2005.004 \text{ kgf}$$

Weld Stress Value [τ]:

$$= fwelds / (Lt(0.49 \cdot L41T + 0.6 \cdot tw1 + 0.49 \cdot L43T))$$

$$= 2005.004 / (118.438(0.49 \cdot 7.071 + 0.6 \cdot 28.0 + 0.49 \cdot 0.0))$$

$$= 83.538 < \text{or } = \text{ to } 1407.129 \text{ Weld Stress Passed}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:



$$= t_n / \ln[1 + t_n/R_n]$$

$$= 44.0 / \ln[1 + 44.0/31.4]$$

$$= 50.228 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 28.0 / \ln[1 + 28.0/856.0]$$

$$= 869.924 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S$$

$$= 1.5 \cdot 1407.129$$

$$= 2110.7 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 1.055 \cdot 50.228 \cdot 65.17$$

$$= 34.5 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 1.055 \cdot 869.924 (62.8 + 44.0)$$

$$= 979.8 \text{ kgf}$$

Discontinuity Force from External Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 1.055 \cdot 869.924 \cdot 31.4$$

$$= 288.1 \text{ kgf}$$

Area Resisting External Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (979.8 + 288.1 + 34.5) / 1.055$$

$$= 1235.0 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 2110.694 / (2 \cdot 1234.969 / 46.759 - 869.924 / 28.0)$$

$$= 97.0 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / [t / R_{xs}]$$

$$= 1407.129 [28.0 / 869.924]$$

$$= 45.3 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(97.024, 45.291)$$

$$= 45.291 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (34.521 + 979.81 + 288.071) / 46.759$$

$$= 27.854 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 1.055 \cdot 869.924 / 28.0$$

$$= 32.8 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 27.854 - 32.765, 32.765)$$

$$= 32.8 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2110.69	kgf/cm ²
Local Primary Membrane Stress	PL	32.77	kgf/cm ²
Maximum Allowable External Pressure	Pmax	45.29	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:



Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

The Drop for this Nozzle is : 3.3508 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 287.3507 mm

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Input, Nozzle Desc: L650

From: 50

Pressure for Reinforcement Calculations	P	45.006	kgf/cm ²
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	1700.00	mm
Design Length of Section	L	18483.8340	mm
Shell Finished (Minimum) Thickness	t	34.0000	mm
Shell Internal Corrosion Allowance	c	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		105.00	cm
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User Entered Minimum Design Metal Temperature		-12.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1505.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1682.00	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		50.8000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm

Flange Material		SA-105	+IT
Flange Type		N/A	

Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	33.2360	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Style of FVC Nozzle		Long Weld Neck	

Corrosion Allowance	can	6.0000	mm
---------------------	-----	--------	----

Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

Flange Class		300
Flange Grade		GR 1.1

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: L650

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	50.800	mm.
Actual Thickness Used in Calculation	12.000	mm

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 62.8 \cdot (\exp(45.006/(1505.0 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 0.9532 + 6.0 + 0.0 = 6.9532 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1505.0 \cdot 1.0 \cdot \ln((2 \cdot 6.0 + 62.8) / 62.8) - 0.0$$

$$= 263.169 - 0.0 = 263.1686 \text{ kgf/cm}^2$$

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 138.20 mm

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	10.800 mm
Min. required shell thickness, internal p + c	: 33.824 mm
Min. required shell thickness, external p + c	: 26.261 mm

Nozzle neck minimum required thickness	: 10.800 mm
--	-------------

Minimum nozzle neck thickness 11.989 mm , is greater than required thickness 10.800 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1505.0/1407.1, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 50.0 - 6.0$$

$$= 44.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 12.0 - 6.0$$

$$= 6.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 1712.0/28.0$$

$$= 61.143 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 1700.0/2 + 6.0$$

$$= 856.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{856.0 \cdot 28.0}, 2 \cdot 31.4)$$

$$= 62.800 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 28.0, 0.0] + \sqrt{31.4 \cdot 44.0}$$

$$= 37.170 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 250.0$$

$$= 250.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(28.0 + 0.0)$$

$$= 224.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles}$$

$$= \min[37.17, 250.0, 224.0] + 28.0$$

$$= 65.170 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 28.0 + ((0.0 \cdot 1.0) / 62.8)$$

$$= 28.000 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot Rn + tn) / (\sqrt{(Di + teff) \cdot teff}))$$

$$= \min(12, (2 \cdot 31.4 + 44.0) / (\sqrt{(1712.0 + 28.0) \cdot 28.0}))$$

$$= 0.484$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 28.0 \cdot 62.8 \cdot \max((0.484/5)^{0.85}, 1)$$

$$= 17.584 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= tn \cdot LH$$

$$= 44.0 \cdot 65.17$$

$$= 28.675 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot Leg^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + frn(A2 + A3) + A41 + A42 + A43$$

$$= 17.584 + 1.0(28.675 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 46.759 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= tn / \ln[1 + tn/Rn]$$

$$= 44.0 / \ln[1 + 44.0/31.4]$$

$$= 50.228 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= teff / \ln[1 + teff/Reff]$$

$$= 28.0 / \ln[1 + 28.0/856.0]$$

$$= 869.924 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 1407.129 \cdot 1.0$$

$$= 2110.7 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot Rxn \cdot LH$$

$$= 45.006 \cdot 50.228 \cdot 65.17$$

$$= 1473.2 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot Rxs(LR + tn)$$

$$= 45.006 \cdot 869.924(62.8 + 44.0)$$

$$= 41814.1 \text{ kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot Rxs \cdot Rnc$$

$$= 45.006 \cdot 869.924 \cdot 31.4$$

$$= 12293.7 \text{ kgf}$$

Area Resisting Internal Pressure [Ap]:



$$= (fS + fY + fN) / P$$

$$= (41814.1 + 12293.7 + 1473.2) / 45.006$$

$$= 1235.0 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap / AT - Rxs / teff)$$

$$= 2110.694 / (2 \cdot 1234.969 / 46.759 - 869.924 / 28.0)$$

$$= 97.0 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t / Rxs]$$

$$= 1407.129 [28.0 / 869.924]$$

$$= 45.3 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(97.024, 45.291)$$

$$= 45.291 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (fN + fS + fY) / AT$$

$$= (1473.216 + 41814.137 + 12293.67) / 46.759$$

$$= 1188.678 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot Rxs / teff$$

$$= 45.006 \cdot 869.924 / 28.0$$

$$= 1398.3 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 1188.678 - 1398.281, 1398.281)$$

$$= 1398.3 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2110.69	kgf/cm ²
Local Primary Membrane Stress	PL	1398.28	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	45.29	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$

$$= (31.4 + 44.0) / 31.4$$

$$= 2.401 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi / 2 \cdot (Rn + tn)$$

$$= \pi / 2 \cdot (31.4 + 44.0)$$

$$= 118.438 \text{ mm}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 44.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi / 4 \cdot P \cdot Rn^2 \cdot ky^2)$$

$$= \min(12294 \cdot 2.4, 1.5 \cdot 1505.0(28.675 + 0.0), \pi / 4 \cdot 45.0 \cdot 31.4^2 \cdot 2.4^2)$$

$$= 2009.570 \text{ kgf}$$

Weld Stress Value [τ]:

$$= fwelds / (Lt(0.49 \cdot L41T + 0.6 \cdot tw1 + 0.49 \cdot L43T))$$

$$= 2009.57 / (118.438(0.49 \cdot 7.071 + 0.6 \cdot 28.0 + 0.49 \cdot 0.0))$$

$$= 83.728 < \text{or } = \text{ to } 1407.129 \text{ Weld Stress Passed}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:



$$= t_n / \ln[1 + t_n/R_n]$$

$$= 44.0 / \ln[1 + 44.0/31.4]$$

$$= 50.228 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 28.0 / \ln[1 + 28.0/856.0]$$

$$= 869.924 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S$$

$$= 1.5 \cdot 1407.129$$

$$= 2110.7 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 1.055 \cdot 50.228 \cdot 65.17$$

$$= 34.5 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 1.055 \cdot 869.924 (62.8 + 44.0)$$

$$= 979.8 \text{ kgf}$$

Discontinuity Force from External Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 1.055 \cdot 869.924 \cdot 31.4$$

$$= 288.1 \text{ kgf}$$

Area Resisting External Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (979.8 + 288.1 + 34.5) / 1.055$$

$$= 1235.0 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 2110.694 / (2 \cdot 1234.969 / 46.759 - 869.924 / 28.0)$$

$$= 97.0 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / [t / R_{xs}]$$

$$= 1407.129 [28.0 / 869.924]$$

$$= 45.3 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(97.024, 45.291)$$

$$= 45.291 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σAvg]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (34.521 + 979.81 + 288.071) / 46.759$$

$$= 27.854 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 1.055 \cdot 869.924 / 28.0$$

$$= 32.8 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 27.854 - 32.765, 32.765)$$

$$= 32.8 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2110.69	kgf/cm ²
Local Primary Membrane Stress	PL	32.77	kgf/cm ²
Maximum Allowable External Pressure	Pmax	45.29	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:



Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

The Drop for this Nozzle is : 3.3508 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 287.3507 mm

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Input, Nozzle Desc: DP150

From: 50

Pressure for Reinforcement Calculations	P	44.822	kgf/cm ²
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	1700.00	mm
Design Length of Section	L	18483.8340	mm
Shell Finished (Minimum) Thickness	t	34.0000	mm
Shell Internal Corrosion Allowance	c	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		285.00	cm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1505.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1682.00	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		50.8000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm

Flange Material		SA-105	+IT
Flange Type		N/A	

Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	33.2360	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Style of FVC Nozzle		Long Weld Neck	

Corrosion Allowance	can	6.0000	mm
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Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

Flange Class		300
Flange Grade		GR 1.1

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: DP150

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	50.800	mm.
Actual Thickness Used in Calculation	12.000	mm

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 62.8 \cdot (\exp(44.822/(1505.0 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 0.9492 + 6.0 + 0.0 = 6.9492 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1505.0 \cdot 1.0 \cdot \ln((2 \cdot 6.0 + 62.8) / 62.8) - 0.0$$

$$= 263.169 - 0.0 = 263.1686 \text{ kgf/cm}^2$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	10.800 mm
Min. required shell thickness, internal p + c	: 33.708 mm
Min. required shell thickness, external p + c	: 26.261 mm

Nozzle neck minimum required thickness	: 10.800 mm
--	-------------

Minimum nozzle neck thickness 11.989 mm , is greater than required thickness 10.800 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1505.0/1407.1, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 50.0 - 6.0$$

$$= 44.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 12.0 - 6.0$$

$$= 6.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D/t$$

$$= 1712.0/28.0$$

$$= 61.143 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 1700.0/2 + 6.0$$

$$= 856.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{856.0 \cdot 28.0}, 2 \cdot 31.4)$$

$$= 62.800 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 28.0, 0.0] + \sqrt{31.4 \cdot 44.0}$$

$$= 37.170 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 250.0$$

$$= 250.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(28.0 + 0.0)$$

$$= 224.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[37.17, 250.0, 224.0] + 28.0$$

$$= 65.170 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 28.0 + ((0.0 \cdot 1.0) / 62.8)$$

$$= 28.000 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot Rn + tn) / (\sqrt{((Di + teff) \cdot teff)}))$$

$$= \min(12, (2 \cdot 31.4 + 44.0) / (\sqrt{((1712.0 + 28.0) \cdot 28.0)}))$$

$$= 0.484$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 28.0 \cdot 62.8 \cdot \max((0.484/5)^{0.85}, 1)$$

$$= 17.584 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= tn \cdot LH$$

$$= 44.0 \cdot 65.17$$

$$= 28.675 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot Leg^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$

$$= 17.584 + 1.0(28.675 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 46.759 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= tn / \ln[1 + tn/Rn]$$

$$= 44.0 / \ln[1 + 44.0/31.4]$$

$$= 50.228 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= teff / \ln[1 + teff/Reff]$$

$$= 28.0 / \ln[1 + 28.0/856.0]$$

$$= 869.924 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 1407.129 \cdot 1.0$$

$$= 2110.7 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot Rxn \cdot LH$$

$$= 44.822 \cdot 50.228 \cdot 65.17$$

$$= 1467.2 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot Rxs \cdot (LR + tn)$$

$$= 44.822 \cdot 869.924 \cdot (62.8 + 44.0)$$

$$= 41643.1 \text{ kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot Rxs \cdot Rnc$$

$$= 44.822 \cdot 869.924 \cdot 31.4$$

$$= 12243.4 \text{ kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P$$

$$= (41643.1 + 12243.4 + 1467.2) / 44.822$$

$$= 1235.0 \text{ cm}^2$$



Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= \text{Sallow} / (2 \cdot \text{Ap} / \text{AT} - \text{Rxs} / \text{teff})$$

$$= 2110.694 / (2 \cdot 1234.969 / 46.759 - 869.924 / 28.0)$$

$$= 97.0 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= \text{S} / \text{Rxs}$$

$$= 1407.129 / [28.0 / 869.924]$$

$$= 45.3 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(\text{Pmax1}, \text{Pmax2})$$

$$= \min(97.024, 45.291)$$

$$= 45.291 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (fN + fS + fY) / \text{AT}$$

$$= (1467.191 + 41643.133 + 12243.394) / 46.759$$

$$= 1183.817 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot \text{Rxs} / \text{teff}$$

$$= 44.822 \cdot 869.924 / 28.0$$

$$= 1392.6 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}})$$

$$= \max(2 \cdot 1183.817 - 1392.562, 1392.562)$$

$$= 1392.6 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2110.69	kgf/cm ²
Local Primary Membrane Stress	PL	1392.56	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	45.29	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (31.4 + 44.0) / 31.4$$

$$= 2.401 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [LT]:

$$= \pi / 2 \cdot (R_n + t_n)$$

$$= \pi / 2 \cdot (31.4 + 44.0)$$

$$= 118.438 \text{ mm}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 44.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi / 4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(12243 \cdot 2.4, 1.5 \cdot 1505.0(28.675 + 0.0), \pi / 4 \cdot 44.8 \cdot 31.4^2 \cdot 2.4^2)$$

$$= 2001.352 \text{ kgf}$$

Weld Stress Value [τ]:

$$= \text{fwelds} / (L_T(0.49 \cdot L41T + 0.6 \cdot t_{w1} + 0.49 \cdot L43T))$$

$$= 2001.352 / (118.438(0.49 \cdot 7.071 + 0.6 \cdot 28.0 + 0.49 \cdot 0.0))$$

$$= 83.386 < \text{or } = \text{to } 1407.129 \text{ Weld Stress Passed}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n / R_n]$$

$$= 44.0 / \ln[1 + 44.0 / 31.4]$$

$$= 50.228 \text{ mm}$$



Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 28.0 / \ln[1 + 28.0/856.0]$$

$$= 869.924 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S$$

$$= 1.5 \cdot 1407.129$$

$$= 2110.7 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 1.055 \cdot 50.228 \cdot 65.17$$

$$= 34.5 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 1.055 \cdot 869.924 (62.8 + 44.0)$$

$$= 979.8 \text{ kgf}$$

Discontinuity Force from External Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 1.055 \cdot 869.924 \cdot 31.4$$

$$= 288.1 \text{ kgf}$$

Area Resisting External Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (979.8 + 288.1 + 34.5) / 1.055$$

$$= 1235.0 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 2110.694 / (2 \cdot 1234.969 / 46.759 - 869.924 / 28.0)$$

$$= 97.0 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / [t / R_{xs}]$$

$$= 1407.129 [28.0 / 869.924]$$

$$= 45.3 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(97.024, 45.291)$$

$$= 45.291 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σAvg]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (34.521 + 979.81 + 288.071) / 46.759$$

$$= 27.854 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 1.055 \cdot 869.924 / 28.0$$

$$= 32.8 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 27.854 - 32.765, 32.765)$$

$$= 32.8 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2110.69	kgf/cm ²
Local Primary Membrane Stress	PL	32.77	kgf/cm ²
Maximum Allowable External Pressure	Pmax	45.29	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.



Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

The Drop for this Nozzle is : 3.3508 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 287.3507 mm

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Input, Nozzle Desc: UC2600

From: 60

Pressure for Reinforcement Calculations	P	44.699	kgf/cm ²
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	1700.00	mm
Design Length of Section	L	18483.8340	mm
Shell Finished (Minimum) Thickness	t	34.0000	mm
Shell Internal Corrosion Allowance	c	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		405.00	cm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1505.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1682.00	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		568.0000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm

Flange Material		SA-105	+IT
-----------------	--	--------	-----

Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	64.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		80.0000	mm

Corrosion Allowance	can	6.0000	mm
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Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	32.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

This is a Manway, Coupling or Access Opening.

Flange Class		300
Flange Grade		GR 1.1

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: UC2600

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	568.000	mm.
Actual Thickness Used in Calculation	16.000	mm

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight



of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 580.0 \cdot (\exp(44.699/(1505.0 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 8.7423 + 6.0 + 0.0 = 14.7423 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1505.0 \cdot 1.0 \cdot \ln((2 \cdot 10.0 + 580.0) / 580.0) - 0.0$$

$$= 51.022 - 0.0 = 51.0218 \text{ kgf/cm}^2$$

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1505.0/1407.1, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 80.0 - 6.0$$

$$= 74.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 16.0 - 6.0$$

$$= 10.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 1712.0/28.0$$

$$= 61.143 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 1700.0/2 + 6.0$$

$$= 856.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{856.0 \cdot 28.0}, 2 \cdot 290.0)$$

$$= 154.816 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 28.0, 0.0] + \sqrt{290.0 \cdot 74.0}$$

$$= 146.492 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 500.0$$

$$= 500.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(28.0 + 0.0)$$

$$= 224.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[146.492, 500.0, 224.0] + 28.0$$

$$= 174.492 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 28.0 + ((0.0 \cdot 1.0) / 154.816)$$

$$= 28.000 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot Rn + tn) / (\sqrt{(Di + teff) \cdot teff}))$$

$$= \min(12, (2 \cdot 290.0 + 74.0) / (\sqrt{(1712.0 + 28.0) \cdot 28.0}))$$

$$= 2.963$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 28.0 \cdot 154.816 \cdot \max((2.963/5)^{0.85}, 1)$$

$$= 43.348 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A2A + A2B$$

$$= 94.72 + 23.597$$

$$= 118.317 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot Leg^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$

$$= 43.348 + 1.0(118.317 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 162.165 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= tn / \ln[1 + tn/Rn]$$

$$= 74.0 / \ln[1 + 74.0/290.0]$$

$$= 325.600 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= teff / \ln[1 + teff/Reff]$$

$$= 28.0 / \ln[1 + 28.0/856.0]$$

$$= 869.924 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 1407.129 \cdot 1.0$$

$$= 2110.7 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot Rxn \cdot LH$$

$$= 44.699 \cdot 325.6 \cdot 174.492$$

$$= 25395.7 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot Rxs \cdot (LR + tn)$$

$$= 44.699 \cdot 869.924 \cdot (154.816 + 74.0)$$

$$= 88975.0 \text{ kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot Rxs \cdot Rnc$$

$$= 44.699 \cdot 869.924 \cdot 290.0$$

$$= 112766.4 \text{ kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P$$

$$= (88975.0 + 112766.4 + 25395.7) / 44.699$$

$$= 5081.5 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:



$$= \text{Sallow} / (2 \cdot \text{Ap}/\text{AT} - \text{Rxs}/\text{teff})$$

$$= 2110.694 / (2 \cdot 5081.452/162.165 - 869.924/28.0)$$

$$= 66.8 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/\text{Rxs}]$$

$$= 1407.129[28.0/869.924]$$

$$= 45.3 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(\text{Pmax1}, \text{Pmax2})$$

$$= \min(66.791, 45.291)$$

$$= 45.291 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / \text{AT}$$

$$= (25395.727 + 88975.008 + 112766.367) / 162.165$$

$$= 1400.655 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot \text{Rxs} / \text{teff}$$

$$= 44.699 \cdot 869.924/28.0$$

$$= 1388.8 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}})$$

$$= \max(2 \cdot 1400.655 - 1388.75, 1388.75)$$

$$= 1412.6 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2110.69	kgf/cm ²
Local Primary Membrane Stress	PL	1412.56	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	45.29	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (\text{Rnc} + \text{tn}) / \text{Rnc}$$

$$= (290.0 + 74.0) / 290.0$$

$$= 1.255 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [L_T]:

$$= \pi/2 \cdot (\text{Rn} + \text{tn})$$

$$= \pi/2 \cdot (290.0 + 74.0)$$

$$= 571.770 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$\text{tc}, \text{L41T} = 7.071, \text{ must be } \geq \min(0.7 \cdot 74.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot \text{Rn}^2 \cdot k_y^2)$$

$$= \min(112766 \cdot 1.26, 1.5 \cdot 1505.0(118.317 + 0.0), \pi/4 \cdot 44.7 \cdot 290.0^2 \cdot 1.26^2)$$

$$= 46514.996 \text{ kgf}$$

Weld Stress Value [τ]:

$$= \text{fwelds} / (L_T(0.49 \cdot \text{L41T} + 0.6 \cdot \text{tw1} + 0.49 \cdot \text{L43T}))$$

$$= 46514.996 / (571.77(0.49 \cdot 7.071 + 0.6 \cdot 28.0 + 0.49 \cdot 0.0))$$

$$= 401.449 < \text{or} = \text{to } 1407.129 \text{ Weld Stress Passed}$$

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:

$$= \text{tn} / \ln[1 + \text{tn}/\text{Rn}]$$

$$= 74.0 / \ln[1 + 74.0/290.0]$$



= 325.600 mm

Shell Radius for Force Calculation [Rxs]:

= $t_{eff} / \ln[1 + t_{eff}/R_{eff}]$
 = $28.0 / \ln[1 + 28.0/856.0]$
 = 869.924 mm

Allowable Local Primary Membrane Stress [Sallow]:

= $1.5 \cdot S$
 = $1.5 \cdot 1407.129$
 = 2110.7 kgf/cm²

Force acting on the Nozzle [fN]:

= $P \cdot R_{xn} \cdot L_H$
 = $1.055 \cdot 325.6 \cdot 174.492$
 = 599.2 kgf

Force acting on the Shell [fS]:

= $P \cdot R_{xs} (L_R + t_n)$
 = $1.055 \cdot 869.924 (154.816 + 74.0)$
 = 2099.2 kgf

Discontinuity Force from External Pressure [fY]:

= $P \cdot R_{xs} \cdot R_{nc}$
 = $1.055 \cdot 869.924 \cdot 290.0$
 = 2660.5 kgf

Area Resisting External Pressure [Ap]:

= $(f_S + f_Y + f_N) / P$
 = $(2099.2 + 2660.5 + 599.2) / 1.055$
 = 5081.5 cm²

Maximum Allowable Working Pressure Candidate [Pmax1]:

= $S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$
 = $2110.694 / (2 \cdot 5081.452 / 162.165 - 869.924 / 28.0)$
 = 66.8 kgf/cm²

Maximum Allowable Working Pressure Candidate [Pmax2]:

= $S / [t / R_{xs}]$
 = $1407.129 [28.0 / 869.924]$
 = 45.3 kgf/cm²

Maximum Allowable Working Pressure [Pmax]:

= $\min(P_{max1}, P_{max2})$
 = $\min(66.791, 45.291)$
 = 45.291 kgf/cm²

Average Primary Membrane Stress [σAvg]:

= $(f_N + f_S + f_Y) / A_T$
 = $(599.169 + 2099.215 + 2660.532) / 162.165$
 = 33.046 kgf/cm²

General Primary Membrane Stress [σCirc]:

= $P \cdot R_{xs} / t_{eff}$
 = $1.055 \cdot 869.924 / 28.0$
 = 32.8 kgf/cm²

Maximum Local Primary Membrane Stress [PL]:

= $\max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$
 = $\max(2 \cdot 33.046 - 32.765, 32.765)$
 = 33.3 kgf/cm²

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2110.69	kgf/cm ²
Local Primary Membrane Stress	PL	33.33	kgf/cm ²
Maximum Allowable External Pressure	Pmax	45.29	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :



Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Flange MDMT including temperature reduction per 3.11:
Flange MDMT with Temp reduction -26 °C

The Drop for this Nozzle is : 81.8828 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 615.8828 mm

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Input, Nozzle Desc: MH3600

From: 70

Pressure for Reinforcement Calculations	P	44.260	kgf/cm ²
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	1700.00	mm
Design Length of Section	L	18483.8340	mm
Shell Finished (Minimum) Thickness	t	34.0000	mm
Shell Internal Corrosion Allowance	c	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		835.00	cm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1505.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1682.00	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		568.0000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm

Flange Material		SA-105	+IT
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Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	64.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		80.0000	mm

Corrosion Allowance	can	6.0000	mm
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Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	32.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

This is a Manway, Coupling or Access Opening.

Flange Class		300
Flange Grade		GR 1.1

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: MH3600

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	568.000	mm.
Actual Thickness Used in Calculation	16.000	mm

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight



of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 580.0 \cdot (\exp(44.26/(1505.0 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 8.6551 + 6.0 + 0.0 = 14.6551 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1505.0 \cdot 1.0 \cdot \ln((2 \cdot 10.0 + 580.0) / 580.0) - 0.0$$

$$= 51.022 - 0.0 = 51.0218 \text{ kgf/cm}^2$$

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1505.0/1407.1, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 80.0 - 6.0$$

$$= 74.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 16.0 - 6.0$$

$$= 10.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 1712.0/28.0$$

$$= 61.143 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 1700.0/2 + 6.0$$

$$= 856.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{856.0 \cdot 28.0}, 2 \cdot 290.0)$$

$$= 154.816 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 28.0, 0.0] + \sqrt{290.0 \cdot 74.0}$$

$$= 146.492 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 500.0$$

$$= 500.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(28.0 + 0.0)$$

$$= 224.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[146.492, 500.0, 224.0] + 28.0$$

$$= 174.492 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 28.0 + ((0.0 \cdot 1.0) / 154.816)$$

$$= 28.000 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot Rn + tn) / (\sqrt{(Di + teff) \cdot teff}))$$

$$= \min(12, (2 \cdot 290.0 + 74.0) / (\sqrt{(1712.0 + 28.0) \cdot 28.0}))$$

$$= 2.963$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 28.0 \cdot 154.816 \cdot \max((2.963/5)^{0.85}, 1)$$

$$= 43.348 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A2A + A2B$$

$$= 94.72 + 23.597$$

$$= 118.317 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot Leg^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$

$$= 43.348 + 1.0(118.317 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 162.165 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= tn / \ln[1 + tn/Rn]$$

$$= 74.0 / \ln[1 + 74.0/290.0]$$

$$= 325.600 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= teff / \ln[1 + teff/Reff]$$

$$= 28.0 / \ln[1 + 28.0/856.0]$$

$$= 869.924 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 1407.129 \cdot 1.0$$

$$= 2110.7 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot Rxn \cdot LH$$

$$= 44.26 \cdot 325.6 \cdot 174.492$$

$$= 25145.9 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot Rxs \cdot (LR + tn)$$

$$= 44.26 \cdot 869.924 \cdot (154.816 + 74.0)$$

$$= 88099.8 \text{ kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot Rxs \cdot Rnc$$

$$= 44.26 \cdot 869.924 \cdot 290.0$$

$$= 111657.1 \text{ kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P$$

$$= (88099.8 + 111657.1 + 25145.9) / 44.26$$

$$= 5081.5 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:



$$= \text{Sallow} / (2 \cdot \text{Ap}/\text{AT} - \text{Rxs}/\text{teff})$$

$$= 2110.694 / (2 \cdot 5081.452/162.165 - 869.924/28.0)$$

$$= 66.8 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/\text{Rxs}]$$

$$= 1407.129[28.0/869.924]$$

$$= 45.3 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(\text{Pmax1}, \text{Pmax2})$$

$$= \min(66.791, 45.291)$$

$$= 45.291 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / \text{AT}$$

$$= (25145.92 + 88099.789 + 111657.125) / 162.165$$

$$= 1386.878 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot \text{Rxs} / \text{teff}$$

$$= 44.26 \cdot 869.924/28.0$$

$$= 1375.1 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}})$$

$$= \max(2 \cdot 1386.878 - 1375.09, 1375.09)$$

$$= 1398.7 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2110.69	kgf/cm ²
Local Primary Membrane Stress	PL	1398.67	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	45.29	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (\text{Rnc} + \text{tn}) / \text{Rnc}$$

$$= (290.0 + 74.0) / 290.0$$

$$= 1.255 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi/2 \cdot (\text{Rn} + \text{tn})$$

$$= \pi/2 \cdot (290.0 + 74.0)$$

$$= 571.770 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$\text{tc}, \text{L41T} = 7.071, \text{ must be } \geq \min(0.7 \cdot 74.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot \text{Rn}^2 \cdot k_y^2)$$

$$= \min(111657 \cdot 1.26, 1.5 \cdot 1505.0(118.317 + 0.0), \pi/4 \cdot 44.3 \cdot 290.0^2 \cdot 1.26^2)$$

$$= 46057.445 \text{ kgf}$$

Weld Stress Value [τ]:

$$= \text{fwelds} / (\text{Lt}(0.49 \cdot \text{L41T} + 0.6 \cdot \text{tw1} + 0.49 \cdot \text{L43T}))$$

$$= 46057.445 / (571.77(0.49 \cdot 7.071 + 0.6 \cdot 28.0 + 0.49 \cdot 0.0))$$

$$= 397.500 < \text{or} = \text{to } 1407.129 \text{ Weld Stress Passed}$$

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

$$= \text{tn} / \ln[1 + \text{tn}/\text{Rn}]$$

$$= 74.0 / \ln[1 + 74.0/290.0]$$



= 325.600 mm

Shell Radius for Force Calculation [Rxs]:

= $t_{eff} / \ln[1 + t_{eff}/R_{eff}]$
 = $28.0 / \ln[1 + 28.0/856.0]$
 = 869.924 mm

Allowable Local Primary Membrane Stress [Sallow]:

= $1.5 \cdot S$
 = $1.5 \cdot 1407.129$
 = 2110.7 kgf/cm²

Force acting on the Nozzle [fN]:

= $P \cdot R_{xn} \cdot L_H$
 = $1.055 \cdot 325.6 \cdot 174.492$
 = 599.2 kgf

Force acting on the Shell [fS]:

= $P \cdot R_{xs} (L_R + t_n)$
 = $1.055 \cdot 869.924 (154.816 + 74.0)$
 = 2099.2 kgf

Discontinuity Force from External Pressure [fY]:

= $P \cdot R_{xs} \cdot R_{nc}$
 = $1.055 \cdot 869.924 \cdot 290.0$
 = 2660.5 kgf

Area Resisting External Pressure [Ap]:

= $(f_S + f_Y + f_N) / P$
 = $(2099.2 + 2660.5 + 599.2) / 1.055$
 = 5081.5 cm²

Maximum Allowable Working Pressure Candidate [Pmax1]:

= $S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$
 = $2110.694 / (2 \cdot 5081.452 / 162.165 - 869.924 / 28.0)$
 = 66.8 kgf/cm²

Maximum Allowable Working Pressure Candidate [Pmax2]:

= $S / [t / R_{xs}]$
 = $1407.129 [28.0 / 869.924]$
 = 45.3 kgf/cm²

Maximum Allowable Working Pressure [Pmax]:

= $\min(P_{max1}, P_{max2})$
 = $\min(66.791, 45.291)$
 = 45.291 kgf/cm²

Average Primary Membrane Stress [σAvg]:

= $(f_N + f_S + f_Y) / A_T$
 = $(599.169 + 2099.215 + 2660.532) / 162.165$
 = 33.046 kgf/cm²

General Primary Membrane Stress [σCirc]:

= $P \cdot R_{xs} / t_{eff}$
 = $1.055 \cdot 869.924 / 28.0$
 = 32.8 kgf/cm²

Maximum Local Primary Membrane Stress [PL]:

= $\max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$
 = $\max(2 \cdot 33.046 - 32.765, 32.765)$
 = 33.3 kgf/cm²

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2110.69	kgf/cm ²
Local Primary Membrane Stress	PL	33.33	kgf/cm ²
Maximum Allowable External Pressure	Pmax	45.29	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :



Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Flange MDMT including temperature reduction per 3.11:
Flange MDMT with Temp reduction -26 °C

The Drop for this Nozzle is : 81.8828 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 615.8828 mm

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Input, Nozzle Desc: DP250

From: 70

Pressure for Reinforcement Calculations	P	44.239	kgf/cm ²
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	1700.00	mm
Design Length of Section	L	18483.8340	mm
Shell Finished (Minimum) Thickness	t	34.0000	mm
Shell Internal Corrosion Allowance	c	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		855.00	cm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1505.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1682.00	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		50.8000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm

Flange Material		SA-105	+IT
-----------------	--	--------	-----

Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	33.2360	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm

Corrosion Allowance	can	6.0000	mm
---------------------	-----	--------	----

Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgmv	30.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

Flange Class		300
Flange Grade		GR 1.1

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: DP250

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	50.800	mm.
Actual Thickness Used in Calculation	12.000	mm

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 62.8 \cdot (\exp(44.239/(1505.0 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 0.9367 + 6.0 + 0.0 = 6.9367 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1505.0 \cdot 1.0 \cdot \ln((2 \cdot 6.0 + 62.8) / 62.8) - 0.0$$

$$= 263.169 - 0.0 = 263.1686 \text{ kgf/cm}^2$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	10.800 mm
Min. required shell thickness, internal p + c	33.342 mm
Min. required shell thickness, external p + c	26.261 mm

Nozzle neck minimum required thickness	10.800 mm
--	-----------

Minimum nozzle neck thickness 11.989 mm , is greater than required thickness 10.800 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1505.0/1407.1, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 50.0 - 6.0$$

$$= 44.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 12.0 - 6.0$$

$$= 6.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 1712.0/28.0$$

$$= 61.143 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 1700.0/2 + 6.0$$

$$= 856.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{856.0 \cdot 28.0}, 2 \cdot 31.4)$$

$$= 62.800 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 28.0, 0.0] + \sqrt{31.4 \cdot 44.0}$$

$$= 37.170 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 250.0$$

$$= 250.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(28.0 + 0.0)$$



$$= 224.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles}$$

$$= \min[37.17, 250.0, 224.0] + 28.0$$

$$= 65.170 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 28.0 + ((0.0 \cdot 1.0) / 62.8)$$

$$= 28.000 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot Rn + tn) / (\sqrt{(Di + teff) \cdot teff}))$$

$$= \min(12, (2 \cdot 31.4 + 44.0) / (\sqrt{(1712.0 + 28.0) \cdot 28.0}))$$

$$= 0.484$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 28.0 \cdot 62.8 \cdot \max((0.484/5)^{0.85}, 1)$$

$$= 17.584 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= tn \cdot LH$$

$$= 44.0 \cdot 65.17$$

$$= 28.675 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot Leg^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + frn(A2 + A3) + A41 + A42 + A43$$

$$= 17.584 + 1.0(28.675 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 46.759 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= tn / \ln[1 + tn/Rn]$$

$$= 44.0 / \ln[1 + 44.0/31.4]$$

$$= 50.228 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= teff / \ln[1 + teff/Reff]$$

$$= 28.0 / \ln[1 + 28.0/856.0]$$

$$= 869.924 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 1407.129 \cdot 1.0$$

$$= 2110.7 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot Rxn \cdot LH$$

$$= 44.239 \cdot 50.228 \cdot 65.17$$

$$= 1448.1 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot Rxs(LR + tn)$$

$$= 44.239 \cdot 869.924(62.8 + 44.0)$$

$$= 41101.6 \text{ kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot Rxs \cdot Rnc$$

$$= 44.239 \cdot 869.924 \cdot 31.4$$

$$= 12084.2 \text{ kgf}$$

Area Resisting Internal Pressure [Ap]:



$$= (fS + fY + fN) / P$$

$$= (41101.6 + 12084.2 + 1448.1) / 44.239$$

$$= 1235.0 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap / AT - Rxs / teff)$$

$$= 2110.694 / (2 \cdot 1234.969 / 46.759 - 869.924 / 28.0)$$

$$= 97.0 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t / Rxs]$$

$$= 1407.129 [28.0 / 869.924]$$

$$= 45.3 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(97.024, 45.291)$$

$$= 45.291 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (fN + fS + fY) / AT$$

$$= (1448.112 + 41101.621 + 12084.186) / 46.759$$

$$= 1168.423 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot Rxs / teff$$

$$= 44.239 \cdot 869.924 / 28.0$$

$$= 1374.5 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 1168.423 - 1374.454, 1374.454)$$

$$= 1374.5 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2110.69	kgf/cm ²
Local Primary Membrane Stress	PL	1374.45	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	45.29	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$

$$= (31.4 + 44.0) / 31.4$$

$$= 2.401 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi / 2 \cdot (Rn + tn)$$

$$= \pi / 2 \cdot (31.4 + 44.0)$$

$$= 118.438 \text{ mm}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 44.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi / 4 \cdot P \cdot Rn^2 \cdot ky^2)$$

$$= \min(12084 \cdot 2.4, 1.5 \cdot 1505.0(28.675 + 0.0), \pi / 4 \cdot 44.2 \cdot 31.4^2 \cdot 2.4^2)$$

$$= 1975.327 \text{ kgf}$$

Weld Stress Value [τ]:

$$= fwelds / (Lt(0.49 \cdot L41T + 0.6 \cdot tw1 + 0.49 \cdot L43T))$$

$$= 1975.327 / (118.438(0.49 \cdot 7.071 + 0.6 \cdot 28.0 + 0.49 \cdot 0.0))$$

$$= 82.301 < \text{ or } = \text{ to } 1407.129 \text{ Weld Stress Passed}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:



$$= t_n / \ln[1 + t_n/R_n]$$

$$= 44.0 / \ln[1 + 44.0/31.4]$$

$$= 50.228 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 28.0 / \ln[1 + 28.0/856.0]$$

$$= 869.924 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S$$

$$= 1.5 \cdot 1407.129$$

$$= 2110.7 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 1.055 \cdot 50.228 \cdot 65.17$$

$$= 34.5 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 1.055 \cdot 869.924 (62.8 + 44.0)$$

$$= 979.8 \text{ kgf}$$

Discontinuity Force from External Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 1.055 \cdot 869.924 \cdot 31.4$$

$$= 288.1 \text{ kgf}$$

Area Resisting External Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (979.8 + 288.1 + 34.5) / 1.055$$

$$= 1235.0 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 2110.694 / (2 \cdot 1234.969 / 46.759 - 869.924 / 28.0)$$

$$= 97.0 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / [t / R_{xs}]$$

$$= 1407.129 [28.0 / 869.924]$$

$$= 45.3 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(97.024, 45.291)$$

$$= 45.291 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (34.521 + 979.81 + 288.071) / 46.759$$

$$= 27.854 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 1.055 \cdot 869.924 / 28.0$$

$$= 32.8 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 27.854 - 32.765, 32.765)$$

$$= 32.8 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2110.69	kgf/cm ²
Local Primary Membrane Stress	PL	32.77	kgf/cm ²
Maximum Allowable External Pressure	Pmax	45.29	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:



Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -26 °C

The Drop for this Nozzle is : 3.3508 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 287.3507 mm

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Input, Nozzle Desc: UC1600

From: 80

Pressure for Reinforcement Calculations	P	44.086	kgf/cm ²
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	1700.00	mm
Design Length of Section	L	18483.8340	mm
Shell Finished (Minimum) Thickness	t	34.0000	mm
Shell Internal Corrosion Allowance	c	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1005.00	cm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1505.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1682.00	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		568.0000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm
Flange Material		SA-105	+IT
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	64.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		80.0000	mm
Corrosion Allowance	can	6.0000	mm
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	32.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
This is a Manway, Coupling or Access Opening.			
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: UC1600

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	568.000	mm.
Actual Thickness Used in Calculation	16.000	mm

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight



of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 580.0 \cdot (\exp(44.086/(1505.0 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 8.6206 + 6.0 + 0.0 = 14.6206 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1505.0 \cdot 1.0 \cdot \ln((2 \cdot 10.0 + 580.0) / 580.0) - 0.0$$

$$= 51.022 - 0.0 = 51.0218 \text{ kgf/cm}^2$$

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 298.80 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 518.82 mm

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1505.0/1407.1, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 80.0 - 6.0$$

$$= 74.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 16.0 - 6.0$$

$$= 10.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 1712.0/28.0$$

$$= 61.143 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 1700.0/2 + 6.0$$

$$= 856.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{R_{eff} \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{856.0 \cdot 28.0}, 2 \cdot 290.0)$$

$$= 154.816 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 28.0, 0.0] + \sqrt{290.0 \cdot 74.0}$$

$$= 146.492 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 500.0$$

$$= 500.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(28.0 + 0.0)$$

$$= 224.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:
 $= \min[LH1, LH2, LH3] + t$ for set-in nozzles
 $= \min[146.492, 500.0, 224.0] + 28.0$
 $= 174.492 \text{ mm}$

Effective Vessel Thickness [teff]:
 $= t + ((A5 \cdot frp) / LR)$
 $= 28.0 + ((0.0 \cdot 1.0) / 154.816)$
 $= 28.000 \text{ mm}$

Parameter [λ]:
 $= \min(12, (2 \cdot Rn + tn) / (\sqrt{(Di + teff) \cdot teff}))$
 $= \min(12, (2 \cdot 290.0 + 74.0) / (\sqrt{(1712.0 + 28.0) \cdot 28.0}))$
 $= 2.963$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:
 $= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$
 $= 28.0 \cdot 154.816 \cdot \max((2.963/5)^{0.85}, 1)$
 $= 43.348 \text{ cm}^2$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:
 $= A2A + A2B$
 $= 94.72 + 23.597$
 $= 118.317 \text{ cm}^2$

Area Contributed by the Outside Fillet Weld [A41]:
 $= 0.5 \cdot Leg41^2$
 $= 0.5 \cdot 10.0^2$
 $= 0.500 \text{ cm}^2$

The total area contributed by A1 through A43 [AT]:
 $= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$
 $= 43.348 + 1.0(118.317 + 0.0) + 0.5 + 0.0 + 0.0$
 $= 162.165 \text{ cm}^2$

Nozzle Radius for Force Calculation [Rxn]:
 $= tn / \ln[1 + tn/Rn]$
 $= 74.0 / \ln[1 + 74.0/290.0]$
 $= 325.600 \text{ mm}$

Shell Radius for Force Calculation [Rxs]:
 $= teff / \ln[1 + teff/Reff]$
 $= 28.0 / \ln[1 + 28.0/856.0]$
 $= 869.924 \text{ mm}$

Allowable Local Primary Membrane Stress [Sallow]:
 $= 1.5 \cdot S \cdot E$
 $= 1.5 \cdot 1407.129 \cdot 1.0$
 $= 2110.7 \text{ kgf/cm}^2$

Force acting on the Nozzle [fN]:
 $= P \cdot Rxn \cdot LH$
 $= 44.086 \cdot 325.6 \cdot 174.492$
 $= 25047.2 \text{ kgf}$

Force acting on the Shell [fS]:
 $= P \cdot Rxs(LR + tn)$
 $= 44.086 \cdot 869.924(154.816 + 74.0)$
 $= 87753.8 \text{ kgf}$

Discontinuity Force from Internal Pressure [fY]:
 $= P \cdot Rxs \cdot Rnc$
 $= 44.086 \cdot 869.924 \cdot 290.0$
 $= 111218.6 \text{ kgf}$

Area Resisting Internal Pressure [Ap]:
 $= (fS + fY + fN) / P$



$$= (87753.8 + 111218.6 + 25047.2) / 44.086$$

$$= 5081.5 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= \text{Sallow} / (2 \cdot \text{Ap}/\text{AT} - \text{Rxs}/\text{teff})$$

$$= 2110.694 / (2 \cdot 5081.452 / 162.165 - 869.924 / 28.0)$$

$$= 66.8 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= \text{S}[t/\text{Rxs}]$$

$$= 1407.129 [28.0 / 869.924]$$

$$= 45.3 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(\text{Pmax1}, \text{Pmax2})$$

$$= \min(66.791, 45.291)$$

$$= 45.291 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / \text{AT}$$

$$= (25047.158 + 87753.773 + 111218.594) / 162.165$$

$$= 1381.431 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot \text{Rxs} / \text{teff}$$

$$= 44.086 \cdot 869.924 / 28.0$$

$$= 1369.7 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}})$$

$$= \max(2 \cdot 1381.431 - 1369.689, 1369.689)$$

$$= 1393.2 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2110.69	kgf/cm ²
Local Primary Membrane Stress	PL	1393.17	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	45.29	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (\text{Rnc} + \text{tn}) / \text{Rnc}$$

$$= (290.0 + 74.0) / 290.0$$

$$= 1.255 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi / 2 \cdot (\text{Rn} + \text{tn})$$

$$= \pi / 2 \cdot (290.0 + 74.0)$$

$$= 571.770 \text{ mm}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$\text{tc}, \text{L41T} = 7.071, \text{ must be } \geq \min(0.7 \cdot 74.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot \text{Sn}(\text{A2} + \text{A3}), \pi / 4 \cdot P \cdot \text{Rn}^2 \cdot k_y^2)$$

$$= \min(111219 \cdot 1.26, 1.5 \cdot 1505.0 (118.317 + 0.0), \pi / 4 \cdot 44.1 \cdot 290.0^2 \cdot 1.26^2)$$

$$= 45876.551 \text{ kgf}$$

Weld Stress Value [τ]:

$$= \text{fwelds} / (\text{Lt} (0.49 \cdot \text{L41T} + 0.6 \cdot \text{tw1} + 0.49 \cdot \text{L43T}))$$

$$= 45876.551 / (571.77 (0.49 \cdot 7.071 + 0.6 \cdot 28.0 + 0.49 \cdot 0.0))$$

$$= 395.939 < \text{ or } = \text{ to } 1407.129 \text{ Weld Stress Passed}$$

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: External Pressure Case:



Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 74.0 / \ln[1 + 74.0/290.0]$$

$$= 325.600 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 28.0 / \ln[1 + 28.0/856.0]$$

$$= 869.924 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S$$

$$= 1.5 \cdot 1407.129$$

$$= 2110.7 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 1.055 \cdot 325.6 \cdot 174.492$$

$$= 599.2 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 1.055 \cdot 869.924 (154.816 + 74.0)$$

$$= 2099.2 \text{ kgf}$$

Discontinuity Force from External Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 1.055 \cdot 869.924 \cdot 290.0$$

$$= 2660.5 \text{ kgf}$$

Area Resisting External Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (2099.2 + 2660.5 + 599.2) / 1.055$$

$$= 5081.5 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 2110.694 / (2 \cdot 5081.452 / 162.165 - 869.924 / 28.0)$$

$$= 66.8 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / (t / R_{xs})$$

$$= 1407.129 [28.0 / 869.924]$$

$$= 45.3 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(66.791, 45.291)$$

$$= 45.291 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σAvg]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (599.169 + 2099.215 + 2660.532) / 162.165$$

$$= 33.046 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 1.055 \cdot 869.924 / 28.0$$

$$= 32.8 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 33.046 - 32.765, 32.765)$$

$$= 33.3 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2110.69	kgf/cm ²
Local Primary Membrane Stress	PL	33.33	kgf/cm ²
Maximum Allowable External Pressure	Pmax	45.29	kgf/cm ²



Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -26 °C

The Drop for this Nozzle is : 81.8828 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 615.8828 mm

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Input, Nozzle Desc: A150

From: 90

Pressure for Reinforcement Calculations	P	43.582	kgf/cm ²
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	1706.00	mm
Design Length of Section	L	18483.8359	mm
Shell Finished (Minimum) Thickness	t	28.0000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		1497.50	cm
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User Entered Minimum Design Metal Temperature		-12.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1505.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1682.00	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		150.0000	mm

Size and Thickness Basis		Minimum	
Nominal Thickness		120	

Flange Material		SA-105	+IT
-----------------	--	--------	-----

Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	38.5105	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm

Corrosion Allowance	can	6.0000	mm
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Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	28.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: A150

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		143.294	mm.
Actual Thickness Used in Calculation		12.490	mm

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 155.294 \cdot (\exp(43.582/(1505.0 \cdot 1.0)) - 1) + 6.0 + 0.0$
 $= 2.2814 + 6.0 + 0.0 = 8.2814 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1505.0 \cdot 1.0 \cdot \ln((2 \cdot 6.49 + 155.294) / 155.294) - 0.0$
 $= 120.82 - 0.0 = 120.8195 \text{ kgf/cm}^2$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	12.220 mm
Min. required shell thickness, internal p + c :	27.867 mm
Min. required shell thickness, external p + c :	16.233 mm

Nozzle neck minimum required thickness :	12.220 mm
--	-----------

Minimum nozzle neck thickness 12.497 mm , is greater than required thickness 12.220 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained :	568.9,	Allowable :	1505.0 kgf/cm ²	Passed
Expansion :	502.4,	Allowable :	3414.8 kgf/cm ²	Passed
Occasional :	250.2,	Allowable :	2001.7 kgf/cm ²	Passed
Shear :	254.9,	Allowable :	1053.5 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[1505.0/1407.1, 1]$
 $= 1.000$

Thickness of Nozzle at Shell [tn]:
 $= \text{hub thickness} - \text{corrosion allowance}$
 $= 50.0 - 6.0$
 $= 44.000 \text{ mm}$

Thickness of Nozzle at Top [tn2]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 12.49 - 6.0$
 $= 6.490 \text{ mm}$

Shell Diameter to Thickness ratio [D/t]:
 $= D_i/t$
 $= 1708.0/27.0$
 $= 63.259 \text{ must be less than } 400.$

Effective Pressure Radius [Reff]:
 $= D_i/2 + \text{corrosion allowance}$
 $= 1706.0/2 + 1.0$
 $= 854.000 \text{ mm}$

Effective Length of Vessel Wall [LR]:
 $= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$
 $= \min(\sqrt{854.0 \cdot 27.0}, 2 \cdot 77.647)$
 $= 151.849 \text{ mm}$

Thickness Limit Candidate [LH1]:
 $= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$
 $= \min[1.5 \cdot 27.0, 0.0] + \sqrt{77.647 \cdot 44.0}$
 $= 58.451 \text{ mm}$

Thickness Limit Candidate [LH2]:
 $= L_{pr1}$
 $= 300.0$



= 300.000 mm

Thickness Limit Candidate [LH3]:

= $8(t + t_e)$
 = $8(27.0 + 0.0)$
 = 216.000 mm

Effective Nozzle Wall Length Outside the Vessel [LH]:

= $\min[LH1, LH2, LH3] + t$ for set-in nozzles
 = $\min[58.451, 300.0, 216.0] + 27.0$
 = 85.451 mm

Effective Vessel Thickness [teff]:

= $t + ((A5 \cdot frp) / LR)$
 = $27.0 + ((0.0 \cdot 1.0) / 151.849)$
 = 27.000 mm

Parameter [λ]:

= $\min(12, (2 \cdot Rn + t_n) / (\sqrt{(D_i + teff) \cdot teff}))$
 = $\min(12, (2 \cdot 77.65 + 44.0) / (\sqrt{(1708.0 + 27.0) \cdot 27.0}))$
 = 0.921

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

= $t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$
 = $27.0 \cdot 151.849 \cdot \max((0.921/5)^{0.85}, 1)$
 = 40.999 cm²

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

= $t_n \cdot LH$
 = $44.0 \cdot 85.451$
 = 37.598 cm²

Area Contributed by the Outside Fillet Weld [A41]:

= $0.5 \cdot Leg^2$
 = $0.5 \cdot 10.0^2$
 = 0.500 cm²

The total area contributed by A1 through A43 [AT]:

= $A1 + frn(A2 + A3) + A41 + A42 + A43$
 = $41.0 + 1.0(37.598 + 0.0) + 0.5 + 0.0 + 0.0$
 = 79.097 cm²

Nozzle Radius for Force Calculation [Rxn]:

= $t_n / \ln[1 + t_n/Rn]$
 = $44.0 / \ln[1 + 44.0/77.647]$
 = 98.006 mm

Shell Radius for Force Calculation [Rxs]:

= $teff / \ln[1 + teff/Reff]$
 = $27.0 / \ln[1 + 27.0/854.0]$
 = 867.429 mm

Allowable Local Primary Membrane Stress [Sallow]:

= $1.5 \cdot S \cdot E$
 = $1.5 \cdot 1407.129 \cdot 1.0$
 = 2110.7 kgf/cm²

Force acting on the Nozzle [fN]:

= $P \cdot Rxn \cdot LH$
 = $43.582 \cdot 98.006 \cdot 85.451$
 = 3649.9 kgf

Force acting on the Shell [fS]:

= $P \cdot Rxs(LR + t_n)$
 = $43.582 \cdot 867.429(151.849 + 44.0)$
 = 74039.4 kgf

Discontinuity Force from Internal Pressure [fY]:



$$= P \cdot Rxs \cdot Rnc$$

$$= 43.582 \cdot 867.429 \cdot 77.647$$

$$= 29354.0 \text{ kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P$$

$$= (74039.4 + 29354.0 + 3649.9) / 43.582$$

$$= 2456.1 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap / AT - Rxs / teff)$$

$$= 2110.694 / (2 \cdot 2456.127 / 79.097 - 867.429 / 27.0)$$

$$= 70.4 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t / Rxs]$$

$$= 1407.129 [27.0 / 867.429]$$

$$= 43.8 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(70.411, 43.799)$$

$$= 43.799 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σAvg]:

$$= (fN + fS + fY) / AT$$

$$= (3649.875 + 74039.383 + 29353.998) / 79.097$$

$$= 1353.312 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot Rxs / teff$$

$$= 43.582 \cdot 867.429 / 27.0$$

$$= 1400.2 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ)$$

$$= \max(2 \cdot 1353.312 - 1400.165, 1400.165)$$

$$= 1400.2 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2110.69	kgf/cm ²
Local Primary Membrane Stress	PL	1400.16	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	43.80	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$

$$= (77.647 + 44.0) / 77.647$$

$$= 1.567 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [LT]:

$$= \pi / 2 \cdot (Rn + tn)$$

$$= \pi / 2 \cdot (77.647 + 44.0)$$

$$= 191.083 \text{ mm}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 44.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi / 4 \cdot P \cdot Rn^2 \cdot ky^2)$$

$$= \min(29354 \cdot 1.57, 1.5 \cdot 1505.0(37.598 + 0.0), \pi / 4 \cdot 43.6 \cdot 77.65^2 \cdot 1.57^2)$$

$$= 5065.258 \text{ kgf}$$

Weld Stress Value [τ]:

$$= fwelds / (LT(0.49 \cdot L41T + 0.6 \cdot tw1 + 0.49 \cdot L43T))$$

$$= 5065.258 / (191.083(0.49 \cdot 7.071 + 0.6 \cdot 27.0 + 0.49 \cdot 0.0))$$



= 134.800 < or = to 1407.129 Weld Stress Passed

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 44.0 / \ln[1 + 44.0/77.647] \\ &= 98.006 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 27.0 / \ln[1 + 27.0/854.0] \\ &= 867.429 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \\ &= 1.5 \cdot 1407.129 \\ &= 2110.7 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot L_H \\ &= 1.055 \cdot 98.006 \cdot 85.451 \\ &= 88.3 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot R_{xs} (L_R + t_n) \\ &= 1.055 \cdot 867.429 (151.849 + 44.0) \\ &= 1791.6 \text{ kgf} \end{aligned}$$

Discontinuity Force from External Pressure [fY]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 1.055 \cdot 867.429 \cdot 77.647 \\ &= 710.3 \text{ kgf} \end{aligned}$$

Area Resisting External Pressure [Ap]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (1791.6 + 710.3 + 88.3) / 1.055 \\ &= 2456.1 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff}) \\ &= 2110.694 / (2 \cdot 2456.127 / 79.097 - 867.429 / 27.0) \\ &= 70.4 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t / R_{xs}] \\ &= 1407.129 [27.0 / 867.429] \\ &= 43.8 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(70.411, 43.799) \\ &= 43.799 \text{ kgf/cm}^2 \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (f_N + f_S + f_Y) / A_T \\ &= (88.32 + 1791.61 + 710.31) / 79.097 \\ &= 32.748 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot R_{xs} / t_{eff} \\ &= 1.055 \cdot 867.429 / 27.0 \\ &= 33.9 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 32.748 - 33.881, 33.881) \\ &= 33.9 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:



Allowed Local Primary Membrane Stress	Sallow	2110.69	kgf/cm ²
Local Primary Membrane Stress	PL	33.88	kgf/cm ²
Maximum Allowable External Pressure	Pmax	43.80	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Flange MDMT including temperature reduction per 3.11:
 Flange MDMT with Temp reduction -64 °C

The Drop for this Nozzle is : 8.7187 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 336.7187 mm

Input Echo, WRC297 Item 1, Description: A150 :

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	1.0000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	1706.000	mm	
Shell Thickness	28.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-516 70		
Vessel Cold S.I. Allowable	Smc	1407.13	kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1407.13	kgf/cm ²
Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance	6.0000	mm	
Nozzle Diameter	143.294	mm	
Nozzle Thickness	50.0000	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-105		
Nozzle Cold S.I. Allowable	SNmc	1682.00	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1505.00	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	43.58	kgf/cm ²
Radial Load	P	-1146.80	kgf
Circumferential Shear	Vc	764.53	kgf
Longitudinal Shear	Vl	764.53	kgf
Circumferential Moment	Mc	321.11	kgf-m.
Longitudinal Moment	Ml	321.11	kgf-m.
Torsional Moment	Mt	458.73	kgf-m.
Include Axial Pressure Thrust		Yes	
Include Pressure Stress Indices per Div. 2		No	
Local Loads applied at end of Nozzle/Attachment		No	

Warning - The ratio, Dn/Tn (4.866) is < 10
 check the Limitations of WRC bulletin 297.

Note: The radial load includes the effect of the pressure thrust force (Pressure times Area). This may lead to overly conservative results in some cases.

Stress Computations at the Edge of the Nozzle:



Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 243.294 + 2 \cdot 1.65 \cdot \sqrt{(867.5(28.0 - 1.0))}$$

$$= 748.340 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1735.000 mm
 Nozzle Outside Diameter (d) = 243.294 mm
 Vessel Thickness used (T) = 27.000 mm
 Nozzle Thickness used (t) = 44.000 mm
 T / t = 0.614
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.124$$

Nr/P = 0.103
 Mr/P = 0.200
 M0/P = 0.061
 N0/P = 0.125
 MrD/Mc = 0.299
 NrDL/Mc = 0.105
 M0d/Mc = 0.097
 N0DL/Mc = 0.043
 MrD/Ml = 0.182
 NrDL/Ml = 0.090
 M0D/Ml = 0.061
 N0DL/Ml = 0.125

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	161	161	161	161
Outplane Bending (P)	468	-468	468	-468
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-22	-22	22	22
Outplane Bending (ML)	-65	65	65	-65
Normal Pressure Stress	1356	1400	1356	1400
Outplane Stress Summary	1897	1136	2075	1049

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	132	132	132	132
Inplane Bending (P)	1549	-1549	1549	-1549
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-16	-16	16	16
Inplane Bending (ML)	-197	197	197	-197
Inplane Pressure Stress	678	678	678	678
Inplane Stress Summary	2146	-557	2573	-918

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	7	7	-7	-7
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	18	18	18	18
Shear Stress Summary	25	25	10	10



Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Two * Max Shear Stress	2149	1694	2574	1968
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Membrane (P)	161	161	161	161
Outplane Bending (P)	468	-468	468	-468
Outplane Membrane (Mc)	-7	-7	7	7
Outplane Bending (Mc)	-105	105	105	-105
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	678	678	678	678

Outplane Stress Summary	1194	469	1421	272
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Inplane Membrane (P)	132	132	132	132
Inplane Bending (P)	1549	-1549	1549	-1549
Inplane Membrane (Mc)	-18	-18	18	18
Inplane Bending (Mc)	-324	324	324	-324
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1356	1400	1356	1400

Inplane Stress Summary	2695	289	3382	-321
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-7	-7	7	7
Torsional Shear (Mt)	18	18	18	18

Shear Stress Summary	10	10	25	25
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Two * Max Shear Stress	2695	469	3382	596
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Hoop Membrane (P)	161	161	161	161
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-22	-22	22	22
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	71	124	71	124

Hoop Stress Summary	210	262	255	307
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	34	34	34	34
Axial Bending (P)	433	-433	433	-433
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-27	-27	27	27
Axial Bending (ML)	-55	55	55	-55
Axial Pressure Stress	35	35	35	35
Axial Stress Summary	420	-334	586	-392

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	4	4	-4	-4
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	11	11	11	11
Shear Stress Summary	15	15	6	6

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	421	597	586	699

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	161	161	161	161
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-7	-7	7	7
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	71	124	71	124
Hoop Stress Summary	225	277	240	292

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	34	34	34	34
Axial Bending (P)	433	-433	433	-433
Axial Membrane (Mc)	-27	-27	27	27
Axial Bending (Mc)	-100	100	100	-100
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	35	35	35	35
Axial Stress Summary	375	-289	631	-436

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-4	-4	4	4
Shear due to Torsion	11	11	11	11
Shear Stress Summary	6	6	15	15



Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	375	567	631	730

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1356	1400	1356	1400	1356	1400	1356	1400
Circ. Pl (SUS)	139	139	183	183	114	114	150	150
Circ. Q (SUS)	403	-403	533	-533	1225	-1225	1873	-1873
Long. Pm (SUS)	678	678	678	678	678	678	678	678
Long. Pl (SUS)	116	116	148	148	154	154	168	168
Long. Q (SUS)	1352	-1352	1746	-1746	363	-363	573	-573
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	7	7	-7	-7	-7	-7	7	7
Shear Q (SUS)	18	18	18	18	18	18	18	18
Pm (SUS)	1356	1400	1356	1400	1356	1400	1356	1400
Pm+Pl (SUS)	1495	1539	1539	1583	1470	1514	1506	1550
Pm+Pl+Q (Total)	2148	1694	2572	1970	2695	469	3379	598

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1400.00	1407.13	Passed
Pm+Pl (SUS)	1583.06	2110.69	Passed
Pm+Pl+Q (TOTAL)	3379.32	4221.39	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	71	124	71	124	71	124	71	124
Circ. Pl (SUS)	139	139	183	183	154	154	168	168
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	35	35	35	35	35	35	35	35
Long. Pl (SUS)	7	7	61	61	7	7	61	61
Long. Q (SUS)	378	-378	488	-488	333	-333	533	-533
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	4	4	-4	-4	-4	-4	4	4
Shear Q (SUS)	11	11	11	11	11	11	11	11
Pm (SUS)	71.0	124.0	71.0	124.0	71.0	124.0	71.0	124.0
Pm+Pl (SUS)	210.1	263.1	254.1	307.1	225.1	278.1	239.1	292.1
Pm+Pl+Q (Total)	421.1	599.8	584.1	699.1	375.3	569.2	629.6	729.6

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	124.00	1505.00	Passed
Pm+Pl (SUS)	307.08	2257.50	Passed
Pm+Pl+Q (TOTAL)	729.62	4515.00	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: MH2600

From: 90

Pressure for Reinforcement Calculations	P	43.605	kgf/cm ²
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	1706.00	mm
Design Length of Section	L	18483.8359	mm
Shell Finished (Minimum) Thickness	t	28.0000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		1475.00	cm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1505.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1682.00	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		568.0000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm

Flange Material		SA-105	+IT
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Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	64.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		80.0000	mm

Corrosion Allowance	can	6.0000	mm
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Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	28.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

This is a Manway, Coupling or Access Opening.

Flange Class		300
Flange Grade		GR 1.1

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: MH2600

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	568.000	mm.
Actual Thickness Used in Calculation	16.000	mm

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight



of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 580.0 \cdot (\exp(43.605/(1505.0 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 8.5252 + 6.0 + 0.0 = 14.5252 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1505.0 \cdot 1.0 \cdot \ln((2 \cdot 10.0 + 580.0) / 580.0) - 0.0$$

$$= 51.022 - 0.0 = 51.0218 \text{ kgf/cm}^2$$

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1505.0/1407.1, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 80.0 - 6.0$$

$$= 74.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 16.0 - 6.0$$

$$= 10.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 1708.0/27.0$$

$$= 63.259 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 1706.0/2 + 1.0$$

$$= 854.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{854.0 \cdot 27.0}, 2 \cdot 290.0)$$

$$= 151.849 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 27.0, 0.0] + \sqrt{290.0 \cdot 74.0}$$

$$= 146.492 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 500.0$$

$$= 500.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(27.0 + 0.0)$$

$$= 216.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[146.492, 500.0, 216.0] + 27.0$$

$$= 173.492 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 27.0 + ((0.0 \cdot 1.0) / 151.849)$$

$$= 27.000 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot Rn + tn) / (\sqrt{(Di + teff) \cdot teff}))$$

$$= \min(12, (2 \cdot 290.0 + 74.0) / (\sqrt{(1708.0 + 27.0) \cdot 27.0}))$$

$$= 3.022$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 27.0 \cdot 151.849 \cdot \max((3.022/5)^{0.85}, 1)$$

$$= 40.999 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A2A + A2B$$

$$= 93.98 + 23.597$$

$$= 117.577 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot Leg^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + frn(A2 + A3) + A41 + A42 + A43$$

$$= 41.0 + 1.0(117.577 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 159.076 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= tn / \ln[1 + tn/Rn]$$

$$= 74.0 / \ln[1 + 74.0/290.0]$$

$$= 325.600 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= teff / \ln[1 + teff/Reff]$$

$$= 27.0 / \ln[1 + 27.0/854.0]$$

$$= 867.429 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 1407.129 \cdot 1.0$$

$$= 2110.7 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot Rxn \cdot LH$$

$$= 43.605 \cdot 325.6 \cdot 173.492$$

$$= 24632.1 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot Rxs \cdot (LR + tn)$$

$$= 43.605 \cdot 867.429 \cdot (151.849 + 74.0)$$

$$= 85425.8 \text{ kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot Rxs \cdot Rnc$$

$$= 43.605 \cdot 867.429 \cdot 290.0$$

$$= 109690.6 \text{ kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P$$

$$= (85425.8 + 109690.6 + 24632.1) / 43.605$$

$$= 5039.5 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:



$$= \text{Sallow} / (2 \cdot \text{Ap}/\text{AT} - \text{Rxs}/\text{teff})$$

$$= 2110.694 / (2 \cdot 5039.51/159.076 - 867.429/27.0)$$

$$= 67.6 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/\text{Rxs}]$$

$$= 1407.129[27.0/867.429]$$

$$= 43.8 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(\text{Pmax1}, \text{Pmax2})$$

$$= \min(67.579, 43.799)$$

$$= 43.799 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / \text{AT}$$

$$= (24632.129 + 85425.766 + 109690.609) / 159.076$$

$$= 1381.410 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot \text{Rxs} / \text{teff}$$

$$= 43.605 \cdot 867.429/27.0$$

$$= 1400.9 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}})$$

$$= \max(2 \cdot 1381.41 - 1400.904, 1400.904)$$

$$= 1400.9 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2110.69	kgf/cm ²
Local Primary Membrane Stress	PL	1400.90	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	43.80	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (\text{Rnc} + \text{tn}) / \text{Rnc}$$

$$= (290.0 + 74.0) / 290.0$$

$$= 1.255 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [L_T]:

$$= \pi/2 \cdot (\text{Rn} + \text{tn})$$

$$= \pi/2 \cdot (290.0 + 74.0)$$

$$= 571.770 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$\text{tc}, \text{L41T} = 7.071, \text{ must be } \geq \min(0.7 \cdot 74.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot \text{Rn}^2 \cdot k_y^2)$$

$$= \min(109691 \cdot 1.26, 1.5 \cdot 1505.0(117.577 + 0.0), \pi/4 \cdot 43.6 \cdot 290.0^2 \cdot 1.26^2)$$

$$= 45376.434 \text{ kgf}$$

Weld Stress Value [τ]:

$$= \text{fwelds} / (L_T(0.49 \cdot \text{L41T} + 0.6 \cdot \text{tw1} + 0.49 \cdot \text{L43T}))$$

$$= 45376.434 / (571.77(0.49 \cdot 7.071 + 0.6 \cdot 27.0 + 0.49 \cdot 0.0))$$

$$= 403.571 < \text{or to } 1407.129 \text{ Weld Stress Passed}$$

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:

$$= \text{tn} / \ln[1 + \text{tn}/\text{Rn}]$$

$$= 74.0 / \ln[1 + 74.0/290.0]$$



= 325.600 mm

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 27.0 / \ln[1 + 27.0/854.0]$$

$$= 867.429 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S$$

$$= 1.5 \cdot 1407.129$$

$$= 2110.7 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 1.055 \cdot 325.6 \cdot 173.492$$

$$= 595.7 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 1.055 \cdot 867.429 (151.849 + 74.0)$$

$$= 2066.0 \text{ kgf}$$

Discontinuity Force from External Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 1.055 \cdot 867.429 \cdot 290.0$$

$$= 2652.9 \text{ kgf}$$

Area Resisting External Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (2066.0 + 2652.9 + 595.7) / 1.055$$

$$= 5039.5 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 2110.694 / (2 \cdot 5039.51 / 159.076 - 867.429 / 27.0)$$

$$= 67.6 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / [t / R_{xs}]$$

$$= 1407.129 [27.0 / 867.429]$$

$$= 43.8 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(67.579, 43.799)$$

$$= 43.799 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σAvg]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (595.736 + 2066.048 + 2652.901) / 159.076$$

$$= 33.410 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 1.055 \cdot 867.429 / 27.0$$

$$= 33.9 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 33.41 - 33.881, 33.881)$$

$$= 33.9 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2110.69	kgf/cm ²
Local Primary Membrane Stress	PL	33.88	kgf/cm ²
Maximum Allowable External Pressure	Pmax	43.80	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :



Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Flange MDMT including temperature reduction per 3.11:
Flange MDMT with Temp reduction -27 °C

The Drop for this Nozzle is : 81.5644 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 609.5644 mm

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Input, Nozzle Desc: E40

From: 100

Pressure for Reinforcement Calculations	P	43.482	kgf/cm ²
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	1706.00	mm
Design Length of Section	L	18483.8359	mm
Shell Finished (Minimum) Thickness	t	28.0000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1595.00	cm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1505.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1682.00	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		40.0000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm
Flange Material		SA-105	+IT
Flange Type		N/A	
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	40.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Style of FVC Nozzle		Long Weld Neck	
Corrosion Allowance	can	6.0000	mm
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	28.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		900	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: E40

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	40.000	mm.
Actual Thickness Used in Calculation	10.000	mm

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 52.0 \cdot (\exp(43.482/(1505.0 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 0.7621 + 6.0 + 0.0 = 6.7621 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1505.0 \cdot 1.0 \cdot \ln((2 \cdot 4.0 + 52.0) / 52.0) - 0.0$$

$$= 215.367 - 0.0 = 215.3666 \text{ kgf/cm}^2$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	9.420 mm
Min. required shell thickness, internal p + c :	27.804 mm
Min. required shell thickness, external p + c :	16.233 mm

Nozzle neck minimum required thickness :	9.420 mm
--	----------

Minimum nozzle neck thickness 10.008 mm , is greater than required thickness 9.420 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained :	622.9,	Allowable :	1505.0 kgf/cm ²	Passed
Expansion :	0.0,	Allowable :	3360.9 kgf/cm ²	Passed
Occasional :	131.2,	Allowable :	2001.7 kgf/cm ²	Passed
Shear :	410.3,	Allowable :	1053.5 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1505.0/1407.1, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 50.0 - 6.0$$

$$= 44.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 10.0 - 6.0$$

$$= 4.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 1708.0/27.0$$

$$= 63.259 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 1706.0/2 + 1.0$$

$$= 854.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{854.0 \cdot 27.0}, 2 \cdot 26.0)$$

$$= 52.000 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 27.0, 0.0] + \sqrt{26.0 \cdot 44.0}$$

$$= 33.823 \text{ mm}$$

Thickness Limit Candidate [LH2]:



$$\begin{aligned}
 &= Lpr1 \\
 &= 300.0 \\
 &= 300.000 \text{ mm}
 \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned}
 &= 8(t + t_e) \\
 &= 8(27.0 + 0.0) \\
 &= 216.000 \text{ mm}
 \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned}
 &= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles} \\
 &= \min[33.823, 300.0, 216.0] + 27.0 \\
 &= 60.823 \text{ mm}
 \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned}
 &= t + ((A5 \cdot frp) / LR) \\
 &= 27.0 + ((0.0 \cdot 1.0) / 52.0) \\
 &= 27.000 \text{ mm}
 \end{aligned}$$

Parameter [λ]:

$$\begin{aligned}
 &= \min(12, (2 \cdot Rn + t_n) / (\sqrt{ (Di + teff) \cdot teff })) \\
 &= \min(12, (2 \cdot 26.0 + 44.0) / (\sqrt{ (1708.0 + 27.0) \cdot 27.0 })) \\
 &= 0.444
 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned}
 &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\
 &= 27.0 \cdot 52.0 \cdot \max((0.444/5)^{0.85}, 1) \\
 &= 14.040 \text{ cm}^2
 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned}
 &= t_n \cdot LH \\
 &= 44.0 \cdot 60.823 \\
 &= 26.762 \text{ cm}^2
 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned}
 &= 0.5 \cdot Leg^2 \\
 &= 0.5 \cdot 10.0^2 \\
 &= 0.500 \text{ cm}^2
 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned}
 &= A1 + frn(A2 + A3) + A41 + A42 + A43 \\
 &= 14.04 + 1.0(26.762 + 0.0) + 0.5 + 0.0 + 0.0 \\
 &= 41.302 \text{ cm}^2
 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned}
 &= t_n / \ln[1 + t_n / Rn] \\
 &= 44.0 / \ln[1 + 44.0 / 26.0] \\
 &= 44.427 \text{ mm}
 \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned}
 &= teff / \ln[1 + teff / Reff] \\
 &= 27.0 / \ln[1 + 27.0 / 854.0] \\
 &= 867.429 \text{ mm}
 \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned}
 &= 1.5 \cdot S \cdot E \\
 &= 1.5 \cdot 1407.129 \cdot 1.0 \\
 &= 2110.7 \text{ kgf/cm}^2
 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned}
 &= P \cdot Rxn \cdot LH \\
 &= 43.482 \cdot 44.427 \cdot 60.823 \\
 &= 1175.0 \text{ kgf}
 \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned}
 &= P \cdot Rxs(LR + t_n) \\
 &= 43.482 \cdot 867.429(52.0 + 44.0) \\
 &= 36209.2 \text{ kgf}
 \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot Rxs \cdot Rnc$$

$$= 43.482 \cdot 867.429 \cdot 26.0$$

$$= 9806.7 \text{ kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P$$

$$= (36209.2 + 9806.7 + 1175.0) / 43.482$$

$$= 1085.3 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap / AT - Rxs / teff)$$

$$= 2110.694 / (2 \cdot 1085.285 / 41.302 - 867.429 / 27.0)$$

$$= 103.3 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/Rxs]$$

$$= 1407.129[27.0/867.429]$$

$$= 43.8 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(103.331, 43.799)$$

$$= 43.799 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σAvg]:

$$= (fN + fS + fY) / AT$$

$$= (1174.964 + 36209.195 + 9806.656) / 41.302$$

$$= 1142.577 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot Rxs / teff$$

$$= 43.482 \cdot 867.429 / 27.0$$

$$= 1397.0 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ)$$

$$= \max(2 \cdot 1142.577 - 1396.962, 1396.962)$$

$$= 1397.0 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2110.69	kgf/cm ²
Local Primary Membrane Stress	PL	1396.96	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	43.80	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$

$$= (26.0 + 44.0) / 26.0$$

$$= 2.692 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi/2 \cdot (Rn + tn)$$

$$= \pi/2 \cdot (26.0 + 44.0)$$

$$= 109.956 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 44.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2)$$

$$= \min(9807 \cdot 2.69, 1.5 \cdot 1505.0(26.762 + 0.0), \pi/4 \cdot 43.5 \cdot 26.0^2 \cdot 2.69^2)$$

$$= 1673.401 \text{ kgf}$$

Weld Stress Value [τ]:

$= \text{fwelds} / (L_t (0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$
 $= 1673.401 / (109.956 (0.49 \cdot 7.071 + 0.6 \cdot 27.0 + 0.49 \cdot 0.0))$
 $= 77.391 < \text{or} = \text{to } 1407.129 \text{ Weld Stress Passed}$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

$= t_n / \ln[1 + t_n/R_n]$
 $= 44.0 / \ln[1 + 44.0/26.0]$
 $= 44.427 \text{ mm}$

Shell Radius for Force Calculation [Rxs]:

$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$
 $= 27.0 / \ln[1 + 27.0/854.0]$
 $= 867.429 \text{ mm}$

Allowable Local Primary Membrane Stress [Sallow]:

$= 1.5 \cdot S$
 $= 1.5 \cdot 1407.129$
 $= 2110.7 \text{ kgf/cm}^2$

Force acting on the Nozzle [fN]:

$= P \cdot R_{xn} \cdot L_H$
 $= 1.055 \cdot 44.427 \cdot 60.823$
 $= 28.5 \text{ kgf}$

Force acting on the Shell [fS]:

$= P \cdot R_{xs} (L_R + t_n)$
 $= 1.055 \cdot 867.429 (52.0 + 44.0)$
 $= 878.2 \text{ kgf}$

Discontinuity Force from External Pressure [fY]:

$= P \cdot R_{xs} \cdot R_{nc}$
 $= 1.055 \cdot 867.429 \cdot 26.0$
 $= 237.8 \text{ kgf}$

Area Resisting External Pressure [Ap]:

$= (f_S + f_Y + f_N) / P$
 $= (878.2 + 237.8 + 28.5) / 1.055$
 $= 1085.3 \text{ cm}^2$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$
 $= 2110.694 / (2 \cdot 1085.285 / 41.302 - 867.429 / 27.0)$
 $= 103.3 \text{ kgf/cm}^2$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$= S / [t / R_{xs}]$
 $= 1407.129 [27.0 / 867.429]$
 $= 43.8 \text{ kgf/cm}^2$

Maximum Allowable Working Pressure [Pmax]:

$= \min(P_{max1}, P_{max2})$
 $= \min(103.331, 43.799)$
 $= 43.799 \text{ kgf/cm}^2$

Average Primary Membrane Stress [σAvg]:

$= (f_N + f_S + f_Y) / A_T$
 $= (28.497 + 878.202 + 237.846) / 41.302$
 $= 27.712 \text{ kgf/cm}^2$

General Primary Membrane Stress [σCirc]:

$= P \cdot R_{xs} / t_{eff}$
 $= 1.055 \cdot 867.429 / 27.0$
 $= 33.9 \text{ kgf/cm}^2$

Maximum Local Primary Membrane Stress [PL]:

$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$
 $= \max(2 \cdot 27.712 - 33.881, 33.881)$
 $= 33.9 \text{ kgf/cm}^2$



Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2110.69	kgf/cm ²
Local Primary Membrane Stress	PL	33.88	kgf/cm ²
Maximum Allowable External Pressure	Pmax	43.80	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

The Drop for this Nozzle is : 2.8771 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 330.8771 mm

Input Echo, WRC297 Item 1, Description: E40 :

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	1.0000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	1706.000	mm	
Shell Thickness	28.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-516 70		
Vessel Cold S.I. Allowable	Smc	1407.13	kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1407.13	kgf/cm ²
Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance	6.0000	mm	
Nozzle Diameter	40.000	mm	
Nozzle Thickness	50.0000	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-105		
Nozzle Cold S.I. Allowable	SNmc	1682.00	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1505.00	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	43.48	kgf/cm ²
Radial Load	P	-382.27	kgf
Circumferential Shear	Vc	254.84	kgf
Longitudinal Shear	Vl	254.84	kgf
Circumferential Moment	Mc	35.68	kgf-m.
Longitudinal Moment	Ml	35.68	kgf-m.
Torsional Moment	Mt	50.97	kgf-m.
Include Axial Pressure Thrust		No	
Include Pressure Stress Indices per Div. 2		No	
Local Loads applied at end of Nozzle/Attachment		No	

Warning - The ratio, Dn/Tn (2.800) is < 10
 check the limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))} \\
 &= 140.0 + 2 \cdot 1.65 \cdot \sqrt{(867.5(28.0 - 1.0))} \\
 &= 645.046 \text{ mm}
 \end{aligned}$$



WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1735.000 mm
 Nozzle Outside Diameter (d) = 140.000 mm
 Vessel Thickness used (T) = 27.000 mm
 Nozzle Thickness used (t) = 44.000 mm
 T / t = 0.614
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.647$$

Nr/P = 0.134
 Mr/P = 0.277
 M0/P = 0.085
 N0/P = 0.157
 MrD/Mc = 0.307
 NrDL/Mc = 0.097
 M0d/Mc = 0.103
 N0DL/Mc = -0.035
 MrD/Ml = 0.246
 NrDL/Ml = 0.099
 M0D/Ml = 0.083
 N0DL/Ml = 0.071

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	8	8	8	8
Outplane Bending (P)	26	-26	26	-26
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-2	-2	2	2
Outplane Bending (ML)	-17	17	17	-17
Normal Pressure Stress	1353	1397	1353	1397
Outplane Stress Summary	1368	1393	1408	1364

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	87	-87	87	-87
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-51	51	51	-51
Inplane Pressure Stress	676	676	676	676
Inplane Stress Summary	716	644	826	548

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	4	4	-4	-4
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	6	6	6	6
Shear Stress Summary	10	10	1	1

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Two * Max Shear Stress 1368 1393 1408 1364

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	8	8	8	8
Outplane Bending (P)	26	-26	26	-26
Outplane Membrane (Mc)	1	1	-1	-1
Outplane Bending (Mc)	-21	21	21	-21
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	676	676	676	676
Outplane Stress Summary	691	681	731	635

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	87	-87	87	-87
Inplane Membrane (Mc)	-3	-3	3	3
Inplane Bending (Mc)	-64	64	64	-64
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1353	1397	1353	1397
Inplane Stress Summary	1380	1377	1515	1256

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-4	-4	4	4
Torsional Shear (Mt)	6	6	6	6
Shear Stress Summary	1	1	10	10

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	1380	1377	1515	1256

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	8	8	8	8
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-2	-2	2	2
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	16	68	16	68
Hoop Stress Summary	22	74	27	79

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside



Axial Membrane (P)	2	2	2	2
Axial Bending (P)	24	-24	24	-24
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-13	-13	13	13
Axial Bending (ML)	-15	15	15	-15
Axial Pressure Stress	8	8	8	8

Axial Stress Summary	7	-11	64	-15
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE				
(Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	3	3	3	3
Shear Stress Summary	6	6	1	1

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE				
(Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	24	87	64	95

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Stresses in the	Cu Left	Cl Left	Du Right	Dl Right
hoop direction)	Outside	Inside	Outside	Inside
Hoop Membrane (P)	8	8	8	8
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	1	1	-1	-1
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	16	68	16	68
Hoop Stress Summary	26	78	23	75

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to	Cu Left	Cl Left	Du Right	Dl Right
pipe cross-section)	Outside	Inside	Outside	Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	24	-24	24	-24
Axial Membrane (Mc)	-13	-13	13	13
Axial Bending (Mc)	-20	20	20	-20
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	8	8	8	8
Axial Stress Summary	2	-6	69	-20

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	3	3	3	3
Shear Stress Summary	1	1	6	6

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside



Two * Max Shear Stress 26 85 70 97

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1353	1397	1353	1397	1353	1397	1353	1397
Circ. Pl (SUS)	6	6	10	10	3	3	9	9
Circ. Q (SUS)	9	-9	43	-43	23	-23	151	-151
Long. Pm (SUS)	676	676	676	676	676	676	676	676
Long. Pl (SUS)	3	3	9	9	9	9	7	7
Long. Q (SUS)	36	-36	138	-138	5	-5	47	-47
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	4	4	-4	-4	-4	-4	4	4
Shear Q (SUS)	6	6	6	6	6	6	6	6
Pm (SUS)	1353	1397	1353	1397	1353	1397	1353	1397
Pm+Pl (SUS)	1359	1403	1363	1407	1356	1400	1362	1406
Pm+Pl+Q (Total)	1368	1394	1406	1364	1379	1377	1513	1255

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1397.00	1407.13	Passed
Pm+Pl (SUS)	1407.02	2110.69	Passed
Pm+Pl+Q (TOTAL)	1513.13	4221.39	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	16	68	16	68	16	68	16	68
Circ. Pl (SUS)	6	6	10	10	9	9	7	7
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	8	8	8	8	8	8	8	8
Long. Pl (SUS)	-11	-11	15	15	-11	-11	15	15
Long. Q (SUS)	9	-9	39	-39	4	-4	44	-44
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	16.0	68.0	16.0	68.0	16.0	68.0	16.0	68.0
Pm+Pl (SUS)	25.3	77.1	27.0	78.1	28.3	80.1	25.0	75.1
Pm+Pl+Q (Total)	23.4	86.6	62.0	94.0	25.0	84.0	67.6	96.5

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
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Pm (SUS)	68.00	1505.00	Passed
Pm+Pl (SUS)	80.10	2257.50	Passed
Pm+Pl+Q (TOTAL)	96.52	4515.00	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: MH1600

From: 100

Pressure for Reinforcement Calculations	P	43.416	kgf/cm ²
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	1706.00	mm
Design Length of Section	L	18483.8359	mm
Shell Finished (Minimum) Thickness	t	28.0000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		1660.00	cm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1505.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1682.00	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		568.0000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm

Flange Material		SA-105	+IT
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Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	64.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		80.0000	mm

Corrosion Allowance	can	6.0000	mm
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Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	28.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

This is a Manway, Coupling or Access Opening.

Flange Class		300
Flange Grade		GR 1.1

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: MH1600

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	568.000	mm.
Actual Thickness Used in Calculation	16.000	mm

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight



of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 580.0 \cdot (\exp(43.416/(1505.0 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 8.4877 + 6.0 + 0.0 = 14.4877 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1505.0 \cdot 1.0 \cdot \ln((2 \cdot 10.0 + 580.0) / 580.0) - 0.0$$

$$= 51.022 - 0.0 = 51.0218 \text{ kgf/cm}^2$$

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1505.0/1407.1, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 80.0 - 6.0$$

$$= 74.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 16.0 - 6.0$$

$$= 10.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 1708.0/27.0$$

$$= 63.259 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 1706.0/2 + 1.0$$

$$= 854.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{854.0 \cdot 27.0}, 2 \cdot 290.0)$$

$$= 151.849 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 27.0, 0.0] + \sqrt{290.0 \cdot 74.0}$$

$$= 146.492 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 500.0$$

$$= 500.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(27.0 + 0.0)$$

$$= 216.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[146.492, 500.0, 216.0] + 27.0$$

$$= 173.492 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 27.0 + ((0.0 \cdot 1.0) / 151.849)$$

$$= 27.000 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot Rn + tn) / (\sqrt{(Di + teff) \cdot teff}))$$

$$= \min(12, (2 \cdot 290.0 + 74.0) / (\sqrt{(1708.0 + 27.0) \cdot 27.0}))$$

$$= 3.022$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 27.0 \cdot 151.849 \cdot \max((3.022/5)^{0.85}, 1)$$

$$= 40.999 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A2A + A2B$$

$$= 93.98 + 23.597$$

$$= 117.577 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot Leg^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$

$$= 41.0 + 1.0(117.577 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 159.076 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= tn / \ln[1 + tn/Rn]$$

$$= 74.0 / \ln[1 + 74.0/290.0]$$

$$= 325.600 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= teff / \ln[1 + teff/Reff]$$

$$= 27.0 / \ln[1 + 27.0/854.0]$$

$$= 867.429 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 1407.129 \cdot 1.0$$

$$= 2110.7 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot Rxn \cdot LH$$

$$= 43.416 \cdot 325.6 \cdot 173.492$$

$$= 24525.3 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot Rxs \cdot (LR + tn)$$

$$= 43.416 \cdot 867.429 \cdot (151.849 + 74.0)$$

$$= 85055.2 \text{ kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot Rxs \cdot Rnc$$

$$= 43.416 \cdot 867.429 \cdot 290.0$$

$$= 109214.8 \text{ kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P$$

$$= (85055.2 + 109214.8 + 24525.3) / 43.416$$

$$= 5039.5 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:



$$= \text{Sallow} / (2 \cdot \text{Ap}/\text{AT} - \text{Rxs}/\text{teff})$$

$$= 2110.694 / (2 \cdot 5039.51/159.076 - 867.429/27.0)$$

$$= 67.6 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/\text{Rxs}]$$

$$= 1407.129[27.0/867.429]$$

$$= 43.8 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(\text{Pmax1}, \text{Pmax2})$$

$$= \min(67.579, 43.799)$$

$$= 43.799 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (fN + fS + fY) / \text{AT}$$

$$= (24525.27 + 85055.172 + 109214.75) / 159.076$$

$$= 1375.417 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot \text{Rxs} / \text{teff}$$

$$= 43.416 \cdot 867.429/27.0$$

$$= 1394.8 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}})$$

$$= \max(2 \cdot 1375.417 - 1394.826, 1394.826)$$

$$= 1394.8 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2110.69	kgf/cm ²
Local Primary Membrane Stress	PL	1394.83	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	43.80	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (290.0 + 74.0) / 290.0$$

$$= 1.255 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi/2 \cdot (R_n + t_n)$$

$$= \pi/2 \cdot (290.0 + 74.0)$$

$$= 571.770 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 74.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(109215 \cdot 1.26, 1.5 \cdot 1505.0(117.577 + 0.0), \pi/4 \cdot 43.4 \cdot 290.0^2 \cdot 1.26^2)$$

$$= 45179.582 \text{ kgf}$$

Weld Stress Value [τ]:

$$= \text{fwelds} / (L_t(0.49 \cdot L41T + 0.6 \cdot t_{w1} + 0.49 \cdot L43T))$$

$$= 45179.582 / (571.77(0.49 \cdot 7.071 + 0.6 \cdot 27.0 + 0.49 \cdot 0.0))$$

$$= 401.821 < \text{or} = \text{to } 1407.129 \text{ Weld Stress Passed}$$

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 74.0 / \ln[1 + 74.0/290.0]$$



= 325.600 mm

Shell Radius for Force Calculation [Rxs]:

= $t_{eff} / \ln[1 + t_{eff}/R_{eff}]$
 = $27.0 / \ln[1 + 27.0/854.0]$
 = 867.429 mm

Allowable Local Primary Membrane Stress [Sallow]:

= $1.5 \cdot S$
 = $1.5 \cdot 1407.129$
 = 2110.7 kgf/cm²

Force acting on the Nozzle [fN]:

= $P \cdot R_{xn} \cdot L_H$
 = $1.055 \cdot 325.6 \cdot 173.492$
 = 595.7 kgf

Force acting on the Shell [fS]:

= $P \cdot R_{xs} (L_R + t_n)$
 = $1.055 \cdot 867.429 (151.849 + 74.0)$
 = 2066.0 kgf

Discontinuity Force from External Pressure [fY]:

= $P \cdot R_{xs} \cdot R_{nc}$
 = $1.055 \cdot 867.429 \cdot 290.0$
 = 2652.9 kgf

Area Resisting External Pressure [Ap]:

= $(f_S + f_Y + f_N) / P$
 = $(2066.0 + 2652.9 + 595.7) / 1.055$
 = 5039.5 cm²

Maximum Allowable Working Pressure Candidate [Pmax1]:

= $S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$
 = $2110.694 / (2 \cdot 5039.51 / 159.076 - 867.429 / 27.0)$
 = 67.6 kgf/cm²

Maximum Allowable Working Pressure Candidate [Pmax2]:

= $S / [t / R_{xs}]$
 = $1407.129 [27.0 / 867.429]$
 = 43.8 kgf/cm²

Maximum Allowable Working Pressure [Pmax]:

= $\min(P_{max1}, P_{max2})$
 = $\min(67.579, 43.799)$
 = 43.799 kgf/cm²

Average Primary Membrane Stress [σAvg]:

= $(f_N + f_S + f_Y) / A_T$
 = $(595.736 + 2066.048 + 2652.901) / 159.076$
 = 33.410 kgf/cm²

General Primary Membrane Stress [σCirc]:

= $P \cdot R_{xs} / t_{eff}$
 = $1.055 \cdot 867.429 / 27.0$
 = 33.9 kgf/cm²

Maximum Local Primary Membrane Stress [PL]:

= $\max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$
 = $\max(2 \cdot 33.41 - 33.881, 33.881)$
 = 33.9 kgf/cm²

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2110.69	kgf/cm ²
Local Primary Membrane Stress	PL	33.88	kgf/cm ²
Maximum Allowable External Pressure	Pmax	43.80	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :



Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Flange MDMT including temperature reduction per 3.11:
Flange MDMT with Temp reduction -27 °C

The Drop for this Nozzle is : 81.5644 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 609.5644 mm

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Input, Nozzle Desc: C100

From: 110

Pressure for Reinforcement Calculations	P	43.209	kgf/cm ²
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	130	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²

Inside Diameter of Elliptical Head	D	1706.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	26.0000	mm
Head Internal Corrosion Allowance	c	1.0000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	0.0000	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1505.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1682.00	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		100.0000	mm

Size and Thickness Basis		Nominal	
Nominal Thickness		160	

Flange Material		SA-105	+IT
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Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	36.5126	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm

Corrosion Allowance	can	6.0000	mm
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Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	26.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: C100

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	87.325	mm.
Actual Thickness Used in Calculation	13.487	mm

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1



Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 99.325 \cdot (\exp(43.209/(1505.0 \cdot 1.0)) - 1) + 6.0 + 0.0$
 $= 1.4465 + 6.0 + 0.0 = 7.4465 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1505.0 \cdot 1.0 \cdot \ln((2 \cdot 7.487 + 99.325) / 99.325) - 0.0$
 $= 211.343 - 0.0 = 211.343 \text{ kgf/cm}^2$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	11.270 mm
Min. required shell thickness, internal p + c :	30.070 mm
Min. required shell thickness, external p + c :	6.078 mm
Nozzle neck minimum required thickness :	11.270 mm

Minimum nozzle neck thickness 11.811 mm , is greater than required thickness 11.270 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained :	556.5,	Allowable :	1505.0 kgf/cm ²	Passed
Expansion :	0.0,	Allowable :	3427.3 kgf/cm ²	Passed
Occasional :	174.7,	Allowable :	2001.7 kgf/cm ²	Passed
Shear :	306.4,	Allowable :	1053.5 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[1505.0/1407.1, 1]$
 $= 1.000$

Thickness of Nozzle at Shell [tn]:
 $= \text{hub thickness} - \text{corrosion allowance}$
 $= 50.0 - 6.0$
 $= 44.000 \text{ mm}$

Thickness of Nozzle at Top [tn2]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 13.487 - 6.0$
 $= 7.487 \text{ mm}$

Effective Pressure Radius [Reff]:
 $= 0.9/6 \cdot D_i [2 + (D_i/(2h))^2]$
 $= 0.9/6 \cdot 1708.0 [2 + (1708.0/(2 \cdot 427.5))^2]$
 $= 1534.804 \text{ mm}$

Effective Length of Vessel Wall [LR]:
 $= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$
 $= \min(\sqrt{1534.804 \cdot 25.0}, 2 \cdot 49.663)$
 $= 99.325 \text{ mm}$

Intermediate value [Cn]:
 $= \min(((t + t_e)/t_n)^{0.35}, 1)$
 $= \min(((25.0 + 0.0)/44.0)^{0.35}, 1)$
 $= \min(0.82, 1)$
 $= 0.820$

Intermediate value [Cp]:
 $= \exp((0.35 \cdot d_i - X_o)/(16t))$
 $= \exp((0.35 \cdot 1708.0 - 93.663)/(16 \cdot 25.0))$
 $= 3.527$

Effective Nozzle Wall Length Outside the Vessel [LH]:
 $= \min(t + t_e + F_p \cdot \sqrt{R_n \cdot t_n}, L_{pr1} + t)$



$$= \min(25.0+0.0+0.82\sqrt{(49.663 \cdot 44.0)}, 300.0+25.0)$$

$$= 63.354 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 25.0 + ((0.0 \cdot 1.0) / 99.325)$$

$$= 25.000 \text{ mm}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR$$

$$= 25.0 \cdot 99.325$$

$$= 24.831 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$

$$= 44.0 \cdot 63.354$$

$$= 27.876 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot Leg^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + frn(A2 + A3) + A41 + A42 + A43$$

$$= 24.831 + 1.0(27.876 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 53.207 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 44.0 / \ln[1 + 44.0/49.663]$$

$$= 69.352 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 25.0 / \ln[1 + 25.0/1534.804]$$

$$= 1547.267 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 1407.129 \cdot 1.0$$

$$= 2110.7 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$

$$= 43.209 \cdot 69.352 \cdot 63.354$$

$$= 1898.5 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} \cdot (LR + t_n) / 2$$

$$= 43.209 \cdot 1547.267 \cdot (99.325 + 44.0) / 2$$

$$= 47910.6 \text{ kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= (P \cdot R_{xs} \cdot R_{nc}) / 2$$

$$= (43.209 \cdot 1547.267 \cdot 49.663) / 2$$

$$= 16601.1 \text{ kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P$$

$$= (47910.6 + 16601.1 + 1898.5) / 43.209$$

$$= 1537.0 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap/AT - R_{xs} / (2 \cdot t_{eff}))$$

$$= 2110.694 / (2 \cdot 1536.956 / 53.207 - 1547.267 / (2 \cdot 25.0))$$

$$= 78.7 \text{ kgf/cm}^2$$



Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot S[t/Rxs]$$

$$= 2 \cdot 1407.129[25.0/1547.267]$$

$$= 45.5 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(78.677, 45.471)$$

$$= 45.471 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (fN + fS + fY) / AT$$

$$= (1898.477 + 47910.551 + 16601.146) / 53.207$$

$$= 1248.147 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot Rxs / (2 \cdot teff)$$

$$= 43.209 \cdot 1547.267 / (2 \cdot 25.0)$$

$$= 1337.1 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 1248.147 - 1337.116, 1337.116)$$

$$= 1337.1 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2110.69	kgf/cm ²
Local Primary Membrane Stress	PL	1337.12	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	45.47	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$

$$= (49.663 + 44.0) / 49.663$$

$$= 1.886 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi/2 \cdot (Rn + tn)$$

$$= \pi/2 \cdot (49.663 + 44.0)$$

$$= 147.125 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 44.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2)$$

$$= \min(16601 \cdot 1.89, 1.5 \cdot 1505.0(31.568 + 0.0), \pi/4 \cdot 43.2 \cdot 49.66^2 \cdot 1.89^2)$$

$$= 2977.115 \text{ kgf}$$

Weld Stress Value [τ]:

$$= fwelds / (Lt(0.49 \cdot L41T + 0.6 \cdot tw1 + 0.49 \cdot L43T))$$

$$= 2977.115 / (147.125(0.49 \cdot 7.071 + 0.6 \cdot 25.0 + 0.49 \cdot 0.0))$$

$$= 109.589 < \text{or} = \text{to } 1407.129 \text{ Weld Stress Passed}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

$$= tn / \ln[1 + tn/Rn]$$

$$= 44.0 / \ln[1 + 44.0/49.663]$$

$$= 69.352 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= teff / \ln[1 + teff/Reff]$$

$$= 25.0 / \ln[1 + 25.0/1534.804]$$

$$= 1547.267 \text{ mm}$$



Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S$$

$$= 1.5 \cdot 1407.129$$

$$= 2110.7 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 1.055 \cdot 69.352 \cdot 63.354$$

$$= 46.3 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} \cdot (L_R + t_n) / 2$$

$$= 1.055 \cdot 1547.267 \cdot (99.325 + 44.0) / 2$$

$$= 1169.4 \text{ kgf}$$

Discontinuity Force from External Pressure [fY]:

$$= (P \cdot R_{xs} \cdot R_{nc}) / 2$$

$$= (1.055 \cdot 1547.267 \cdot 49.663) / 2$$

$$= 405.2 \text{ kgf}$$

Area Resisting External Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (1169.4 + 405.2 + 46.3) / 1.055$$

$$= 1537.0 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot A_p / A_T - R_{xs} / (2 \cdot t_{eff}))$$

$$= 2110.694 / (2 \cdot 1536.956 / 53.207 - 1547.267 / (2 \cdot 25.0))$$

$$= 78.7 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot S / [t / R_{xs}]$$

$$= 2 \cdot 1407.129 [25.0 / 1547.267]$$

$$= 45.5 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(78.677, 45.471)$$

$$= 45.471 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (46.336 + 1169.357 + 405.186) / 53.207$$

$$= 30.464 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / (2 \cdot t_{eff})$$

$$= 1.055 \cdot 1547.267 / (2 \cdot 25.0)$$

$$= 32.6 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 30.464 - 32.635, 32.635)$$

$$= 32.6 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2110.69	kgf/cm ²
Local Primary Membrane Stress	PL	32.64	kgf/cm ²
Maximum Allowable External Pressure	Pmax	45.47	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.



Flange MDMT including temperature reduction per 3.11:
 Flange MDMT with Temp reduction -65 °C

The Drop for this Nozzle is : 2.8211 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 328.8211 mm

Input Echo, WRC107/537 Item 1, Description: C100 :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	1.0000	mm
Vessel Diameter	Dv	3070.800	mm
Vessel Thickness	Tv	26.000	mm
Design Temperature	T1	130.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	1407.13	kgf/cm²
Vessel Hot S.I. Allowable	Smh	1407.13	kgf/cm²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	6.0000	mm
Nozzle Diameter	Dn	87.325	mm
Nozzle Thickness	Tn	50.000	mm
Nozzle Material		SA-105	
Nozzle UNS Number		K03504	
Nozzle Cold S.I. Allowable	SNmc	1682.00	kgf/cm²
Nozzle Hot S.I. Allowable	SNmh	1505.00	kgf/cm²
Design Internal Pressure	Dp	43.209	kgf/cm²
Include Pressure Thrust		No	

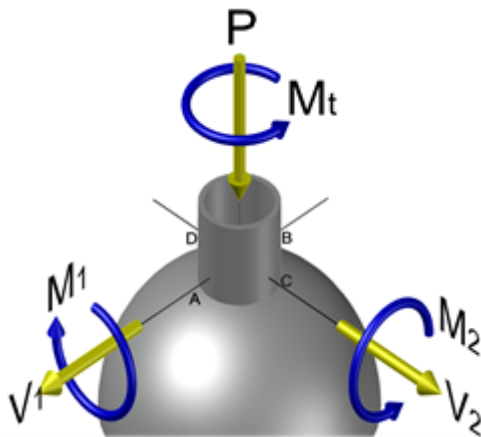
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-764.5	kgf
Longitudinal Shear (SUS)	(V1) V1	509.7	kgf
Circumferential Shear (SUS)	(Vc) V2	509.7	kgf
Circumferential Moment (SUS)	(Mc) M1	142.7	kgf-m.
Longitudinal Moment (SUS)	(Ml) M2	142.7	kgf-m.
Torsional Moment (SUS)	Mt	203.9	kgf-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 187.325 + 2 \cdot 1.65 \cdot \sqrt{(1548.9 (26.0 - 1.0))}$$

$$= 836.700 \text{ mm}$$

WRC 107 Stress Calculation for SUSTained loads:

Radial Load	P	-764.5	kgf
Circumferential Shear	(VC) V2	509.7	kgf
Longitudinal Shear	(VL) V1	509.7	kgf
Circumferential Moment	(MC) M1	142.7	kgf-m.
Longitudinal Moment	(ML) M2	142.7	kgf-m.
Torsional Moment	MT	203.9	kgf-m.

Dimensionless Param: U = 0.48 TAU = 5.00 (1.63) RHO = 0.57

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 1	0.07736	(A,B,C,D)
$M(x) / P$	SP 1	0.09585	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.11253	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 1	0.25146	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.11253	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 1	0.25146	(A,B,C,D)
$N(y) * T / P$	SP 1	0.12619	(A,B,C,D)
$M(y) / P$	SP 1	0.03084	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.09547	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 1	0.09120	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.09547	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 1	0.09120	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (kgf/cm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb.	P	9.5	9.5	9.5	9.5	9.5	9.5	9.5	9.5
Rad. Bend.	P	70.4	-70.4	70.4	-70.4	70.4	-70.4	70.4	-70.4
Rad. Memb.	MC	0.0	0.0	0.0	0.0	-13.1	-13.1	13.1	13.1
Rad. Bend.	MC	0.0	0.0	0.0	0.0	-175.1	175.1	175.1	-175.1
Rad. Memb.	ML	-13.1	-13.1	13.1	13.1	0.0	0.0	0.0	0.0
Rad. Bend.	ML	-175.1	175.1	175.1	-175.1	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-108.3	101.1	268.0	-222.9	-108.3	101.1	268.0	-222.9



Tang. Memb. P	15.4	15.4	15.4	15.4	15.4	15.4	15.4	15.4
Tang. Bend. P	22.6	-22.6	22.6	-22.6	22.6	-22.6	22.6	-22.6
Tang. Memb. MC	0.0	0.0	0.0	0.0	-11.1	-11.1	11.1	11.1
Tang. Bend. MC	0.0	0.0	0.0	0.0	-63.5	63.5	63.5	-63.5
Tang. Memb. ML	-11.1	-11.1	11.1	11.1	0.0	0.0	0.0	0.0
Tang. Bend. ML	-63.5	63.5	63.5	-63.5	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-36.5	45.2	112.6	-59.6	-36.5	45.2	112.6	-59.6
Shear VC	6.9	6.9	-6.9	-6.9	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-6.9	-6.9	6.9	6.9
Shear MT	14.8	14.8	14.8	14.8	14.8	14.8	14.8	14.8
Tot. Shear	21.7	21.7	7.9	7.9	7.9	7.9	21.7	21.7
Str. Int.	114.4	108.6	268.3	223.3	109.2	102.2	270.9	225.7

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (kgf/cm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	1338.5	1338.5	1338.5	1338.5	1338.5	1338.5	1338.5	1338.5
Rad.	Pl (SUS)	-3.6	-3.6	22.5	22.5	-3.6	-3.6	22.5	22.5
Rad.	Q (SUS)	-104.7	104.7	245.4	-245.4	-104.7	104.7	245.4	-245.4
Long.	Pm (SUS)	1338.5	1338.5	1338.5	1338.5	1338.5	1338.5	1338.5	1338.5
Long.	Pl (SUS)	4.4	4.4	26.5	26.5	4.4	4.4	26.5	26.5
Long.	Q (SUS)	-40.9	40.9	86.1	-86.1	-40.9	40.9	86.1	-86.1
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	6.9	6.9	-6.9	-6.9	-6.9	-6.9	6.9	6.9
Shear	Q (SUS)	14.8	14.8	14.8	14.8	14.8	14.8	14.8	14.8
Pm (SUS)		1338.5	1338.5	1338.5	1338.5	1338.5	1338.5	1338.5	1338.5
Pm+Pl (SUS)		1346.9	1346.9	1370.2	1370.2	1346.9	1346.9	1370.2	1370.2
Pm+Pl+Q (Total)		1308.1	1447.1	1606.9	1279.3	1302.9	1440.7	1609.4	1281.7

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1338.51	1407.13	Passed
Pm+Pl (SUS)	1370.23	2110.69	Passed
Pm+Pl+Q (TOTAL)	1609.44	4780.43	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

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